



NGA.STND.0024-1_1.5
2025-06-10

NGA STANDARDIZATION DOCUMENT

SENSOR INDEPENDENT COMPLEX DATA (SICD)

Volume 1

Design & Implementation Description Document

Specification of the design of SICD data products
for SAR complex images.

(2025-06-10)

Version 1.5

The National Geospatial-Intelligence Agency's Office of General Counsel has determined that the standards and information set forth in this document are not subject to the export restrictions of the Arms Export Control Act and the International Traffic in Arms Regulations.

FOREWORD

The suite of NGA.STND.0024 standardization documents defines the Sensor Independent Complex Data (SICD) format for complex image data products generated by Synthetic Aperture Radar (SAR) systems and their data processing elements.

A SAR complex image is an intermediate data product. The real utility is in the products and measurements that may be derived from it. The quality of the pixel array (resolution, signal-to-noise ratio, etc.), along with the set of metadata provided, are critical in generating the derived products. The “sensor independence” of the SICD product refers to the ability of the allowed pixel array and metadata options to accurately describe the image products from many sensors and data processing systems. Sensor independence does NOT mean that all products have the same format for the pixel array or the same set of metadata parameters.

The SICD documentation has been organized into three volumes and one XML schema. The three volumes are summarized below. The SICD XML schema document defines the XML metadata structure for the XML metadata instance included in a given product.

Volume 1 Design & Implementation Description Document

Contains the description needed by producers of SAR complex image products to design a SICD product and the set of metadata that describe it.

Volume 2 File Format Description Document

Defines the placement of SICD data products in the allowed image file formats. Also provides the description needed by users of SICD products to read and properly extract the SICD data components from a SICD product file.

Volume 3 Image Projections Description Document

Describes the SICD sensor model and the correct projections from image location to ground point and from ground point to image location for all SICD products.

The companion suite of NGA.STND.0025 standardization documents defines the Sensor Independent Derived Data (SIDDD) format for image data products that may be derived from a SICD data product. The NGA.STND.0024 and NGA.STND.0025 documents and associated XML artifacts are available in the National System for Geospatial-Intelligence (NSG) Standards Registry (<https://nsgreg.nga.mil>).

REVISION HISTORY

Date	Version	Description
2011-09-28	1.0	Initial publication.
2014-09-30	1.1	Table 3.7: Parameter descriptions updated for improved clarity. Table 3.13: MatchCollection parameter is block now OPTIONAL (was required). Section 4.7.3 Area Imaged. Description updated for improved clarity.
2018-12-13	1.2.1	Table 1-1 Applicable Documents updated. Table 3-7 Radar Collection Parameters: TxRcvPolarization updated for additional allowable values. Table 3-8 Image Formation Parameters: TxRcvPolarizationProc updated for additional allowable values.
2021-11-30	1.3.0	Table 3-7 Radar Collection Parameters: TxPolarization and TxRcvPolarization updated for additional allowable values. Table 3-8 Image Formation Parameters: TxRcvPolarizationProc updated for additional allowable values. Table 3-12 Error Parameters: Optional structure added for Unmodeled Error statistics. Section 4.12.3 added updated to provide guidance on the use of the Unmodeled error statistics. Added Table 4-1 Provides descriptions of the polarization types. 10/7/2022: Document re-released with minor administrative corrections – all equations & symbols now display correctly.
2023-10-26	1.4.0 (1 of 2)	Improved support for images formed from Bistatic collections. Changes include: • Sect 1: Support for Bistatic images updated and clarified. • Table 3-1 For Bistatic images: IlluminatorName & CollectType = “BISTATIC” are required. • Table 3-6 For Bistatic images: GRPPoly, TxAPCPoly, RcvAPC are required. • Table 3-9 Added Bistatic SCP COA parameters that specify the Tx & Rcv platform geometry. • Table 3-11 TwoWay antenna structure restricted to use for Monostatic images. • Table 3-12 Modified Error Statistics structure to fully support Monostatic & Bistatic images. Updated guidance on use of monostatic Component statistics and added bistatic Component statistics. • Table 2-1 Added XML Type MTX

SICD Volume 1 Design & Implementation Description
NGA.STND.0024-1_1.5 (2025-06-10)

Date	Version	Description
2023-10-26	1.4.0 (2 of 2)	• Sect 4.6 Reference Position Parameters – Revised to provide specific guidance computing APC, GRP, and ARP position polynomials. • Sect 4.15.1 & 4.15.2 Added guidance for computing PFA specific metadata for Bistatic images and supporting figures. • Table 3-12 Added optional Adjustable Parameter Offsets for Monostatic images and for Bistatic images. • Section 4.9.1 Change to allow SCP COA grazing angle values less than zero. • Table 3-9 GrazeAng & IncidenceAng allowed range of values changed: $-90 < \text{GrazeAng} < 90$ • Sect 4-9 Modified grazing angle computation to correctly compute values of GRAZ < 0. • Change to Sect 4.4.2 & Fig 4.4-2 to rename Image Plane Point from IPP to IP_PT to avoid confusion with Inter-Pulse Period. • Added List of Abbreviation. • Modified highlighted portions of Tables 3-1 thru 3-16 for clarity on required vs. optional structures. No change to defined data structure.
2024-05-01	1.4.0	• No changes to this document. • V1.4.0 XML schema corrected to show APOs are optional. Corrected schema is dated 2024-05-01 and matches Table 3-12 of this document.
2025-06-10	1.5	• Version number reduced to two-digit format per NITFS Technical Board direction. • Table 3-3 Allow repeated children in GeoInfo block to match CPHD. • Table 3-7 Added optional SegmentPolygon and Polygon blocks under Area/Plane. • Sect 4.5 Required time span for IPP Polynomial functions reduced to the slow time period processed to form the image. • Sect 4.12 Revised to include both monostatic and bistatic images. • Notation standardized for size/index attributes in XML parameter tables.
2026-02-28	1.5	• Administrative changes made following the GEOINT Standards Working Group review. • Typographical errors corrected. No change to SICD V1.5 format design and definition. • Reference added to Table 1-1: NGA.STND.0036_WGS84 • Correction made to the guidance in section 4.16.3 (6) for computing Rdot_COA(rg, az) to match the more precise guidance specified in SICD Vol 3, Sect 4.3 (5). • No changes made to the SICD V1.5 XML schema.

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LIST OF ABBREVIATIONS

Abbreviation & Acronyms	
Refer to NGA.STND.0024-1 SICD Volume 1 and NGA.STND.0024-3 SICD Volume 3	
Abbreviation	Phrase in Full – Reference for SICD specific description.
ACF	Antenna Coordinate Frame – See Volume 1: Section 4.11
APC	Antenna Phase Center – See Volume 1: Section 4.6
APO	Adjustable Parameter Offset – See Volume 3: Section 1.3
ARP	Aperture Reference Point – See Volume 1: Section 4.6
CCW	Counterclockwise
CA	Closest Approach – See Volume 1: Section 4.16.3
COA	Center of Aperture – See Volume 1: Section 4.6
CST	Collection Start Time
DCA	Doppler Cone Angle
DEM	Digital Elevation Model – See Volume 3: Section 10
DRSF	Doppler Rate Scale Factor – See Volume 1: Section 4.16.3
EB	Electrical Boresight – See Volume 1: Section 4.11
ECF or ECEF	Earth Centered Fixed (ECF) or Earth Centered Earth Fixed (ECEF) – Used interchangeably to refer to the ECEF coordinate frame w/ ECF primarily used in XML parameter names.
ENU	Enumeration Type – See Volume 1: Table 2-1
ETP	Earth Tangent Plane – See Volume 1: Section 4.9.1
FP	Focus Plane – See Volume 1: Section 4.15
GCP	Ground Center Point – See Volume 3: Section 7
GPC	Ground Plane Coordinates
GPN	Ground Plane Normal – See Volume 3: Section 5
GREF	Ground Plane Reference Point – See Volume 3: Section 5
GRP	Ground Reference Point – See Volume 1: Section 4.6
HAE	Height Above the Ellipsoid – Measured relative to the WGS 84 reference ellipsoid.
HMSL	Height Above Mean Sea Level
Hz	Hertz
I2S	Image-to-Scene – See Volume 3: Section 1
IP	Image Plane – See Volume 1: Section 4.15
IFA	Image Formation Algorithm
INCA	Imaging Near Closest Approach – See Volume 1: Section 4.16
IPP	Inter-Pulse Period – See Volume 1: Section 4.5
IRW	Impulse Response Width
Kap	Polar Aperture Spatial Frequency – See Volume 1: Section 4.15.2
KSF	Polar Aperture Spatial Frequency Scale Factor – See Volume 1: Section 4.15.2
LLH	Latitude, Longitude, Height Above the Ellipsoid

Abbreviation & Acronyms	
Refer to NGA.STND.0024-1 SICD Volume 1 and NGA.STND.0024-3 SICD Volume 3	
Abbreviation	Phrase in Full – Reference for SICD specific description.
ML	Main Lobe or Mainlobe
NITF	National Imagery Transmission Format
NSG	National System for Geospatial-Intelligence
PFA	Polar Format Algorithm
PHD	Phase History Data
POI	Plane of Incidence – See Volume 1: Section 4.9
RCS	Radar Cross Section
RIC	Radial, In-Track, Cross-Track – See Volume 3: Section 3.4
RMA	Range Migration Algorithm
RMAT	Range Migration – Along Track
RMCR	Range Migration – Cross Range
S2I	Scene-to-Image – See Volume 3: Section 1
SAR	Synthetic Aperture Radar
SCP	Scene Center Point – See Volume 1: Section 4.2
SICD	Sensor Independent Complex Data
SIDD	Sensor Independent Derived Data
UTC	Coordinated Universal Time
VARP	Aperture Reference Point Velocity
WGS 84	World Geodetic System 1984
XML	Extensible Markup Language

1 Introduction

Synthetic Aperture Radar (SAR) systems are capable of high-fidelity ground imaging using many types of platforms and a variety of radar technologies. The platforms carrying SAR systems range from low flying aircraft to earth orbiting satellites. The radar systems can operate in a wide range of collection geometries. The radar operating frequencies range from several hundred megahertz to tens of gigahertz. The SAR image resolution can vary from many meters to a fraction of a meter. The imaged area can vary in size from a fraction of a square kilometer to hundreds of square kilometers. The majority of SAR systems use a single radar platform to transmit and receive (monostatic imaging) although a relatively small number of systems use a pair of radar platforms operating as separate transmitter and passive receiver (bistatic imaging).

These systems all employ the fundamental SAR imaging principles. Resolution in range is achieved by controlling the bandwidth of the radar waveform. Resolution in cross range is achieved by coherently processing observations from a span of viewing angles to effectively synthesize an array with fine angular resolution. The range resolution (ΔR) is inversely proportional to the bandwidth (BW) of the radar waveform. The azimuth resolution (ΔA), expressed as a distance in cross range, is a function of the center frequency of the transmitted waveform (f_{CTR}) and the variation in viewing angle ($\Delta\theta$) to a given point being imaged. The basic expressions for ΔR and ΔA are given in the equations below.

$$\Delta R = \frac{c}{2 \bullet BW} \qquad \Delta A = \frac{c}{2 \bullet f_{CTR} \bullet \Delta\theta}$$

The majority of SAR imaging systems operate in one of two common SAR ground imaging modes. The first is known as Spotlight mode. In the Spotlight mode, the scene being imaged is illuminated throughout the collection period. The second is known as Stripmap mode. In the most common Stripmap mode, the scene being imaged is a strip of ground that is oriented parallel to the trajectory of the platform. The radar is controlled to illuminate the portion of the strip at broadside. Individual ground points are imaged as they pass through the radar beam. A more general form of the Stripmap mode allows for the strip being imaged to be at an arbitrary orientation relative to the ground track. The radar control required for this mode is more complex due to the dynamic nature of the radar imaging geometry. This more general form of stripmap mode is referred to as Dynamic Stripmap.

A SAR and its associated data processing systems are capable of producing many types of image products. The image formation processing exploits the coherent nature of the collected SAR signals to form the image. The initial image formed is usually a two-dimensional complex pixel grid that represents the radar reflectivity of the scene that was imaged. The complex pixel values contain information about both the amplitude and phase of the reflected radar signals. The mapping of the three-dimensional scene reflectivity to the two-dimensional image is a function of both the SAR imaging geometry and the image

formation processing. For most complex images, the spatial sampling of the image is closely related to the natural coordinates of the radar imaging geometry (for example, range and cross range).

The SAR complex images have great utility in that many other image products may be derived from them. The derived products include human viewable images that are accurately geo-located. The complex image retains information regarding the amplitude and phase of the scene reflectivity within the processed frequency band and the span of viewing angles. Many useful products that exploit the phase of the scene reflectivity may be derived from the complex image. For example, a pair of complex images taken at different times may be used to detect small scale changes in the scene that are not noticeable in the viewable products.

The Sensor Independent Complex Data (SICD) product has been developed to store and transmit complex image products from a wide range of SAR sensors and data processing systems. A SICD product contains the complex image pixel data and a set of metadata describing both the radar collection and the image formation processing. The SICD format has been designed to support images for the following types of collections.

- (1) Support for images from many sensors including both airborne and spaceborne SAR sensors.
- (2) Support for images collected with any of the common imaging modes including Spotlight mode, side looking Stripmap mode, or Dynamic Stripmap mode.
- (3) Support for monostatic images formed using any one of several different image formation algorithms. Full support for bistatic images formed using the Polar Format Algorithm.

What Is Meant By “Sensor Independent” Complex Image Products?

A SAR complex image is an intermediate data product. The real utility is in the products and measurements that may be derived from it. The quality of the pixel array (resolution, SNR, etc.) along with the set of metadata provided are critical in generating the derived products. The “sensor independence” of the SICD product refers to the ability of the allowed pixel array and metadata options to accurately describe the image products from many sensors and data processing systems. Sensor independence does NOT mean that all products have the same format for the pixel array or the same set of metadata parameters.

The SICD product pixel array allows for only a limited number of pixel formats and pixel array orientations. The pixel format options allow for file size versus dynamic range to be balanced while limiting the number of possibilities an exploitation application must be able to interpret. The orientation of the pixel array is also limited so that an exploitation application will not need to be capable of handling all possible orientations. Variety in the input array orientation does not add utility to any derived product.

All SICD products are supplied with metadata that may logically be divided into two types: (1) product identification parameters and (2) SAR science parameters. The product identification parameters identify the SAR platform that collected the image, the time and date of the collection, etc. The product identification parameters may also include parameters related to the tasking request that initiated the collection. The SAR science parameters are those used to describe the collection and processing of the complex image. In general, the SAR science parameters are those used along with the image pixel array to produce the derived data products.

The product identification parameters are grouped as a single block within the metadata structure. The parameter block includes a minimal set of required “text-based” fields and optional fields may be included as needed by the systems that store and disseminate the image products. The product identification parameters are NOT sensor independent. Product identification parameters (both the number and their definition) are ALWAYS specific to the SAR sensor and the tasking, processing and product dissemination systems. They are, however, readily identified in all products independent of the SAR system that produced the product.

The SAR science parameters provided with an image are also dependent on the SAR system that produced the image (i.e., both the SAR sensor and the image formation processor). The SAR science parameters include required data blocks and may also include a number of optional data blocks. The required blocks are common to all images and are needed for fundamental exploitation. For advanced SAR systems, many additional parameters may be provided via the optional data blocks. The optional parameters, when available, allow for the full exploitation of a well-calibrated SAR sensor/image formation processing system.

The SICD products are “sensor independent” in that SAR science parameters are selected from an independent set of parameters. Parameter definitions are independent of both the sensor and the image formation processor. Sensor independence means that a parameter has the same definition for all sensors and processing systems that choose to include it.

SICD Design Documents

The SICD product design description documents are contained in this and several other documents. The applicable documents and references are listed in Table 1-1.

Table 1-1 Applicable Documents & References	
Number	Title & Website
NGA.STND.0024-2_1.5	Sensor Independent Complex Data (SICD), Volume 2, File Format Description Document, Version 1.5 https://nsgreg.nga.mil/ntb.jsp
NGA.STND.0024-3_1.5	Sensor Independent Complex Data (SICD), Volume 3, Image Projections Description Document, Version 1.5 https://nsgreg.nga.mil/ntb.jsp
NGA.STND.0024-4_1.5	NGA.STND.0024-4_1.5_Schema_2026_02_28.xsd https://nsgreg.nga.mil/ntb.jsp
NGA.STND.0036_1.0.0	Department of Defense World Geodetic System 1984: Its Definition and Relationships with Local Geodetic Systems, Version 1.0.0, 2014-07-08 https://nsgreg.nga.mil/doc/view?i=4085

1.1 Product Design Definition

A SICD product contains a single SAR complex image and a set of metadata that describes how the data was collected, how the data was processed, and the parameters needed for exploitation. The image data is a single two-dimensional grid of complex valued pixels. The image data has been processed such that the fundamental 2-dimensional impulse response varies smoothly across the image. Such an image will have a smoothly varying amplitude and phase response across the image prior to any spatially variant image-based processing that may be applied (e.g., image patch-based focus corrections).

A SAR imaging collection may be characterized across multiple dimensions. The dimensions include the time span of the collection, the collected RF bandwidth and the dimensions of the area that was imaged. Additionally, a particular platform may simultaneously collect multiple radar channels (e.g., multiple polarizations). A SICD product shall contain only a single image. Multiple images from the same collection are placed in separate SICD products. The SICD metadata will describe the image data contained in the product with respect to which radar channel, the portion of the collection time that was processed and the portion of the RF bandwidth that was processed. Optional metadata may be included to also describe the entire collection (all channels collected, total area imaged, etc.).

The SICD product definition for a particular SAR sensor and data processing system starts with an assessment of the collection types, collection modes, and image formation processing algorithms used to produce the complex image products. Also, to be considered are the derived image products to be produced and the types of exploitation to be supported. From these considerations, a set of metadata parameters are selected from the available SICD metadata parameters. This set of metadata parameters is referred to as a metadata profile. The metadata profile includes both the product identification parameters and the SAR science parameters. The metadata profile will include all required parameters, conditional parameters (as appropriate) and the optional parameters that may apply.

1.2 SICD Data Components

A SICD product consists of a pair of data components. The first component is a block of complex image pixel data. The image pixel block is described in Sections 1.2.1 and 1.2.2 below. The second component is a block of metadata expressed using the eXtensible Markup Language (XML), Version 1.0. The metadata contains the selected set of parameters that describe the imaging collection and the data processing that formed the image. A description of the XML implementation and a complete list of metadata parameters is contained in Sections 2, 3 and 4 below. The precise XML implementation is defined in the SICD XML Schema. A given XML instance shall be well-formed and valid as specified by the XML schema.

1.2.1 Image Pixel Array

The SICD image pixel data block is a single two-dimensional array of complex numbers. The image pixel array contains NumRows rows by NumCols columns.

The SICD image pixel data shall be stored in one of three allowed formats. The selected pixel format is specified by parameter PixelType. The allowed pixel types are as follows.

PixelType = RE32F_IM32F

Each pixel is stored as a pair of numbers that represent the real and imaginary components. Each component is stored in a 32-bit IEEE binary32 floating point format (4 bytes per component, 8 bytes per pixel).

PixelType = RE16I_IM16I

Each pixel is stored as a pair of numbers that represent the real and imaginary components. Each component is stored in a 16-bit signed integer in 2's complement format (2 bytes per component, 4 bytes per pixel).

PixelType = AMP8I_PHS8I

Each pixel is stored as a pair of numbers that represent the amplitude and phase components. Each component is stored in an 8-bit unsigned integer (1 byte per component, 2 bytes per pixel).

1.2.2 Image Arrays: Full Image & Sub-Image

The image pixel array contained in a SICD product may be the initial image array as produced by the image formation processing system or a sub-image extracted from the initial image. The initial image is referred to as the “full image”. A sub-image, sometimes referred to as an “image chip”, is a single two-dimensional portion of the full image. The full image pixel array is NumRowsFI rows by NumColsFI columns. An example full image array is shown in Figure 1.2-1. The rows of the full image pixel array are indexed as follows.

Full image row index: $\text{row} = 0, 1, 2, \dots, \text{NumRowsFI} - 1$.

Full image column index: $\text{col} = 0, 1, 2, \dots, \text{NumColsFI} - 1$.

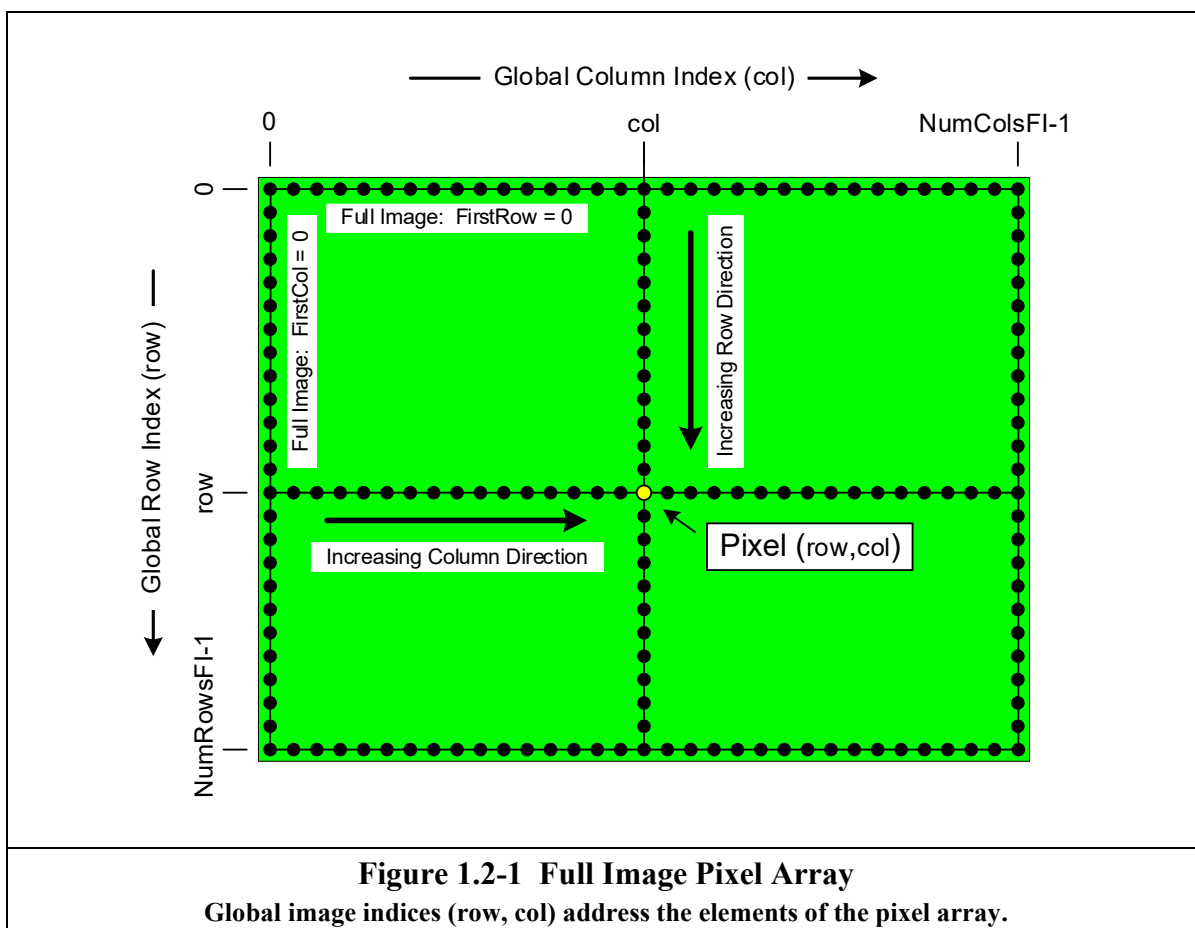


Image pixel indices (row, col) are referred to as the “global” image indices. The first row of the full image is indexed row = 0. The first column of the full image is indexed col = 0. Parameters NumRowsFI and NumColsFI are contained in the XML metadata for all image products.

Image size parameters NumRows and NumCols always denote the size of the image pixel array contained in the product file. The correct exploitation of a sub-image is dependent upon the relative position of the sub-image within the full image array. An example sub-image array and the full image array from which it was extracted are shown in Figure 1.2-2. The first row of the sub-image (indexed rows = 0) is extracted from row = FirstRow of the full image. The first column of the sub-image (indexed cols = 0) is extracted from col = FirstCol of the full image. FirstRow and FirstCol are contained in the XML metadata for all image products. For a full image product, parameters FirstRow = 0 and FirstCol = 0.

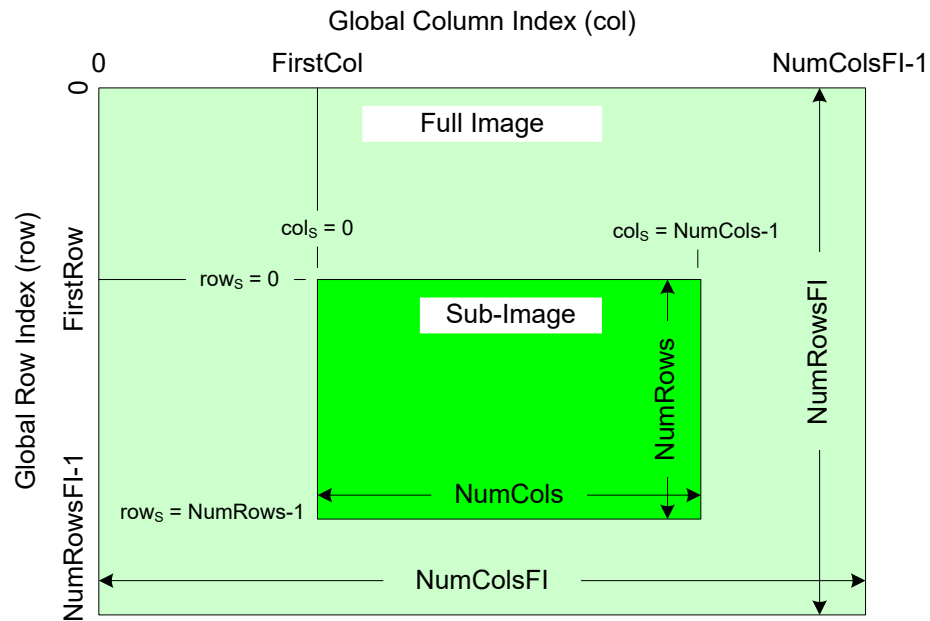


Figure 1.2-2 Full Image & Sub-Image Pixel Arrays

The sub-image pixel array relative position in the full image array is given by parameters **FirstRow** and **FirstCol**.

1.3 Container File Formats

See NGA.STND.0024-2 SICD File Format Description Document for allowed file formats.

1.4 Program Specific Implementation Documentation

For the SAR sensor and data processing system that will generate SICD products, a Program Specific Implementation Document should be developed to document the selected metadata profile. The implementation document will serve as a guide for those developing the products as well as for users of the products. The metadata profile will specify the product identification parameters as well as the selected SAR science parameters.

The SICD implementation document should address a number of implementation specific items. This list below is provided as a guide for developing implementation documents.

- (1) Identify the SAR platforms for which the image products will be produced and a brief description of their attributes. The description may include imaging modes, number of collected channels, etc.
- (2) Identify the Image Formation Algorithms (IFAs) that are used to produce the products. Proper exploitation is highly dependent upon the IFA used to produce the image.
- (3) Specify the product identification metadata parameters to be included and the guidance for each parameter. Product identification parameters are selected from the Collection & Image Creation Information Parameters. See Table 3-1.
- (4) Specify product file naming conventions. Product file names will usually identify the SAR platform and the specific imaging collection. For collections with multiple receive channels, the channel(s) used to form the image may also be identified.
- (5) Specify the implementation specific values for the header and sub-headers of the SICD container file. These parameters are explicitly identified in the SICD File Format Description Document. See Table 1-1 above.
- (6) Specify the set of SAR science parameters that are included. Parameters that will only be included in some products should be identified and the criteria for including them (e.g., for a particular collection mode or IFA).
- (7) For products that use a reference frequency when specifying frequency values, the values of the reference frequency indices and the reference frequencies should be specified. See Table 3-7 below.
- (8) If the Matched Collection Parameters block is to be included, the types of matched collections to be identified are selected. The criteria for match selection and the optional match parameters to be included (if any) are specified. See Table 3-13 below.

2 SICD XML Metadata

2.1 XML Metadata Structure

2.1.1 Field Name

The “Field Name” column lists the whole path to a particular field in the XML metadata tree. The tree structure is denoted “xxx.yyy.zzz” where “xxx” is the parent tag of “yyy” and “yyy” is the parent tag of “zzz”. A particular child tag is only valid if it has been placed in the proper location within the parent tag. For every instance of a tag in the file, there will also be an instance of its child tags. Below is an XML example of “xxx.yyy.zzz”. Note: zzz holds a double value in the example below.

```
<xxx>
  <yyy>
    <zzz>3.14159</zzz>
  </yyy>
</xxx>
```

2.1.2 Required vs. Optional

The Required / Optional (Req / Opt) column outlines the usage rules of a particular field. There are four options: “R”, “O”, “C”, and “OC”. The “R” value means that the field must appear in the XML tree. The “O” value means that the field is optional. The “C” value means the field is one of a required set of choices. The included field is included based upon the values of other fields. The “OC” value means the field is one of an optional set of choices that, if included, are based on the value of other fields. There is one more caveat to the rules for this column usage. If a tag is marked as “R” and its parent tag is marked as “O” or “C”, then the tag is only required if its parent tag is included.

2.1.3 Types

The “Type” column defines the XML Type of parameter contained in a particular field. The parameter types are listed in Table 2-1.

2.1.4 Description

The “Description” column explains the meaning of a particular field or block. A block groups its following fields under the same section. An example is parameter SICD.SCP (See Table 3-3). The fields ECF and LLH represent the earth center fixed and geodetic coordinates of the Scene Center Point.

- SICD.GeoData.SCP → Is a Block
- SICD.GeoData.SCP.ECF → ECF is a parameter under the SCP, representing SCP information
- SICD.GeoData.SCP.LLH → LLH is a parameter under the SCP, representing SCP information.

Table 2-1 XML Parameter Types		
Type	Meaning	Example XML Parameter
TXT	Value is a string of characters	Field: SICD.CollectionInfo.CollectorName Value: "SARPlatformIdentifier"
ENU	Value can be a string of characters or an integer. There is a certain allowed set of character strings or integer values.	Field: SICD.CollectionInfo.CollectType Value: "MONOSTATIC" or "BISTATIC"
BOOL	Value is a Boolean type. The Boolean type is used to specify true or false.	Field: SICD.PFA.Comp.Applied Value: "false"
INT	Value is an integer. It may be a positive or negative value with an optional positive sign ("+") when positive.	Field: SICD.ImageData.NumRows Value: 1000
DBL	Value is a real-valued decimal (base 10) number that when converted to binary format should be converted to a 64-bit floating point type (e.g., IEEE binary64 floating point). It may be a positive or negative value with an optional positive sign ("+") when positive. The value is represented in the scientific notation (The E23.7 notation) with 16 digits of precision.	CollectDuration is: 0.44444 seconds Field: SICD.Timeline.CollectDuration Value: 4.44440000000000E-01
XDT	Value represents the dateTime XML type. The dateTime form is: "YYYY-MM-DDThh:mm:ss.s+" • YYYY indicates the year • MM indicates the month • DD indicates the day • "T" indicates the start of the required time section • hh indicates hour • mm indicates minute • ss indicates second • s+ indicates fractional seconds * The seconds should be followed by a Z to indicate the time is in UTC (Coordinated Universal Time) *All components are required.	Field: SICD.ImageCreation.DateTime Value: 2008-08-06T20:27:55.123456Z
RC	Identifies a parent tag that consists of a required row and column component. The values of each component are integers, "INT". • "Row" represents row • "Col" represents column	Field: SICD.ImageData.SCPPixel.Row Value: 500 Field: SICD.ImageData.SCPPixel.Col Value: 500
CMPLX	Identifies a parent tag that consists of a required real and imaginary component. The values of each component are floating point type, "DBL".	Field: SICD.ImageFormation.PolarizationCalibration.Distortion.F1.Real

Table 2-1 XML Parameter Types		
Type	Meaning	Example XML Parameter
	“Real” represents the real part. “Imag” represents the imaginary part.	Value: 1.0500000000000000E00 Field: SICD.ImageFormation.PolarizationCalibration.Distortion.F1.Imag Value: 8.0100000000000000E-01
LS	Identifies a parent tag with children Line and Sample. The children are each of type DBL.	Field: SICD.RadarCollection.Area.Plane.Polygon.Vertex.Line Value: 4.4444000000000000E01 Field: SICD.RadarCollection.Area.Plane.Polygon.Vertex.Sample Value: 4.4444000000000000E01
XYZ	Identifies a parent tag that consists of a x, y and z component. The values of each component are floating point type, “DBL”. “X” represents the x component “Y” represents the y component “Z” represents the z component	Field: SICD.Grid.Row.UVectECF.X Value: 4.4444000000000000E01 Field: SICD.Grid.Row.UVectECF.Y Value: 4.4444000000000000E01 Field: SICD.Grid.Row.UVectECF.Z Value: 4.4444000000000000E01
LLH	Identifies a parent tag that consists of a geodetic latitude, longitude and height above ellipsoid component. The values of each component are floating point type, “DBL”. “Lat” represents the latitude point “Lon” represents the longitude point “HAE” represents the height point	Field: SICD.GeoData.SCP.LLH.Lat Value: 4.4444000000000000E01 Field: SICD.GeoData.SCP.LLH.Lon Value: 4.4444000000000000E01 Field: SICD.GeoData.SCP.LLH.HAE Value: 4.4444000000000000E01
LL	Identifies a parent tag that consists of a geodetic latitude and longitude component. The values of each component are floating point type, “DBL”. “Lat” represents the latitude point “Lon” represents the longitude point	Field: SICD.GeoData.ImageCorners.ICP.Lat Attribute: “index = 1:FRFC” Value: 4.4444000000000000E01 Field: SICD.GeoData.ImageCorners.ICP.Lon Attribute: “index = 1:FRFC” Value: 4.4444000000000000E01
POLY	Identifies a parent tag that consists of a set of coefficients for a one-dimensional polynomial function. The values of each component are floating point type, “DBL”.	M = 2 Parent Tag: SICD.Timeline.IPP.Set.IPPPoly Attribute: “order1 = 2”

Table 2-1 XML Parameter Types

Type	Meaning	Example XML Parameter
	“Coef” represents a coefficient A one-dimensional polynomial input variable 1 (Var1). Variable 1 of order1=M, where M equals the maximum value of the exponent1 attribute. The parent tag POLY will have an order1 attribute. Each component “Coef” will have an exponent1 attribute. The attribute represents the exponents of that coefficient. Zero coefficients may be omitted. Total number of possible coefficients is: (M + 1) $Z(\text{Var1}) = \sum_{m=0}^M c_m \cdot (\text{Var1})^m$	Field: SICD.Timeline.IPP.Set.IPPPoly.Coef Attribute “exponent1 = 0” Value: 4.444400000000000E01 Field: SICD.Timeline.IPP.Set.IPPPoly.Coef Attribute “exponent1 = 1” Value: 4.444400000000000E01 Field: SICD.Timeline.IPP.Set.IPPPoly.Coef Attribute “exponent1 = 2” Value: 4.444400000000000E01
2D_POLY	Identifies a parent tag that consists of a set of coefficients for a two-dimensional polynomial function. The values of each component are floating point type, “DBL”. “Coef” represents a coefficient Two-dimensional polynomial inputs variable 1 (Var1) and variable 2 (Var2). Variable 1 of order1= M, where M equals the maximum value of the exponent1 attribute. Variable 2 of order2= N, where N equals the maximum value of the exponent2 attribute. The parent tag POLY will have an order1 and order2 attribute. Each component “Coef” will have an exponent1 and exponent2 attribute. The attributes represent the exponents of that coefficient. Total number of possible coefficients is:	M = 2 and N = 1 Parent Tag: SICD.Grid.TimeCOAPoly Attribute: “order1 = 2”, “order2 = 1” Field: SICD.Grid.TimeCOAPoly.Coef Attribute: “exponent1 = 0” “exponent2 = 0” Value: 4.444400000000000E01 Field: SICD.Grid.TimeCOAPoly.Coef Attribute: “exponent1 = 0” “exponent2 = 1” Value: 4.444400000000000E01 Field: SICD.Grid.TimeCOAPoly.Coef Attribute: “exponent1 = 1” “exponent2 = 0” Value: 4.444400000000000E01

Table 2-1 XML Parameter Types		
Type	Meaning	Example XML Parameter
	$(M + 1) * (N + 1)$ $Z(\text{Var1}, \text{Var2}) = \sum_{m=0}^M \sum_{n=0}^N c_{m,n} \cdot (\text{Var1})^m \cdot (\text{Var2})^n$ *If a coefficient has a zero value it may be omitted.	Field: SICD.Grid.TimeCOAPoly.Coeff Attribute: “exponent1 = 1” “exponent2 = 1” Value: 4.444400000000000E01 Field: SICD.Grid.TimeCOAPoly.Coeff Attribute: “exponent1 = 2” “exponent2 = 0” Value: 4.444400000000000E01 Field: SICD.Grid.TimeCOAPoly.Coeff Attribute: “exponent1 = 2” “exponent2 = 1” Value: 4.444400000000000E01
XYZ_POLY	Identifies a parent tag that consists of a x, y and z component. Each component is a POLY type. Each parent tag, X,Y,Z will have the order1 attribute. Each component “Coef” will have the exponent1 attribute. “X.Coeff” represents a x component coefficient “Y.Coeff” represents a y component coefficient “Z.Coeff” represents a z component coefficient A one-dimensional polynomial with same input variable 1 (Var1). Total number of possible coefficients is: $3 \times (M + 1)$. $\mathbf{V}(\text{Var1}) = \begin{bmatrix} X(\text{Var1}) \\ Y(\text{Var1}) \\ Z(\text{Var1}) \end{bmatrix} = \begin{bmatrix} \sum_{m=0}^M c_{x,m} \cdot (\text{Var1})^m \\ \sum_{m=0}^M c_{y,m} \cdot (\text{Var1})^m \\ \sum_{m=0}^M c_{z,m} \cdot (\text{Var1})^m \end{bmatrix}$	M = 1 Parent tag: SICD.Position.GRPPoly.X Parent tag: SICD.Position.GRPPoly.Y Parent tag: SICD.Position.GRPPoly.Z All have an attribute: “order1=1” Field: SICD.Position.GRPPoly.X.Coeff Attribute: “exponent1 = 0” Value: 4.444400000000000E01 Field: SICD.Position.GRPPoly.X.Coeff Attribute: “exponent1 = 1” Value: 4.444400000000000E01 Field: SICD.Position.GRPPoly.Y.Coeff Attribute: “exponent1 = 0” Value: 4.444400000000000E01 Field: SICD.Position.GRPPoly.Y.Coeff Attribute: “exponent1 = 1” Value: 4.444400000000000E01 Field: SICD.Position.GRPPoly.Z.Coeff Attribute: “exponent1 = 0” Value: 4.444400000000000E01

Table 2-1 XML Parameter Types		
Type	Meaning	Example XML Parameter
	*If a coefficient has a zero value it may be omitted.	Field: SICD.Position.GRPPoly.Z.Coef Attribute: "exponent1 = 1" Value: 4.444400000000000E01
MTX	Identifies a parent tag that consists of a set of entries for a two-dimensional matrix. The values of each child tag are floating point type, "DBL". "Entry" represents an entry in the matrix The parent tag will have attributes "size1" and "size2". Total number entries less than or equal to NumEntries = size1 x size2. Each entry "Entry" will have attributes "index1" and "index2". The value of index1 may be from 1 to size1. The value of index2 may be from 1 to size2. Matrix entries equal to zero may be omitted.	Parent Tag: SICD.ErrorStatistics.TxPV Attribute: size1="6" and size2="6" indicates complex type contains the entries of a 6 x 6 square matrix. Field: SICD.ErrorStatistics.TxPV.Entry Attributes: index1="1" index2="2" indicate the entry is the (1,2) entry of the matrix.

2.1.5 Units

The "Units" column defines the measurement units assigned to a particular field. Units are seconds (sec), meters (m), seconds per meter (sec/m), meters per second (m/sec), Hertz (Hz), Hertz per second (Hz/sec), degrees (deg), decimal degrees (dd), cycles (cyc), cycles per meter (cyc/m), cycles per meter squared (cyc/m**2, etc), radians (rad), radians per second (rad/sec), radians per second squared (rad/sec**2, etc), samples per second (samples/sec) and decibel (dB).

2.1.6 Rpt

The "Repeatable" column indicates if a tag may be repeated. There are two values: "Y" and "N". "Y" means that a tag may appear more than once and "N" means that a tag may appear only once.

2.1.7 Attribute

The "Attribute" column is used to associate name=value pairs with elements. It is part of an element and provides additional information about that element. There are several different types of attributes. SICD is using the Enumeration and ID type. Enumeration type allows a specific list of values that the attribute must match. The values are listed in the table along with the attribute. ID type allows a unique identifier for each element. The identifier being used can either have an integer dataType or a string dataType. The string dataType allows a

user to add sensor specific metadata parameters. The allowed values and ranges for the specified parameter are also defined in this column.

The attributes that are Enumeration Type from the Attribute column in Tables 3-1 through 3-15:

- index = “1:FRFC”, ..., “4:LRLC”

The attributes that are ID types from the Attribute column in Tables 3-1 through 3-15:

- name = “xxx”, string
- index = “x”, integer
- size= “x”, integer
- order1 = “M”, integer
- order2 = “N”, integer
- exponent1= “0 to M”
- exponent2= “0 to N”

Example XML of Attributes “size” and “index”

Field: SICD.Grid.Row.WgtFunct.Wgt, Attribute: size= “x” and index=“x”

```
<WgtFunct size=“4”>
  <Wgt index= “1”> value here </Wgt>
  <Wgt index= “2”> value here </Wgt>
  <Wgt index= “3”> value here </Wgt>
  <Wgt index= “4”> value here </Wgt>
</WgtFunct>
```

3 XML Metadata Parameter List

The SICD metadata parameters are divided into blocks of related parameters. Blocks are marked as Required, Optional or Conditional. Required blocks may contain both optional and conditional parameters. Some blocks contain fields that permit sensor specific parameter additions called Parameters.

3.1 XML Metadata Block Summary

(1) Collection & Image Creation Information Required

Parameters that identify the collection including specific platform(s) and imaging collection. Includes product classification and release parameters as applicable. See parameters listed in Table 3-1.

(2) Image Data Parameters Required

Parameters that describe the image pixel array. Includes the dimensions of the image pixel array in the product, the original full image pixel array and areas of zero-filled pixels if applicable. See parameters listed in Table 3-2.

(3) Image Geographic Reference Parameters Required

Parameters that describe the geographic coordinates of the region covered by the image. Geographic coordinates of the SCP (Scene Center Point) and the image corner points are included. See parameters listed in Table 3-3.

(4) Image Grid Parameters Required

Parameters that describe the spatial sampling represented by the sample grid. Includes the type of sample grid, the sample spacings and resolutions. Includes parameters describing the two-dimensional spatial frequency domain. See parameters listed in Table 3-4.

(5) Collection Timeline Parameters Required

Parameters that describe the imaging collection timeline. Includes the collection start time, collection duration and the radar pulse repetition interval / inter-pulse period parameters. See parameters listed in Table 3-5.

(6) Reference Position Parameters Required

Parameters that describe the platform and ground reference positions in ECF coordinates versus time. Includes the effective Aperture Reference Point position and the Ground Reference Point position. The individual transmit and receive phase center positions are optional parameters. See parameters listed in Table 3-6.

(7) Radar Collection Parameters

Required

Parameters that describe the radar collection including area covered by the collection and the finest SAR resolutions that may be achieved. Transmitted waveform parameters include the transmitted bandwidth and polarization(s). Receive parameters include the number of channels and the polarization(s). Transmitted frequency values may be expressed as offsets from a reference frequency. See parameters listed in Table 3-7.

(8) Image Formation Parameters

Required

Parameters that describe the image formation processing parameters that produced the image. Parameters indicate which receive channel(s) were processed. Parameters describe the portion of the collection time and bandwidth processed. See parameters listed in Table 3-8.

(9) SCP Center Of Aperture Parameters

Required

Parameters that describe the Center of Aperture time and collection geometry for the Scene Center Point (SCP). Parameters include the ARP position and velocity, the imaging side of track and range to the SCP. Parameters provided to support search and discovery. See parameters listed in Table 3-9.

(10) Radiometric Parameters

Optional

Parameters that enable the conversion of pixel power level to radar reflectivity parameters. Parameters may be included to convert pixel power to common clutter reflectivity parameters (such as σ_0) or point target radar cross section. See parameters listed in Table 3-10.

(11) Antenna Parameters

Optional

Parameters that describe the transmit and receive antenna patterns. An option is also included to describe an effective two-way antenna for monostatic operation. Parameters include antenna orientation, mainlobe pointing and beam shape as a function of time. See parameters listed in Table 3-11.

(12) Error Estimation Parameters

Optional

Parameters that describe the errors in radar collection parameters (e.g., position and velocity) that allow estimation of image geolocation parameters. See parameters listed in Table 3-12.

(13) Collection Match Parameters

Optional

Parameters that describe other known collections that are matched to the collection from which this image was produced. See parameters listed in Table 3-13.

(14) Range & Azimuth Compression Parameters Conditional

Parameters that describe the images formed using simple Range & Azimuth Compression – the simplest image formation algorithm. The resulting image is a simple range, Doppler image. See parameters listed in Table 3-14.

(15) Polar Format Algorithm Parameters Conditional

Parameters that describe the image formation processing for images formed using the Polar Format Algorithm. Parameters include the image formation plane used, the focus plane used, the mapping of platform position to polar angle, and transmitted frequency to aperture frequency scale factors. See parameters listed in Table 3-15.

(16) Range Migration Algorithm Parameters Conditional

Parameters that describe the image formation processing for images formed with any one of several Range Migration Algorithms. Parameters include the reference slant plane and portion of the spatial frequency domain processed. Parameters commonly used for imaging near closest approach may be used when applicable. See parameters listed in Table 3-16.

The XML structure is specified by the XML schema and is described in Tables 3-1 through 3-16 below. Each table describes one or more of the primary branches. In the following tables, each branch is highlighted by color to indicate when the branch is required, optional, or may be included as one of an optional set of choices.

BLUE

Required branch to be included in all instances that also include the parent node of the branch. Examples include:

Table 3-1 / SICD / CollectionInfo

Table 3-5 / SICD / Timeline / IPP / Set

YELLOW

Optional branch that may be included in all instances that also include the parent node of the branch. Examples include:

Table 3-1 / SICD / ImageCreation

Table 3-2 / SICD / ImageData / ValidData

ORANGE

Optional branch that may be included by a choice from two or more branches. Examples include:

Table 3-12 / SICD / ErrorStatistics / CompositeSCP

/ BistaticCompositeSCP

3.2 XML Metadata Parameter Lists

3.2.1 XML: CollectionInfo & Image Creation

Table 3-1 Collection & Image Creation Information Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							-		
	CollectionInfo			R		This block contains general information about the collection.	-	N	
		CollectorName		R	TXT	Radar platform identifier that collected the Initial PHD. For Bistatic collections, list the Receive-Only platform.	-	N	
		IlluminatorName		O	TXT	Radar platform identifier that provided the illumination. Required for images with CollectType = BISTATIC.	-	N	
		CoreName		R	TXT	Collection & imaging data set identifier. Uniquely identifies imaging collections per Program Specific Implementation Doc.	-	N	
		CollectType		O	ENU	Collection type identifier. Required for images formed from Bistatic collections. Encouraged to be included in all products.	-	N	Allowed values: “MONOSTATIC”, “BISTATIC”
		RadarMode		R			-	N	
			ModeType	R	ENU	Radar imaging mode.	-	N	Allowed Values: “SPOTLIGHT”, “STRIPMAP”, “DYNAMIC STRIPMAP”
			ModeID	O	TXT	Radar imaging mode per Program Specific Implementation Document.	-	N	

Table 3-1 Collection & Image Creation Information Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		Classification	R	TXT	Contains the human-readable banner. Contains classification, file control & handling, file releasing, and/or proprietary markings. Specified per Program Specific Implementation Document.	-	N	Default value: "UNCLASSIFIED"
		InformationSecurityMarking	O	TXT	Included if ISM metadata provided via container file metadata structure.		N	
		CountryCode	O	TXT	List of country codes for region covered by the image.	-	Y	
		Parameter	O	TXT	Free format field that can be used to pass forward the radar collection information.	-	Y	name = "xxx", name is a descriptive identifier for this information
SICD								
	ImageCreation		O		This block contains general information about the image creation.		N	
		Application	O	TXT	Name and version of the application used to create the image.	-	N	
		DateTime	O	XDT	Date and time the image creation application processed the image (UTC).	-	N	
		Site	O	TXT	The creation site of this SICD product	-	N	
		Profile	O	TXT	Identifies what profile was used to create this SICD product.	-	N	

3.2.2 XML: ImageData

Table 3-2 Image Data Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
	ImageData	R		This block describes image pixel data	-	N	
	PixelType	R	ENU	Indicates the pixel type and binary format of the data. PixelType = "RE32F_IM32F": - Real & Imaginary components. Each component a 32 bit IEEE binary32 floating point. PixelType = "RE16I_IM16I": - Real & Imaginary components. Each component a 16 bit signed integer, 2's complement. PixelType = "AMP8I_PHS8I": - Amplitude & Phase components. Each component an 8 bit unsigned integer.	-	N	Allowed values: "RE32F_IM32F", "RE16I_IM16I", "AMP8I_PHS8I"
	AmpTable	O		Amplitude lookup table for PixelType = "AMP8I_PHS8I".	-	N	size = "256"
	Amplitude	R	DBL	Amplitude table entries.	-	Y	index = "0", "1", ..., "255"
	NumRows	R	INT	Number of rows in the product. Includes zero filled pixels (if present).	-	N	
	NumCols	R	INT	Number of columns in the product. Includes zero filled pixels (if present).	-	N	
	FirstRow	R	INT	Global row index of the first row in the product (non-negative integer value). Set equal to 0 in full image product.	-	N	

Table 3-2 Image Data Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		FirstCol	R	INT	Global column index of the first column in the product. Non-negative integer value. Set equal to 0 in the full image product.	-	N	
		FullImage	R		Original full image product	-	N	
		NumRows	R	INT	Number of rows in the original full image product. Includes zero filled pixels (if present).	-	N	
		NumCols	R	INT	Number of columns in the original full image product. Includes zero filled pixels (if present).	-	N	
		SCPPixel	R	RC	Scene Center Point pixel global row & column index. Located near the center of the full image.	-	N	
		ValidData	O		Indicates the full image includes both valid data and some zero filled pixels. Simple polygon encloses the valid data (may include some zero filled pixels for simplification). Vertices in clockwise order. Attribute "size" = NumVertices.	-	N	size = NumVertices NumVertices $\geq$ 3
		Vertex	R	RC	Polygon vertex pixel global row & column index. Vertex 1 determined by: (1) minimum row index, (2) minimum column index if 2 vertices with minimum row index. 1 st and last vertices of polygon connect.	-	Y	index = "1", "2", ..., NumVertices

3.2.3 XML: GeoData

Table 3-3 Image Geographic Reference Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD									
	GeoData			R		This block describes the geographic coordinates of the region covered by the image.	-	N	
		EarthModel		R	ENU	Identifies the earth model used for latitude, longitude and height parameters. All height values are Height Above The Ellipsoid (HAE).	-	N	Allowed Values: “WGS_84”
		SCP		R		Scene Center Point (SCP) in full (global) image. This is the precise location.	-	N	
			ECF	R	XYZ	Scene Center Point position in ECF coordinates.	m	N	
			LLH	R	LLH	Scene Center Point geodetic latitude, longitude and height. -90.0 ≤ Lat ≤ 90.0, -180.0 ≤ Lon ≤ 180.0	dd dd m	N	
		ImageCorners		R		Image corners points projected to the ground/surface level. Points may be projected to the same height as the SCP if ground/surface height data is not available. The corner positions are approximate geographic locations and not intended for analytical use.	-	N	

Table 3-3 Image Geographic Reference Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			ICP	R	LL	Image Corner Point (ICP) data for the 4 corners in product. ICPs indexed x = 1, 2, 3, 4, clockwise. x = 1 ⇔ First row, First column x = 2 ⇔ First row, Last column x = 3 ⇔ Last row, Last column x = 4 ⇔ Last row, First column	dd	Y	index = “1:FRFC” or “2:FRLC” or “3:LRLC” or “4:LRFC”
		ValidData		O		Indicates the full image includes both valid data and some zero filled pixels. Attribute “size” = NumVertices. Where the SICD XML has a GeoData.ValidData block, the SICD XML shall also have a ImageData.ValidData block.	-	N	size = NumVertices NumVertices ≥ 3
			Vertex	R	LL	Valid data points projected to the ground/surface level. Points may be projected to the same height as the SCP if ground/surface height data is not available. The vertex positions are approximate geographic locations and not intended for analytical use. Vertices in same order as SICD.ImageData.ValidData.Vertex. -90.0 ≤ Lat ≤ 90.0, -180.0 ≤ Lon ≤ 180.0		Y	index = “1”, “2”, ..., NumVertices
		GeoInfo		O		Parameters describing geographic features.	-	Y	name = “xxx”, name is a descriptive identifier for this information
			Desc	O	TXT	Used to specify a name and description of a geographic feature.	-	Y	name = “xxx”, name is a descriptive identifier for this information
			Point	O	LL	Used to specify a single point. -90.0 ≤ Lat ≤ 90.0, -180.0 ≤ Lon ≤ 180.0	dd	Y	

Table 3-3 Image Geographic Reference Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			Line	O		Used to specify a “linear” feature with connected line segments. The size attribute represents the number of endpoints (NumEndpoints).	-	Y	size = NumEndpoints NumEndpoints ≥ 2
			Endpoint	R	LL	Line segment endpoints.	dd	Y	index = “1”, “2”, ..., NumEndpoints
			Polygon	O		Used to specify an area with a polygon. The size attribute represents the number of endpoints (NumVertices).	-	Y	size = NumVertices NumVertices ≥ 3
			Vertex	R	LL	Polygon vertices indexed clockwise. 1 st and last vertices of polygon connect. -90.0 ≤ Lat ≤ 90.0, -180.0 ≤ Lon ≤ 180.0	dd	Y	index = “1”, “2”, ..., NumVertices
			GeoInfo	O		GeoInfo block(s) may be embedded within a GeoInfo block.	-	Y	name = "xxx", name is a descriptive identifier for this information

3.2.4 XML: Grid

Table 3-4 Image Grid Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD								
	Grid		R		This block of parameters describes the image sample grid.	-	N	
		ImagePlane	R	ENU	Defines the type of image plane that the best describes the sample grid. Precise plane defined by Row Direction and Column Direction unit vectors.	-	N	Allowed values: "GROUND", "SLANT", "OTHER"
		Type	R	ENU	Defines the type of spatial sampling grid represented by the image sample grid. Row coordinate first, column coordinate second. RGAZIM: Grid for a simple range, Doppler image. Also, the natural grid for images formed with the Polar Format Algorithm. RGZERO: A grid for images formed with the Range Migration Algorithm. Used only for imaging near closest approach (i.e. near zero Doppler). XRGYCR & XCTYAT – Slant plane grids for images formed with the Range Migration Algorithm. Used for any imaging geometry. PLANE – Arbitrary plane with orientation other than the specific XRGYCR & XCTYAT above.	-	N	Allowed values: "RGAZIM", "RGZERO", "XRGYCR", "XCTYAT", "PLANE". RGAZIM <=> Range & azimuth relative to the ARP at a reference time. RGZERO <=> Range from ARP trajectory at zero Doppler and "azimuth" aligned with the strip being imaged. XRGYCR <=> Orthogonal slant plane grid oriented range and cross range relative to the ARP at a reference time. XCTYAT <=> Orthogonal slant plane grid with X oriented cross track. PLANE <=> Uniformly sampled in an arbitrary plane along directions U & V.

Table 3-4 Image Grid Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		TimeCOAPoly	R	2D_POLY	Time of Center Of Aperture (t_COA) polynomial as a function of image coordinates. The polynomial is a function of image row coordinate (variable 1) and column coordinate (variable 2). Note: Coefficient (0,0) is the SCP COA time.	s, s/m, s/m ² , s/m ³ , etc.	N	order1 = "M" order2 = "N"
Parameters are provided for the Row Direction and the Column Direction. Row Direction ⇔ Increasing Row Index Direction. Column Direction ⇔ Increasing Column Index Direction.								
Row Direction Parameters								
		Row	R		Parameters describing increasing row direction image coordinate.	-	N	
		UVectECF	R	XYZ	Unit vector in the increasing row direction (ECF) at the SCP pixel.	-	N	
		SS	R	DBL	Sample spacing in the increasing row direction. Precise spacing at the SCP.	m	N	
		ImpRespWid	R	DBL	Half power impulse response width in the increasing row direction. Measured at the SCP.	m	N	
		Sgn	R	ENU	Integer sign of the exponent in the DFT to transform the row dimension to spatial frequency dimension (Krow).	-	N	Allowed values: "-1", "+1"
		ImpRespBW	R	DBL	Spatial bandwidth in Krow used to form the impulse response in the row direction. Measured at the center of support for the SCP.	cyc/m	N	

Table 3-4 Image Grid Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			KCtr	R	DBL	Center spatial frequency in the Krow dimension. Corresponds to the zero frequency of the DFT in the row direction.	cyc/m	N	
			DeltaK1	R	DBL	Minimum row offset from KCtr of the spatial frequency support for the image.	cyc/m	N	
			DeltaK2	R	DBL	Maximum row offset from KCtr of the spatial frequency support for the image.	cyc/m	N	
			DeltaKCOAPoly	O	2D_POLY	Offset from KCtr of the center of support in the row spatial frequency (Krow). The polynomial is a function of image row coordinate (variable 1) and column coordinate (variable 2).	cyc/m, $^*/m^2$, $^*/m^3$, etc.	N	order1 = "M" order2 = "N"
			WgtType	O		Parameters describing aperture weighting type applied in the spatial frequency domain (Krow) to yield the impulse response in the row direction.	-	N	
			WindowName	R	TXT	Type of aperture weighting applied in the spatial frequency domain (Krow) to yield the impulse response in the row direction.	-	N	Example: "UNIFORM", "TAYLOR", "UNKNOWN", "HAMMING"
			Parameter	O	TXT	Free format field that can be used to pass forward the weighting parameter information.	-	Y	name = "xxx", name is a descriptive identifier for this information
			WgtFunct	O		Sampled aperture amplitude weighting function applied in Krow to form the SCP impulse response in the row direction. Attribute "size" equals the number of weights (NW).	-	N	size = NW $NW \geq 2$
			Wgt	R	DBL	Sampled amplitude values that span the ImpRespBW of the SCP.	-	Y	index = "1", "2", ..., NW

Table 3-4 Image Grid Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
Column Direction Parameters								
		Col	R		Parameters describing increasing column direction image coordinate.	-	N	
		UVectECF	R	XYZ	Unit vector in the increasing column direction (ECF) at the SCP pixel.	-	N	
		SS	R	DBL	Sample spacing in the increasing column direction. Precise spacing at the SCP.	m	N	
		ImpRespWid	R	DBL	Half power impulse response width in the increasing column direction. Measured at the SCP.	m	N	
		Sgn	R	ENU	Integer sign of the exponent in the DFT to transform the column dimension to spatial frequency dimension (Kcol).	-	N	Allowed values: “-1”, “+1”
		ImpRespBW	R	DBL	Spatial bandwidth in Kcol used to form the impulse response in the column direction. Measured at the center of support for the SCP.	cyc/m	N	
		KCtr	R	DBL	Center spatial frequency in the Kcol dimension. Corresponds to the zero frequency of the DFT in the column direction.	cyc/m	N	
		DeltaK1	R	DBL	Minimum column offset from KCtr of the spatial frequency support for the image.	cyc/m	N	
		DeltaK2	R	DBL	Maximum column offset from KCtr of the spatial frequency support for the image.	cyc/m	N	
		DeltaKCOAPoly	O	2D_POLY	Offset from KCtr of the center of support in the column spatial frequency (Kcol). The polynomial is a function of image row coordinate (variable 1) and column coordinate (variable 2).	cyc/m, */m ² , */m ³ , etc	N	order1 = “M” order2 = “N”

Table 3-4 Image Grid Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			WgtType	O	TXT	Parameters describing aperture weighting type applied in the spatial frequency domain (Kcol) to yield the impulse response in the column direction.	-	N	
			WindowName	R	TXT	Type of aperture weighting applied in the spatial frequency domain (Kcol) to yield the impulse response in the column direction.	-	N	Example: "UNIFORM", "TAYLOR", "UNKNOWN", "HAMMING"
			Parameter	O	TXT	Free format field that can be used to pass forward the weighting parameter information.	-	Y	name = "xxx", name is a descriptive identifier for this information
			WgtFunct	O		Sampled aperture amplitude weighting function applied in Kcol to form the SCP impulse response in the column direction. Attribute "size" equals the number of weights (NW).	-	N	size = NW NW ≥ 2
			Wgt	R	DBL	Sampled amplitude values that span the ImpRespBW of the SCP.	-	Y	index = "1", "2", ..., NW

3.2.5 XML: Timeline

Table 3-5 Collection Timeline Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
	Timeline	R		This block describes the imaging collection timeline.	-	N	
	CollectStart	R	XDT	Collection date and start time (UTC). Time reference for times measured from collection start (i.e. slow time t = 0).	-	N	
	CollectDuration	R	DBL	Duration of collection period.	sec	N	
	IPP	O		Describes Inter-Pulse Period (IPP) parameters. The size attribute represents the number of IPP sets (NSets).	-	N	size = NSets NSets ≥ 1
	Set	R		A set of IPP parameters.	-	Y	index= "1", "2", ..., NSets
	TStart	R	DBL	Start time for the period relative to Collection Start.	sec	N	
	TEnd	R	DBL	End time for the period relative to Collection Start.	sec	N	
	IPPStart	R	INT	Starting IPP index for the period described.	-	N	
	IPPEnd	R	INT	Ending IPP index for the period described.	-	N	
	IPPPoly	R	POLY	IPP index polynomial coefficients yield IPP index as a function of time t (variable 1), starting TStart to TEnd.	-	N	order1 = "M"

3.2.6 XML: Position

Table 3-6 Reference Position Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
	Position	R		This block describes the platform and ground reference positions verse time.		N	
	ARPPoly	R	XYZ_POLY	Aperture Reference Point (ARP) position polynomial in ECF as a function of time t (variable 1). Time t = 0 at Collection Start.	m, m/sec, etc.	N	See Section 4.6.3.
	GRPPoly	O	XYZ_POLY	Ground Reference Point (GRP) position polynomial in ECF as a function of time t (variable 1). Time t = 0 at Collection Start. Required for CollectType = BISTATIC.	m, m/sec, etc.	N	See Section 4.6.2
	TxAPCPoly	O	XYZ_POLY	Transmit Aperture Phase Center (APC) polynomial in ECF as a function of time t (variable 1). Time t = 0 at Collection Start. Required for CollectType = BISTATIC.	m, m/sec, etc.	N	See Section 4.6.1
	RcvAPC	O		Receive Aperture Phase Center (APC). The size attribute represents the number of Receive APC (NumRcvAPCs) polynomials below. Required for CollectType = BISTATIC.		N	size = NumRcvAPCs NumRcvAPCs ≥ 1 See Section 4.6.1
	RcvAPCPoly	R	XYZ_POLY	Receive Aperture Phase Center polynomial in ECF as a function of time t (variable 1). Time t = 0 at collection start.	m, m/sec, etc.	Y	index = "1", "2", ..., NumRcvAPCs

3.2.7 XML: RadarCollection

Table 3-7 Radar Collection Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD								
	RadarCollection		R		This block describes the radar collection information.		N	
		TxFrequency	R		Transmitted frequency range of values.	-	N	
		Min	R	DBL	Minimum transmitted RF frequency. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
		Max	R	DBL	Maximum transmitted RF frequency. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
		RefFreqIndex	O	INT	Indicates all RF frequency values are expressed as offsets from a reference frequency (Ref_Freq). If RefFreqIndex included ⇔ All RF frequency values are offsets from a reference frequency (Ref_Freq (Hz)). Ref_Freq value defined per Program Specific Implementation Document. See Section 1.4.	-	N	
		Waveform	O		Transmit and receive demodulation waveform parameters. The size attribute represents the number of waveforms transmitted. (NumWaveforms).	-	N	size = NumWaveforms
		WFParameters	R		Parameters for waveform.	-	Y	index = "1", "2",, NumWaveforms
		TxPulseLength	O	DBL	Transmit pulse length.	sec	N	

Table 3-7 Radar Collection Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				TxRFBandwidth	O	DBL	Transmit RF bandwidth of the transmit pulse.	Hz	N	
				TxFreqStart	O	DBL	Transmit Start frequency for Linear FM waveform. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
				TxFMRate	O	DBL	Transmit FM rate for Linear FM waveform.	Hz/sec	N	
				RcvDemodType	O	ENU	Receive demodulation used when Linear FM waveform is used on transmit. “STRETCH” = De-ramp on Receive demodulation. “CHIRP” = No De-ramp On Receive demodulation.	-	N	Allowed values: “STRETCH”, “CHIRP”
				RcvWindowLength	O	DBL	Receive window duration.	sec	N	
				ADCSampleRate	O	DBL	Analog-to-Digital Converter sampling rate.	samps/sec	N	
				RcvIFBandwidth	O	DBL	Receive IF bandwidth.	Hz	N	
				RcvFreqStart	O	DBL	Receive demodulation start frequency. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
				RcvFMRate	O	DBL	Receive FM rate. Set to 0 for RcvDemodType = “CHIRP”.	Hz/sec	N	

Table 3-7 Radar Collection Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		TxPolarization		R	ENU	Transmit polarization type. See Table 4-1.		N	Allowed values: “V”, “H”, “X”, “Y”, “S”, “E”, “RHC”, “LHC”, “UNKNOWN”, “SEQUENCE”, “OTHER*” Use of “OTHER*” defined per Program Specific Implementation Document where * = 0 or more characters that does not contain “:” (0x3A).
		TxSequence		O		Indicates the transmit signal steps through a repeating sequence of waveforms and/or polarizations. One step per Inter-Pulse Period. The size attribute represents the number of steps in the sequence (NumTxSteps).	-	N	size = NumTxSteps NumTxSteps ≥ 1
			TxStep	R		Transmit sequence step index.	-	Y	index = “1”, “2”, ..., NumTxSteps
			WFIndex	O	INT	Waveform number for this step. See Waveform parameters.		N	

Table 3-7 Radar Collection Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			TxPolarization	O	ENU	Transmit signal polarization for this step. See Table 4-1.	-	N	Allowed values: "V", "H", "X", "Y", "S", "E", "RHC", "LHC", "UNKNOWN", "OTHER*" Use of "OTHER*" defined per Program Specific Implementation Document where * = 0 or more characters that does not contain "." (0x3A).
		RcvChannels		R		Receive data channel parameters. The size attribute represents the number of receive data channels (NumRcvChans).	-	N	size = NumRcvChans
			ChanParameters	R		Parameters for data channel.	-	Y	index = "1", "2", ..., NumRcvChans

Table 3-7 Radar Collection Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			TxRcvPolarization	R	ENU	Indicates the combined transmit and receive polarization for the channel. See Table 4-1.	-	N	Allowed values include the form TX:RCV that is formed from one TX value and one RCV value. Allowed TX values: "V", "H", "X", "Y", "S", "E", "RHC", "LHC", "OTHER" Allowed RCV values: "V", "H", "X", "Y", "S", "E", "RHC", "LHC", "OTHER", Allowed values also include: "OTHER", "UNKNOWN" Use of "OTHER*" defined per Program Specific Implementation Document where * = 0 or more characters that does not contain "." (0x3A).
			RcvAPCIndex	O	INT	Index of the Receive Aperture Phase Center (Rcv APC). Only include if Receive APC position polynomial(s) are included. See Table 3-5.	-	N	
		Area		O		Parameters describing the imaged area covered by the collection.	-	N	
		Corner		R		Set of 4 corner points in LLH.	-	N	

Table 3-7 Radar Collection Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				ACP	R	LLH	Corner point parameters indexed clockwise. -90.0 ≤ Lat ≤ 90.0, -180.0 ≤ Lon ≤ 180.0	dd dd m	Y	index = “1”, “2”, “3”, “4”
			Plane		O		Parameters describing a rectangular area in a geo-located display plane.	-	N	
				RefPt	R		Reference Point parameters.	-	N	Optional: name = “xxx”
				ECF	R	XY Z	Reference Point in ECF.	m	N	
				Line	R	DBL	Reference Point line index.	-	N	
				Sample	R	DBL	Reference Point sample index.	-	N	
				XDir	R		X Direction parameters of the geo-located plane. X direction is the increasing line direction.	-	N	
				UVectECF	R	XYZ	Unit vector in the X direction (ECF).	m	N	
				LineSpacing	R	DBL	Line spacing in the X direction.	m	N	
				NumLines	R	INT	Number of lines in the X direction.	-	N	
				FirstLine	R	INT	First line index.	-	N	
				YDir	R		Y Direction parameters of the geo-located plane. Y direction is the increasing sample direction.	-	N	
				UVectECF	R	XYZ	Unit vector in the Y direction (ECF).	m	N	
				SampleSpacing	R	DBL	Sample spacing in the Y direction.	m	N	
				NumSamples	R	INT	Number of samples in the Y direction.	-	N	
				FirstSample	R	INT	First sample index.	-	N	

Table 3-7 Radar Collection Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				SegmentList	O		Segments that correspond to the RadarCollection.Area.Plane defined above. (XDir/YDir). The size attribute represents the number of segments (NumSegs).		N	size = NumSegs
				Segment	R		Defines the data boundaries of the segment.		Y	index = "1", "2", ..., NumSegs
				StartLine	R	INT	Start (first) line of the segment.	-	N	
				StartSample	R	INT	Start (first) sample of the segment.	-	N	
				EndLine	R	INT	End (last) line of the segment.	-	N	
				EndSample	R	INT	End (last) sample of the segment.	-	N	
				Identifier	R	TXT	Identifier for the segment data boundary defined above.	-	N	
				SegmentPolygon	O		Simple polygon that describes a portion of the segment rectangle with vertices in clockwise order. Attribute "size" = NumVertices.		N	size = NumVertices NumVertices ≥ 3
				SV	R	LS	Vertex specified in line and sample. First and last vertices connect.	-	Y	index = "1", "2", ..., NumVertices
				Orientation	O	ENU	Describes the shadow intent of the display plane defined above.	-	N	Allowed values: "UP", "DOWN" "LEFT", "RIGHT", "ARBITRARY"
				Polygon	O		Simple polygon bounded by image area rectangle that refines the imaged area with vertices in clockwise order. Attribute "size" = NumVertices.		N	size = NumVertices NumVertices ≥ 3
				Vertex	R	LS	Vertex specified in line and sample. First and last vertices connect.	-	Y	index = "1", "2", ..., NumVertices

Table 3-7 Radar Collection Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		Parameter	O	TXT	Free format field that can be used to pass forward the radar collection information.	-	Y	name = "xxx" "xxx" is a descriptive identifier for this information

3.2.8 XML: ImageFormation

Table 3-8 Image Formation Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD									
	ImageFormation			R		This block describes the image formation process.	-	N	
		RcvChanProc		R		Parameters of the received processed channel	-	N	
			NumChanProc	R	INT	Number of receive data channels processed to form the image.	-	N	
			PRFScaleFactor	O	DBL	Factor indicating the ratio of the effective PRF to the actual PRF.	-	N	
			ChanIndex	R	INT	Index of a data channel that was processed.	-	Y	

Table 3-8 Image Formation Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
	TxRcvPolarizationProc	R	ENU	Indicates the combined transmit and receive polarization processed to form the image. See Table 4-1.	-	N	Allowed values include the form TX:RCV that is formed from one TX value and one RCV value. Allowed TX values: "V", "H", "X", "Y", "S", "E", "RHC", "LHC", "OTHER*" Allowed RCV values: "V", "H", "X", "Y", "S", "E", "RHC", "LHC", "OTHER*" Allowed values also include: "OTHER", "UNKNOWN" Use of "OTHER*" defined per Program Specific Implementation Document where * = 0 or more characters that does not contain "." (0x3A).
	TStartProc	R	DBL	Earliest slow time value for data processed to form the image from CollectionStart	sec	N	
	TEndProc	R	DBL	Latest slow time value for data processed to form the image from CollectionStart.	sec	N	

Table 3-8 Image Formation Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		TxFrequencyProc	R		The range of transmit frequency processed to form the image.	-	N	
		MinProc	R	DBL	Minimum transmit frequency processed to form the image. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
		MaxProc	R	DBL	Maximum transmit frequency processed to form the image. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
		SegmentIdentifier	O	TXT	Identifier that describes the image that was processed. Shall be included when SICD.RadarCollection.Area.Plane.SegmentList is included.	-	N	
		ImageFormAlgo	R	ENU	Image formation algorithm used. PFA ⇔ Polar Format Algorithm. RMA ⇔ Range Migration (Omega-K, Chirp Scaling, Range-Doppler) RGAZCOMP ⇔ Simple range, Doppler compression.	-	N	Allowed values: "PFA", "RMA", "RGAZCOMP", "OTHER"

Table 3-8 Image Formation Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		STBeamComp	R	ENU	Parameter indicating if slow time beam shape compensation has been applied.	-	N	Allowed values: “NO” = No ST beam shape compensation. “GLOBAL” = Global ST beam shape compensation applied. “SV” = Spatially variant beam shape compensation applied.
		ImageBeamComp	R	ENU	Parameter indicating if image domain beam shape compensation has been applied.	-	N	Allowed values: “NO” = No image domain beam shape compensation. “SV” = Spatially variant image domain beam shape compensation applied.
		AzAutofocus	R	ENU	Parameter indicating if azimuth autofocus correction has been applied.	-	N	Allowed values: “NO” = No azimuth autofocus applied. “GLOBAL” = Global azimuth autofocus applied. “SV” = Spatially variant azimuth autofocus applied.
		RgAutofocus	R	ENU	Parameter indicating if range autofocus correction has been applied.	-	N	Allowed Values: “NO” = No range autofocus applied. “GLOBAL” = Global range autofocus applied. “SV” = Spatially variant range autofocus applied.

Table 3-8 Image Formation Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		Processing	O		Parameters to describe types of specific processing that may have been applied such as additional compensations.		Y	
		Type	R	TXT	Text identifier for the type of processing algorithm."		N	
		Applied	R	BOOL	Parameter indicating if the processing has been applied.		N	Allowed values: "false", "true"
		Parameter	O	TXT	Free format field that can be used to pass forward processing information.		Y	name = "xxxx", Name is a descriptive identifier for this information.
		PolarizationCalibration	O		Polarization calibration parameter block.	-	N	
		DistortCorrectionApplied	R	BOOL	Parameter indicating if polarization calibration parameters have been applied.	-	N	true ⇔ Polarization channel distortion correction processing applied. false ⇔ Not applied.
		Distortion	R				N	
Polarization distortion model related the observed scattering matrix {O} to the true scattering matrix {S}. $\underbrace{\begin{bmatrix} O_{HH} & O_{VH} \\ O_{HV} & O_{VV} \end{bmatrix}}_{\text{Observed Scattering Matrix}} = A \cdot \underbrace{\begin{bmatrix} 1 & Q1 \\ Q2 & F1 \end{bmatrix}}_{\text{Receive Distortion Matrix}} \cdot \underbrace{\begin{bmatrix} S_{HH} & S_{VH} \\ S_{HV} & S_{VV} \end{bmatrix}}_{\text{True Scattering Matrix}} \cdot \underbrace{\begin{bmatrix} 1 & Q4 \\ Q3 & F2 \end{bmatrix}}_{\text{Transmit Distortion Matrix}}$								
		CalibrationDate	O	XDT	Date of the calibration measurement.	-	N	
		A	R	DBL	Absolute amplitude scale factor.	-	N	

Table 3-8 Image Formation Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				F1	R	CMPLX	Receive distortion element (2,2).	-	N	
				Q1	R	CMPLX	Receive distortion element (1,2).	-	N	
				Q2	R	CMPLX	Receive distortion element (2,1).	-	N	
				F2	R	CMPLX	Transmit distortion element (2,2).	-	N	
				Q3	R	CMPLX	Transmit distortion element (2,1).	-	N	
				Q4	R	CMPLX	Transmit distortion element (1,2).	-	N	
				GainErrorA	O	DBL	Gain estimation error standard deviation (in dB) for parameter A.	dB	N	
				GainErrorF1	O	DBL	Gain estimation error standard deviation (in dB) for parameter F1.	dB	N	
				GainErrorF2	O	DBL	Gain estimation error standard deviation (in dB) for parameter F2.	dB	N	
				PhaseErrorF1	O	DBL	Phase estimation error standard deviation (in radians) for parameter F1.	rad	N	
				PhaseErrorF2	O	DBL	Phase estimation error standard deviation (in radians) for parameter F2.	rad	N	

3.2.9 XML: SCPCOA

Table 3-9 SCP Center Of Aperture Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
	SCPCOA	R		Parameters describing the COA for the Scene Center Point.	-	N	
	SCPTIME	R	DBL	Center Of Aperture time for the SCP t_COA_SCP (from collection start).	sec	N	
Time dependent parameters computed for the ARP at the SCP COA time (t_COA_SCP).							
	ARPPos	R	XYZ	ARP position at t_COA_SCP in ECF.	m	N	
	ARPVel	R	XYZ	ARP Velocity at t_COA_SCP in ECF.	m/s	N	
	ARPAcc	R	XYZ	ARP Acceleration at t_COA_SCP in ECF.	m/s ²	N	
	SideOfTrack	R	ENU	Side of track parameter. Allowed: L or R	-	N	Allowed: "L", "R"
	SlantRange	R	DBL	Slant range from the ARP to the SCP.	m	N	
	GroundRange	R	DBL	Ground Range from the ARP nadir to the SCP. Distance measured along spherical earth model passing through the SCP.	m	N	
	DopplerConeAng	R	DBL	The Doppler Cone Angle to SCP at t_COA_SCP.	deg	N	
Parameters computed for the ARP position and velocity relative to the Earth Tangent Plane (ETP) at the SCP.							
	GrazeAng	R	DBL	Grazing Angle between the SCP Line of Sight (LOS) and Earth Tangent Plane (ETP)	deg	N	Range: [-90:90]
	IncidenceAng	R	DBL	Incidence Angle between the SCP LOS and ETP normal.	deg	N	Range: [0:180]
	TwistAng	R	DBL	Angle between cross range in the ETP and cross range in the slant plane.	deg	N	Range: [-90:90]
	SlopeAng	R	DBL	Slope Angle from the ETP to the slant plane at t_COA_SCP.	deg	N	Range: [0:90]

Table 3-9 SCP Center Of Aperture Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		AzimAng		R	DBL	Angle from north to the line from the SCP to the ARP Nadir at COA. Measured clockwise in the ETP. AzimuthAng = 45 ➔ ARP is northeast of the SCP at COA.	deg	N	Range: [0:360]
		LayoverAng		R	DBL	Angle from north to the layover direction in the ETP at COA. Measured clockwise in the ETP. LayoverAng = 45 ➔ A point above the ETP will layover along the line northeast of the SCP. Layover Distance = Height above ETP x tan(Slope Angle).	deg	N	Range: [0:360]
Where the SICD CollectType is BISTATIC, the SICD XML shall include an SCPCOA.Bistatic block.									
		Bistatic		O		Parameters that describe bistatic geometry at COA for the SCP.		N	
			BistaticAng	R	DBL	Bistatic angle computed from SCP COA geometry.	deg	N	Range: [0:180]
			BistaticAngRate	R	DBL	Instantaneous rate of change of the bistatic angle.	deg/s	N	
			TxPlatform	R		Parameters that describe the Transmit platform geometry relative to SCP at time $t = tx_{COA}^{SCP}$.		N	
			Time	R	DBL	Transmit time $t = tx_{COA}^{SCP}$	sec	N	
			Pos	R	XYZ	Transmit APC position at $t = tx_{COA}^{SCP}$	m	N	
			Vel	R	XYZ	Transmit APC velocity at $t = tx_{COA}^{SCP}$	m/s	N	
			Acc	R	XYZ	Transmit APC acceleration at $t = tx_{COA}^{SCP}$	m/s ²	N	

Table 3-9 SCP Center Of Aperture Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				SideOfTrack	R	ENU	Side of Track for the Tx platform.	-	N	Allowed: "L", "R"
				SlantRange	R	DBL	Slant range from the Tx APC to the SCP	m	N	
				GroundRange	R	DBL	Ground range from the Tx APC to the SCP	m	N	
				DopplerConeAng	R	DBL	Doppler Cone Angle to SCP	deg	N	
				GrazeAng	R	DBL	Grazing Angle for the Tx APC to the SCP LOS relative to the SCP ETP.	deg	N	Range: [-90:90]
				IncidenceAng	R	DBL	Incidence Angle for Tx APC to SCP LOS and SCP ETP.	deg	N	Range: [0:180]
				AzimAng	R	DBL	Angle from north for the line from SCP to the Tx APC projected onto the SCP ETP. Measured clockwise in the ETP.	deg	N	Range: [0:360]
				RcvPlatform	R		Parameters that describe the Receive platform geometry relative to SCP at time $t = tr_{COA}^{SCP}$.		N	
				Time	R	DBL	Receive time $t = tr_{COA}^{SCP}$	sec	N	
				Pos	R	XYZ	Receive APC position at $t = tr_{COA}^{SCP}$	m	N	
				Vel	R	XYZ	Receive APC velocity at $t = tr_{COA}^{SCP}$	m/s	N	
				Acc	R	XYZ	Receive APC acceleration at $t = tr_{COA}^{SCP}$	m/s ²	N	
				SideOfTrack	R	ENU	Side of Track for the Rcv platform	-	N	Allowed: "L", "R"
				SlantRange	R	DBL	Slant range from the Rcv APC to the SCP	m	N	
				GroundRange	R	DBL	Ground range from the Rcv APC to the SCP	m	N	
				DopplerConeAng	R	DBL	Doppler Cone Angle to SCP	deg	N	

Table 3-9 SCP Center Of Aperture Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				GrazeAng	R	DBL	Grazing Angle for the Rcv APC to the SCP LOS relative to the SCP ETP.	deg	N	Range: [-90:90]
				IncidenceAng	R	DBL	Incidence Angle for Rcv APC to SCP LOS and SCP ETP.	deg	N	Range: [0:180]
				AzimAng	R	DBL	Angle from north for the line from SCP to the Rcv APC projected onto the SCP ETP. Measured clockwise in the ETP.	deg	N	Range: [0:360]

3.2.10 XML: Radiometric

Table 3-10 Radiometric Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
SICD									
	Radiometric			O		Radiometric calibration parameters.	-	N	
Image Pixel Power: $P_{wrn}(\text{row,col}) = \text{Real}(S_{II}(\text{row,col}))^2 + \text{Imag}(S_{II}(\text{row,col}))^2$.									
		NoiseLevel			O		Noise Level Structure	-	N
			NoiseLevelType	R	ENU	Parameter to indicate that the noise power polynomial yields either absolute power level or power level relative to the SCP pixel location.	-	N	Allowed values: "ABSOLUTE", "RELATIVE"
			NoisePoly	R	2D_POLY	Polynomial coefficients that yield thermal noise power (in dB) in a pixel as a function of image row coordinate (variable 1) and column coordinate (variable 2).	dB, dB/m, etc.	N	order1 = "M" order2 = "N"
		RCSSFPoly			O	2D_POLY	Polynomial coefficients that yield a scale factor to convert pixel power to RCS (sqm) as a function of image row coordinate (variable 1) and column coordinate (variable 2). Scale factor computed for a target at HAE = SCP_HAE.	m ² , m, 1, 1/m, etc.	N
		SigmaZeroSFPoly			O	2D_POLY	Polynomial coefficients that yield a scale factor to convert pixel power to clutter parameter Sigma-Zero (σ^0) as a function of image row coordinate (variable 1) and column coordinate (variable 2). Scale factor computed for a clutter cell at HAE = SCP_HAE.	1, 1/m, 1/m ² etc.	N

Table 3-10 Radiometric Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes
		BetaZeroSFPoly	O	2D_POLY	Polynomial coefficients that yield a scale factor to convert pixel power to radar brightness or Beta-Zero (β_0) as a function of image row coordinate (variable 1) and column coordinate (variable 2). Scale factor computed for a clutter cell at HAE = SCP_HAE.	1, 1/m, 1/m ² etc.	N	order1 = "M" order2 = "N"
		GammaZeroSFPoly	O	2D_POLY	Polynomial coefficients that yield a scale factor to convert pixel power to clutter parameter Gamma-Zero (γ_0) as a function of image row coordinate (variable 1) and column coordinate (variable 2). Scale factor computed for a clutter cell at HAE = SCP_HAE.	1, 1/m, 1/m ² etc.	N	order1 = "M" order2 = "N"

3.2.11 XML: Antenna

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
SICD								N	
	Antenna			O		Parameters that describe the antenna illumination patterns during the collection.		N	
Parameters may be provided separately for the transmit (Tx) antenna and the receive (Rcv) antenna. Where the SICD XML has Tx and Rcv patterns, the SICD XML shall include Tx APC and Rcv APC position polynomials. See Table 3-5.									
		Tx		O		Transmit Antenna Parameters		N	
			XAxisPoly	R	XYZ_POLY	Antenna X-Axis unit vector in ECF as a function of time (variable 1). Time t = 0 at collection start.	1, 1/s, etc.	N	
			YAxisPoly	R	XYZ_POLY	Antenna Y-Axis unit vector in ECF as a function of time (variable 1). Time t = 0 at collection start.	1, 1/s, etc.	N	
			FreqZero	R	DBL	RF frequency (f0) used to specify the array pattern and EB steering direction cosines. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
			EB	O		Electrical boresight (EB) steering directions for an electronically steered array.		N	
			DCXPoly	R	POLY	Electrical boresight steering X-axis direction cosine (DCX) as a function of slow time (variable 1).	1, 1/s, etc.	N	order1 = "M"

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
			DCYPoly	R	POLY	Electrical boresight steering Y-axis direction cosine (DCY) as a function of slow time (variable 1).	1, 1/s, etc.	N	order1 = "M"
			Array	R		Array pattern polynomials that define the shape of the mainlobe.		N	
			GainPoly	R	2D_POLY	One-way signal gain (in dB) as a function of Δ DCX (variable 1) and Δ DCY (variable 2). Gain relative to gain at Δ DCX = 0 and Δ DCY = 0. Constant coefficient = 0.0 always.	dB	N	order1 = "M" order2 = "N"
			PhasePoly	R	2D_POLY	One-way signal phase (in cycles) as a function of Δ DCX (variable 1) and Δ DCY (variable 2). Phase relative to phase at Δ DCX = 0 and Δ DCY = 0. Constant coefficient = 0.0 always.	cycles	N	order1 = "M" order2 = "N"
			Elem	O		Element array pattern polynomials for electronically steered arrays.		N	
			GainPoly	R	2D_POLY	One-way signal gain (in dB) as a function of DCX (variable 1) and DCY (variable 2). Gain relative to gain at DCX = 0 and DCY = 0. Constant coefficient = 0.0 always.	dB	N	order1 = "M" order2 = "N"
			PhasePoly	R	2D_POLY	One-way signal phase (in cycles) as a function of DCX (variable 1) and DCY (variable 2). Phase relative to phase at DCX = 0 and DCY = 0. Constant coefficient = 0.0 always.	cycles	N	order1 = "M" order2 = "N"

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
			GainBSPoly	O	POLY	Gain polynomial (dB) vs. frequency for boresight (BS) at DCX = 0 and DCY = 0. Frequency ratio ((f-f0)/f0) input variable (variable 1). Constant coefficient = 0.0 always. GainBS(f) = c1*((f-f0)/f0) + c2*((f-f0)/f0)**2 + etc.	dB	N	order1 = "M"
			EBFreqShift	O	BOOL	Parameter indicating the EB shifts with frequency for an electronically steered array. false ⇔ No shift with frequency. true ⇔ Shift with frequency per ideal array theory.		N	Allowed values: "false", "true"
			MLFreqDilation	O	BOOL	Parameter indicating the mainlobe (ML) width changes with frequency. false ⇔ No change with frequency. true ⇔ Change with frequency per ideal array theory.		N	Allowed values: "false", "true"
		Rcv		O		Receive Antenna Parameters		N	
			XAxisPoly	R	XYZ_POLY	Antenna X-Axis unit vector in ECF as a function of time (variable 1). Time t = 0 at collection start.	1, 1/s, etc.	N	
			YAxisPoly	R	XYZ_POLY	Antenna Y-Axis unit vector in ECF as a function of time (variable 1). Time t = 0 at collection start.	1, 1/s, etc.	N	
			FreqZero	R	DBL	RF frequency (f0) used to specify the array pattern and EB steering direction cosines. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
			EB	O		Electrical boresight (EB) steering directions for an electronically steered array.		N	
			DCXPoly	R	POLY	Electrical boresight steering X-axis direction cosine (DCX) as a function of slow time (variable 1).	1, 1/s, etc.	N	order1 = "M"
			DCYPoly	R	POLY	Electrical boresight steering Y-axis direction cosine (DCY) as a function of slow time (variable 1).	1, 1/s, etc.	N	order1 = "M"
			Array	R		Array pattern polynomials that define the shape of the mainlobe.		N	
			GainPoly	R	2D_POLY	One-way signal gain (in dB) as a function of Δ DCX (variable 1) and Δ DCY (variable 2). Gain relative to gain at Δ DCX = 0 and Δ DCY = 0. Constant coefficient = 0.0 always.	dB	N	order1 = "M" order2 = "N"
			PhasePoly	R	2D_POLY	One-way signal phase (in cycles) as a function of Δ DCX (variable 1) and Δ DCY (variable 2). Phase relative to phase at Δ DCX = 0 and Δ DCY = 0. Constant coefficient = 0.0 always.	cycles	N	order1 = "M" order2 = "N"
			Elem	O		Element array pattern polynomials for electronically steered arrays.		N	
			GainPoly	R	2D_POLY	One-way signal gain (in dB) as a function of DCX (variable 1) and DCY (variable 2). Gain relative to gain at DCX = 0 and DCY = 0. Constant coefficient = 0.0 always.	dB	N	order1 = "M" order2 = "N"

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
			PhasePoly	R	2D_POLY	One-way signal phase (in cycles) as a function of DCX (variable 1) and DCY (variable 2). Phase relative to phase at DCX = 0 and DCY = 0. Constant coefficient = 0.0 always.	cycles	N	order1 = "M" order2 = "N"
			GainBSPoly	O	POLY	Gain polynomial (dB) vs. frequency for boresight (BS) at DCX = 0 and DCY = 0. Frequency ratio ((f-f0)/f0) input variable (variable 1). Constant coefficient = 0.0 always. $\text{GainBS}(f) = c1*((f-f0)/f0) + c2*((f-f0)/f0)**2 + \text{etc.}$	dB	N	order1 = "M"
			EBFreqShift	O	BOOL	Parameter indicating the EB shifts with frequency for an electronically steered array. false ⇔ No shift with frequency. true ⇔ Shift with frequency per ideal array theory.		N	Allowed values: "false", "true"
			MLFreqDilation	O	BOOL	Parameter indicating the mainlobe (ML) width changes with frequency. false ⇔ No change with frequency. true ⇔ Change with frequency per ideal array theory.		N	Allowed values: "false", "true"
For an equivalent Two-Way pattern, the ARP position versus time is the effective phase center. Where the SICD CollectType is BISTATIC, the SICD XML shall not include a TwoWay pattern.									
			TwoWay	O		Two-way Antenna Parameters		N	
			XAxisPoly	R	XYZ_POLY	Antenna X-Axis unit vector in ECF as a function of time (variable 1). Time t = 0 at collection start.	1, 1/s, etc.	N	

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
			YAxisPoly	R	XYZ_POLY	Antenna Y-Axis unit vector in ECF as a function of time (variable 1). Time t = 0 at collection start.	1, 1/s, etc.	N	
			FreqZero	R	DBL	RF frequency (f0) used to specify the array pattern and EB steering direction cosines. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
			EB	O		Electrical boresight (EB) steering directions for an electronically steered array.		N	
			DCXPoly	R	POLY	Electrical boresight steering X-axis direction cosine (DCX) as a function of slow time (variable 1).	1, 1/s, etc.	N	order1 = "M"
			DCYPoly	R	POLY	Electrical boresight steering Y-axis direction cosine (DCY) as a function of slow time (variable 1).	1, 1/s, etc.	N	order1 = "M"
			Array	R		Array pattern polynomials that define the shape of the mainlobe.		N	
			GainPoly	R	2D_POLY	Two-way signal gain (in dB) as a function of ΔDCX (variable 1) and ΔDCY (variable 2). Gain relative to gain at ΔDCX = 0 and ΔDCY = 0. Constant coefficient = 0.0 always.	dB	N	order1 = "M" order2 = "N"
			PhasePoly	R	2D_POLY	Two-way signal phase (in cycles) as a function of ΔDCX (variable 1) and ΔDCY (variable 2). Phase relative to phase at ΔDCX = 0 and ΔDCY = 0. Constant coefficient = 0.0 always.	cycles	N	order1 = "M" order2 = "N"

Table 3-11 Antenna Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes
			Elem	O		Element array pattern polynomials for electronically steered arrays.		N	
			GainPoly	R	2D_POLY	Two-way signal gain (in dB) as a function of DCX (variable 1) and DCY (variable 2). Gain relative to gain at DCX = 0 and DCY = 0. Constant coefficient = 0.0 always.	dB	N	order1 = "M" order2 = "N"
			PhasePoly	R	2D_POLY	Two-way signal phase (in cycles) as a function of DCX (variable 1) and DCY (variable 2). Phase relative to phase at DCX = 0 and DCY = 0. Constant coefficient = 0.0 always.	cycles	N	order1 = "M" order2 = "N"
			GainBSPoly	O	POLY	Gain polynomial (dB) vs. frequency for boresight (BS) at DCX = 0 and DCY = 0. Frequency ratio ((f-f ₀)/f ₀) input variable (variable 1). Constant coefficient = 0.0 always. GainBS(f) = c1*((f-f ₀)/f ₀) + c2*((f-f ₀)/f ₀)**2 + etc.	dB	N	order1 = "M"
			EBFreqShift	O	BOOL	Parameter indicating the EB shifts with frequency for an electronically steered array. false ⇔ No shift with frequency. true ⇔ Shift with frequency per ideal array theory.		N	Allowed values: "false", "true"
			MLFreqDilation	O	BOOL	Parameter indicating the mainlobe (ML) width changes with frequency. false ⇔ No change with frequency. true ⇔ Change with frequency per ideal array theory.		N	Allowed values: "false", "true"

3.2.12 XML: ErrorStatistics

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD									
	ErrorStatistics			O		Parameters used to compute error statistics within the SICD sensor model. May also include Adjustable Parameter Offsets (APOs) to compensate for image projection errors.		N	
Where the SICD CollectType is not BISTATIC, the SICD XML shall not include BistaticCompositeSCP. Where the SICD CollectType is BISTATIC, the SICD XML shall not include CompositeSCP.									
		CompositeSCP		OC		Composite projection error statistics for the Scene Center Point for a Monostatic image. Errors in the SCP COA slant plane.		N	
			Rg	R	DBL	Range error (ΔRg) standard deviation.	m	N	
			Az	R	DBL	Azimuth error (ΔAz) standard deviation.	m	N	
			RgAz	R	DBL	Range error and azimuth error correlation coefficient.		N	
		BistaticCompositeSCP		OC		Composite projection error statistics for the SCP for a Bistatic image.		N	
			RAvg	R	DBL	Average range error (ΔR_Avg) standard deviation.	m	N	
			RdotAvg	R	DBL	Average range rate error ($\Delta Rdot_Avg$) standard deviation.	m/s	N	
			RAvgRdotAvg	R	DBL	Average range error and average range rate error correlation coefficient.		N	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
Where the SICD CollectType is not BISTATIC, the SICD XML shall not include BistaticComponents. Where the SICD CollectType is BISTATIC, the SICD XML shall not include Components.									
		Components			OC		Error statistics for the primary components of image projection error for MONOSTATIC images.		N
			PosVelErr		R		ARP position and velocity error statistics for the image.		N
			Frame		R	ENU	Coordinate frame used for expressing P,V errors statistics. For an RIC frame, RIC to ECF defined at SCP COA time.		N Allowed values: "ECF", "RIC_ECF", "RIC_ECI". RIC = Radial, In-Track, Cross-Track. Radial – From earth center through the platform position.
			P1		R	DBL	Position coordinate 1 standard deviation.	m	N
			P2		R	DBL	Position coordinate 2 standard deviation.	m	N
			P3		R	DBL	Position coordinate 3 standard deviation.	m	N
			V1		R	DBL	Velocity coordinate 1 standard deviation.	m/sec	N
			V2		R	DBL	Velocity coordinate 2 standard deviation.	m/sec	N
			V3		R	DBL	Velocity coordinate 3 standard deviation.	m/sec	N
			CorrCoefs		O		Correlation Coefficient parameters		N
				P1P2	R	DBL	P1, P2 correlation coefficient.		N
				P1P3	R	DBL	P1, P3 correlation coefficient.		N
				P1V1	R	DBL	P1, V1 correlation coefficient.		N
				P1V2	R	DBL	P1, V2 correlation coefficient.		N
				P1V3	R	DBL	P1, V3 correlation coefficient.		N
				P2P3	R	DBL	P2, P3 correlation coefficient.		N

Table 3-12 Error Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				P2V1	R	DBL	P2, V1 correlation coefficient.		N	
				P2V2	R	DBL	P2, V2 correlation coefficient.		N	
				P2V3	R	DBL	P2, V3 correlation coefficient.		N	
				P3V1	R	DBL	P3, V1 correlation coefficient.		N	
				P3V2	R	DBL	P3, V2 correlation coefficient.		N	
				P3V3	R	DBL	P3, V3 correlation coefficient.		N	
				V1V2	R	DBL	V1, V2 correlation coefficient.		N	
				V1V3	R	DBL	V1, V3 correlation coefficient.		N	
				V2V3	R	DBL	V2, V3 correlation coefficient.		N	
				PositionDecorr	O		Platform position error decorrelation function.		N	
				CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).		N	
				DecorrRate	R	DBL	Error decorrelation rate. Simple linear decorrelation rate (DCR). $\Delta t = t_2 - t_1 $ $CC(\Delta t) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR * \Delta t))$	1/sec	N	
				RadarSensor	R		Radar sensor error statistics.		N	
				RangeBias	R	DBL	Range bias error (ΔR_Bias) standard deviation.	m	N	
				ClockFreqSF	O	DBL	Payload control clock frequency error (Δf_Clk) expressed as a scale factor (f_Clk_SF) standard deviation. $f_Clk_SF = \Delta f_Clk / f_Clk_0$ Frequency f_Clk_0 = Clock frequency used in the PHD processing.		N	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			TransmitFreqSF	O	DBL	Transmit frequency scale factor standard deviation. $SF = \Delta f/f_0$. V1.4.0: Deprecated – Not used in SICD error model.	-	N	
			RangeBiasDecorr	O		Range bias decorrelation rate.		N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N	
			DecorrRate	R	DBL	Error decorrelation rate. Simple linear decorrelation rate (DCR). $\Delta t = t_2 - t_1 $ $CC(\Delta t) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR * \Delta t))$	1/sec	N	
			TropoError	O		Troposphere delay error statistics.		N	
			TropoRangeVertical	O	DBL	Troposphere two-way delay error for normal incidence standard deviation. Expressed as a range error. $\Delta R = \Delta T * c/2$. V1.4.0: Deprecated – Use TropoRangeSlant	m	N	
			TropoRangeSlant	O	DBL	Troposphere two-way delay error for the SCP line of sight at COA standard deviation. Expressed as a range error. $\Delta R = \Delta T * c/2$.	m	N	
			TropoRangeDecorr	O				N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N	
			DecorrRate	R	DBL	Error decorrelation rate. Simple linear decorrelation rate (DCR). $\Delta t = t_2 - t_1 $ $CC(\Delta t) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR * \Delta t))$	1/sec	N	
			IonoError	O		Ionosphere delay error statistics.		N	

Table 3-12 Error Parameters											
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values	
				lonoRangeVertical	O	DBL	Ionosphere two-way delay error for normal incidence standard deviation. Expressed as a range: $\Delta R = \Delta T \cdot c/2$.	m	N		
				lonoRangeRateVertical	O	DBL	Ionosphere two-way delay rate of change error for normal incidence standard deviation. Expressed as a range rate error. $\Delta R_{dot} = \Delta T_{dot} \times c/2$.	m/sec	N		
				lonoRgRgRateCC	R	DBL	Ionosphere range error and range rate error correlation coefficient.	-	N		
				lonoRangeVertDecorr	O		Ionosphere range error decorrelation rate.		N		
				CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N		
				DecorrRate	R	DBL	Error decorrelation rate. Simple linear decorrelation rate (DCR). $\Delta t = t_2 - t_1 $ $CC(\Delta t) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR \cdot \Delta t))$	1/sec	N		
		BistaticComponents			OC		Error statistics for the primary components of image projection error for BISTATIC images.		N		
			PosVelErr		R		Position & Velocity Error statistics for transmit and receive platforms.		N		
				TxFrma	R	ENU	Coordinate frame used for TxAPC P,V error statistics. For an RIC frame, RIC to ECEF defined at SCP COA time.		N	Allowed values: "ECF", "RIC_ECF", "RIC_ECI".	
				TxPVCov	R	MTX	TxAPC position & velocity error covariance matrix.		N	size1 = "6" and size2 = "6"	
				RcvFrame	R	ENU	Coordinate frame used for RcvAPC P,V errors statistics. For an RIC frame, RIC to ECEF defined at SCP COA time.		N	Allowed values: "ECF", "RIC_ECF", "RIC_ECI".	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			RcvPVCov	R	MTX	RcvAPC position & velocity error covariance matrix.		N	size1 = "6" and size2 = "6"
			TxRcvPVXCov	R	MTX	Tx PV error/ Rcv PV error cross covariance matrix.		N	size1 = "6" and size2 = "6"
			RadarSensor	R		Radar sensor error statistics for transmit timing & receive timing.		N	
			TxRcvTimeFreq	R	MTX	Transmit & receive timing error covariance matrix.		N	size1 = "4" and size2 = "4"
			TxRcvTimeFreqDecorr	O		Decorrelation rate parameters for transmit & receive timing errors. Simple linear decorrelation model: $\Delta t = t_2 - t_1 $ $CC(\Delta t) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR * \Delta t))$			
			TxTimeDecorr	R		Transmit time error ($\Delta t_x(t)$) decorrelation parameters.		N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N	
			DecorrRate	R	DBL	Error decorrelation rate (DCR).	1/sec	N	
			TxClockFreqDecorr	R		Transmit clock frequency error SF (f_Clk_X_SF) decorrelation parameters.		N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N	
			DecorrRate	R	DBL	Error decorrelation rate (DCR).	1/sec	N	
			RcvTimeDecorr	R		Receive time error ($\Delta t_r(t)$) decorrelation parameters.		N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N	
			DecorrRate	R	DBL	Error decorrelation rate (DCR).	1/sec	N	

Table 3-12 Error Parameters										
Field Name					Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
				RcvClockFreqDecorr	R		Receive clock frequency error SF (f_Clk_R_SF) decorrelation parameters.		N	
				CorrCoefZero	R	DBL	Error correlation coefficient for zero time difference (CC0).	-	N	
				DecorrRate	R	DBL	Error decorrelation rate (DCR).	1/sec	N	
			AtmosphericError		R		Atmospheric delay compensation error statistics. For spaceborne sensors, the atmospheric delay estimate includes the tropospheric delay and the ionospheric delay estimates.		N	
				TxSCP	R	DBL	Atmospheric delay error on transmit (ΔTD_Atm_X) standard deviation. Error for the TxAPC-to-SCP path at $t = t_{x_COA}^{SCP}$.	sec	N	
				RcvSCP	R	DBL	Atmospheric delay error on receive (ΔTD_Atm_R) standard deviation. Error for the RcvAPC-to-SCP path at $t = t_{r_COA}^{SCP}$.	sec	N	
				TxRcvCC	R	DBL	Transmit atmospheric delay error (ΔTD_Atm_X) / Receive atmospheric delay error (ΔTD_Atm_R) correlation coefficient.	-	N	
		Unmodeled			O		Unmodeled Error statistics at the SCP pixel location (xrow, ycol) = (0, 0).		N	Only included when also including monostatic or bistatic Component error statistics.
				Xrow	R	DBL	Estimated unmodeled error standard deviation in the xrow dimension of the complex image.	m	N	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			Ycol	R	DBL	Estimated unmodeled error standard deviation in the ycol dimension of the complex image.	m	N	
			XrowYcol	R	DBL	Estimated row error and column error correlation coefficient.		N	
			UnmodeledDecorr	O		Unmodeled error decorrelation parameters.		N	
			Xrow	R		Decorrelation parameters in the xrow dimension.		N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero distance in the xrow dimension between the point pair (CC0).		N	
			DecorrRate	R	DBL	Error decorrelation rate. Simple linear decorrelation rate (DCR). $\Delta xrow = xrow2 - xrow1 $ $CC(\Delta xrow) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR * \Delta xrow))$	1/m	N	
			Ycol	R		Decorrelation parameters in the ycol dimension.		N	
			CorrCoefZero	R	DBL	Error correlation coefficient for zero distance in the ycol dimension between the point pair (CC0).		N	
			DecorrRate	R	DBL	Error decorrelation rate. Simple linear decorrelation rate (DCR). $\Delta ycol = ycol2 - ycol1 $ $CC(\Delta ycol) = \text{Min}(1.0, \text{Max}(0.0, CC0 - DCR * \Delta ycol))$	1/m	N	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
		AdditionalParms		O		Additional user defined errors parameters.		N	
			Parameter	R	TXT	Free format field that can be used to include additional parameters.	~	Y	name = "xxx", name is a descriptive identifier for this information
Where the SICD CollectType is not BISTATIC, the SICD XML shall not include BistaticAdjustableParameterOffsets. Where the SICD CollectType is BISTATIC, the SICD XML shall not include AdjustableParameterOffsets.									
		AdjustableParameterOffsets		OC		Adjustable Parameter Offsets that may be included for a MONOSTATIC image. See SICD Vol 3 Image Projections DD.		N	
			ARPPosSCPCOA	R	XYZ	ARP Position offset at t = t_COA_SCP in ECEF coordinates.	m	N	
			ARPVel	R	XYZ	ARP Velocity offset in ECEF coordinates that is constant for the image.	m/sec	N	
			TxTimeSCPCOA	R	DBL	Transmit time offset (Δtx_{COA}^{SCP}). Assumed constant for the collect.	sec	N	
			RcvTimeSCPCOA	R	DBL	Receive time offset (Δtr_{COA}^{SCP}). Assumed constant for the collect. Time offsets used to compute range offset: $\Delta R_ARP = (c/2) \cdot (\Delta tr_{COA}^{SCP} - \Delta tx_{COA}^{SCP})$	sec	N	
			APOError	O	MTX	APO Error Covariance matrix for the set of monostatic APOs.		N	size1 = "8" and size2 = "8"
			CompositeSCP	O		Composite projection error statistics for the SCP pixel projected the SCP ETP using the provided APOs.		N	
			Rg	R	DBL	Range error (ΔRg) standard deviation.	m	N	
			Az	R	DBL	Azimuth error (ΔAz) standard deviation.	m	N	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			RgAz	R	DBL	Range error and azimuth error correlation coefficient.		N	
			BistaticAdjustableParameterOffsets	OC		Adjustable Parameter Offsets that may be included for a BISTATIC image. See SICD Vol 3 Image Projections DD.		N	
			TxPlatform	R		Transmit / Illuminator platform Adjustable Parameter Offsets		N	
			APCPosSCPCOA	R	XYZ	Transmit APC Position offset at $t = t_COA_SCP$ in ECEF coordinates.	m	N	
			APCVel	R	XYZ	Transmit APC Velocity offset in ECEF coordinates that is constant for the image.	m/sec	N	
			TimeSCPCOA	R	DBL	Transmit time offset (Δt_{COA}^{SCP}) at $t = t_COA_SCP$.	sec	N	
			ClockFreqSF	R	DBL	Transmit payload control clock frequency offset (Δf_Clk_X) expressed as a scale factor ($f_Clk_X_SF$). $\Delta f_Clk_X = f_Clk_X_SF \times f_Clk_X_0$ Frequency $f_Clk_X_0$ = Clock frequency used in the PHD processing.		N	
			RcvPlatform	R		Receive / Collector platform Adjustable Parameter Offsets		N	
			APCPosSCPCOA	R	XYZ	Receive APC Position offset at $t = t_COA_SCP$ in ECEF coordinates.	m	N	
			APCVel	R	XYZ	Receive APC Velocity offset in ECEF coordinates that is constant for the image.	m/sec	N	
			TimeSCPCOA	R	DBL	Receive time offset (Δt_{COA}^{SCP}) at $t = t_COA_SCP$.	sec	N	

Table 3-12 Error Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			ClockFreqSF	R	DBL	Receive payload control clock frequency offset (Δf_Clk_R) expressed as a scale factor ($f_Clk_R_SF$). $\Delta f_Clk_R = f_Clk_R_SF \times f_Clk_R_0$ Frequency $f_Clk_R_0$ = Clock frequency used in the PHD processing.		N	
			APOError	O	MTX	APO Error Covariance matrix for the set of monostatic APOs.		N	size1 = "16" and size2 = "16"
			BistaticCompositeSCP	O		Composite projection error statistics for the SCP pixel projected the SCP ETP using the provided APOs.		N	
			RAvg	R	DBL	Average range error (ΔR_Avg) standard deviation.	m	N	
			RdotAvg	R	DBL	Average range rate error ($\Delta Rdot_Avg$) standard deviation.	m/s	N	
			RAvgRdotAvg	R	DBL	Average range error and average range rate error correlation coefficient.		N	

3.2.13 XML: MatchInfo

Table 3-13 Matched Collection Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
	MatchInfo	O		Information about other collections that are matched to the current collection. The current collection is the collection from which this SICD product was generated.		N	
Note: The use of Matched Collection Parameters is per program specific implementation. Match Types, Type IDs, and match parameters are defined in the Program Specific Implementation Document.							
		NumMatchTypes	R	INT	Number of types of matched collections.		N
		MatchType	R		Block containing information about match type.		Y index = "1", "2", ..., NumMatchTypes
		TypeID	R	TXT	Text string identifying the match type. Examples: "COHERENT", "STEREO"		N
		CurrentIndex	O	INT	Collection sequence index for the current collection.		N
		NumMatchCollections	R	INT	Number of matched collections for this match type.		N May be set to 0.
		MatchCollection	O		Block containing information about match collection.		Y index = "1", "2", ..., NumMatchCollections
		CoreName	R	TXT	Text string that uniquely identifies the matching collection.		N
		MatchIndex	O	INT	Collection sequence index for the match collection.		N
		Parameter	O	TXT	Relevant match parameter. Attribute name identifies the parameter.		Y name = "xxx"

3.2.14 XML: RgAzComp

Table 3-14 Range & Azimuth Compression Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
Where the SICD ImageFormAlgo is RGAZCOMP, the SICD XML shall include a RgAzComp block.							
	RgAzComp		C		Parameters included for a Range, Doppler image.		N
		AzSF	R	DBL	Scale factor that scales image coordinate az = ycol (meters) to a delta cosine of the Doppler Cone Angle at COA. $\Delta\cos DCA_{COA} = AzSF \times az$	1/m	N
		KazPoly	R	POLY	Polynomial function that yields azimuth spatial frequency (Kaz = Kcol) as a function of slow time (variable 1). Slow Time (sec) -> Azimuth spatial frequency (cycles/meter). Time relative to collection start.	cyc/m, cyc/m/s, cyc./m/s ² , etc.	N order1 = "M"

3.2.15 XML: PFA

Table 3-15 Polar Format Algorithm Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
Where the SICD ImageFormAlgo is PFA, the SICD XML shall include a PFA block.							
	PFA	C		Parameters included when the image is formed using the Polar Formation Algorithm.		N	
	FPN	R	XYZ	Focus Plane unit normal (ECF). Unit vector FPN points away from the center of the Earth.	-	N	
	IPN	R	XYZ	Image Formation Plane unit normal (ECF). Unit vector IPN points away from the center of the Earth.	-	N	
	PolarAngRefTime	R	DBL	Polar image formation reference time. Polar Angle = 0 at the reference time. Measured relative to collection start. Note: Reference time is typically set equal to the SCP COA time but may be different.	sec	N	
	PolarAngPoly	R	POLY	Polynomial function that yields Polar Angle (radians) as function of time (variable 1) relative to Collection Start. Time (sec) -> Polar Angle (radians).	rad, rad/sec, rad/sec ² , etc.	N	order1 = "M"
	SpatialFreqSFPoly	R	POLY	Polynomial that yields the Spatial Frequency Scale Factor (KSF) as a function of Polar Angle (variable 1). Polar Angle(radians) -> KSF (dimensionless). Used to scale RF frequency (fx, Hz) to aperture spatial frequency (Kap, cycles/m).	1, 1/rad, 1/rad ² , etc.	N	order1 = "M"

Table 3-15 Polar Format Algorithm Parameters

Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
					Kap = $f_x \times (2/c) \times \text{KSF}$. Kap is the effective spatial frequency in the polar aperture.			
		Krg1	R	DBL	Minimum range spatial frequency (Krg) output from the polar to rectangular resampling.	cyc/m	N	
		Krg2	R	DBL	Maximum range spatial frequency (Krg) output from the polar to rectangular resampling.	cyc/m	N	
		Kaz1	R	DBL	Minimum azimuth spatial frequency (Kaz) output from the polar to rectangular resampling.	cyc/m	N	
		Kaz2	R	DBL	Maximum azimuth spatial frequency (Kaz) output from the polar to rectangular resampling.	cyc/m	N	
		STDeskew	O		Parameters to describe image domain ST Deskew processing.		N	
		Applied	R	BOOL	Parameter indicating if ST Deskew Phase function has been applied.		N	Allowed values: "false", "true"
		STDSPHasePoly	R	2D_POLY	Slow time deskew phase function to perform the ST / Kaz shift. Two-dimensional phase (cycles) polynomial function of image range coordinate (variable 1) and azimuth coordinate (variable 2).	cyc, cyc/m, cyc/m ² , etc.	N	order1 = "M" order2 = "N"

3.2.16 XML: RMA

Table 3-16 Range Migration Algorithm Parameters							
Field Name		Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
SICD							
Where the SICD ImageFormAlgo is RMA, the SICD XML shall include an RMA block.							
	RMA	C		Parameters included when the image is formed using the Range Migration Algorithm.		N	
	RMAAlgoType	R	ENU	Identifies the type of migration algorithm used. OMEGA_K: Algorithms that employ Stolt interpolation of the Kxt dimension. $K_x = (K_f^2 - K_y^2)^{0.5}$ CSA: Wave number algorithm that process two-dimensional chirp signals. RG_DOP: Range-Doppler algorithms that employ RCMC in the compressed range domain.	-	N	Allowed values: "OMEGA_K", "CSA", "RG_DOP"
	ImageType	R	ENU	Identifies the specific RM image type / metadata type supplied. RMAT: Range Migration w/ Along Track motion compensation. RMCR: Range Migration w/ Cross Range motion compensation. INCA: Imaging Near Closest Approach. Special RM processing used for imaging near closest approach.	-	N	Allowed values: "RMAT", "RMCR", "INCA"

Table 3-16 Range Migration Algorithm Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
Where the SICD RMA.ImageType is RMAT, the SICD XML shall include an RMA.RMAT block.								
		RMAT	C		Parameters for RMA with Along Track (RMAT) motion compensation.		N	
		PosRef	R	XYZ	Platform reference position (ECF) used to establish the reference slant plane.	m	N	
		VelRef	R	XYZ	Platform reference velocity vector (ECF) used to establish the reference slant plane and the Along Track direction.	m/sec	N	
		DopConeAngRef	R	DBL	Reference Doppler Cone Angle (degrees).	deg	N	
Where the SICD RMA.ImageType is RMCR, the SICD XML shall include an RMA.RMCR block.								
		RMCR	C		Parameters for RMA with Cross Range (RMCR) motion compensation.		N	
		PosRef	R	XYZ	Platform reference position (ECF) used to establish the reference slant plane and the range direction in the image.	m	N	
		VelRef	R	XYZ	Platform reference velocity vector (ECF) used to establish the reference slant plane.	m/sec	N	
		DopConeAngRef	R	DBL	Reference Doppler Cone Angle (degrees).	deg	N	

Table 3-16 Range Migration Algorithm Parameters								
Field Name			Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
Where the SICD RMA.ImageType is INCA, the SICD XML shall include an RMA.INCA block.								
		INCA	C		Parameters for Imaging Near Closest Approach (INCA) image description.		N	
		TimeCAPoly	R	POLY	Polynomial function that yields Time of Closest Approach (sec) as function of image column (azimuth) coordinate (m). Time t = 0 at Collection Start.	sec, sec/m, sec/m ² , etc.	N	order1 = "M"
		R_CA_SCP	R	DBL	Range at Closest Approach (R_CA) for the SCP.	m	N	
		FreqZero	R	DBL	RF frequency (f ₀) used for computing Doppler Centroid values. Typical f ₀ set equal to center transmit frequency. Note: Value may be relative to Ref_Freq. See Section 4.7.1.	Hz	N	
		DRateSFPoly	R	2D_POLY	Polynomial function that yields Doppler Rate scale factor (DRSF) as a function of image location. Yields DRSF as a function of image range coordinate (variable 1) and azimuth coordinate (variable 2). Used to compute Doppler Rate at closest approach.	1, 1/m, 1/m ² , etc.	N	order1 = "M" order2 = "N"
		DopCentroidPoly	O	2D_POLY	Polynomial function that yields Doppler Centroid value as a function of image location (fdop_DC). The fdop_DC is the Doppler frequency at the peak signal response. The polynomial is a function of image range coordinate (variable 1) and azimuth coordinate (variable 2). Only used for Stripmap and Dynamic Stripmap collections.	Hz, Hz/m, Hz/m ² , etc.	N	order1 = "M" order2 = "N"

Table 3-16 Range Migration Algorithm Parameters									
Field Name				Req / Opt	Type	Description	Units	Rpt	Attributes/Allowed Values
			DopCentroidCOA	O	BOOL	Flag indicating that the COA is at the peak signal (fdop_COA = fdop_DC). DopCentroidCOA = "true" ⇔ Pixel COA at peak signal for all pixels. DopCentroidCOA = "false" ⇔ Pixel COA not at peak signal for some pixels. Only used for Stripmap and Dynamic Stripmap.	-	N	

4 XML Metadata Parameter Description

4.1 Collection & Image Creation Parameters

The Collection Information parameters identify the specific radar platform that collected the imaging data and the data set identifier. The Image Creation parameters identify the image formation application, the time and the location where the image formation processing occurred. Refer to Table 3-1.

The radar platform identifiers (parameters `CollectorName` and `IlluminatorName`) are defined per program specific convention. The specific collection identifier (parameter `CoreName`) is also defined per program specific convention. These identifiers may be composed of any number of concatenated fields as needed. Optional parameters `IlluminatorName` and `CollectType` are provided to clearly identify bistatic imaging collections.

The radar imaging mode is identified by the sensor independent parameter `ModeType` with allowed values `SPOTLIGHT`, `STRIPMAP` and `DYNAMIC STRIPMAP`. Optional parameter `ModeID` is used to include a program specific mode identifier. The inclusion of the parameter `ModeID` is recommended.

4.2 Image Data Parameters

The Image Data parameters describe the two dimensional pixel array. Refer to Table 3-2. The image pixel array is described by the pixel type and array size. Image pixels are indexed by global indices (row, col). See Section 1.2.1. The image array may be the original image produced by the image formation processor (referred as the full image) or a sub-image extracted from the original image. The first row of the full image is always row 0. The first column of the original image is always indexed column 0.

An important concept in the SICD metadata approach is that of a Scene Center Point (SCP) and a Scene Center Point pixel (SCP pixel). The SCP is a point located near the center of the scene covered by the full image. The image formation processing places the image of the Scene Center Point in the SCP pixel (parameter `SCPPixel`). The SCP pixel is located near the center of the Full Image. For error-free collection and processing, the image impulse response of a point scatterer located at the SCP is centered on the SCP pixel. An example full image pixel array is shown in Figure 4.2-1. The SCP position parameters are defined in Table 3-3. See parameters `SCP.ECF` and `SCP.LLH`.

The SCP position and the SCP Pixel indices are central to the proper interpretation and exploitation of a SICD image product. The SCP position and pixel are used as the reference for image pixel grid to geo-referenced mapping (e.g., image grid to ground plane resampling). The SCP pixel is also used as the reference for computing the image domain to spatial frequency domain transformations. See Section 4.4. The image formation algorithm may use the SCP explicitly in the image formation processing (such as in the Polar Format

Algorithm). For these images, the SCP position and pixel location are defined as part of the image formation process. For algorithms that do not use a SCP explicitly, an SCP position and pixel are computed for the SICD product. The process for defining the SCP pixel and position is to select a pixel close to the center of the image. The ground point in the scene that images to the center of the SCP pixel is then computed. The computation is based upon the precise Center of Aperture used for the selected pixel. The SCP point lies along the Center of Aperture iso-range and iso-range rate contour for the selected pixel. The ground height should correspond to the estimated ground height along the range, range rate contour.

A SICD product allows the pixel array elements to be stored using one of the three common complex image pixel formats. Parameter PixelType indicates the pixel type and binary format used. All pixels in a given product are stored with the same pixel type. Pixel formats of 2 bytes per pixel, 4 bytes per pixel and 8 bytes per pixel are supported. The three pixel types allow flexibility in balancing product file sizes and image dynamic range.

The complex image value of a given pixel is denoted $S_{II}(\text{row}, \text{col})$. For pixel types RE32F_IM32F and RE16I_IM16I, the real and imaginary components are stored in the file.

$$S_{II}(\text{row}, \text{col}) = \text{Real}(S_{II}(\text{row}, \text{col})) + j \cdot \text{Imag}(S_{II}(\text{row}, \text{col}))$$

For pixel type AMP8I_PHS8I, the amplitude and phase components are stored as unsigned 8-bit integers. For a given pixel, the amplitude component is denoted AMP(row,col) and the phase component is denoted PHS(row,col). Each component takes on values from 0 to 255. The amplitude component may be a direct representation of the pixel amplitude or may be used as an index into an amplitude look-up table (parameter AmpTable(0:255)). If no amplitude look-up table is provided, pixel component AMP(row,col) is the pixel amplitude. Pixel amplitude $A_{II}(\text{row}, \text{col})$ is determined as shown below. Pixel phase $P_{II}(\text{row}, \text{col})$ in cycles is computed from the pixel phase component as shown below. The complex image pixel $S_{II}(\text{row}, \text{col})$ is computed using the normal convention where pixel phase is the $\arctan(\text{Imag}(S_{II}(\text{row}, \text{col})) / \text{Real}(S_{II}(\text{row}, \text{col})))$.

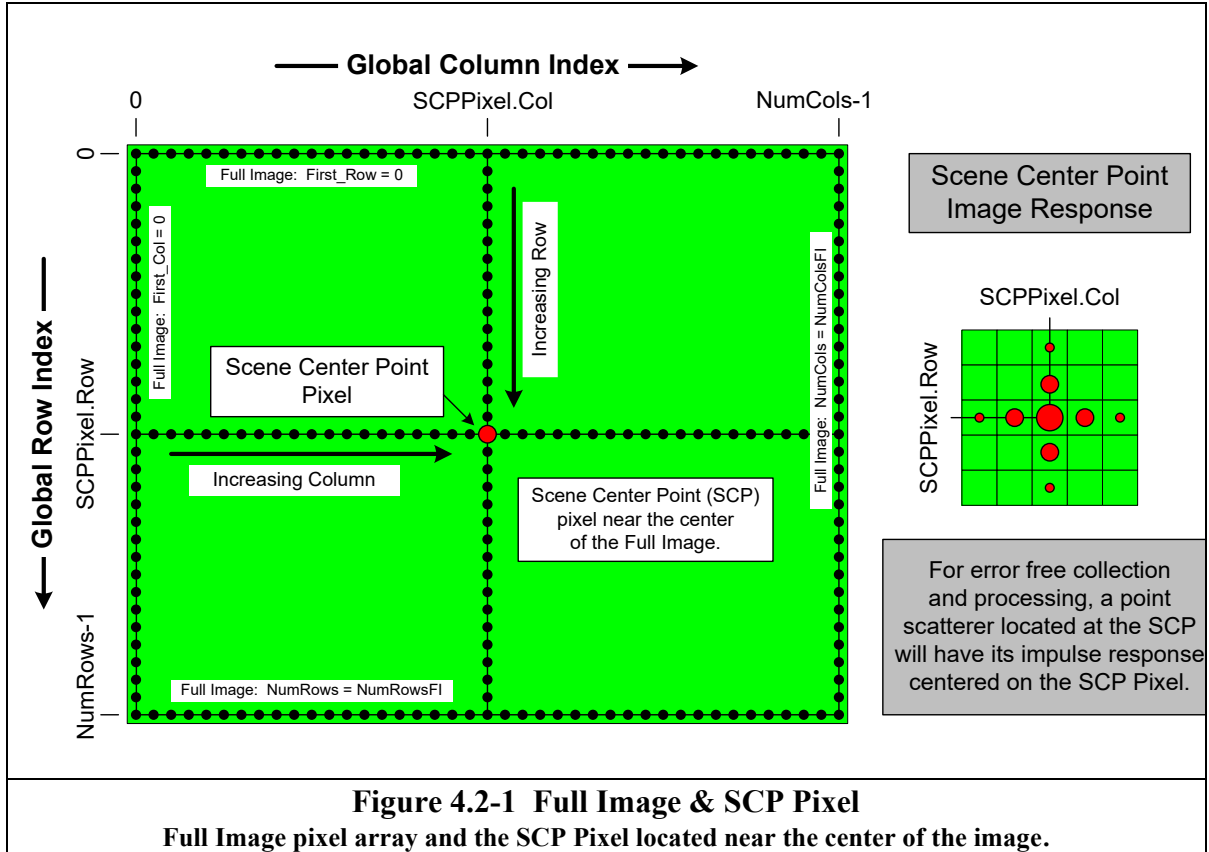
$$A_{II}(\text{row}, \text{col}) = \begin{cases} \text{AmpTable}(\text{AMP}(\text{row}, \text{col})), & \text{if AmpTable(0:255) is included} \\ \text{AMP}(\text{row}, \text{col}), & \text{if AmpTable(0:255) is omitted} \end{cases}$$

$$P_{II}(\text{row}, \text{col}) = \frac{1}{256} \cdot \text{PHS}(\text{row}, \text{col})$$

$$S_{II}(\text{row}, \text{col}) = A_{II}(\text{row}, \text{col}) \cdot \cos(2\pi \cdot P_{II}(\text{row}, \text{col})) \\ + j \cdot A_{II}(\text{row}, \text{col}) \cdot \sin(2\pi \cdot P_{II}(\text{row}, \text{col}))$$

The impact of the binary pixel format on product utility is dependent on many variables including resolution, image signal to noise ratio and scene content. For most SAR systems

and scenes, complex image products using 4 bytes per pixel will have negligible loss in quality relative to products using 8 bytes per pixel. Products using 2 bytes per pixel will have a loss in quality that will impact the utility of derived products. The use of an amplitude look-up table can reduce the impact. In general, the 2 bytes per pixel format should only be used when minimizing product size is a high priority.



4.3 Geographic Reference Parameters

The Image Geographic Reference Parameters describe mapping of image pixel locations to geo-located positions. Refer to Table 3-3. The Scene Center Point (SCP) position is specified as well as the approximate image corner points positions. Optional parameters allow for identifying and locating additional ground points that may be needed to support proper exploitation of the image product.

For SICD products, all geo-located positions shall be expressed using the World Geodetic System 1984 (WGS 84). Parameter GeoData.EarthModel = WGS_84 is included for clarity. All latitude values are Geodetic Latitude and all height values are relative to the WGS 84 reference ellipsoid (HAE = Height Above Ellipsoid). Values of latitude and longitude shall be specified in decimal degrees within the following range of values.

$$-90.0 \leq \text{Latitude} \leq 90.0$$

$$-180.0 \leq \text{Longitude} \leq 180.0$$

The SCP position is the key reference point for relating image pixel grid location to geo-located positions. The SCP position is specified in Earth Centered Fixed (ECF) coordinates (parameter Geodata.SCP.ECF) and in WGS 84 Latitude, Longitude and HAE (parameter Geodata.SCP.LLH). These two positions are identical and redundant information but are provided to support both exploitation as well as search and discovery. A difference between the two positions (if any) is due to rounding of parameters expressed in ASCII format. For all precise computations, the parameter SCP.ECF is to be used.

4.4 Image Grid Parameters

The Image Grid parameters describe the spatial sampling represented by the rows and columns of the image array. Refer to Table 3-4. The parameters define the Image Plane that passes through the SCP position. The parameters describe sample spacing, impulse response width and the spatial frequency domain support for the row and column directions. For all images, the image orientation represents a view from above the earth. For most image grids, the increasing row direction is in an increasing range direction. For the common display convention (row 0 at the top of the display and pixel (0,0) at the upper left), most images will be displayed such that the radar illumination is from top of the display and the radar shadows are downward.

The SAR collection and image formation processing produces a two-dimensional image of the three-dimensional scene. The image may be described as a projection of the three-dimensional scene onto a two-dimensional Image Plane. The spatial sampling of the image grid may be described as a two-dimensional sampling in the Image Plane. Samples along a column of the image are treated as uniformly spaced positions in the plane in the “increasing row” direction. Samples along a row of the image are treated as uniformly spaced positions in the plane in the “increasing column” direction. The increasing row direction and increasing column direction are defined by two vectors that lie in the plane. While this is not precisely true for many SAR complex images, such a description provides a useful approximation for relating image pixels to geo-located positions.

4.4.1 Image Indices & Image Coordinates

The image array locations are specified using two sets of pixel based indices: (row,col) and (irow,icol). The image array locations are also specified using distance based coordinates (xrow,ycol). Indices (row,col) take on only integer values. Indices (irow,icol) and image coordinates (xrow,ycol) take on continuous values. See Figure 4.4-1.

(1) Global Row, Column Indices: (row,col)

The global row, column indices are used to index the pixels of the image array. The origin (0,0) is located at the first row (row = 0) and the first column of the first row (col = 0) of the Full Image array. In all descriptions that follow, the (row,col) indices are only used to index individual elements of the pixel array. The global row, column indices take on INTEGER values only.

Global Row Index: $\text{row} = 0, 1, 2, \dots, \text{NumRowsFI} - 1$

Global Column Index: $\text{col} = 0, 1, 2, \dots, \text{NumColsFI} - 1$

(2) SCP Pixel-Centered Image Indices: (irow,icol)

The SCP pixel-centered image indices are continuous valued indices that are also used to index locations in the image pixel array. The origin (0,0) is located at the CENTER of the SCP pixel. The image indices (irow,icol) are related to global row, column indices (row,col) as follows.

SCP Pixel-Centered Row Index: $irow = row - SCPPixel.Row$

SCP Pixel-Centered Column Index: $icol = col - SCPPixel.Col$

In the descriptions that follow, SCP pixel-centered indices ($irow, icol$) are used to address the image array as a continuous two-dimensional space. The image pixel values are considered to be the discrete samples of a continuous, two-dimensional image signal at the integer values of indices ($irow, icol$).

(3) SCP Centered Image Coordinates: ($xrow, ycol$)

The SCP centered image coordinates are continuous valued distances that treat the image array as equal spaced samples on an orthogonal grid. The origin (0,0) is located at the CENTER of the SCP pixel. Image coordinate $xrow$ is equal to $irow$ scaled by the row spacing (Row_SS). Image coordinate $ycol$ is equal to $icol$ scaled by the column spacing (Col_SS).

$Row_SS = Grid.Row.SS$

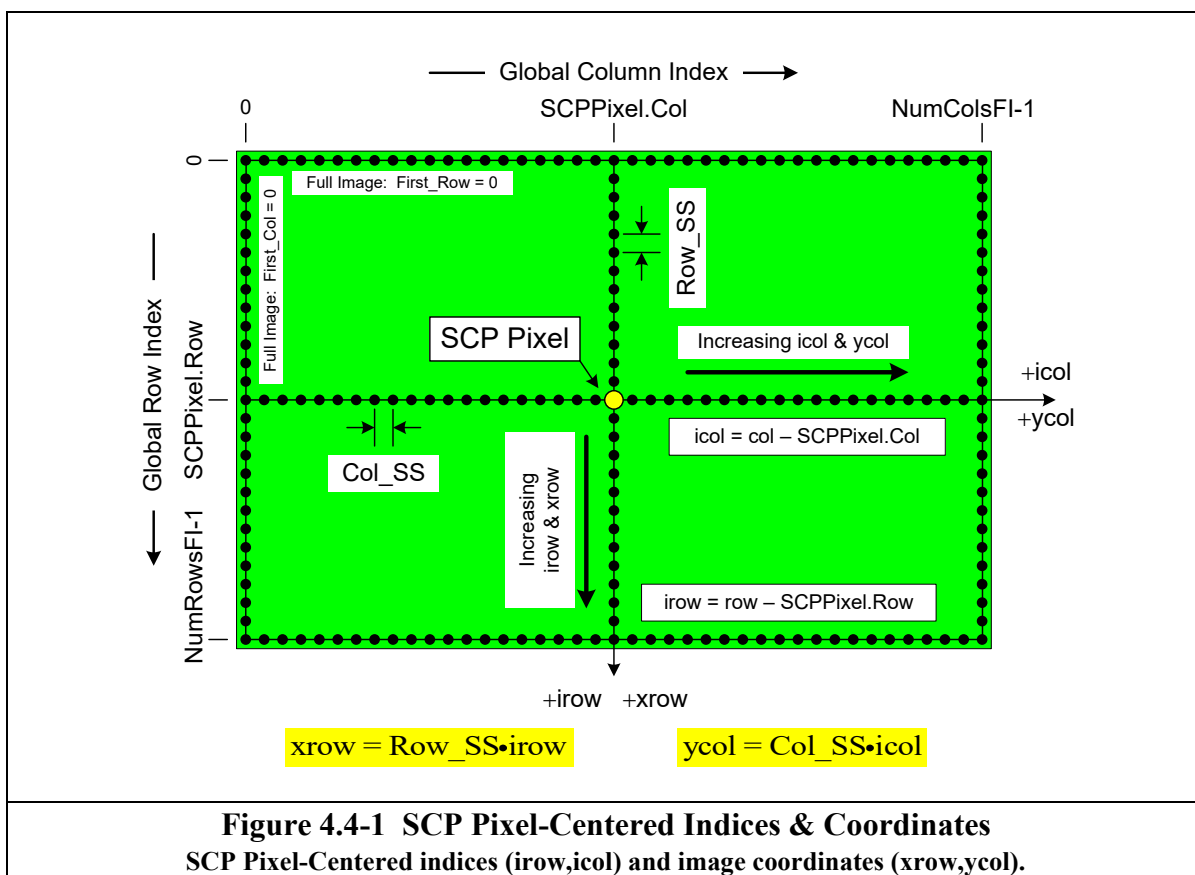
$Col_SS = Grid.Col.SS$

$xrow = Row_SS \cdot irow$

$ycol = Col_SS \cdot icol$

The most common practice in the SAR image formation processing and image exploitation is to use “pixel-centered” image indexing. For some common imagery formats, the specific image indices (row, col) or (r, c) when associated with the Common Coordinate System (CCS) are considered to have the origin at the upper left corner of the pixel (0,0). The image array sample, $S_{II}(row, col)$, is considered to be the image sample at location ($row + 0.5, col + 0.5$). This potential source of confusion is avoided by the use of the precisely defined SCP pixel-centered indices ($irow, icol$).

- (1) **In all SICD descriptions, global indices (row, col) are only used to index the discrete elements of the pixel array. Indices (row, col) take on INTEGER values only.**
- (2) **The SCP Pixel-Centered indices ($irow, icol$) are used to reference the pixel array as a continuous 2-dimensional space. An index value with zero fractional part references the center of the pixel.**



4.4.2 Image Plane Parameters

The Image Plane in the SICD product contains the SCP and the image grid sample directions at the SCP pixel. For some SAR image formation algorithms, the Image Plane is explicitly defined as part of the processing. For others, the Image Plane unit vectors are computed from the image formation processing parameters that were used. Algorithms that do not use an image formation plane explicitly typically produce slant plane images. A qualitative description of the Image Plane is provided as aid to users (parameter Grid.ImagePlane). Allowed values are SLANT, GROUND and OTHER.

The Image Plane is defined by the SCP and two unit vectors that lie in the plane. The unit vector in the row direction represents the change in Image Plane position moving from the SCP pixel in the increasing row direction. The unit vector in the column direction represents the change in Image Plane position moving from the SCP pixel in the increasing column direction. Unit vectors **uRow** and **uCol** are in ECF coordinates.

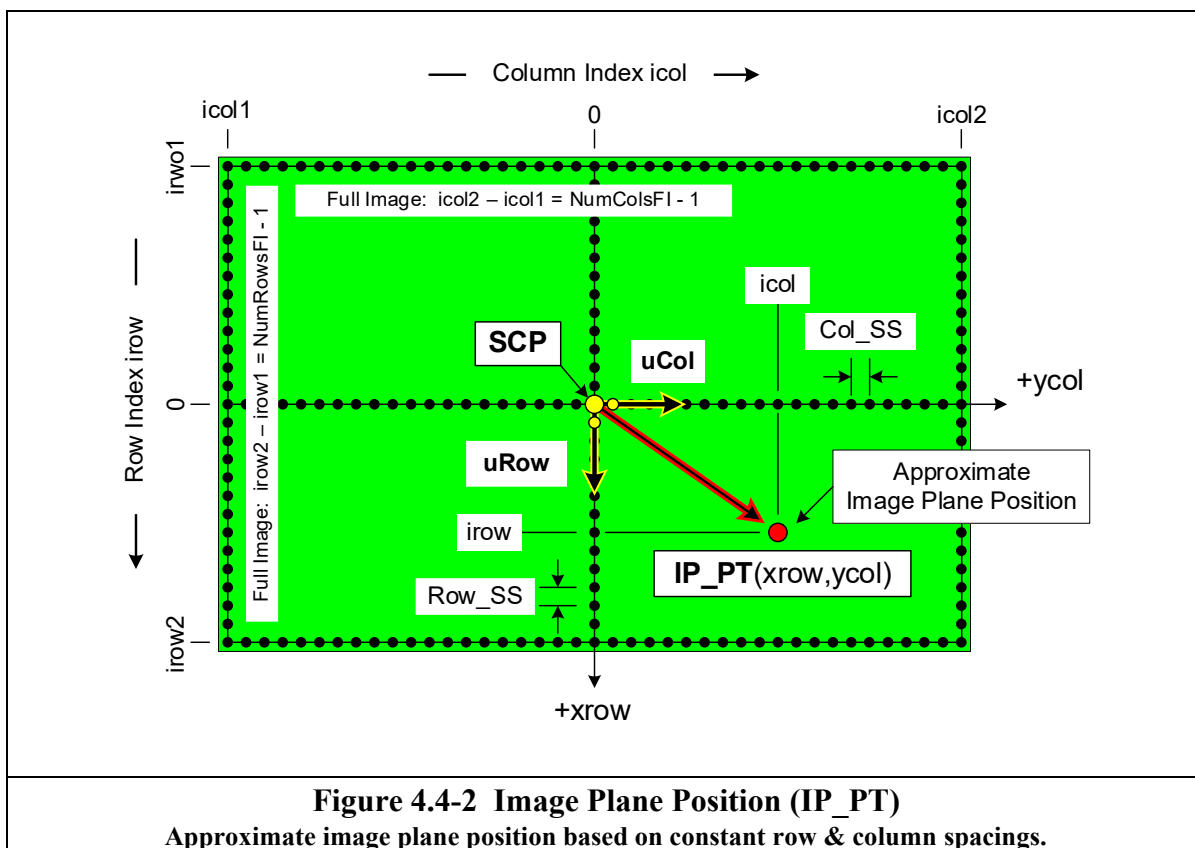
$$\mathbf{uRow} = \text{Grid.Row.UVectECF}$$

$$\mathbf{uCol} = \text{Grid.Col.UVectECF}$$

The image sample spacings and half power impulse response widths are specified at the SCP pixel. The image grid sample spacings (Row_SS and Col_SS) represent the precise row and

column spacings at the SCP pixel and with respect to a displacement in the Image Plane. The row direction sample spacing (Row_SS) represents the displacement in the Image Plane along **uRow**. The column direction sample spacing (Col_SS) represents the displacement in the Image Plane along **uCol**. The image plane position, **IP_PT**(xrow, ycol), in ECEF coordinates is computed as shown below. See Figure 4.4-2.

$$\mathbf{IP_PT}(\text{xrow}, \text{ycol}) = \mathbf{SCP} + \text{xrow} \cdot \mathbf{uRow} + \text{ycol} \cdot \mathbf{uCol}$$



4.4.3 Center Of Aperture Time

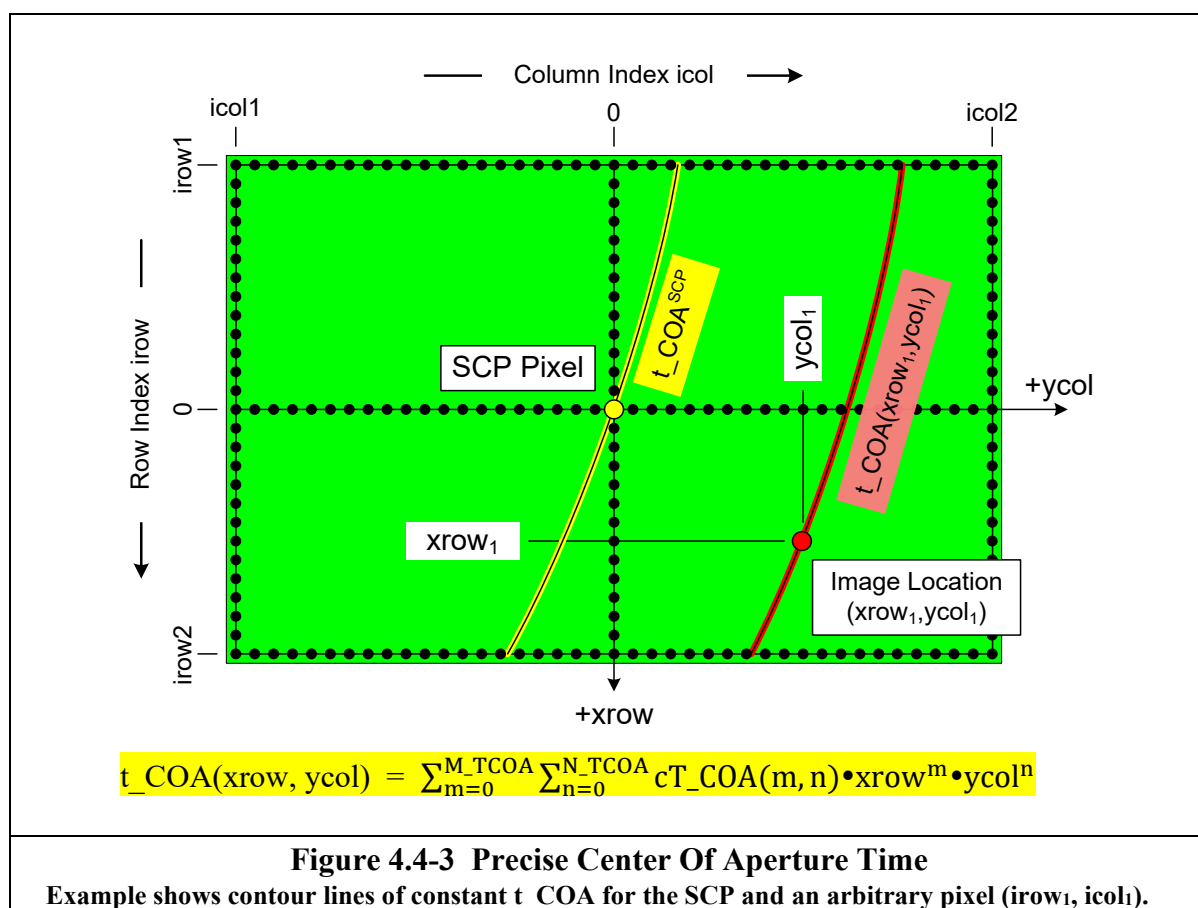
Accurate exploitation of the SAR image is dependent upon accurate knowledge of the Center Of Aperture (COA) time as a function of image location. The Center Of Aperture time (denoted t_{COA}) is used to determine many parameters including the precise range and range rate contour that projects a particular image location to a geo-located position. The t_{COA} is provided as a two-dimensional polynomial of image coordinates (xrow, ycol). The t_{COA} for image location (xrow, ycol) is computed as shown below.

$$\text{Grid.TimeCOAPoly}(m,n) = cT_{COA}(m,n), \quad m = 0 \text{ to } M_{TCOA}, \quad n = 0 \text{ to } N_{TCOA}$$

$$t_COA(xrow, ycol) = \sum_{m=0}^{M_TCOA} \sum_{n=0}^{N_TCOA} cT_COA(m, n) \cdot xrow^m \cdot ycol^n$$

Important to note is the t_COA computed above is the accurate value of t_COA for the image sample. Even though image coordinates ($xrow$, $ycol$) may be approximate displacements, the t_COA computed is the time to be used for accurate exploitation. The choice of the orders M_TCOA and N_TCOA are made such that the resulting t_COA values accurately match those achieved by the image formation processing. While the precise accuracy needed is dependent upon many factors, for most systems and derived products, computed t_COA values within 10 milliseconds of the achieved value in the processing are acceptable.

For all images, coefficient $cT_COA(0,0)$ is the t_COA for the SCP pixel. Images with constant t_COA (typically from spotlight collections) may be stored with only the $cT_COA(0,0)$ coefficient and order parameters $M_TCOA = N_TCOA = 0$. An example image for which t_COA varies across the image is shown in Figure 4.4-3. Shown in the figure are two contours of constant t_COA . For most images, the contours will be approximately linear and equally spaced for even increments in t_COA .



4.4.4 Image Grid Types

The type of image sample grid is a fundamental attribute of the complex image. The spatial sampling of the complex image is usually oriented along the natural radar coordinates range and cross range. The exact grid is a function of both the collection mode and the image formation algorithm. The image sample grid type shall be indicated by parameter Grid.Type. The allowed Grid Types are listed below

- (1) RGAZIM ⇔ Range & Azimuth
- (2) RGZERO ⇔ Range & Zero Doppler
- (3) XRGYCR ⇔ Cartesian Grid Oriented Range, Cross Range
- (4) XCTYAT ⇔ Cartesian Grid Oriented Cross Track, Along Track
- (5) PLANE ⇔ Arbitrary Image Plane w/ Axes U,V

Grid.Type = RGAZIM ⇔ Range & Azimuth

Image samples are oriented in range and azimuth with respect to the ARP position at an image formation reference time. The image formation reference time (t_{REF}) is often set to the SCP Center Of Aperture time. The image is oriented shadows downward and a view from above the earth. See Figures 4.4-4 and 4.4-5. Image rows are samples along constant (or nearly constant) range contours. Image row spacing is usually constant in range. Image columns are samples in azimuth. The column spacing is usually constant (or nearly constant) in some azimuth angle related quantity (e.g., cosine of the Doppler cone angle). The RGAZIM grid is the natural grid that results from simple Range and Azimuth Compression image formation algorithm (IFA). See Section 4.14. The Range & Azimuth grid is also the natural grid that results using the Polar Format IFA. See Section 4.15.

For the RGAZIM grid, the row coordinate is referred to as range and the column coordinate is referred to as azimuth. Pixel-centered indices (irg, iaz) are equal to general image indices (irow, icol) and image coordinates (rg, az) are equal to general image coordinates (xrow, ycol). Spatial frequency parameters (Krg, Kaz) are equal to general spatial frequency parameters (Krow, Kcol). Image coordinate values are in meters and spatial frequency values are in cycles/meter.

$$\begin{aligned}
 \text{irg} &= \text{irow} = \text{row} - \text{SCPPixel.Row} & \text{iaz} &= \text{icol} = \text{col} - \text{SCPPixel.Col} \\
 \text{rg} &= \text{xrow} = \text{Rg_SS} \cdot \text{irg} & \text{az} &= \text{ycol} = \text{Az_SS} \cdot \text{iaz} \\
 \mathbf{uRG} &= \mathbf{uRow} & \mathbf{uAZ} &= \mathbf{uCol} \\
 \text{Krg} &= \text{Krow} & \text{Kaz} &= \text{Kcol}
 \end{aligned}$$

Grid.Type = RGZERO ⇔ Range & Zero Doppler

Image samples are oriented in range offset from the trajectory of the ARP and in azimuth parallel to the trajectory. The image is oriented shadows downward and a view from above the earth. See Figures 4.4-6 and 4.4-7. Image rows are samples along constant range contours relative to the ARP trajectory. Image row spacing is constant in range. The image columns are samples along a contour of closest approach (i.e., zero Doppler). The image column spacing may be uniform in distance in the scene or uniform in time of closest approach. The RGZERO grid is the natural grid that results for images formed using the Range Migration Algorithm IFA adapted to imaging near closest approach (INCA). See Section 4.16.2.

For the RGZERO grid, the row coordinate is referred to as range and the column coordinate is referred to as azimuth. Pixel-centered indices (irg, iaz) are equal to general image indices (irow, icol) and image coordinates (rg, az) are equal to general image coordinates (xrow, ycol). Spatial frequency parameters (Krg, Kaz) are equal to general spatial frequency parameters (Krow, Kcol). Image coordinate values are in meters and spatial frequency values are in cycles/meter.

$$\begin{aligned} \text{irg} &= \text{irow} = \text{row} - \text{SCPPixel.Row} & \text{iaz} &= \text{icol} = \text{col} - \text{SCPPixel.Col} \\ \text{rg} &= \text{xrow} = \text{Rg_SS} \cdot \text{irg} & \text{az} &= \text{ycol} = \text{Az_SS} \cdot \text{iaz} \\ \mathbf{uRG} &= \mathbf{uRow} & \mathbf{uAZ} &= \mathbf{uCol} \\ \text{Krg} &= \text{Krow} & \text{Kaz} &= \text{Kcol} \end{aligned}$$

Grid.Type = XRGYCR ⇔ Range & Cross Range

The XRGYCR grid is a slant plane grid that is oriented range and cross range. The slant plane is defined by the ARP position and velocity at a reference time (**POS_{REF}** and **VEL_{REF}**) and the SCP. See Figure 4.4-8. The +XRG direction is oriented from the ARP to the SCP (in the increasing range direction). The +YCR direction is orthogonal to the +XRG direction (in the cross range direction). The rows of the image are along lines of constant XRG and evenly spaced. The columns of the image are along lines of constant YCR and evenly spaced.

For the XRGYCR grid, the row coordinate is referred to as range and the column coordinate is referred to as cross range. Pixel-centered indices (ixrg, icyr) are equal to general image indices (irow, icol) and image coordinates (xrg, ycr) are equal to general image coordinates (xrow, ycol). Spatial frequency parameters (Kxrg, Kycr) are equal to general spatial frequency parameters (Krow, Kcol). Image coordinate values are in meters and spatial frequency values are in cycles/meter.

$$\begin{aligned} \text{ixrg} &= \text{irow} = \text{row} - \text{SCPPixel.Row} & \text{icyr} &= \text{icol} = \text{col} - \text{SCPPixel.Col} \\ \text{xrg} &= \text{xrow} = \text{Row_SS} \cdot \text{ixrg} & \text{ycr} &= \text{ycol} = \text{Col_SS} \cdot \text{icyr} \end{aligned}$$

$$\mathbf{uXRG} = \mathbf{uRow}$$

$$Kxrg = Krow$$

$$\mathbf{uYCR} = \mathbf{uCol}$$

$$Kycr = Kcol$$

Grid.Type = XCTYAT ⇔ Cross Track & Along Track

The XCTYAT grid is a slant plane grid that is oriented cross track and along track. The slant plane is defined by the ARP position and velocity at a reference time ($\mathbf{POS}_{REF}$ and $\mathbf{VEL}_{REF}$) and the SCP. See Figure 4.4-8. The +XCT direction is orthogonal to the $\mathbf{VEL}_{REF}$ and in the increasing distance from the trajectory. The +YAT direction is orthogonal to the +XCT direction and parallel to the $\mathbf{VEL}_{REF}$. For right looking collections, the +YAT direction is in the same direction as $\mathbf{VEL}_{REF}$. For left looking collections, the +YAT is in the opposite direction as $\mathbf{VEL}_{REF}$. The rows of the image are along lines of constant XCT and evenly spaced. The columns of the image are along lines of constant YAT and evenly spaced. The grid is the natural output for image formation using the RMA.

For the XCTYAT grid, the row coordinate is referred to as cross track and the column coordinate is referred to as along track. Pixel-centered indices (ixct, iyat) are equal to general image indices (irow, icol) and image coordinates (xct, yat) are equal to general image coordinates (xrow, ycol). Spatial frequency parameters (Kxct, Kyat) are equal to general spatial frequency parameters (Krow, Kcol). Image coordinate values are in meters and spatial frequency values are in cycles/meter.

$$ixct = irow = row - SCPPixel.Row \quad iyat = icol = col - SCPPixel.Col$$

$$xct = xrow = Row_SS \cdot ixct \quad yat = ycol = Col_SS \cdot iyat$$

$$\mathbf{uXCT} = \mathbf{uRow}$$

$$Kxct = Krow$$

$$\mathbf{uYAT} = \mathbf{uCol}$$

$$Kyat = Kcol$$

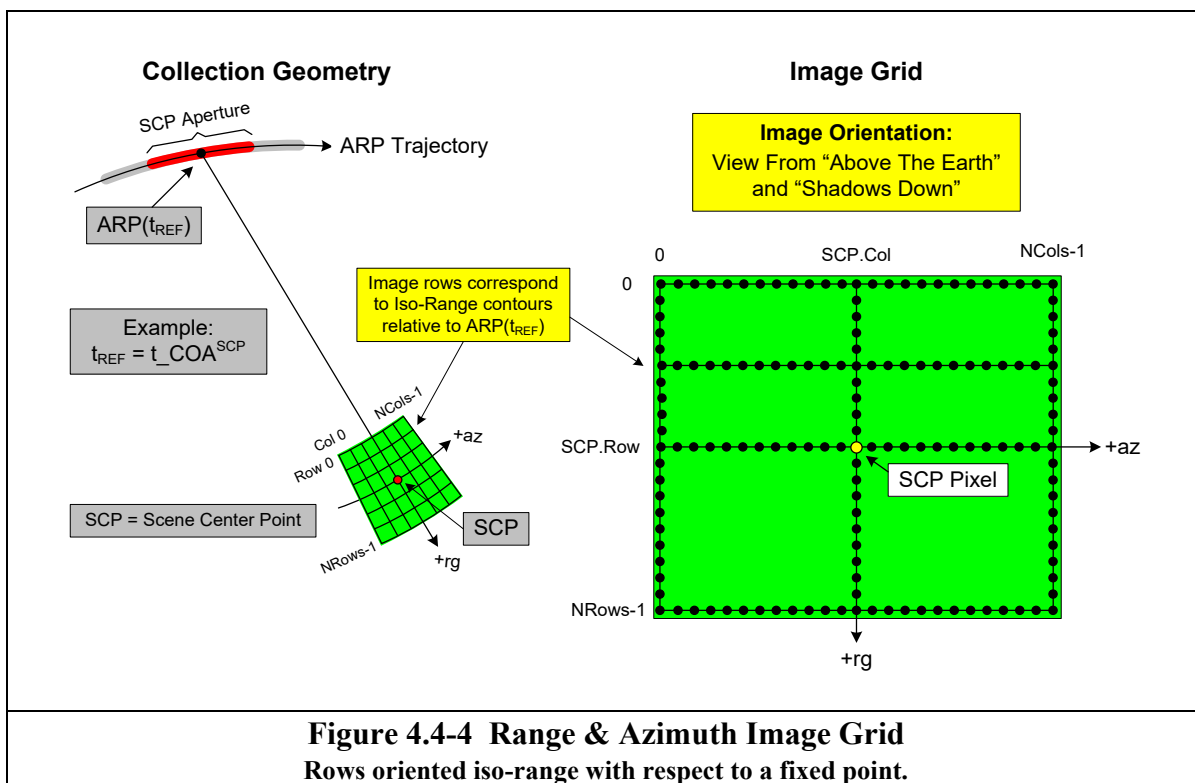
Grid.Type = PLANE ⇔ Arbitrary Image Plane

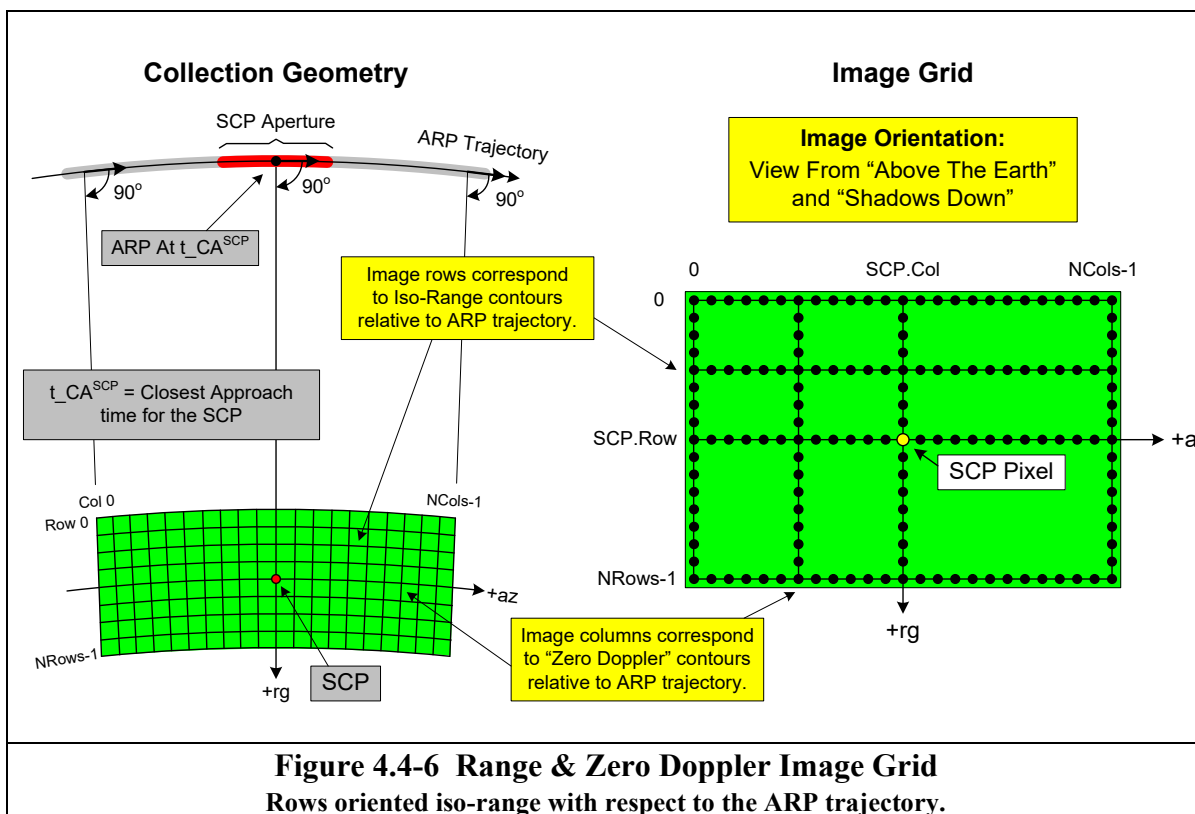
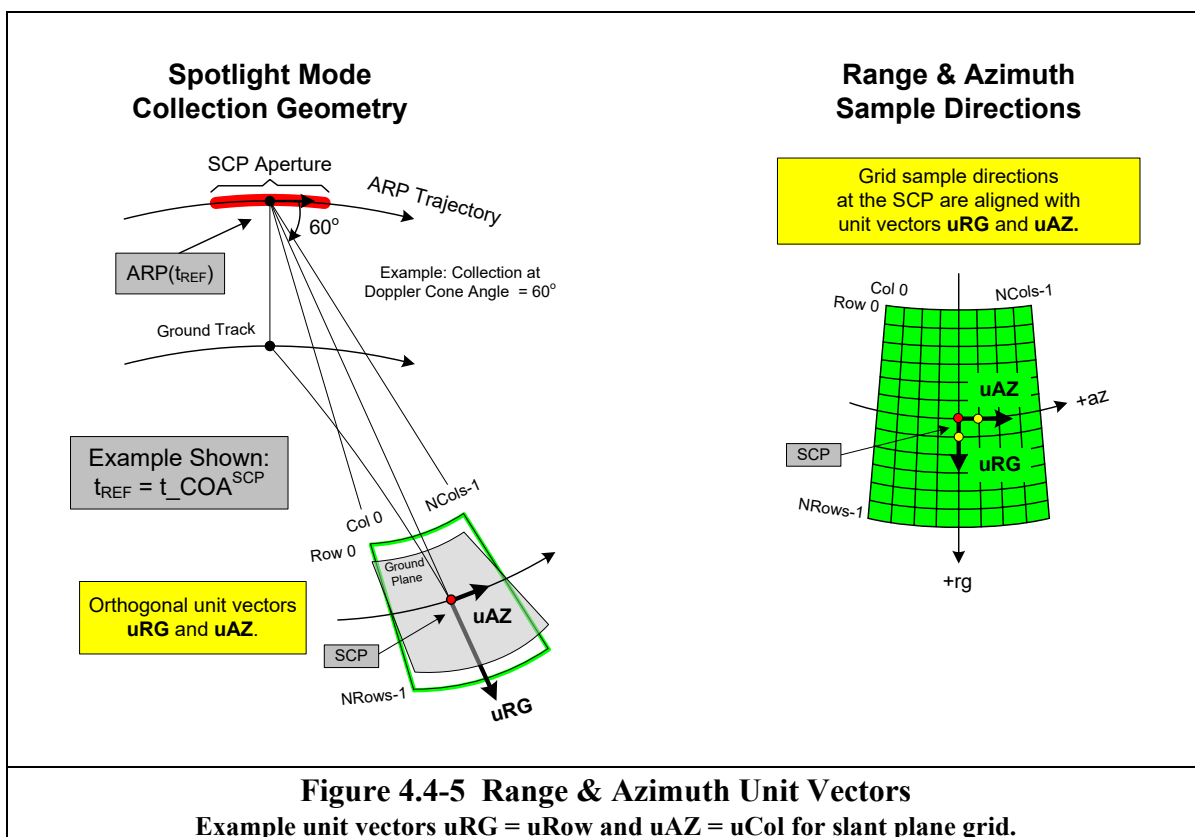
The PLANE grid type is provided for images that do not meet the precise definition of the other grid types. The samples are uniformly spaced in a geolocated image plane that contains the SCP. The +U coordinate is equal to the row coordinate (xrow). The +V coordinate is equal to the column coordinate (ycol). The rows of the image are along lines of constant U coordinate and are evenly spaced. The columns of the image are along lines of constant V coordinate and are evenly spaced. For an arbitrary image grid, these may be an approximation to the true sampling of the rows and columns. The restrictions are that the image be a view from above the earth and that the shadows are most closely aligned with the +U direction.

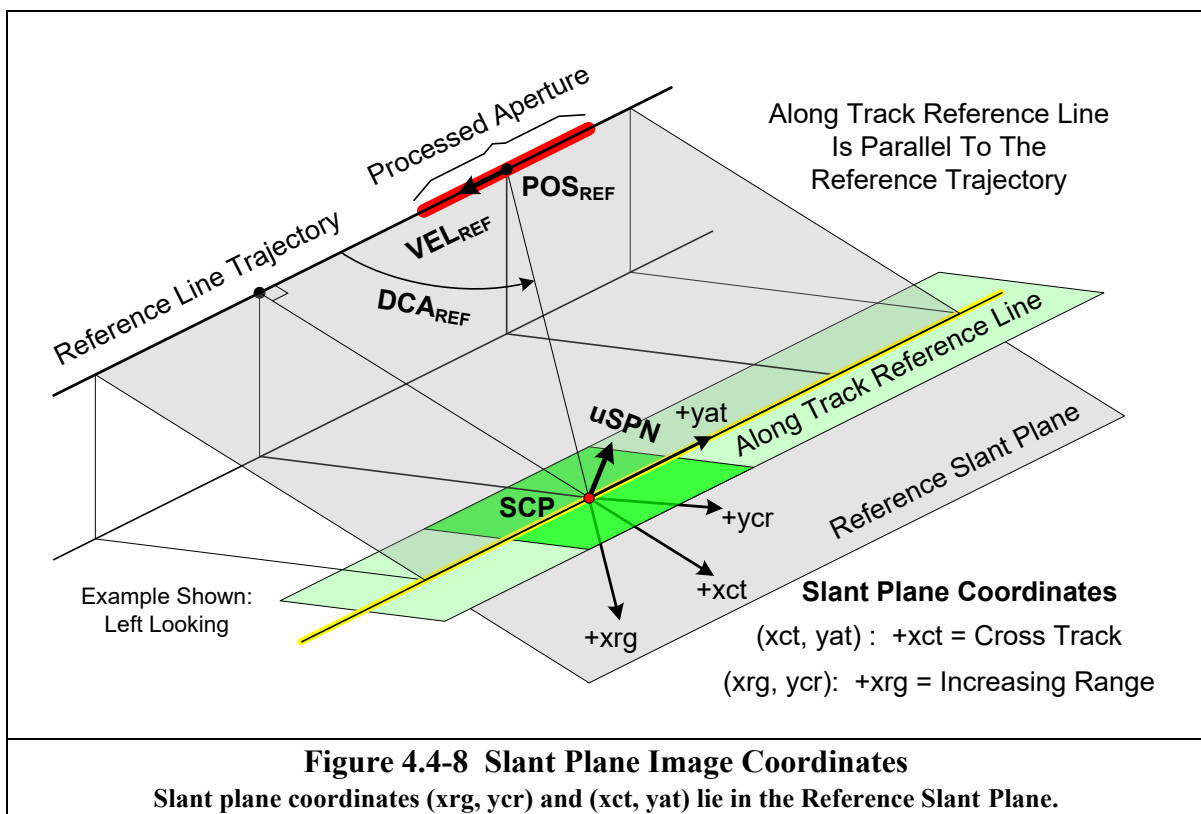
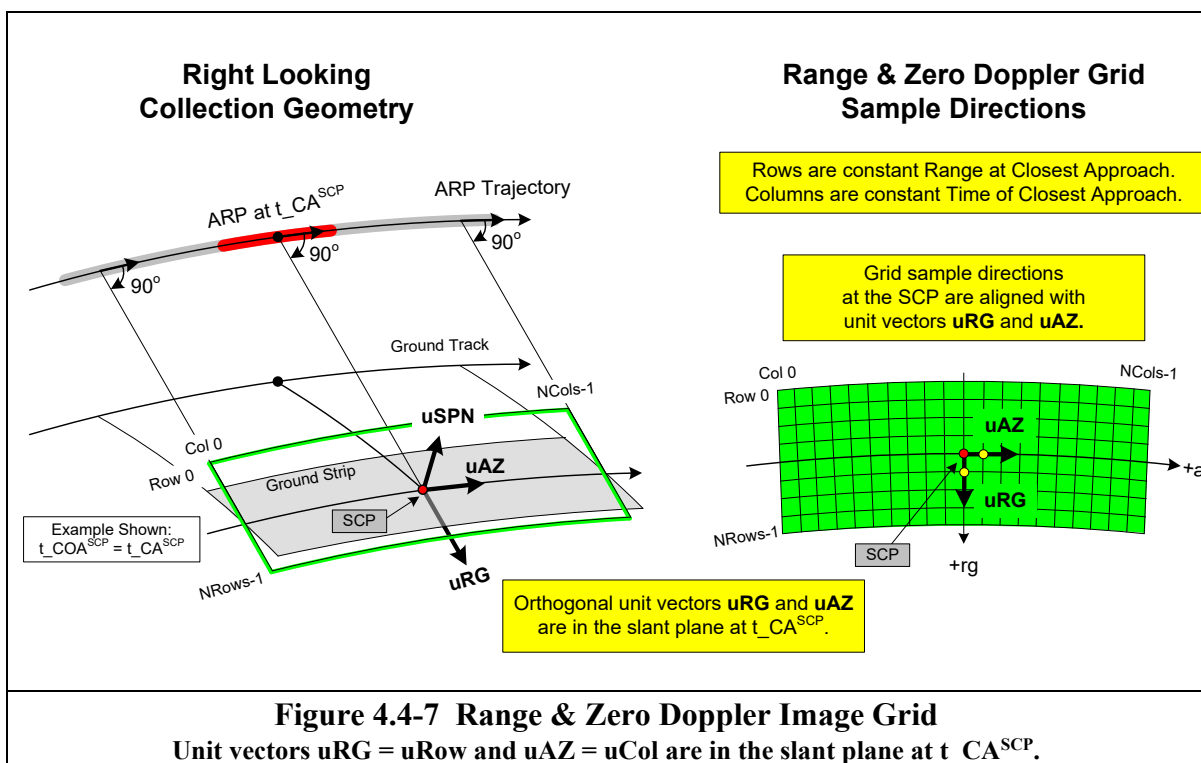
For the PLANE grid, the row coordinate is referred to as +U and the column coordinate is referred to as +V. Pixel-centered indices (iu, iv) are equal to general image indices (irow, icol) and image coordinates (u, v) are equal to general image coordinates (xrow, ycol).

Spatial frequency parameters (K_u , K_v) are equal to general spatial frequency parameters (K_{row} , K_{col}). Image coordinate values are in meters and spatial frequency values are in cycles/meter.

$$\begin{aligned} iu &= irow = row - SCPPixel.Row & iv &= icol = col - SCPPixel.Col \\ u &= xrow = Row_SS \cdot iu & v &= ycol = Col_SS \cdot iv \\ \mathbf{uU} &= \mathbf{uRow} & \mathbf{uV} &= \mathbf{uCol} \\ K_u &= K_{row} & K_v &= K_{col} \end{aligned}$$







4.4.5 Image Spatial Frequency Domain

The two-dimensional Fourier Transform (FT) of the complex image pixel array yields the spatial frequency domain. The characteristics of the spatial frequency domain are a fundamental attribute of the complex image. For all SAR complex images, the attributes of the spatial frequency domain were central to the image formation processing that produced the complex image.

The image domain is a two-dimensional space that is indexed by image pixel-centered indices (irow, icol) and image spatial coordinates (xrow,ycol). The samples of the complex image are considered to be samples of the continuous image signal at integer values of the image indices. For the NumRows x NumCols image grid, the image indices (irow, icol) span the following values. Image row coordinates of the first and last rows are xrow1 and xrow2. Image column coordinates of the first and last column are ycol1 and ycol2. See Figure 4.4-9.

$$\begin{aligned}
 irow1 &\leq irow \leq irow2 & icol1 &\leq icol \leq icol2 \\
 irow1 &= FirstRow - SCPPixel.Row & icol1 &= FirstCol - SCPPixel.Col \\
 irow2 &= irow1 + NumRows - 1 & icol2 &= icol1 + NumCols - 1 \\
 xrow1 &= Row_SS \cdot irow1 & ycol1 &= Col_SS \cdot ycol1 \\
 xrow2 &= Row_SS \cdot irow2 & ycol2 &= Col_SS \cdot ycol2
 \end{aligned}$$

The image domain impulse response widths (IRWs) are key attributes of the image. The half power impulse response widths are Row_IRW and Col_IRW. For most images, the impulse response is constant or varies only slightly within the image. The half power impulse response width is the “focused” response width at the SCP that is the goal of the processing. For images requiring post-image formation focusing (i.e. autofocus), the actual IRW may be degraded.

$$Row_IRW = Row.ImpRespWid \qquad Col_IRW = Col.ImpRespWid$$

The transform of the row coordinate is the row spatial frequency coordinate (xrow $\Leftrightarrow$ Krow). The transform of the column coordinate is the azimuth spatial frequency coordinate (ycol $\Leftrightarrow$ Kcol). The required spatial frequency parameters provided with all image products are as follows. All spatial frequency parameters are in cycles/meter.

$$\begin{aligned}
 R2K &= Row.Sgn & C2K &= Col.Sgn \\
 Krow_IRBW &= Row.ImpRespBW & Kcol_IRBW &= Col.ImpRespBW \\
 Krow_Ctr &= Row.KCtr & Kcol_Ctr &= Col.KCtr \\
 \Delta Krow1 &= Row.DeltaK1 & \Delta Kcol1 &= Col.DeltaK1 \\
 \Delta Krow2 &= Row.DeltaK2 & \Delta Krow2 &= Col.DeltaK2
 \end{aligned}$$

The sampled image signal is $S_{II}(irow, icol)$. The sampled spatial frequency signal is $S_{KK}(nrow, ncol)$. For all SICD products, the sampled spatial frequency is computed as follows. The expression for $S_{KK}(nrow, ncol)$ is highlighted as it is critical to both producers and users of SICD products.

For $nrow = -\frac{1}{2}NFR, \dots, 0, \dots, \frac{1}{2}NFR$:

For $ncol = -\frac{1}{2}NFC, \dots, 0, \dots, \frac{1}{2}NFC$:

$$S_{KK}(nrow, ncol) = \sum_{irow=irow1}^{irow2} \sum_{icol=icol1}^{icol2} S_{II}(irow, icol) \cdot \exp \left\{ j2\pi \cdot \left[R2K \cdot \frac{irow \cdot nrow}{NFR} + C2K \cdot \frac{icol \cdot ncol}{NFC} \right] \right\}$$

- (1) Parameters R2K and C2K define the signs of the exponent in the Discrete Fourier Transform (DFT). For all images, the values of R2K and C2K shall be equal (i.e. $R2K = C2K = +1$ or $R2K = C2K = -1$). The value of R2K (Row.Sgn) is the sign of exponent that transforms the increasing xrow dimension to yield increasing RF frequency.
- (2) The “zero point” on the input signal is the SCP pixel ($S_{II}(0,0)$). Parameters NFR and NFC are the sizes of the transformed signal and are chosen by the image exploitation application.

For convenience, NFR and NFC are assumed to be even integers. Also for convenience, index nrow spans $NFR + 1$ values and index ncol spans $NFC + 1$ values.

The center of the spatial frequency is $S_{KK}(0,0)$. The spatial frequencies of the center point are (Krow_Ctr, Kcol_Ctr). See Figures 4.4-10 and 4.4-11. Spatial frequency Krow_Ctr is always positive and related to the center of the processed RF spectrum. For many images, spatial frequency parameter Kcol_Ctr is equal to 0. The spatial frequency coordinates (Krow, Kcol) are related to the spatial frequency indices (nrow, ncol) as follows. Offset frequencies $\Delta Krow$ and $\Delta Kcol$ are offsets relative to $S_{KK}(0,0)$.

$$Krow_SS = \frac{1}{Row_SS \cdot NFR}$$

$$Kcol_SS = \frac{1}{Col_SS \cdot NFC}$$

$$\Delta Krow(nrow) = Krow_SS \cdot nrow$$

$$\Delta Kcol(ncol) = Kcol_SS \cdot ncol$$

$$Krow(nrow) = Krow_Ctr + \Delta Krow(nrow) \quad Kcol(nrow) = Kcol_Ctr + \Delta Kcol(ncol)$$

Offset frequencies $\Delta Krow$ and $\Delta Kcol$ span the following ranges of values.

$$-\frac{1}{2 \cdot Row_SS} \leq \Delta Krow \leq \frac{1}{2 \cdot Row_SS}$$

$$-\frac{1}{2 \cdot Col_SS} \leq \Delta Kcol \leq \frac{1}{2 \cdot Col_SS}$$

The span of spatial frequencies containing valid data are indicated by parameters $\Delta Krow1$, $\Delta Krow2$, $\Delta Kcol1$ and $\Delta Kcol2$. The parameters indicate the support in the spatial frequency domain for the entire image. Individual image pixels may be formed from a subset of the spatial frequency support for the image.

The spatial frequency support (often referred to as the spatial frequency aperture) for the SCP is described by bandwidth parameters $Krow_IRBW$ and $Kcol_IRBW$. An example spatial frequency support is shown in Figure 4.4-12. The bandwidths are measured at the center of the support in the $Krow$ and $Kcol$ dimensions. For the example shown, the spatial frequency support is a rectangle and centered at $S_{KK}(0,0)$. For images with non-centered spatial frequency support, optional polynomial functions may be included to describe the variation in spatial frequency support as a function of position in the image. For images that have variable aperture centers in row spatial frequency, two-dimensional polynomial coefficients $c\Delta KrowCOA(m,n)$ may be included (order M_KROW and order N_KROW). For images that have variable aperture centers in column spatial frequency, two-dimensional polynomial coefficients $c\Delta KcolCOA(m,n)$ may be included (order M_KCOL and order N_KCOL).

For $m = 0$ to M_KROW & $n = 0$ to N_KROW :

$$c\Delta KrowCOA(m,n) = Row.DeltaKCOAPoly(m,n)$$

For $m = 0$ to M_KCOL & $n = 0$ to N_KCOL :

$$c\Delta KcolCOA(m,n) = Col.DeltaKCOAPoly(m,n)$$

For a given target at image coordinates $(xrow^{TGT}, ycol^{TGT})$, the center of support is assumed to be at $\Delta KrowCOA = 0$ unless the $\Delta KrowCOA$ polynomial is included. The center of support is assumed to be at $\Delta KcolCOA = 0$ unless the $\Delta KcolCOA$ polynomial is included. When the polynomial(s) are provided (either one or both), the center of support relative to $S_{KK}(0,0)$ is then computed as shown below.

$$\Delta KrowCOA^{TGT} = \sum_{m=0}^{M_KROW} \sum_{n=0}^{N_KROW} c\Delta KrowCOA(m,n) \cdot (xrow^{TGT})^m \cdot (ycol^{TGT})^n$$

$$\Delta KcolCOA^{TGT} = \sum_{m=0}^{M_KCOL} \sum_{n=0}^{N_KCOL} c\Delta KcolCOA(m,n) \cdot (xrow^{TGT})^m \cdot (ycol^{TGT})^n$$

A target with non-centered support is shown in Figure 4.4-13. Due to the circular nature of the sampled spectrum, values of $\Delta KrowCOA$ computed from the polynomial may need to be adjusted by an integer multiple of $1/Row_SS$ to yield the effective $\Delta KrowCOA$. Similarly, values of $\Delta KcolCOA$ computed from the polynomial may need to be adjusted by an integer multiple of $1/Col_SS$ to yield the effective $\Delta KcolCOA$. Constant coefficients $c\Delta KrowCOA(0,0)$ and $c\Delta KcolCOA(0,0)$ are offset frequencies for the SCP.

$$\Delta K_{row}COA^{SCP} = c\Delta K_{row}COA(0,0) \quad \Delta K_{col}COA^{SCP} = c\Delta K_{col}COA(0,0)$$

Amplitude weighting applied in the spatial frequency domain to control the properties of the range and azimuth impulse responses may be described using the optional weighting parameters. Text fields may be included as well as a set of samples that define the shape of the window.

For $m = 1$ to NWROW:

$$K_{row_Wgt}(m) = Row.WgtFunct.Wgt(m)$$

For $n = 1$ to NWCOL:

$$K_{col_Wgt}(n) = Col.WgtFunct.Wgt(n)$$

The parameters provided describe the weighting applied to SCP spatial frequency support. See Figure 4.4-12. The row weights represent uniformly spaced samples of the weighting function applied to the K_{row_IRBW} . Weight $K_{row_Wgt}(m)$ is applied at:

$$\Delta K_{row}(m) = \Delta K_{row}COA^{SCP} + \left(\frac{m-1}{NWROW-1} - \frac{1}{2} \right) \cdot K_{row_IRBW}$$

The column weights represent uniformly spaced samples of the weighting function applied to the K_{col_IRBW} . Weight $K_{col_Wgt}(n)$ is applied at:

$$\Delta K_{col}(n) = \Delta K_{col}COA^{SCP} + \left(\frac{n-1}{NWCOL-1} - \frac{1}{2} \right) \cdot K_{col_IRBW}$$

For all other points in the image, it is assumed that the same weighting function(s) are applied and centered on the center of support for each target. For the example target shown in Figure 4.4-13, the weighting function is centered at $\Delta K_{row}COA^{TGT}$ and $\Delta K_{col}COA^{TGT}$.

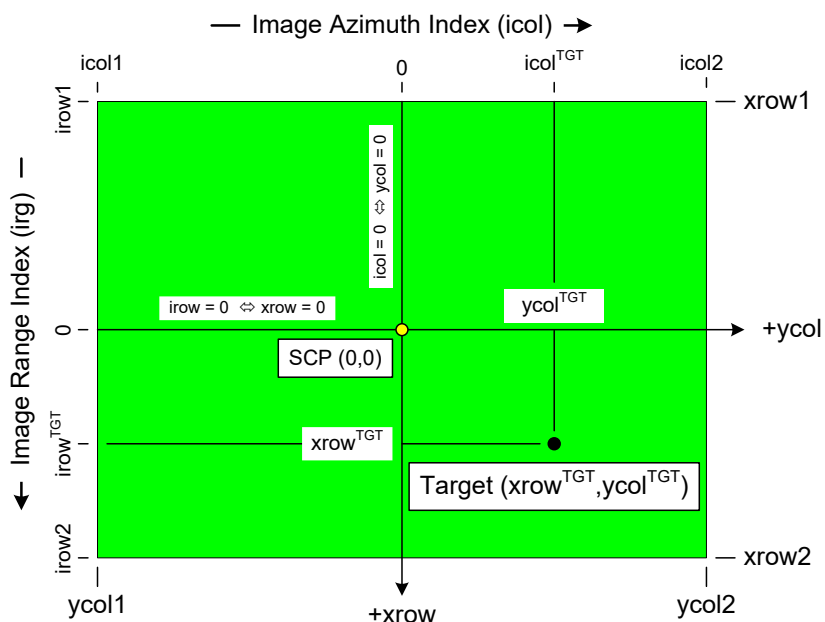


Figure 4.4-9 Image Domain Extent

Image domain extent of the NumRows x NumCols image pixel array.

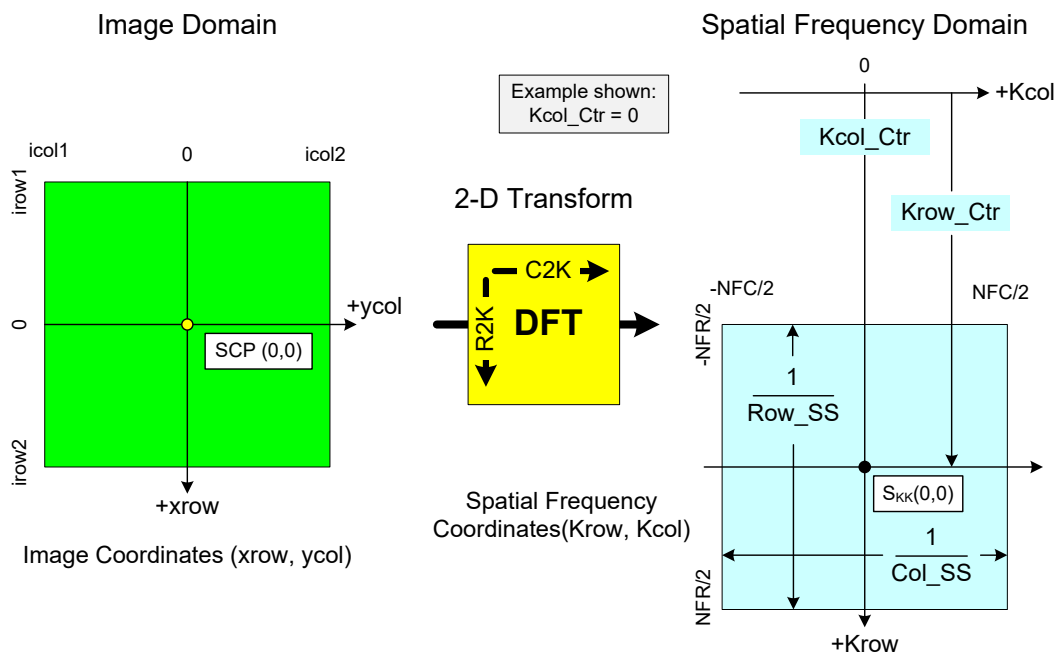
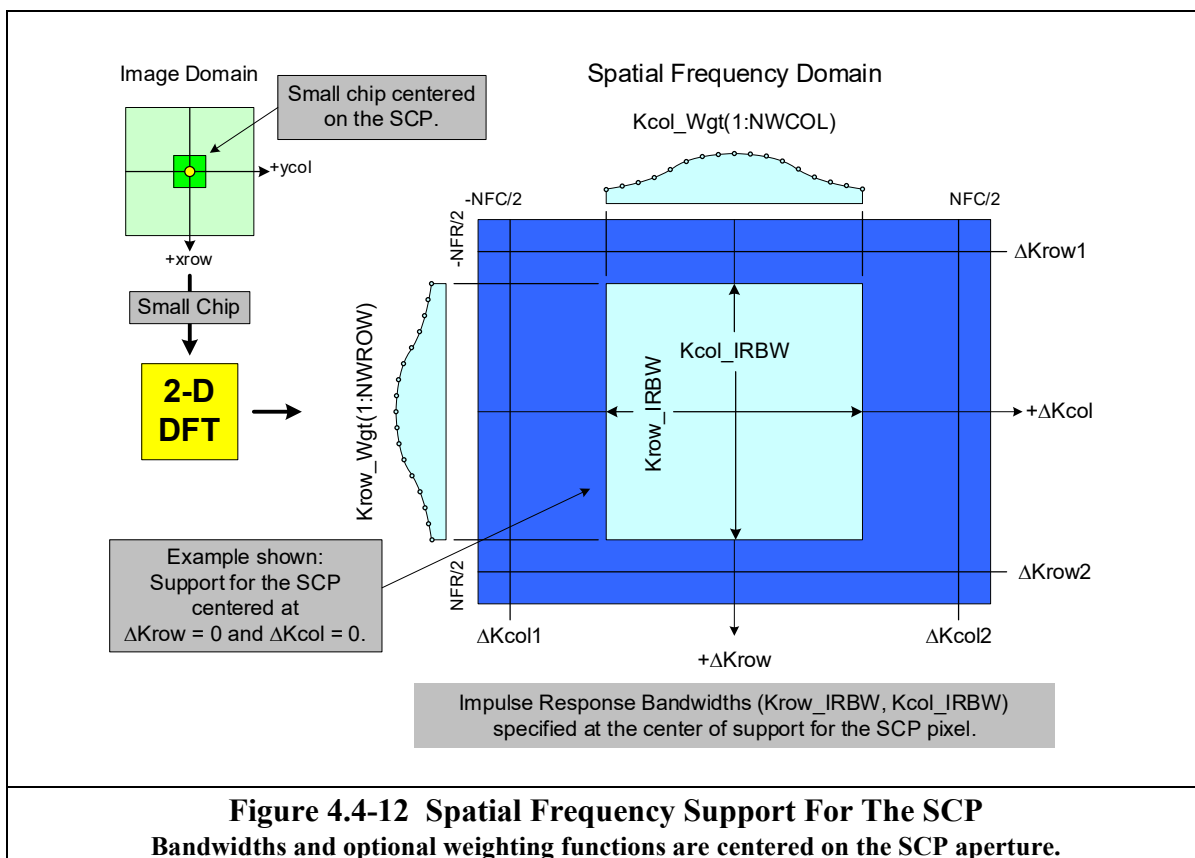
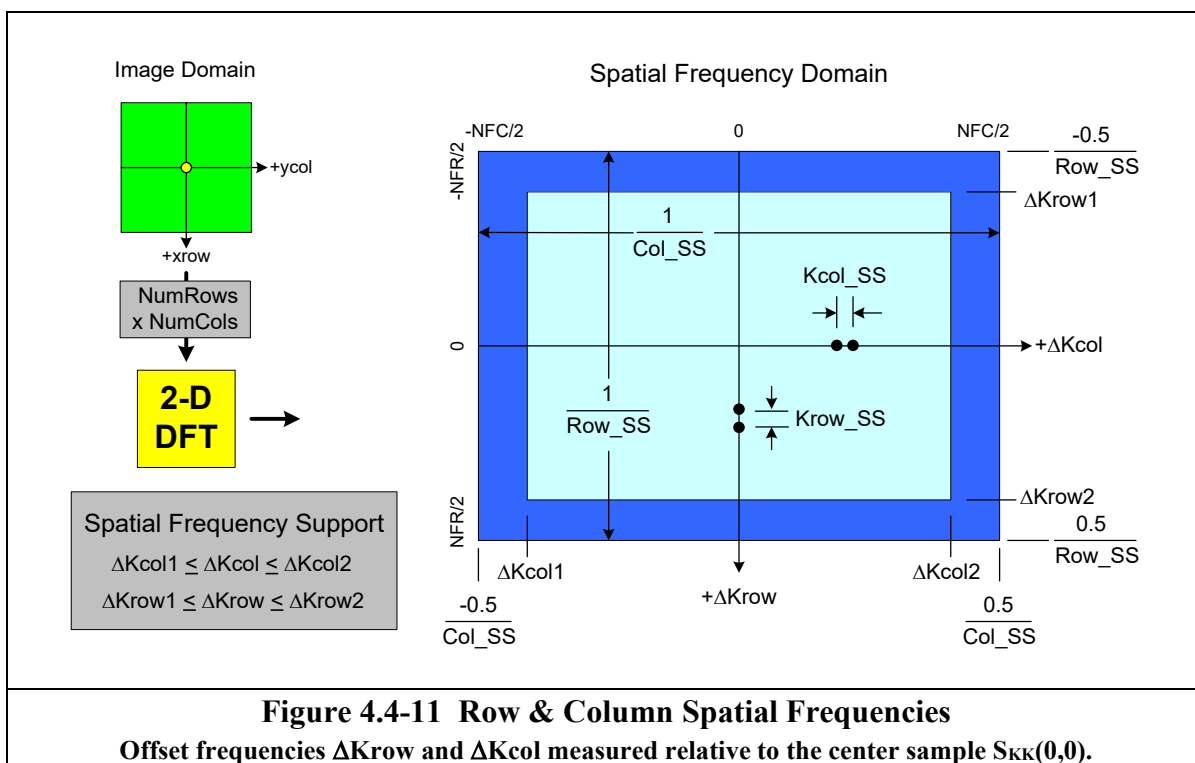
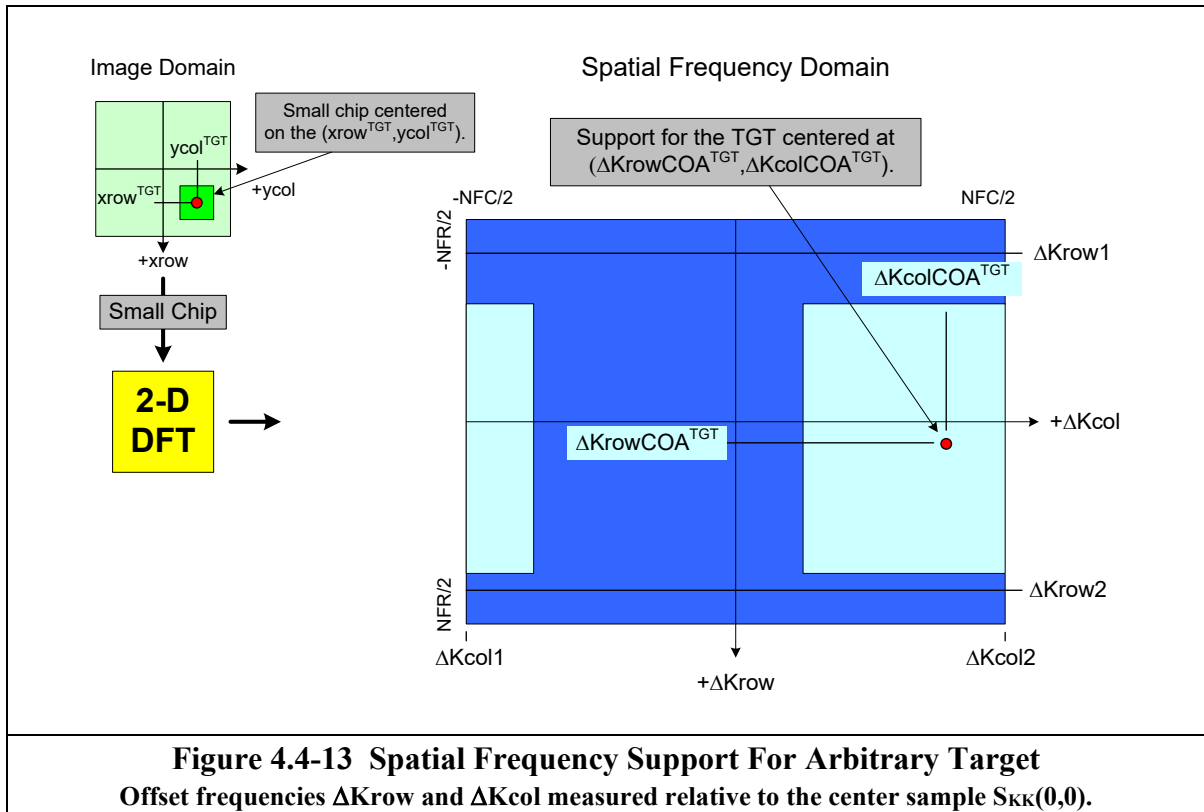


Figure 4.4-10 Image To Spatial Frequency Transform

Spatial frequency centered at $Krow = Krow_Ctr$ and $Kcol = Kcol_Ctr$.

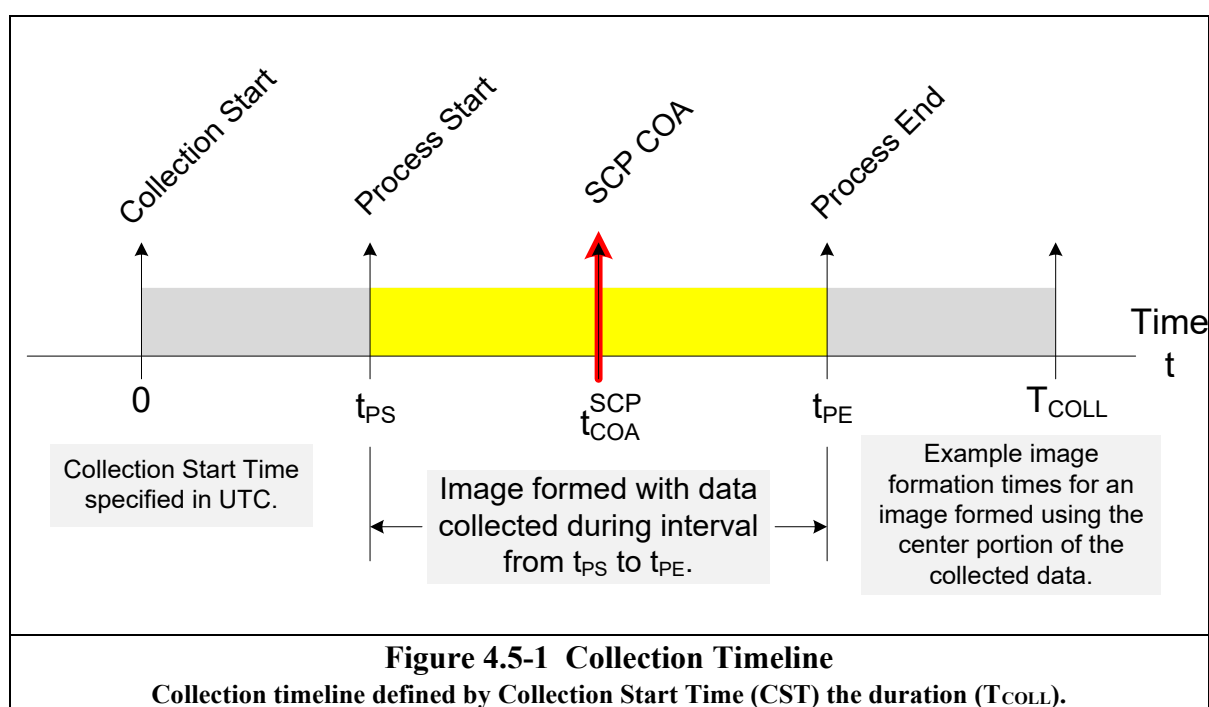




4.5 Collection Timeline Parameters

The parameters that describe the imaging collection timeline are summarized below. Refer to Table 3-5.

The SAR imaging collection timeline is specified by the Collection Start Time and the imaging collection duration. The Collection Start time, denoted CST, shall be specified in Coordinated Universal Time (UTC). The collection duration, denoted T_{COLL} , shall be specified in seconds. For a given collection, the CST and T_{COLL} are independent of the image products that are produced. An example collection timeline is shown in Figure 4.5-1. The CST is the reference for all time-based parameters. Parameters based on time in seconds relative to CST include the position versus time polynomials and COA time (t_{COA}) versus image grid location.



The accurate exploitation of a given image product is not directly dependent upon the CST or the T_{COLL} . Exploitation processing makes use of time parameters expressed in seconds relative to CST. Shown in Figure 4.5-1 are times associated with an example image product. The image formation processing used data collected in the period from t_{PS} to t_{PE} . Times t_{PS} and t_{PE} are included in the Image Formation parameters (see Table 3-8). The image COA times are included in the Image Grid parameters (see Table 3-4).

XML: ImageFormation.TStartProc = t_{PS}

XML: ImageFormation.TEndProc = t_{PE}

The processing times, t_{PS} and t_{PE} , as well as the COA times are defined to be “reference times” in that they are used with the Aperture Reference Point (ARP) position versus time

polynomial. For a given vector of the collected Phase History Data (PHD), the reference time is computed as follows. For the pulse with transmit center time at $t = t_{xc}$ and for which the center of the echo from the center collected range swath is received at time $t = t_{rc}$, the reference time is $t_{ref} = (t_{xc} + t_{rc})/2$. The ARP position associated with the vector is approximately the mid-point between the transmit phase center at time t_{xc} and the receive phase center at time t_{rc} .

Parameters that describe the radar transmit and receive sequence may also be included. These optional parameters are recommended to be included in all products. The timeline is a sequence of Inter-Pulse Periods (IPPs). An IPP is often referred to as a Pulse Repetition Interval (PRI). The IPP includes one transmitted pulse and one listening window where the echoes are received and digitized to form a PHD vector. The IPP rate is referred to as the Pulse Repetition Frequency (PRF). While precise knowledge of the IPP sequence timing is needed for image formation processing, an approximate description will adequately support the analysis of the image products.

The IPP sequence shall contain the earliest and latest slow time values for data processed to form the image, t_{ps} and t_{pe} . The IPP sequence is described by one or more sets of parameters. Each set of parameters spans a portion of the collection time. The sets of parameters are indexed by $idx = 1$ to N_{Sets} . The parameters included in each set are the following.

$t_s(idx)$ = Start time of the interval (in sec).	XML: IPP.Set.TStart
$t_e(idx)$ = End time of the interval (in sec).	XML: IPP.Set.TEnd
$k_s(idx)$ = IPP index of the first IPP in interval.	XML: IPP.Set.IPPStart
$k_e(idx)$ = IPP index of the last IPP in the interval.	XML: IPP.Set.IPPEnd
IPPPoly(idx) = IPP index polynomial, $(cT2IPP(idx,m))$.	XML: IPP.Set.IPPPoly

The IPP index polynomial converts time t in seconds relative to CST to IPP index value. For a given interval, time $t_s(idx)$ is the start of IPP $k_s(idx)$ and time $t_e(idx)$ is the end of IPP $k_e(idx)$. The IPP Index polynomial provides an estimate of the IPP index value for times from $t_s(idx)$ to $t_e(idx)$,

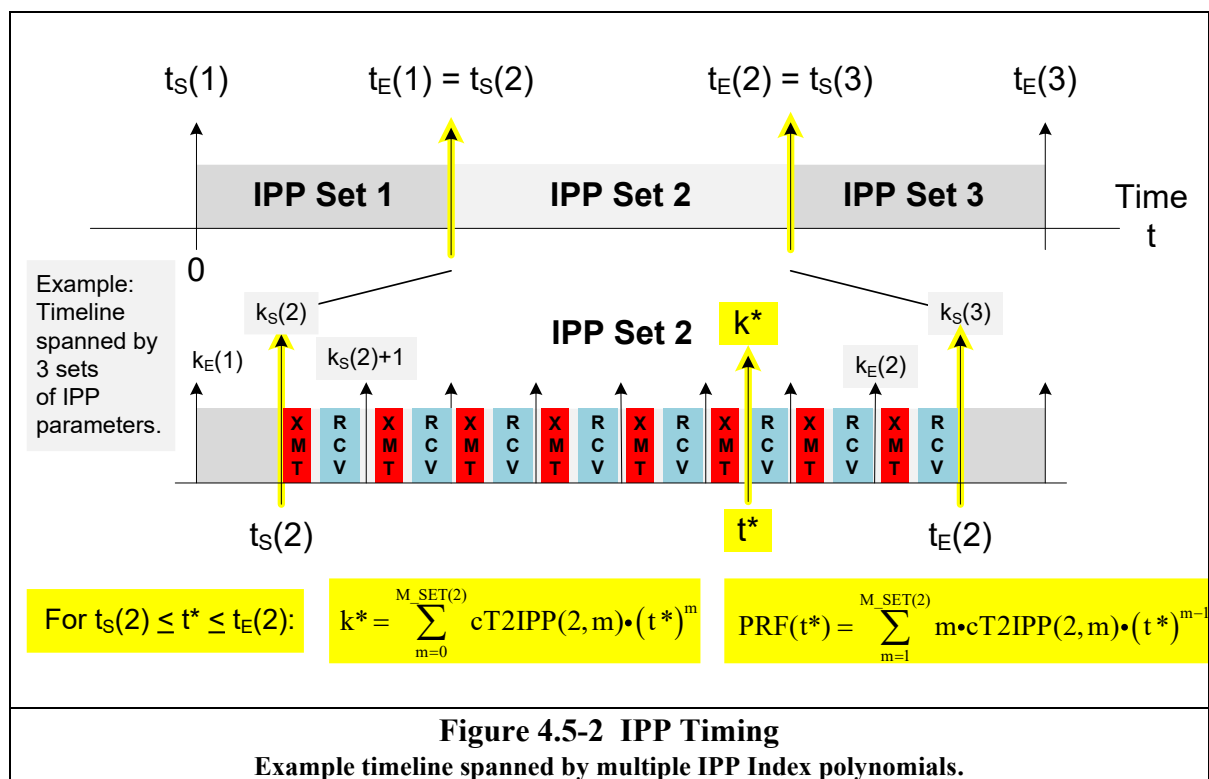
An example timeline spanned by three sets of IPP parameters is shown in Figure 4.5-2. At a given time t^* , the corresponding IPP index k^* is computed by selecting the polynomial that spans the time of interest. For time t^* in the second interval, $t_s(2) \leq t^* < t_e(2)$, index k^* is computed as follows. The instantaneous PRF at t^* may also be computed.

$$k^* = \sum_{m=0}^{M_{SET}(2)} cT2IPP(2, m) \cdot (t^*)^m$$

$$PRF(t^*) = \frac{d}{dt} k(t^*) = \sum_{m=1}^{M_{SET}(2)} m \cdot cT2IPP(2, m) \cdot (t^*)^{m-1}$$

The index k^* takes on an integer value at the start of the IPP. For example, $k^* = 1000.0$ indicates that t^* is the start time of IPP 1000. For most SAR systems, the duration of any given IPP is actually an integer number of system clock cycles. For these systems, the polynomial fit can only approximate a varying IPP duration. For these systems, the

estimated IPP index and PRF are usually very accurate. The estimated PRF, for example, will usually accurately predict the location of ambiguities in range and azimuth.

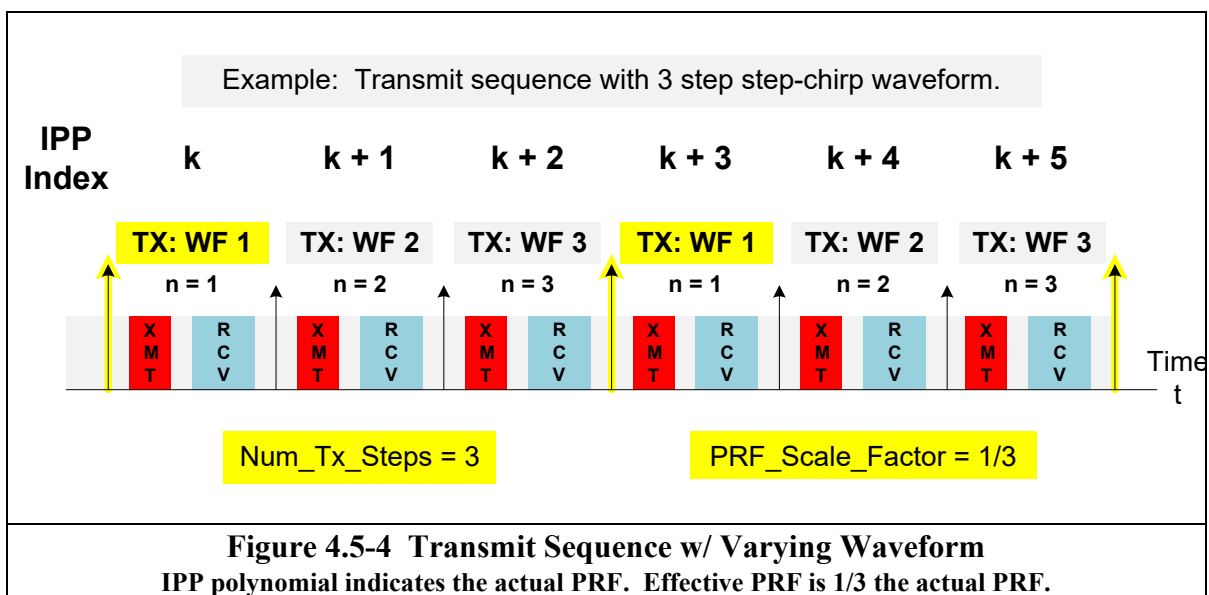
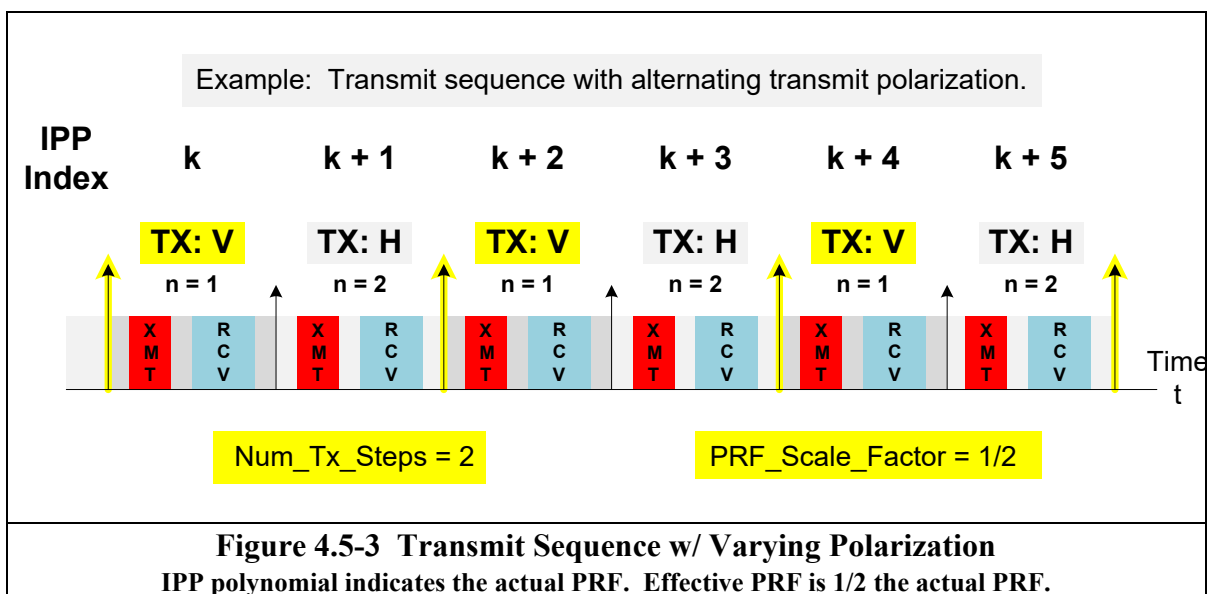


The IPP timing parameters are intended to reflect the true timing of the SAR system during a collection. For systems that used a short, repeating sequence on transmit, the IPP parameters will describe the individual steps of the sequence. The repeating transmit sequence is described with Radar Collection structure TxSequence. See Table 3-7. Shown in Figure 4.5-3 is an example collection with alternating transmit polarizations (vertical polarization on one IPP and horizontal polarization on the next). Radar collection parameter Num_Tx_Steps = 2. For an image formed with data collected using only a single transmit polarization, the Image Formation parameter PRF_Scale_Factor = 1/2 indicating the Effective PRF is 1/2 of the actual PRF. See Table 3-8.

XML: RadarCollection.TxSequence size = 2 ⇔ NumTxSteps = 2
 XML: ImageFormation.RcvChanProc.PRFScaleFactor = 1/2

Shown in Figure 4.5-4 is an example collection with a step-chirp set of waveforms. The transmitted pulse sequences through 3 distinct waveforms. Radar collection parameter Num_Tx_Steps = 3. For an image formed with the collected data, the Image Formation parameter PRF_Scale_Factor = 1/3 indicating the Effective PRF is 1/3 of the actual PRF.

XML: RadarCollection.TxSequence size = 3 ⇔ NumTxSteps = 3
 XML: ImageFormation.RcvChanProc.PRFScaleFactor = 1/3



4.6 Reference Position Parameters

The parameters that specify the positions that vary with time during an imaging collection are summarized below. Refer to Table 3-6.

The position metadata that specifies the imaging geometry and is required for subsequent processing is provided in all products. The position metadata includes the SAR platform position(s) as well as the position for points located in the imaged scene. The position versus time metadata that may be included are for the following types of points.

- (1) Antenna Phase Center (APC): An APC is a point located on or near the physical antenna used during the collection. A unique APC position may be specified for the antenna aperture used on transmit and on receive.
- (2) Ground Reference Point (GRP): The GRP is a point in the imaged scene that is used to specify the instantaneous collection geometry. For a given image, the GRP may be a fixed point or a moving point.
- (3) Aperture Reference Point (ARP): The ARP is a single point that is used to specify the effective position versus time for the collected signal data. The ARP displacement during imaging collection defines the synthetic aperture used to form the image.

For all points the positions are specified as polynomial functions of time. The input time values are in seconds relative to the Collection Start Time. The computed positions are in ECEF coordinates. The position versus time metadata provided is combined with the SCP position to fully specify the collection geometry.

4.6.1 APC Positions vs. Time

The APC position vs. time may be provided for each antenna aperture used on transmit and on receive to collect the Phase History Data (PHD). The position polynomial may be computed via a curve fit to the measured or predicted APC position during the collection. For an orbiting platform, the position polynomial function may provide a precise description of the APC trajectory. For an airborne platform, the position vs. time polynomial may only provide an approximation to the APC position trajectory. However, for use in the subsequent processing of the complex image, the polynomial fit to the true APC trajectory provides an acceptable representation of the APC trajectory.

The APC position metadata elements are optional for monostatic images and are required for bistatic images. The following guidance is provided for how the APC position metadata is to be provided in a given product.

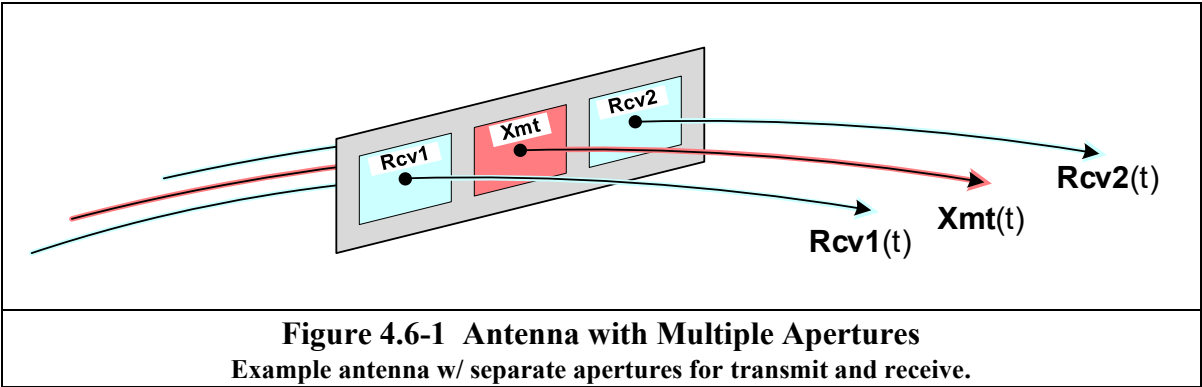
- (1) A single Transmit APC (TxAPC) position and one or more Receive APC (RcvAPC) positions may be included in a product.
- (2) For monostatic collections that use separate aperture locations on transmit and receive or use multiple Receive APCs, the TxAPC and RcvAPC position metadata provides unique information. For an image formed from such a

collection, the APC position metadata is optional but is encouraged to be included.

- (3) TxAPC and RcvAPC position metadata shall be included for bistatic products. The APC position metadata is required to support precise image projection computations.
- (4) TxAPC and RcvAPC position metadata shall be included for image products that include one-way antenna pattern metadata. See Table 3-11.
- (5) For each APC position, the polynomial function should provide an accurate position estimate for the entire slow time period for which the PHD was collected and used to form the image. See Table 3-8, Image Formation parameters TStartProc and TEndProc.
- (6) The order of the APC position polynomials is a design choice that depends on the dynamics of the platform, the duration of the collections and the imaging geometry.

An example antenna with separate transmit and receive apertures is shown in Figure 4.6-1. The transmit aperture is shown in red and the transmit APC position is denoted **Xmt(t)**. The two receive apertures are shown in blue and the receive APC positions are denoted **Rcv1(t)** and **Rcv2(t)**. For the example shown, the APC position XML is populated as shown below. The size attribute of the parameter RcvAPC indicates the number of receive APCs included is 2. The index attribute of each RcvAPCPoly identifies the receive APC number.

XML: Position / TxAPCPoly	Xmt(t)
XML: Position / RcvAPC	size = 2 ⇔ NumRcvAPCs = 2
XML: Position / RcvAPC / RcvAPCPoly	index = 1 ⇔ Rcv1(t)
XML: Position / RcvAPC / RcvAPCPoly	index = 2 ⇔ Rcv2(t)

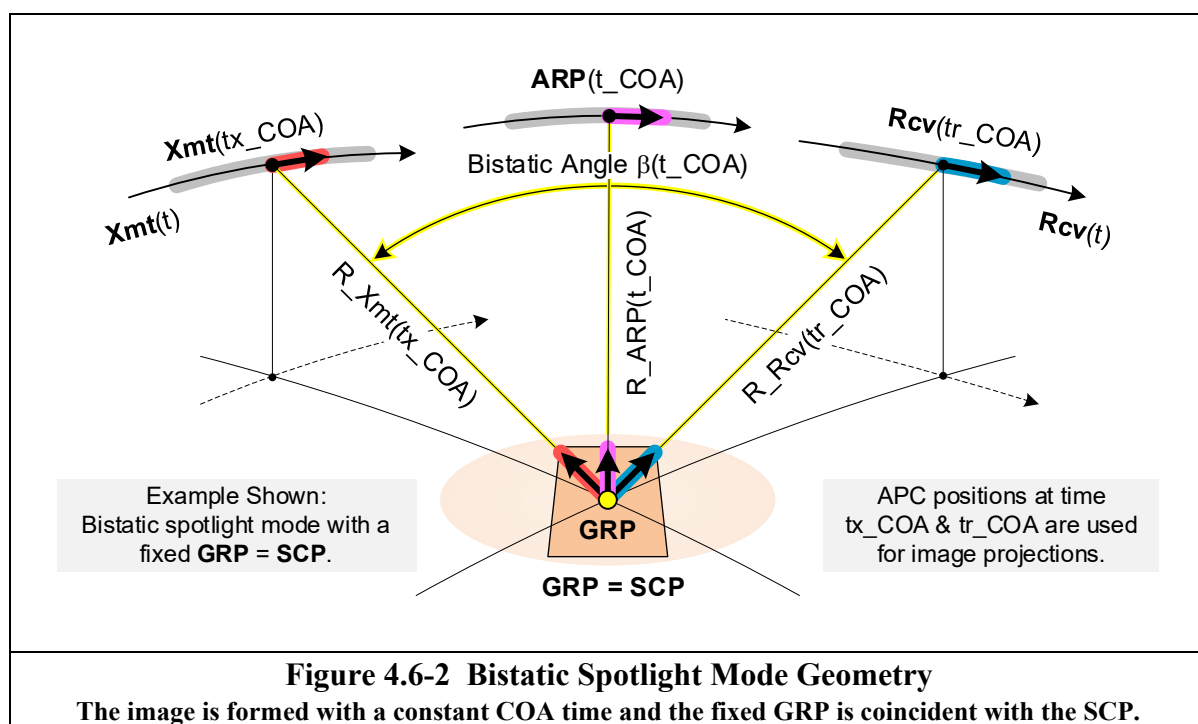


4.6.2 GRP Position vs. Time

The GRP is a point in the imaged scene that may be provided to specify the nominal center of the illuminated area as a function of slow time. For a Spotlight mode collection, the GRP is set to a fixed point. For a Stripmap mode or a Dynamic Stripmap mode, the GRP is typically a moving point that moves along the center of the imaged strip. The GRP position vs. time nominally moves at the same rate as the center of the Illuminator main lobe antenna footprint.

For a monostatic image, the GRP position metadata is optional. For spotlight modes, it provides limited additional information. For stripmap modes, providing a GRP that tracks the center of illumination along a local level surface is useful. The GRP provides an easy method for users of the product to compute the position and speed of the scanning antenna mainlobe footprint as a function of time.

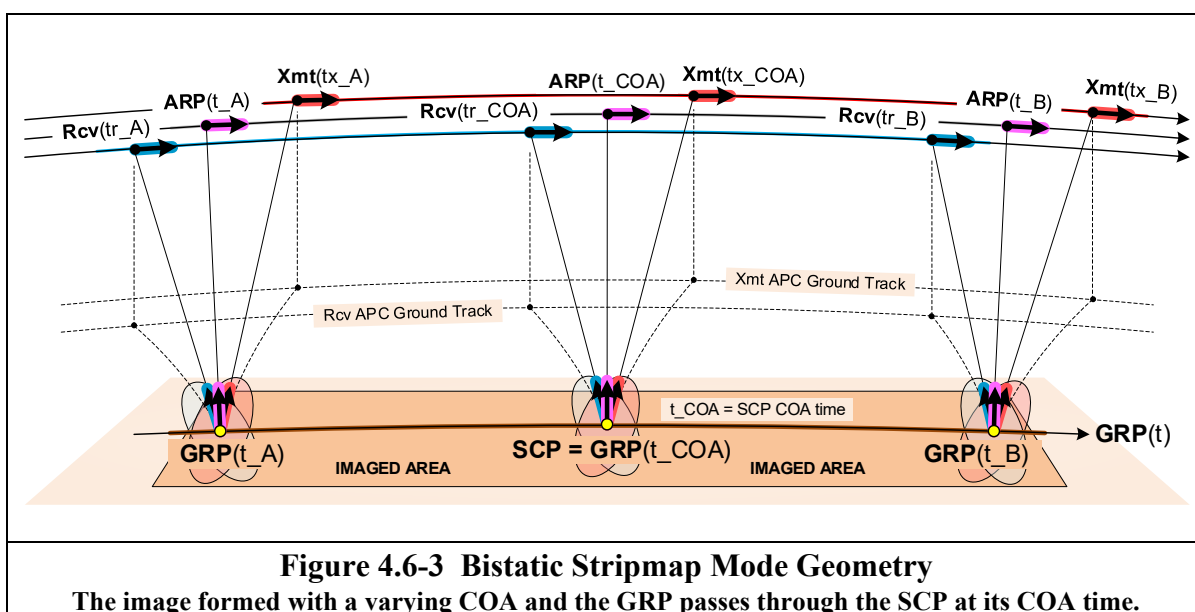
For a bistatic image, the GRP position metadata is required. The GRP is used to compute the bistatic geometry as a function of slow time. The bistatic geometry is then used, for example, to support precise image projection computations. For an image formed from a bistatic spotlight mode collection, the image is typically formed with a fixed COA time for all pixels. For such an image, the GRP is a fixed point and is coincident with the SCP. Shown in Figure 4.6-2 is an example bistatic spotlight mode collection. The transmit APC position vs. time is $\mathbf{Xmt}(t)$ and the receive APC position versus time is $\mathbf{Rcv}(t)$. Also shown is the ARP position vs. time. The highlighted positions $\mathbf{Xmt}(tx_COA)$ and $\mathbf{Rcv}(tr_COA)$ are the APC positions that are used for precise image projection computations.



For an image formed from a bistatic stripmap mode, the image is typically formed with a varying COA time. The COA time vs. image location is typically computed such that the COA for points in the scene nominally occurs at the time of maximum of two-way antenna gain. For such an image, the GRP moves along the center line of the imaged strip. For points along the GRP trajectory, the GRP passes through the point at approximately its COA time. Shown in Figure 4.6-3 is an example of a bistatic stripmap collection. In the example shown, the two platforms are flying parallel to the imaged strip. Shown at the center of the image area is the SCP for the image. The COA time for the SCP is denoted t_{COA} . The moving GRP passes through the SCP at $t = t_{COA}$. Also shown is the GRP at an earlier time t_A and a later time t_B (i.e., $t_A < t_{COA} < t_B$).

For a bistatic image, the GRP position polynomial provided with the product is to be computed based on the following guidance.

- (1) For an image formed with a constant COA time for all pixels, the GRP is a fixed point and is coincident with SCP.
- (2) For an image formed with a varying COA time, the GRP is a moving point that passes through the SCP at the SCP COA time.
- (3) For a moving GRP, the GRP trajectory is nominally parallel to the surface of constant HAE at the SCP.
- (4) For a moving GRP, the GRP passes through a given point on its trajectory at approximately the COA time for the point. The image formation COA time function, $t_{COA}(xrow, ycol)$, and the GRP position polynomial are coupled. Refer to Section 4.4.3.



4.6.3 ARP Position vs. Time

The Aperture Reference Point (ARP) position polynomial is required for all images. The ARP is computed based upon the APCs of the antenna apertures used to collect the PHD. The ARP provides a single reference position vs. time for the subsequent processing of the image array. Hence, the name Aperture Reference Point. A method for computing the ARP position polynomial is provided below.

For a monostatic image, the ARP position vs. time is all that is needed for the subsequent processing of the image array. The ARP is a point that is located on or near the APCs that were used to collect the PHD that was used to form the image. For a collection that uses a single aperture on transmit and receive, the ARP is coincident with the APC. For a monostatic image, the ARP position is computed directly from the APC position(s). For the method shown below, the APC polynomials are assumed to be computed even if they are not included in the product metadata

For a bistatic image, the ARP position vs. time is useful for many types of processing and exploitation of the image array. At a given time, the ARP is a virtual point that is nominally located between the two platforms. For the example collections shown in Figure 4.6-2 and Figure 4.6-3, the ARP position is shown between the transmit and receive positions. For a bistatic image, the ARP position polynomial is computed from APC positions and the GRP position.

Method for Computing the ARP Position Polynomial

For a given image, the ARP position vs. time may be computed per the following method. The ARP position polynomial is computed from the APC positions that span the slow time period when the PHD used to form the image was collected. The slow time period is from:

$$t_Start_Proc \leq t \leq t_End_Proc$$

Refer to Table 3-8, ImageFormation parameters TStartProc & TEndProc.

- (1) Determine/compute the APC position vs. time polynomials for the transmit APC and for the receive APCs that were used to collect the PHD that was used to form the image.

Transmit APC: **Xmt_Poly(t)** Receive APCs: **Rcv1_Poly(t), Rcv2_Poly(t)**

For most images, a single receive APC is used to collect the PHD. For the example above, two receive APCs were used as per the example shown in Figure 4.6-1.

- (2) Compute the receive APC “centroid” position vs. time, denoted **Rcv(t)**, that is the average of the APC positions of the receive apertures used to collect the PHD that was used to form the image.

Receive APC Centroid: **Rcv_Poly(t) = 1/2 • { Rcv1_Poly(t) + Rcv2_Poly(t) }**

- (3) For a monostatic image, compute the ARP position vs. time as the average of the transmit APC position and the receive APC centroid position.

For a Monostatic image: $\mathbf{ARP_Poly}(t) = \frac{1}{2} \bullet \{ \mathbf{Xmt_Poly}(t) + \mathbf{Rcv_Poly}(t) \}$

For a bistatic image, the ARP polynomial may be computed as described in Steps (4) and (5).

- (4) Compute a set of NT equally spaced times, denoted $\{t(n), n = 0, 1, 2, \dots, NT-1\}$, that span the period t_Proc_Start to t_Proc_End . For each of the times, compute the following parameters. Refer to Figure 4.6-4.

- (a) Compute the GRP position at time $t(n)$, $\mathbf{G}(n)$:

$$\mathbf{G}(n) = \mathbf{GRP_Poly}(t(n))$$

- (b) For a transmit signal that is incident at $\mathbf{G}(n)$ at time $t(n)$, compute an estimate of the time it is transmitted, $tx(n)$. Compute the transmit APC position at time $tx(n)$, $\mathbf{X}(n)$.

$$\mathbf{X0}(n) = \mathbf{Xmt_Poly}(t(n)) \quad R_X0_G(n) = |\mathbf{X0}(n) - \mathbf{G}(n)|$$

$$tx(n) = t(n) - (1/c) \bullet R_X0_G(n) \quad \mathbf{X}(n) = \mathbf{Xmt_Poly}(tx(n))$$

- (c) Compute the range from $\mathbf{X}(n)$ to $\mathbf{G}(n)$, $R_X_G(n)$, and the unit vector that points from $\mathbf{G}(n)$ to $\mathbf{X}(n)$, $\mathbf{uX}(n)$.

$$R_X_G(n) = |\mathbf{X}(n) - \mathbf{G}(n)| \quad \mathbf{uX}(n) = \frac{1}{R_X_G(n)} \bullet \{ \mathbf{X}(n) - \mathbf{G}(n) \}$$

- (d) For a transmit signal that is incident at $\mathbf{G}(n)$ at time $t(n)$, compute an estimate of the time it is received at the receive APC centroid, $tr(n)$. Compute the receive APC centroid position at time $tr(n)$, $\mathbf{R}(n)$.

$$\mathbf{R0}(n) = \mathbf{Rcv_Poly}(t(n)) \quad R_R0_G(n) = |\mathbf{R0}(n) - \mathbf{G}(n)|$$

$$tr(n) = t(n) + (1/c) \bullet R_R0_G(n) \quad \mathbf{R}(n) = \mathbf{Rcv_Poly}(tr(n))$$

- (e) Compute the range from $\mathbf{R}(n)$ to $\mathbf{G}(n)$, $R_R_G(n)$, and the unit vector that points from $\mathbf{G}(n)$ to $\mathbf{R}(n)$, $\mathbf{uR}(n)$.

$$R_R_G(n) = |\mathbf{R}(n) - \mathbf{G}(n)| \quad \mathbf{uR}(n) = \frac{1}{R_R_G(n)} \bullet \{ \mathbf{R}(n) - \mathbf{G}(n) \}$$

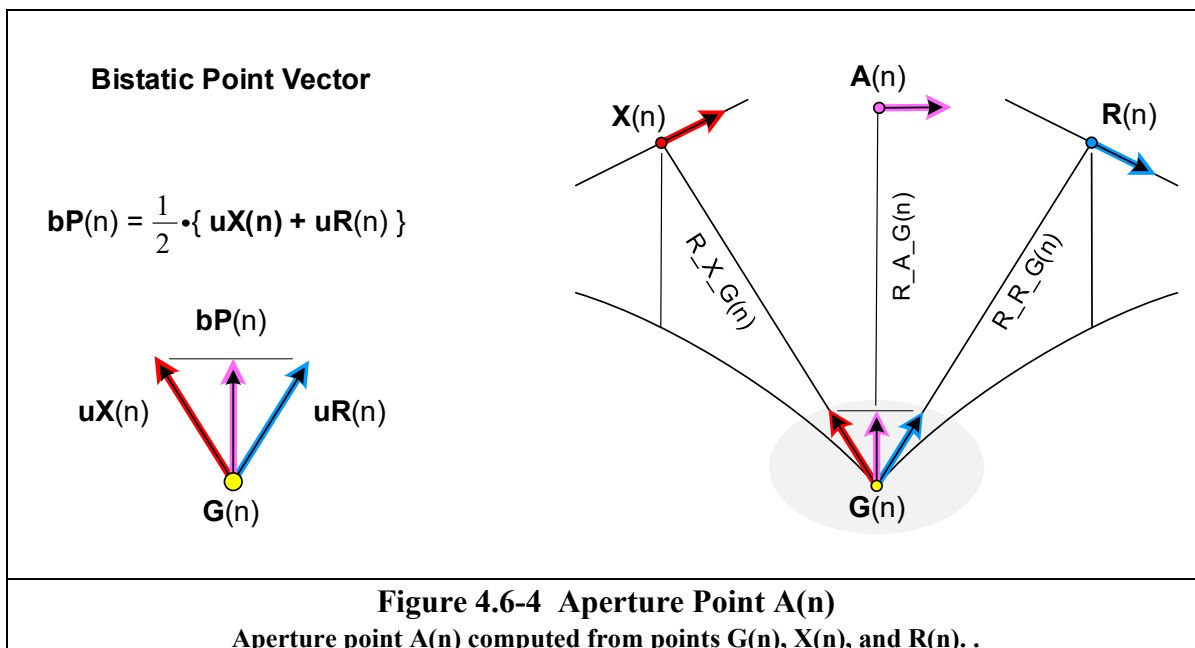
- (e) Compute the bistatic pointing vector, $\mathbf{bP}(n)$, as the average of unit vectors $\mathbf{uX}(n)$ and $\mathbf{uR}(n)$. Compute the range, $R_A_G(n)$, as the average of ranges $R_X_G(n)$ and $R_R_G(n)$.

$$\mathbf{bP}(n) = \frac{1}{2} \bullet \{ \mathbf{uX}(n) + \mathbf{uR}(n) \}$$

$$R_A_G(n) = \frac{1}{2} \bullet \{ R_X_G(n) + R_R_G(n) \}$$

- (f) Use the bistatic pointing vector, $\mathbf{bP}(n)$, and the average range, $R_A_G(n)$, to compute aperture point, $\mathbf{A}(n)$.

$$\mathbf{A}(n) = \mathbf{G}(n) + R_A_G(n) \cdot \left\{ \frac{1}{|\mathbf{bP}(n)|} \cdot \mathbf{bP}(n) \right\}$$



- (5) Use the set of times, $\{t(n)\}$, and the set of aperture points, $\{\mathbf{A}(n)\}$, to compute the ARP position polynomial, $\mathbf{ARP_Poly}(t)$. The ARP polynomial is computed using a least squares curve fit. Choose the order of the $\mathbf{ARP_Poly}(t)$ to be equal the highest order of transmit polynomial and the order of the receive APC centroid polynomial.

For a Bistatic image: $\{t(n)\} \ \& \ \{\mathbf{A}(n)\} \rightarrow \text{LS Fit} \rightarrow \mathbf{ARP_Poly}(t)$

4.6.4 COA Positions & Velocities

The method to be used by applications that input SICD products to compute COA time, position, and velocity parameters is specified below. The COA positions and velocities are fundamental to describing how the complex image was formed. They are also critical for use in the subsequent processing of derived products that are formed from the complex image array. For all SICD products, the metadata needed is included in product. The computed parameters are dependent upon the Collect Type (MONOSTATIC or BISTATIC) used to collect the PHD.

The method input is an image grid location specified in image coordinates, (xrow, ycol). The method outputs one or more times, positions, and velocities. All times are in seconds relative to Collection Start Time (CST). All positions and velocities are in ECEF coordinates. All position components are in meters and all velocity components are in meters/second.

Method for Computing COA Positions & Velocities

For the image grid location, (xrow, ycol), compute COA time, t_{COA} , using the COA Time polynomial. Image coordinates xrow and ycol are in meters. Refer to Sect 4.4.3.

$$t_{COA} = t_{COA}(xrow, ycol) = \sum_{m=0}^{M_{TCOA}} \sum_{n=0}^{N_{TCOA}} cT_COA(m, n) \cdot xrow^m \cdot ycol^n$$

For Collect Type = MONOSTATIC:

(M1) Compute the ARP position and velocity at time t_{COA} .

$$\mathbf{ARP}_{COA} = \mathbf{ARP}(t_{COA}) = \mathbf{ARP_Poly}(t_{COA})$$

$$\mathbf{VARP}_{COA} = \mathbf{VARP}(t_{COA}) = \frac{d}{dt} \mathbf{ARP_Poly}(t_{COA})$$

For Collect Type = BISTATIC:

(B1) Compute the GRP position at time t_{COA} .

$$\mathbf{GRP}_{COA} = \mathbf{GRP}(t_{COA}) = \mathbf{GRP_Poly}(t_{COA})$$

(B2) Compute the transmit time t_{XCOA} . For a radar signal incident at the GRP at time t_{COA} , time t_{XCOA} is an estimate of the time it was transmitted.

$$\begin{aligned} \mathbf{X0} &= \mathbf{Xmt}(t_{COA}) = \mathbf{Xmt_Poly}(t_{COA}) & R_X0 &= | \mathbf{X0} - \mathbf{GRP}_{COA} | \\ t_{XCOA} &= t_{COA} - (1/c) \cdot R_X0 \end{aligned}$$

(B3) Compute the transmit APC position and velocity at time t_{XCOA} .

$$\mathbf{Xmt}_{COA} = \mathbf{Xmt}(t_{XCOA})$$

$$\mathbf{VXmt}_{COA} = \mathbf{VXmt}(t_{XCOA}) = \frac{d}{dt} \mathbf{Xmt_Poly}(t_{XCOA})$$

(B4) Compute the receive time t_{RCOA} . For a radar signal incident at the GRP at time t_{COA} , time t_{RCOA} is an estimate of the time it would be received at the receive APC centroid.

$$\begin{aligned} \mathbf{R0} &= \mathbf{Rcv}(t_{COA}) = \mathbf{Rcv_Poly}(t_{COA}) & R_R0 &= | \mathbf{R0} - \mathbf{GRP}_{COA} | \\ t_{RCOA} &= t_{COA} + (1/c) \cdot R_R0 \end{aligned}$$

(B5) Compute the receive APC centroid position and velocity at time t_{RCOA} .

$$\mathbf{Rcv}_{COA} = \mathbf{Rcv}(t_{RCOA})$$

$$\mathbf{VRcv}_{COA} = \mathbf{VRcv}(t_{RCOA}) = \frac{d}{dt} \mathbf{Rcv_Poly}(t_{RCOA})$$

The COA Position and Velocity set for image location (xrow, ycol) is:

$$\mathbf{MONOSTATIC\ PV\ Set:} \quad \{ t_{COA}, \mathbf{ARP}_{COA}, \mathbf{VARP}_{COA} \}$$

$$\begin{aligned} \mathbf{BISTATIC\ PV\ Set:} \quad & \{ t_{COA}, t_{XCOA}, t_{RCOA}, \\ & \mathbf{Xmt}_{COA}, \mathbf{VXmt}_{COA}, \mathbf{Rcv}_{COA}, \mathbf{VRcv}_{COA} \} \end{aligned}$$

4.7 Radar Collection Parameters

The parameters that describe the imaging collection are summarized below. Refer to Table 3-7. The Radar Collection parameters are independent of the particular image that has been produced. While many parameters are not needed for exploitation of the image contained in the product, they are available to provide context that may be useful.

4.7.1 Transmit Parameters

(1) Transmit Frequency

The minimum and maximum transmit frequency of the collection are provided (XML: TxFrequency.Min and TxFrequency.Max). For most SAR systems, a fixed band of frequencies is transmitted on all pulses. For systems that vary the bandwidth within the collection, the minimum frequency is the overall minimum frequency and the maximum is the overall maximum frequency.

An option is provided to express frequency values as offsets relative to a reference frequency, denoted Ref_Freq. Such a choice is defined as part of the product definition and is specified in the Program Specific Implementation Document. See Section 1.4. Products that have frequency values expressed as offsets are indicated by including the optional parameter Ref_Freq_Index (XML: RadarCollection.RefFreqIndex). The index value(s) and the corresponding reference frequency value(s) are selected as part of the product definition.

For a product that includes Ref_Freq_Index, the true frequency values are obtained by adding the value of the reference frequency to the parameter values contained in the product file.

$$\begin{aligned} \underbrace{\text{Tx.Frequency.Min(True)}}_{\text{True Value}} &= \underbrace{\text{Tx.Frequency.Min(Product)}}_{\text{Value Included In The Product}} + \underbrace{\text{Ref_Freq}}_{\text{Defined In Program Implementation Document}} \\ \underbrace{\text{Tx.Frequency.Max(True)}}_{\text{True Value}} &= \underbrace{\text{Tx.Frequency.Max(Product)}}_{\text{Value Included In The Product}} + \underbrace{\text{Ref_Freq}}_{\text{Defined In Program Implementation Document}} \end{aligned}$$

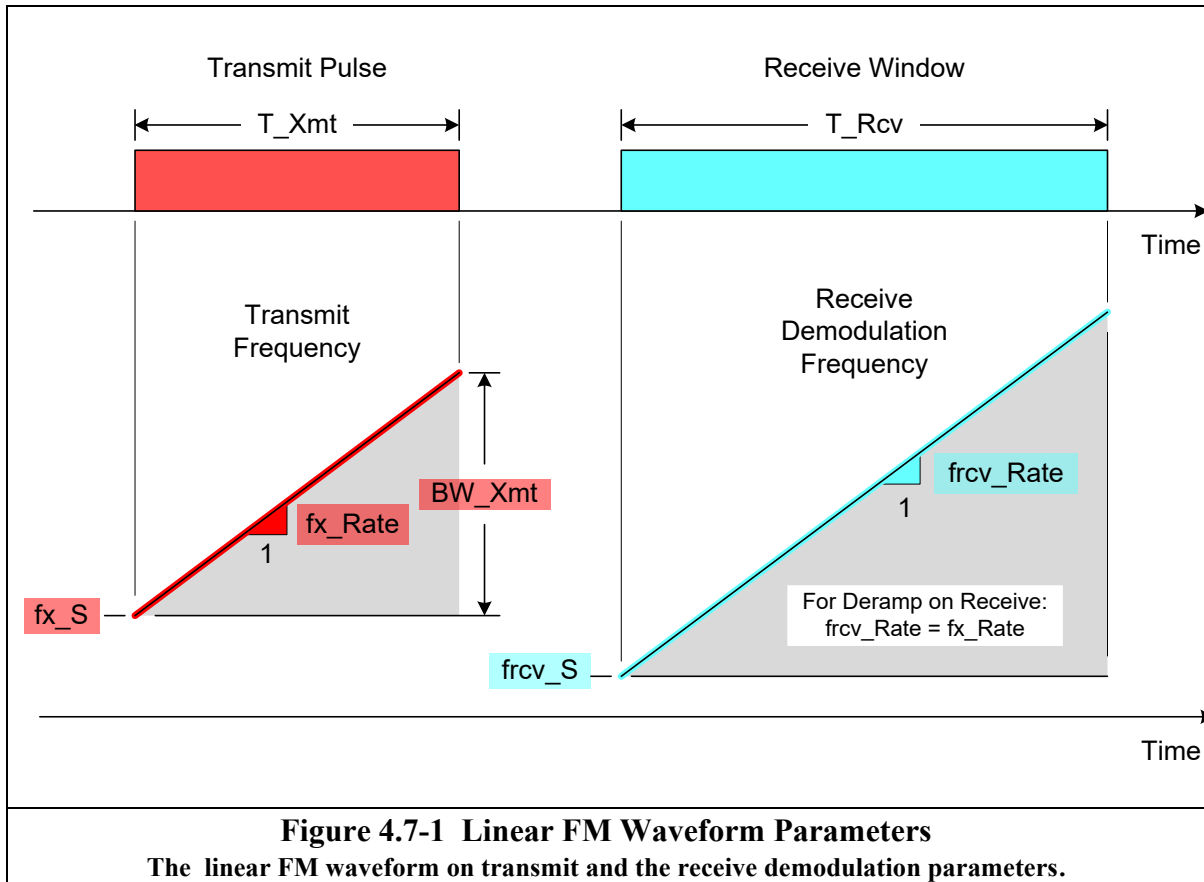
Note: Frequency parameters whose values will be expressed as offsets when Ref_Freq_Index is included are identified in the tables that define them. See Tables 3-7, 3-8, 3-11 and 3-16.

(2) Waveform Parameters

The parameters that describe the transmitted waveform (i.e. pulse length, bandwidth, etc) and the receive demodulation may be included. A complete set of parameters may be included for systems that transmit a linear FM waveform. For most imaging collections, the same waveform is transmitted on every pulse. For such a system, the number of waveforms is set equal to one (size attribute NumWaveforms = 1). For systems that transmit a repeating sequence of waveforms, each waveform in the sequence can be described. The sequencing of the waveforms is specified with the transmit sequence parameters described below. For

example, a collection that employs a step-chirp waveform can be described with NumWaveforms set equal to the number of steps in the sequence. See Figure 4.5-4 for an example timeline.

An example of a system that uses a linear FM waveform with de-ramp on receive demodulation, often referred to as “stretch” demodulation, is shown in Figure 4.7-1. The parameters associated with the transmit waveform, the receive window and receive demodulation are shown.



(3) Transmit Polarization

The transmit polarization type is included with all products. For most imaging collections, the same polarization type is transmitted on every pulse. For such a collection, the transmit polarization is identified with a single parameter (XML: RadarCollection.TxPolarization). For collections that employ a repeating sequence of transmit polarizations, the value is set to “SEQUENCE”. The most common transmit polarization sequence is alternating horizontal and vertical polarization on a per pulse basis. An example timeline for such a sequence is shown in Figure 4.5-3. The sequence of polarizations is then specified with the transmit sequence structure described below.

The set of transmit polarization types is specified in Table 3-7 and in the XML schema. For linear polarization types H, V, and S, the E-field orientation is dependent on the APC position and the scene point selected. The choice of the scene point and the APC position used is a product design choice. Logical choices for the scene point are the antenna electrical boresight ground point or the SCP. A logical choice for the APC position is at the SCP COA time. For each polarization type, Table 4-1 provides a description and guidance.

Table 4-1 Polarization Types		
Type	Polarization Description	Guidance
H	Horizontal Polarization: E-Field is linearly polarized and is nominally aligned with the horizontal direction in the imaged scene to within a tolerance.	Horizontal direction is in the ETP at the selected scene point. The H direction is cross range in the ETP. See Figure 4.9-2.
V	Vertical Polarization: E-Field is linearly polarized and is nominally aligned with the vertical direction in the imaged scene to within a tolerance.	Vertical direction defined using the normal to the ETP at the selected scene point. The V lies in the Plane of Incidence. See Figure 4.9-4.
X	ACF X-Axis Polarization: E-Field is linearly polarized and is aligned with the Antenna Coordinate Frame X-axis. The ACF unit vectors provide the E-field orientation vs. time. See Table 3-11.	Use only when the Antenna metadata is included in the product.
Y	ACF Y-Axis Polarization: E-Field is linearly polarized and is aligned with the Antenna Coordinate Frame Y-axis. The ACF unit vectors provide the E-field orientation vs. time. See Table 3-11.	Use only when the Antenna metadata is included in the product.
S	Slant Polarization: E-field is linearly polarized and has components in both the H and V directions. The H and V components are nominally in-phase or 180 degrees out-of-phase.	The range of E-field orientations and tolerances for H, V, and S are product design choices specified per a Program Specific Implementation Document.
E	Elliptical Polarization: E-field includes components in both H and V directions. The H and V components may have any relative phase.	General type that may be used when other types are not representative.
RHC	Right Hand Circular Polarization: E-field H and V components are nominally equal amplitude and are nominally ± 90 degrees out-of-phase. Right hand rule points in the direction of propagation.	Tolerances on the relative amplitude and phase between H and V components is per a Program Specific Implementation Document.
LHC	Left Hand Circular Polarization: E-field H and V components are nominally equal amplitude and are nominally ± 90 degrees out-of-phase. Left-hand rule points in the direction of propagation.	Tolerances on the relative amplitude and phase between H and V components is per a Program Specific Implementation Document.

(4) Transmit Sequences

For an imaging collection that employed a repeating sequence of waveforms and/or transmit polarizations, the sequence may be described as shown in the example below. The number of steps in the sequence is NumTxSteps. The most commonly used transmit sequences vary either the waveform or the transmit polarization. Hence, the TxSequence structure only includes variation in these two parameters. The example below shows a sequence with 3 steps with the waveform varying.

XML: RadarCollection.TxSequence	size = 3	↔ NumTxSteps = 3
TxStep	index = 1	↔ TxStep 1
WFIndex	= 1	↔ WF 1 Transmitted
TxStep	index = 2	↔ TxStep 2
WFIndex	= 2	↔ WF 2 Transmitted
TxStep	index = 3	↔ TxStep 3
WFIndex	= 3	↔ WF 3 Transmitted

A sequence with NumTxSteps = 1 is allowed. This allows the products from SAR systems that use a sequence for some but not all collections to include the TxSequence structure in all image products.

4.7.2 Receive Data Channel Parameters

The receive data channels are described (XML: RcvChannels). The number of receive data channels is parameter NumRcvChans. The transmit/receive polarization for each channel is indicated. Also, for each data channel, the receive antenna phase center (APC) can also be identified. The receive APC index associates the receive data channel with a receive APC position included in the product. See Table 3-5. For many imaging collections, only one receive channel is collected with a single transmit/receive polarization. For these collections, the number of received data channels is set equal to 1 and the Tx:Rcv polarizations provided.

For imaging collections with multiple transmit/receive polarizations, the NumRcvChans indicates the number of unique transmit/receive polarization combinations collected. Two example collections demonstrate this. Consider an imaging collection with a fixed transmit polarization (e.g., TxPol = V) on all pulses and simultaneous collection of two receiver channels, one with RcvPol = V and one with RcvPol = H. The number of receive data channels is set equal to 2 and the Tx:Rcv polarizations are set to V:V and V:H. For this example the two data channels came from separate receive phase centers.

XML: RadarCollection.RcvChannels	size = 2	↔ NumRcvChans = 2
ChanParameters	index = 1	↔ Channel 1 Parameters
TxRcvPolarization	= V:V	

RcvAPC = 1
 ChanParameters index = 2 ⇔ Channel 2 Parameters
 TxRcvPolarization = V:H
 RcvAPC = 2

Consider a second collection with alternating transmit polarization on a pulse-to-pulse basis (TxPol = V and TxPol = H) and simultaneous collection of two receiver channels (one with RcvPol = V and one with RcvPol = H). The data from each receiver channel is then separated to form two data channels available for image formation processing. For this collection, the number of receive data channels is set equal to 4 and the Tx:Rcv polarizations are set to V:V, V:H, H:V and H:H. For this example, receive data channels 1 and 2 were collected with receive APC 1 and receive data channels 3 and 4 were collected with receive APC 2.

XML: RadarCollection.RcvChannels size = 4 ⇔ NumRcvChans = 4
 ChanParameters index = 1 ⇔ Channel 1 Parameters
 TxRcvPolarization = V:V
 RcvAPC = 1
 ChanParameters index = 2 ⇔ Channel 2 Parameters
 TxRcvPolarization = H:V
 RcvAPC = 1
 ChanParameters index = 3 ⇔ Channel 3 Parameters
 TxRcvPolarization = V:H
 RcvAPC = 2
 ChanParameters index = 4 ⇔ Channel 4 Parameters
 TxRcvPolarization = H:H
 RcvAPC = 2

4.7.3 Area Imaged

The area imaged by the collection can be provided. This information is most useful when the area covered by the collection is larger than the area covered by the image contained in the product. This is most commonly the case for stripmap and dynamic stripmap mode collections. The imaged area is first identified by four area corner points, denoted ACP(i) and indexed $i = 1$ to 4.

The imaged area also may be further specified by a rectangular region in a georeferenced image display plane. The plane is specified by a reference point, denoted RPT, and two orthogonal unit vectors that lie in the plane, denoted $\mathbf{uX}$ and $\mathbf{uY}$. The imaged area is a rectangle aligned with the X and Y unit vectors.

$$\mathbf{uX} = \text{XDir.UVectECF}$$

$$\mathbf{uY} = \text{YDir.UVectECF}$$

The imaged area is specified by a two-dimensional array of grid locations in the plane. The array of grid locations is indexed by lines and samples. An example image area is shown in Figure 4.7-2. The imaged area is specified by the following parameters.

$$L1 = \text{XDir.FirstLine}$$

$$S1 = \text{YDir.FirstSample}$$

$$NL = \text{XDir.NumLines}$$

$$NS = \text{YDir.NumSamples}$$

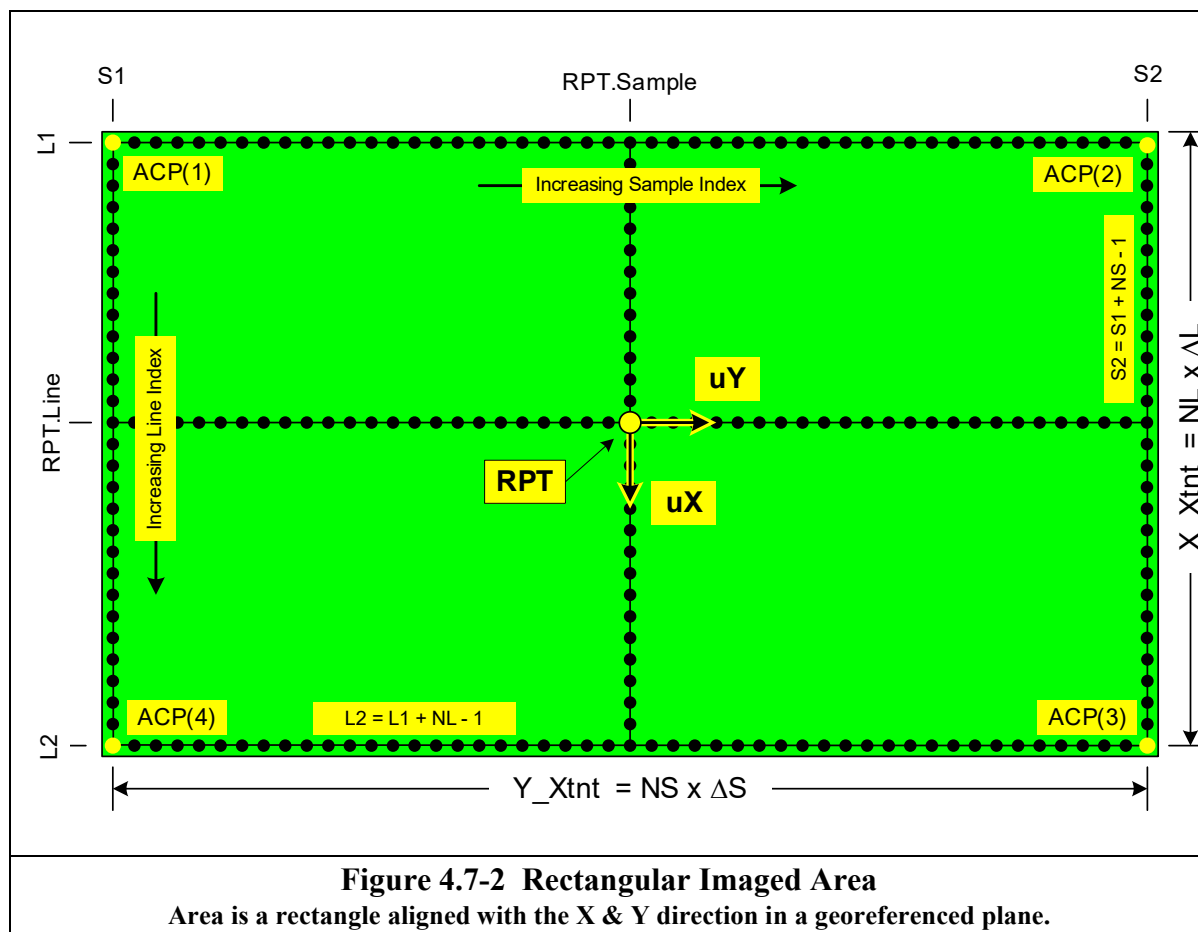
$$\Delta L = \text{XDir.LineSpacing}$$

$$\Delta S = \text{YDir.SampleSpacing}$$

$$L2 = L1 + NL - 1 \text{ (last line)}$$

$$S2 = S1 + NS - 1 \text{ (last sample)}$$

The lines are indexed by $L = L1$ to $L2$ (inclusive). The line index increases in the $\mathbf{uX}$ direction. The extent in the X direction is $NL \times \Delta L$ and is located by the values of $L1$ and the RPT line index. The samples are indexed by $S = S1$ to $S2$ (inclusive). The sample index increases in the $\mathbf{uY}$ direction. The extent in the Y direction is $NS \times \Delta S$ and is located by the values of $S1$ and the RPT sample index.



The imaged area may be further restricted by a polygon. Where the SICD XML contains a `SICD.RadarCollection.Area.Plane.Polygon`, the polygon shall be specified by 3 or more vertices in line and sample in clockwise order bounded by the rectangular imaged area.

4.8 Image Formation Parameters

The parameters that describe the image formation processing are summarized below. Refer to Table 3-8. The image formation parameters describe the data processed and the image formation algorithms that were applied to form the image contained in the product file.

4.8.1 Processed Data Parameters

(1) Receive Data Channels Processed

The receive data channel(s) that were processed to form the image are listed. For most imaging collections, the image formation processing is performed on an individual data channel (parameter NumChanProc = 1). The channel index values are defined in the Radar Collection structure (XML: RadarCollection.ReceiveChannels). See Table 3-7. The combined transmit/receive polarization for the image contained in the product is also provided (XML: TxRcvPolarizationProc).

The optional PRF scale factor parameter (XML: RcvChanProc.PRFScaleFactor) is provided as appropriate to indicate the effective PRF for the image formed is different than the true PRF. The effective PRF is useful in determining the location of the ambiguous returns in range and/or azimuth that may appear in the image. As described in Section 4.5, the collection timeline parameters provide the true PRF of the collection, PRF(t). At time t*, the true PRF is PRF(t*). Effective PRF is computed by multiplying by the scale factor, PRF_SF.

$$\text{Effective_PRF}(t^*) = \text{PRF}(t^*) \cdot \text{PRF_Scale_Factor}$$

(2) Phase History Data Processed

The portion of the collected Phase History Data (PHD) processed is provided. The processed portion of slow time is denoted t_{PS} to t_{PE}. See Section 4.5 and Figure 4.5-1. The portion of the collected frequency band processed is also provided (XML: TxFrequencyProc).

4.8.2 Processing Algorithms Applied

(1) Image Formation Algorithm

The Image Formation Algorithm is identified for the most commonly used image formation algorithms (XML: ImageFormAlgo). Currently, three image formation algorithms may be identified.

- (1) Range & Azimuth Compression (RGAZCOMP)
- (2) Polar Format Algorithm (PFA)
- (3) Range Migration Algorithm (RMA)

For an image formed with one of these three algorithms, additional parameters specific to the algorithm are also included. The Range & Azimuth Compression algorithm is the simplest approach and is described in detail in Section 4.14. For images formed with the Polar Format Algorithm, the parameters provided accommodate image formation for an arbitrary image formation plane and are described in Section 4.15. For images formed with the Range Migration Algorithm, the primary variations of the algorithm (Omega-K, Chirp Scaling and Range-Doppler) are accommodated. The RMA specific parameters are described in Section 4.16.

(2) Slow Time Beam Compensation

The parameter is provided to indicate if the signal modulation in slow time due to the antenna beamshape has been compensated (XML: STBeamComp). The beamshape compensation is applied for most stripmap and dynamic stripmap collections in order to achieve the desired azimuth impulse response. The compensation applied may be the same for the entire image (a global compensation) or a compensation that varies across the scene may be applied (a spatially variant compensation).

(3) Image Domain Beam Compensation

The parameter is provided to indicate if the variation in signal intensity across the image due to antenna beamshape has been compensated (XML: ImageBeamComp). For complex images that have not been compensated for beamshape, the antenna beamshape parameters may be provided that allow the compensation to be computed and applied when creating derived products.

(4) Auto Focus Corrections

Two parameters are provided to indicate if image-derived focus corrections have been applied (XML: AzAutoFocus and RgAutoFocus). Azimuth auto focus corrections are required for many SAR systems in order to achieve a well focused image. Range auto focus corrections may also be required. For both types of correction, the same correction may be applied to the entire image (a global correction) or a correction that varies across the scene may be applied (a spatially variant correction).

4.8.3 Polarization Parameters

For systems that simultaneously collect multiple polarizations, a complete set of polarization distortion parameters may be provided with the image product. The distortion parameters are usually computed from dedicated calibration collections. The date and time of the calibration collections and/or the effective date for the distortion parameters may be provided.

The parameters provided are the elements of the following model that relates the true scattering matrix $\{\mathbf{S}\}$ to the observed scattering matrix $\{\mathbf{O}\}$ for a fully polarimetric system. An example collection from such a system is shown in the second example of Section 4.7.2. The distortion parameters provided are the elements A, F1, F2 and Q1 through Q4 in the following matrix equation. Additionally, the estimated error statistics may be provided for the parameters A, F1 and F2 that provide accuracy estimates of the co-pole distortion corrections.

$$\underbrace{\begin{bmatrix} O_{HH} & O_{VH} \\ O_{HV} & O_{VV} \end{bmatrix}}_{\text{Observed Scattering Matrix}} = A \cdot \underbrace{\begin{bmatrix} 1 & Q1 \\ Q2 & F1 \end{bmatrix}}_{\text{Receive Distortion Matrix}} \cdot \underbrace{\begin{bmatrix} S_{HH} & S_{VH} \\ S_{HV} & S_{VV} \end{bmatrix}}_{\text{True Scattering Matrix}} \cdot \underbrace{\begin{bmatrix} 1 & Q4 \\ Q3 & F2 \end{bmatrix}}_{\text{Transmit Distortion Matrix}}$$

4.9 SCP Center Of Aperture Parameters

The SCP Center Of Aperture (COA) parameters describe the imaging geometry at COA for the SCP. Refer to Table 3-9. The SCP COA parameters are provided to allow direct search and discovery on common SAR geometry parameters. The SCP COA parameters include two sets of parameters. The first set of parameters are computed for all images. They are computed using the ARP position, velocity, and acceleration at the SCP COA. They shall be computed as specified in Section 4.9.1. The second set of parameters is only computed for bistatic images (i.e., for CollectType = BISTATIC). They are computed using the transmit and receive APC positions and velocities at the SCP COA. The bistatic parameters shall be computed as specified in Section 4.9.2.

4.9.1 COA Parameters for All Images

The SCP COA time (t_{COA}) is the precise COA time for the SCP pixel and is equal to the constant coefficient of the COA time polynomial (see Grid.TimeCOAPoly). The ARP position, velocity and acceleration shall be computed by evaluating the ARP position versus time polynomial and its derivatives. Time t_{COA} is in seconds from collection start time. An example SCP COA geometry is shown in Figure 4.9-1. Parameters in bold are vector parameters.

SCP = SCP position in ECF coordinates.

ARP_{COA} = ARP position in ECF coordinates at time t_{COA} .

VARP_{COA} = ARP velocity in ECF coordinates at time t_{COA} .

AARP_{COA} = ARP acceleration in ECF coordinates at time t_{COA} .

The SCP COA parameters slant range (R_{COA}) and ground range (R_{gCOA}) are provided for discovery. The ground range is a simple estimate measured along a spherical earth model passing through the SCP. Parameters ARP_DEC_{COA} and SCP_DEC are the distances from the ECF origin (EC). Compute parameters as follows.

$$R_{COA} = |\mathbf{SCP} - \mathbf{ARP}_{COA}| \quad \text{XML: SlantRange} = R_{COA}$$

$$\text{ARP_DEC}_{COA} = |\mathbf{ARP}_{COA}| \quad \mathbf{uARP}_{COA} = \frac{1}{\text{ARP_DEC}_{COA}} \cdot \mathbf{ARP}_{COA}$$

$$\text{SCP_DEC} = |\mathbf{SCP}| \quad \mathbf{uSCP} = \frac{1}{\text{SCP_DEC}} \cdot \mathbf{SCP}$$

$$\text{EA}_{COA} = \cos^{-1}(\mathbf{uARP}_{COA} \cdot \mathbf{uSCP})$$

$$R_{gCOA} = \text{SCP_DEC} \cdot \text{EA}_{COA} \quad \text{XML: GroundRange} = R_{gCOA}$$

The Side Of Track and the Doppler Cone Angle at COA (DCA_{COA}) are provided for discovery. The Side Of Track is set to “L” or “R”. The DCA is always a positive angle independent of the side of track. Unit vector $\mathbf{uLOS}_{COA}$ is from the ARP position at COA to the SCP. Parameter LOOK is set to +1 or -1 as shown below. Compute parameters as follows. The DCA_{COA} is in decimal degrees.

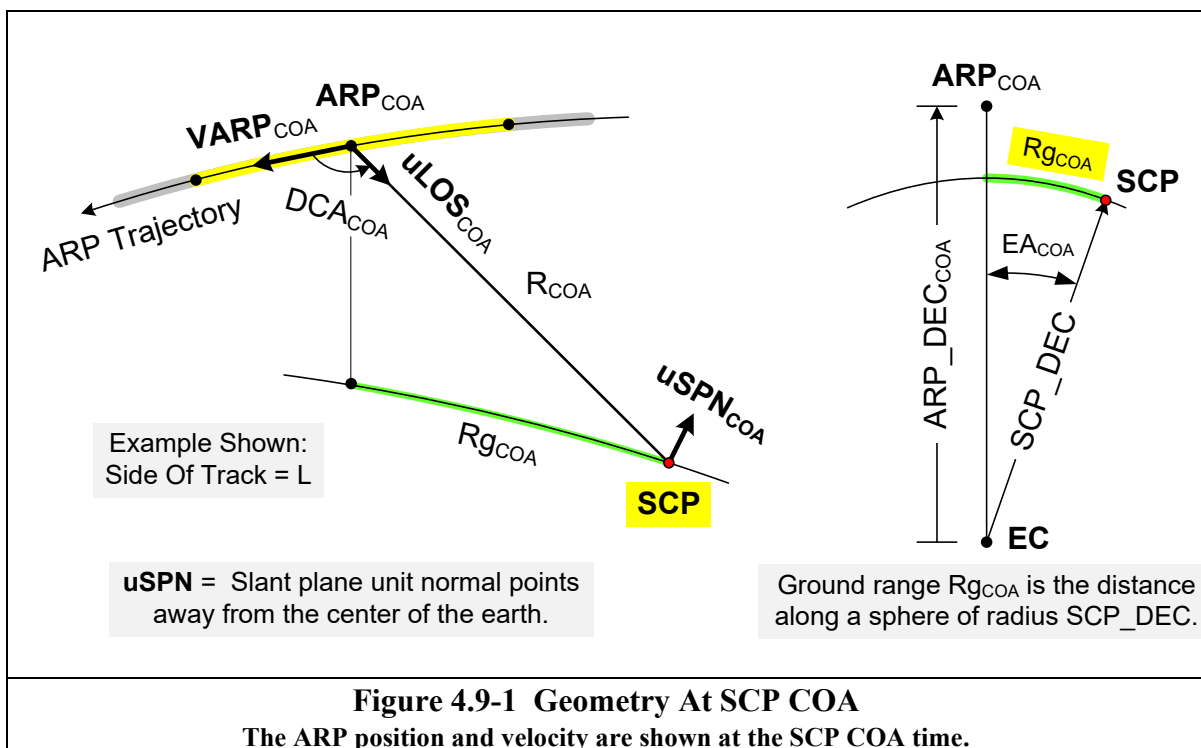
$$VM_{COA} = |\mathbf{VARP}_{COA}| \quad \mathbf{uVARP}_{COA} = \frac{1}{VM_{COA}} \cdot \mathbf{VARP}_{COA}$$

$$\mathbf{uLOS}_{COA} = \frac{1}{R_{COA}} \cdot (\mathbf{SCP} - \mathbf{ARP}_{COA}) \quad \mathbf{LEFT}_{COA} = \mathbf{uARP}_{COA} \times \mathbf{uVARP}_{COA}$$

$$DCA_{COA} = \underbrace{\cos^{-1}(\mathbf{uVARP}_{COA} \cdot \mathbf{uLOS}_{COA})}_{\substack{\text{Angle on the interval} \\ 0 \text{ to } 180 \text{ degrees.}}} \quad \text{XML: DopplerConeAng} = DCA_{COA}$$

$$\text{SideOfTrack} = \begin{cases} \text{L}, & \mathbf{LEFT}_{COA} \cdot \mathbf{uLOS}_{COA} > 0 \\ \text{R}, & \text{otherwise} \end{cases} \quad \text{XML: SideOfTrack}$$

$$\text{LOOK} = \begin{cases} +1, & \mathbf{LEFT}_{COA} \cdot \mathbf{uLOS}_{COA} > 0 \\ -1, & \text{otherwise} \end{cases}$$



The Earth Tangent Plane (ETP) at the SCP is the plane tangent to the surface of constant height above the WGS 84 ellipsoid (HAE) that contains the SCP. See Figure 4.9-2. The ETP is an approximation to the ground plane at the SCP. Define ground plane coordinates (GPX, GPY, GPZ) with origin at the SCP. The +GPZ axis is normal to the plane and in the direction of increasing HAE. The GPX and GPY axes lie in the plane. The +GPX axis is defined by the ARP ETP nadir at COA ($\mathbf{AETP}_{COA}$). The ground plane coordinates (GPC) are a right-handed system. The GPC unit vectors in ECF coordinates ($\mathbf{uGPX}$, $\mathbf{uGPY}$, $\mathbf{uGPZ}$) are computed as follows. Parameters SCP_Lat and SCP_Lon are the SCP WGS 84 geodetic latitude and longitude.

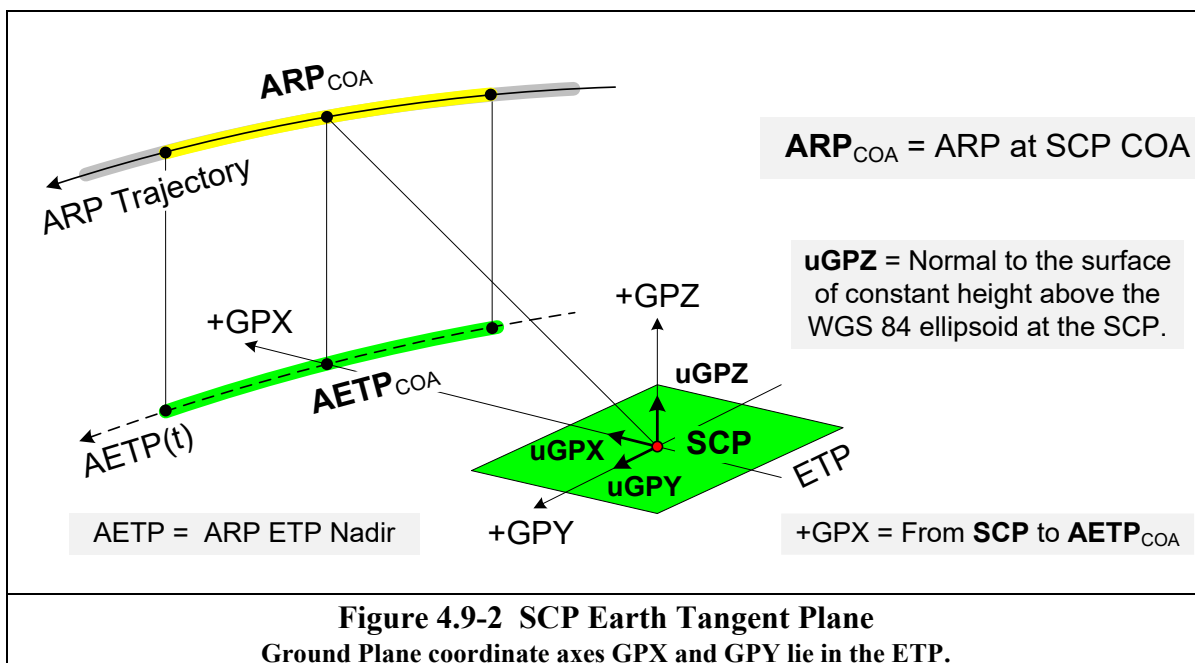
$$\mathbf{uGPZ} = \begin{bmatrix} \cos(\text{SCP_Lon}) \cdot \cos(\text{SCP_Lat}) \\ \sin(\text{SCP_Lon}) \cdot \cos(\text{SCP_Lat}) \\ \sin(\text{SCP_Lat}) \end{bmatrix} \quad \text{ETP Unit Normal}$$

$$\text{ARP_GPZ}_{COA} = (\mathbf{ARP}_{COA} - \mathbf{SCP}) \cdot \mathbf{uGPZ} \quad \text{ARP GPZ coordinate at COA}$$

$$\mathbf{AETP}_{COA} = \mathbf{ARP}_{COA} - \mathbf{uGPZ} \cdot \text{ARP_GPZ}_{COA}$$

$$\text{ARP_GPX}_{COA} = |\mathbf{AETP}_{COA} - \mathbf{SCP}| \quad \text{ARP GPX coordinate at COA}$$

$$\mathbf{uGPX} = \frac{1}{\text{ARP_GPX}_{COA}} \cdot (\mathbf{AETP}_{COA} - \mathbf{SCP}) \quad \mathbf{uGPY} = \mathbf{uGPZ} \times \mathbf{uGPX}$$



The ARP grazing angle (GRAZ) and incidence angle (INCD) are provided for discovery. See Figure 4.9-3. The cosine and sine of the grazing angle (cosGRAZ and sinGRAZ) are computed from the ARP GP coordinates at COA. Angles are in decimal degrees.

$$\cos\text{GRAZ} = \frac{\text{ARP_GPX}_{\text{COA}}}{R_{\text{COA}}}$$

$$\sin\text{GRAZ} = \frac{\text{ARP_GPZ}_{\text{COA}}}{R_{\text{COA}}}$$

$$\text{GRAZ} = \underbrace{\sin^{-1}(\sin\text{GRAZ})}_{\substack{\text{Angle on the interval} \\ -90 \text{ to } 90 \text{ degrees}}}$$

$$\text{XML: GrazeAng} = \text{GRAZ}$$

$$\text{INCD} = 90.0 - \text{GRAZ}$$

$$\text{XML: IncidenceAng} = \text{INCD}$$

The SAR slant plane (SP) is a plane defined at an instant in time and with respect to a point in the scene being imaged. The SCP slant plane at COA is the plane that contains the ARP position and velocity at COA and the SCP. Define slant plane coordinates (SPX, SPY, SPZ) with origin at the SCP. The +SPZ axis is normal to the plane and in the direction of increasing HAE. The SPX and SPY axes lie in the plane. The +SPX axis is the along the line from the SCP to ARP at COA. The slant plane coordinates (SPC) are a right-handed system. The SPC unit vectors in ECF coordinates (**uSPX**, **uSPY**, **uSPZ**) are computed as follows.

$$\text{SPZ} = \text{LOOK} \cdot (\mathbf{uVARP}_{\text{COA}} \times \mathbf{uLOS}_{\text{COA}})$$

$$\mathbf{uSPZ} = \frac{1}{|\text{SPZ}|} \cdot \text{SPZ}$$

$$\mathbf{uSPX} = -\mathbf{uLOS}_{\text{COA}}$$

$$\mathbf{uSPY} = \mathbf{uSPZ} \times \mathbf{uSPX}$$

The slant plane coordinates are defined such that the ARP position and velocity at COA are as shown below. The ARP velocity at COA lies in the plane. For a left looking geometry, $\text{VARP_SPY}_{\text{COA}} > 0$. For a right looking geometry, $\text{VARP_SPY}_{\text{COA}} < 0$.

$$\mathbf{ARP}_{\text{COA}} = \underbrace{\begin{bmatrix} R_{\text{COA}} \\ 0 \\ 0 \end{bmatrix}}_{\substack{\text{Slant Plane} \\ \text{Coordinates}}}$$

$$\mathbf{VARP}_{\text{COA}} = \underbrace{\begin{bmatrix} \text{VARP_SPX}_{\text{COA}} \\ \text{VARP_SPY}_{\text{COA}} \\ 0 \end{bmatrix}}_{\substack{\text{Slant Plane} \\ \text{Coordinates}}}$$

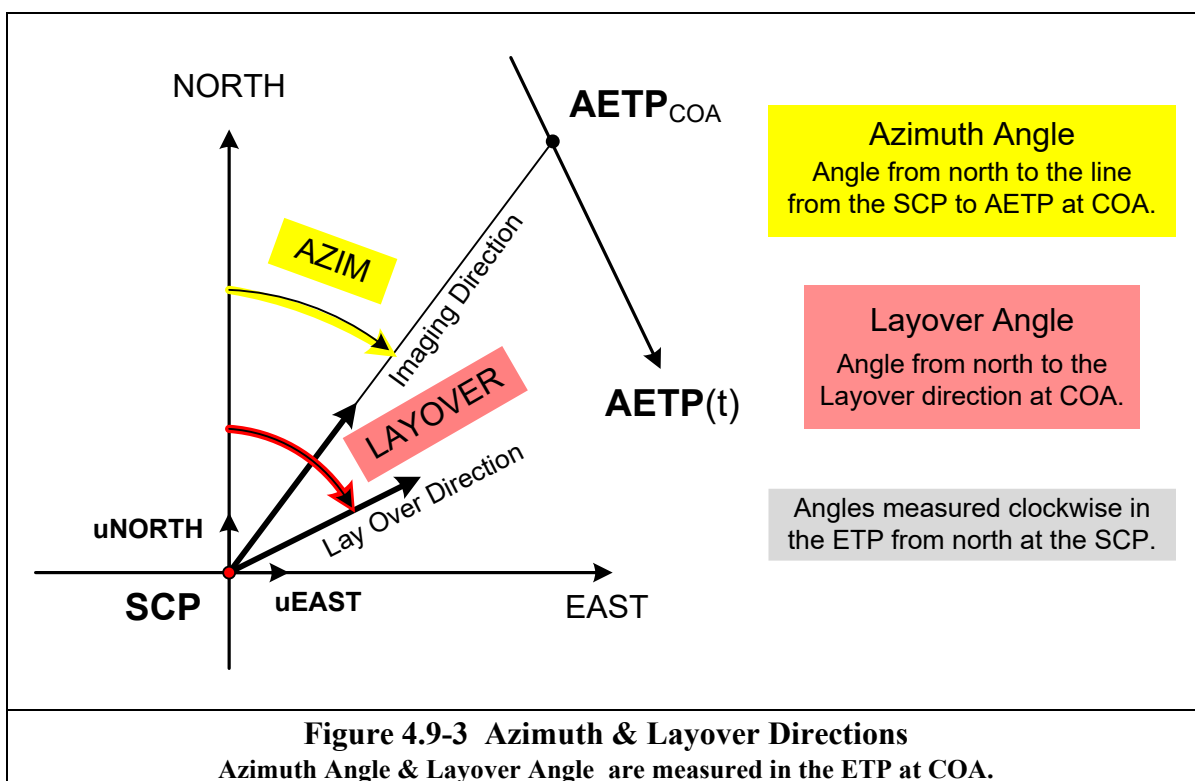
The SCP slant plane slope angle (SLOPE) is provided for discovery. The slope angle is the angle between the ETP normal and the SCP SP normal. The slope angle is expressed as a positive value. The slope angle is computed as follows.

$$\text{SLOPE} = \underbrace{\cos^{-1}(\mathbf{uGPZ} \cdot \mathbf{uSPZ})}_{\substack{\text{Angle on the interval} \\ 0 \text{ to } 90 \text{ degrees}}}$$

$$\text{XML: SlopeAng} = \text{SLOPE}$$

The imaging direction and the layover direction are useful discovery parameters when selecting images of a given scene. The imaging direction is given by the Azimuth Angle (AZIM) and the layover direction is given by the Layover Angle (LAYOVER). See Figure 4.9-3. Both angles are measured in the ETP. Both angles are measured clockwise from north at the SCP. The azimuth angle indicates the direction of the incident illumination at COA. The layover angle indicates the direction a point above the SCP will layover at COA. The unit vectors that lie in the ETP in the north and east directions, **uNORTH** and **uEAST**, are computed as follows. Unit vectors **uNORTH** and **uEAST** are in ECF coordinates.

$$\mathbf{uEAST} = \begin{bmatrix} -\sin(\text{SCP_Lon}) \\ \cos(\text{SCP_Lon}) \\ 0 \end{bmatrix} \quad \mathbf{uNORTH} = \mathbf{uGPZ} \times \mathbf{uEAST}$$



The azimuth angle is computed by using the unit vector in the +GPX direction (**uGPX**).

$$AZ_NORTH = \mathbf{uNORTH} \bullet \mathbf{uGPX}$$

$$AZ_EAST = \mathbf{uEAST} \bullet \mathbf{uGPX}$$

$$AZIM = \tan^{-1} \left(\frac{AZ_EAST}{AZ_NORTH} \right)$$

$$\text{XML: AzimAng} = AZIM$$

Four quadrant ArcTan Function.
 Angle on the interval 0 to 360 degrees.

The layover direction vector in the ETP at COA, denoted $\mathbf{LODIR}_{COA}$, is computed by projecting the ETP unit normal ($\mathbf{uGPZ}$) into the ETP along the slant plane unit normal ($\mathbf{uSPZ}$). The projection distance is $1/\cos(\text{SLOPE})$. The layover angle is computed from the layover direction vector.

$$\mathbf{LODIR}_{COA} = \mathbf{uGPZ} - \frac{1}{\cos(\text{SLOPE})} \mathbf{uSPZ}$$

$$\text{LO_NORTH} = \mathbf{uNORTH} \bullet \mathbf{LODIR}_{COA} \quad \text{LO_EAST} = \mathbf{uEAST} \bullet \mathbf{LODIR}_{COA}$$

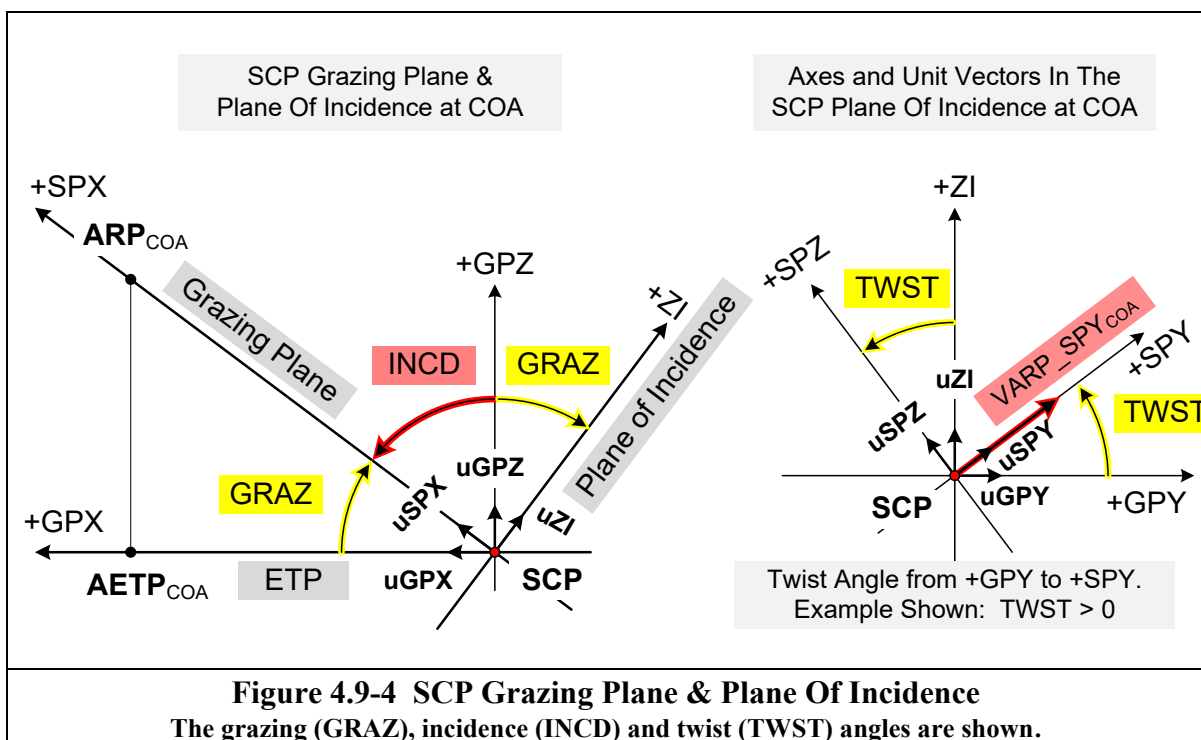
$$\text{LAYOVER} = \underbrace{\tan^{-1} \left(\frac{\text{LO_EAST}}{\text{LO_NORTH}} \right)}_{\substack{\text{Four quadrant ArcTan Function.} \\ \text{Angle on the interval 0 to 360 degrees.}}} \quad \text{XML: LayoverAng} = \text{LAYOVER}$$

The Grazing Plane and the Plane of Incidence are two additional planes also defined at an instant in time and with respect to a point in the scene being imaged. The SCP Grazing Plane (GZP) at COA is the plane that contains the ARP position at COA, the SCP and the GPY axis. The SCP Plane of Incidence (POI) at COA is the plane that contains the SCP, the GPY axis and the Grazing Plane normal. See Figure 4.9-4. The GZP is used as the image formation plane for some processing systems. The POI is useful when describing the polarization of the transmitted waveform. The Grazing Plane unit normal vector is denoted $\mathbf{uZI}$. Vector $\mathbf{uZI}$ in ECF coordinates is computed as follows.

$$\mathbf{uZI} = \mathbf{uGPZ} \bullet \cos \text{GRAZ} - \mathbf{uGPX} \bullet \sin \text{GRAZ}$$

The twist angle (TWST) is provided for discovery. The twist angle, sometimes referred to as the tilt angle, is the angle in the POI from the +GPY axis and to the +SPY axis. See Figure 4.9-3. The twist angle is computed as follows.

$$\text{TWST} = \underbrace{-\sin^{-1}(\mathbf{uGPY} \bullet \mathbf{uSPZ})}_{\substack{\text{Angle on the interval} \\ -90 \text{ to } 90 \text{ degrees.}}} \quad \text{XML: TwistAng} = \text{TWST}$$



Note: The twist or tilt angle in other descriptions may use a different convention for determining positive and negative angles. For the convention defined above, the twist angle will tend to be positive for left looking and closing geometries or right looking and opening geometries. The twist angle will go to zero near broadside for both left and right looking collections.

The Direction Cosine Matrix that relates positions expressed in the Ground Plane coordinates to positions expressed in Slant Plane coordinates is shown below. The coordinates systems are related by two Euler rotations, $R_2(-\text{GRAZ})$ and $R_1(\text{TWST})$. The first rotation, $R_2(-\text{GRAZ})$, is a negative rotation about the +GPY axis. The second rotation, $R_1(\text{TWST})$, is a positive rotation about the +SPX axis.

$$\mathbf{P}_{\text{SPC}} = \text{DCM}_{\text{GPC}}^{\text{SPC}} \cdot \mathbf{P}_{\text{GPC}}$$

$$\text{DCM}_{\text{GPC}}^{\text{SPC}} = R_1(\text{TWST}) \cdot R_2(-\text{GRAZ})$$

$$\underbrace{\begin{bmatrix} \text{P_SPX} \\ \text{P_SPY} \\ \text{P_SPZ} \end{bmatrix}}_{\text{Position in SP Coordinates}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\text{TWST} & \sin\text{TWST} \\ 0 & -\sin\text{TWST} & \cos\text{TWST} \end{bmatrix}}_{R_1(\text{TWST})} \cdot \underbrace{\begin{bmatrix} \cos\text{GRAZ} & 0 & \sin\text{GRAZ} \\ 0 & 1 & 0 \\ -\sin\text{GRAZ} & 0 & \cos\text{GRAZ} \end{bmatrix}}_{R_2(-\text{GRAZ})} \cdot \underbrace{\begin{bmatrix} \text{P_GPX} \\ \text{P_GPY} \\ \text{P_GPZ} \end{bmatrix}}_{\text{Position in GP Coordinates}}$$

4.9.2 Additional COA Parameters for Bistatic Images

The parameters to be computed for a bistatic image are specified below. The parameters include the transmit and receive APC positions and velocities that make up the SCP COA bistatic geometry. The transmit APC position vs. time is $\mathbf{Xmt}(t)$ and the receive APC centroid position vs. time is $\mathbf{Rcv}(t)$. See Section 4.6.

- (1) Compute times t_{xCOA} and t_{rCOA} that are used to compute the bistatic geometry for the SCP. For a signal incident at the SCP at the time t_{COA} , time t_{xCOA} is the approximate time it would have been transmitted. Time t_{rCOA} is the approximate time it would have been received. Refer to Figure 4.9-5.

$$t_{xCOA} = t_{COA} - (1/c) \bullet |\mathbf{Xmt}(t_{COA}) - \mathbf{SCP}|$$

$$t_{rCOA} = t_{COA} + (1/c) \bullet |\mathbf{Rcv}(t_{COA}) - \mathbf{SCP}|$$

Note: For a bistatic image, times t_{xCOA} and t_{rCOA} are, in general, computed using the GRP position at time t_{COA} . However, for all bistatic images, the GRP and the SCP are always coincident at SCP COA time. See Section 4.6.2 and Section 4.6.4.

- (2) Compute the transmit APC position, velocity, and acceleration at time t_{xCOA} , $\mathbf{Xmt}_{COA}$, $\mathbf{VXmt}_{COA}$, and $\mathbf{AXmt}_{COA}$. Also compute the range from the SCP to the transmit APC, R_Xmt_SCP , and the unit vector that points from the SCP to transmit APC, $\mathbf{uXmt}_{COA}$.

$$\mathbf{Xmt}_{COA} = \mathbf{Xmt}(t_{xCOA}) \quad \mathbf{VXmt}_{COA} = \frac{d}{dt} \mathbf{Xmt}(t_{xCOA})$$

$$\mathbf{AXmt}_{COA} = \frac{d^2}{dt^2} \mathbf{Xmt}(t_{xCOA})$$

$$R_Xmt_SCP = |\mathbf{Xmt}_{COA} - \mathbf{SCP}| \quad \mathbf{uXmt}_{COA} = \frac{1}{R_Xmt_SCP} \bullet (\mathbf{Xmt}_{COA} - \mathbf{SCP})$$

- (3) Compute the range rate of the SCP relative to the transmit APC, $Rdot_Xmt_SCP$. Compute the derivative of the unit vector from the SCP to transmit APC, $\mathbf{uXmtDot}_{COA}$.

$$Rdot_Xmt_SCP = \mathbf{uXmt}_{COA} \bullet \mathbf{VXmt}_{COA}$$

$$\mathbf{uXmtDot}_{COA} = \frac{1}{R_Xmt_SCP} \bullet (\mathbf{VXmt}_{COA} - Rdot_Xmt_SCP \bullet \mathbf{uXmt}_{COA})$$

- (4) Compute the receive APC position, velocity, and acceleration at time t_{rCOA} , $\mathbf{Rcv}_{COA}$, $\mathbf{VRcv}_{COA}$, and $\mathbf{ARcv}_{COA}$. Also compute the range from the SCP relative to the receive APC, R_Rcv_SCP , and the unit vector that points from the SCP to receive APC, $\mathbf{uRcv}_{COA}$.

$$\mathbf{Rcv}_{COA} = \mathbf{Rcv}(t_{rCOA}) \quad \mathbf{VRcv}_{COA} = \frac{d}{dt} \mathbf{Rcv}(t_{rCOA})$$

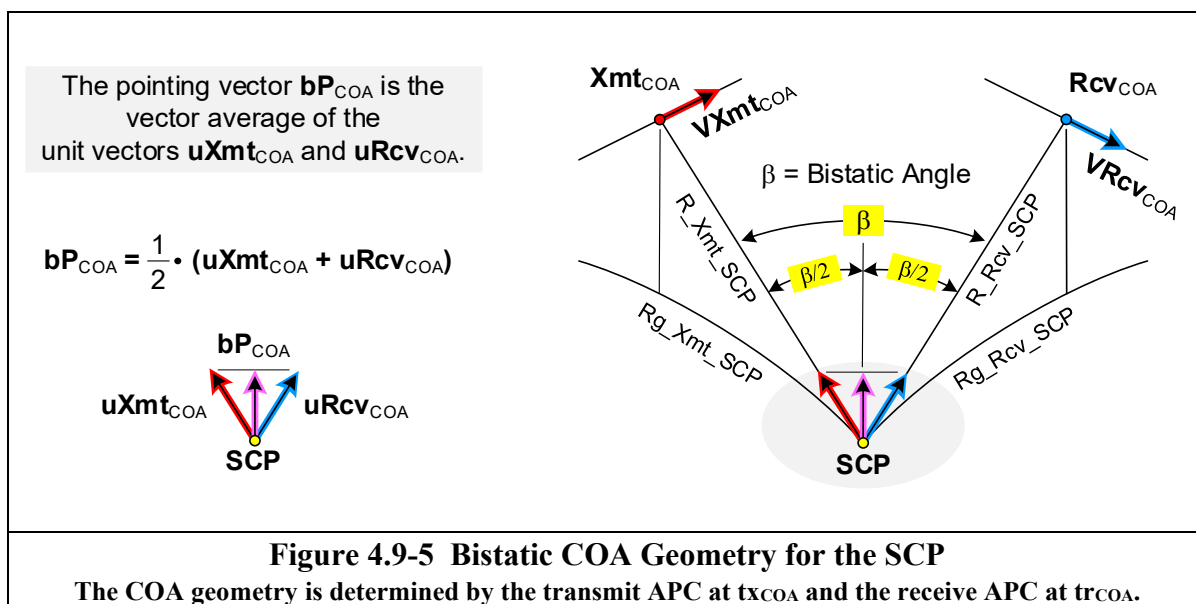
$$\mathbf{ARcv}_{COA} = \frac{d^2}{dt^2} \mathbf{Rcv}(t_{rCOA})$$

$$R_Rcv_SCP = |\mathbf{Rcv}_{COA} - \mathbf{SCP}| \quad \mathbf{uRcv}_{COA} = \frac{1}{R_Rcv_SCP} \bullet (\mathbf{Rcv}_{COA} - \mathbf{SCP})$$

- (5) Compute the range rate of the SCP relative to the receive APC, $R_{dot_Rcv_SCP}$.
 Compute the derivative of the unit vector from the SCP to receive APC, $\mathbf{uRcvDot}_{COA}$.

$$R_{dot_Rcv_SCP} = \mathbf{uRcv}_{COA} \bullet \mathbf{VRcv}_{COA}$$

$$\mathbf{uRcvDot}_{COA} = \frac{1}{R_{Rcv_SCP}} \bullet (\mathbf{VRcv}_{COA} - R_{dot_Rcv_SCP} \bullet \mathbf{uRcv}_{COA})$$



- (6) Compute bistatic pointing vector, $\mathbf{bP}_{COA}$, and its derivative, $\mathbf{bPDot}_{COA}$, from the unit vectors and their derivatives.

$$\mathbf{bP}_{COA} = \frac{1}{2} \bullet (\mathbf{uXmt}_{COA} + \mathbf{uRcv}_{COA})$$

$$\mathbf{bPDot}_{COA} = \frac{1}{2} \bullet (\mathbf{uXmtDot}_{COA} + \mathbf{uRcvDot}_{COA})$$

- (7) Compute the magnitude of the pointing vector, bP_Mag_{COA} , and the bistatic angle, $Bistat_Ang_{COA}$. The bistatic angle is in degrees on the interval $0 \leq Bistat_Ang_{COA} \leq 180.0$.

$$bP_Mag_{COA} = |\mathbf{bP}_{COA}| \quad Bistat_Ang_{COA} = 2 \bullet \arccos(bP_Mag_{COA})$$

- (8) Compute the bistatic angle rate, $Bistat_Ang_Rate_{COA}$, that is the rate of change of the bistatic angle. For the magnitude of the pointing vector equal to 0 or equal to 1, set the rate equal to zero. The bistatic angle rate is in degrees/sec.

For $0.0 < bP_Mag_{COA} < 1.0$:

$$Bistat_Ang_Rate_{COA} = - \left(\frac{180}{\pi} \right) \bullet \left(\frac{4}{\sin(Bistat_Ang_{COA})} \right) \bullet (\mathbf{bP}_{COA} \bullet \mathbf{bPDot}_{COA})$$

For $bP_Mag_{COA} = 0.0$ or $bP_Mag_{COA} = 1.0$: $Bistat_Ang_Rate_{COA} = 0$

(9) The bistatic angle and bistatic angle rate at COA are both included in the SCPCOA / Bistatic branch of the XML.

Bistat_Ang _{COA}	➔	XML: BistaticAng	degrees
Bistat_Ang_Rate _{COA}	➔	XML: BistaticAngRate	degrees/sec

TxPlatform Parameters

The parameters that specify the transmit APC position at SCP COA are included in the SCPCOA / Bistatic / TxPlatform branch of the XML instance. The TxPlatform parameters include the time, position, velocity, and acceleration computed in steps (1) & (2) above: t_{XCOA} , $\mathbf{Xmt}_{COA}$, $\mathbf{VXmt}_{COA}$, and $\mathbf{AXmt}_{COA}$. The other transmit platform parameters are computed as shown in steps (10) through (15).

(10) Compute the ground range from the transmit APC to the SCP, Rg_Xmt_SCP . Refer to Figure 4.9-1.

$$Xmt_DEC = | \mathbf{Xmt}_{COA} | \quad \mathbf{uEC_Xmt}_{COA} = \frac{1}{Xmt_DEC} \bullet \mathbf{Xmt}_{COA}$$

$$EA_Xmt_{COA} = \text{acos}(\mathbf{uEC_Xmt}_{COA} \bullet \mathbf{uSCP})$$

$$Rg_Xmt_SCP = SCP_DEC \bullet EA_Xmt_{COA}$$

(11) Compute the Side of Track of the SCP relative the relative to the transmit APC trajectory. If the velocity of the transmit APC is zero, set the Side of Track to L.

$$\mathbf{LEFT_Xmt} = \mathbf{uEC_Xmt}_{COA} \times \mathbf{VXmt}_{COA}$$

$$\text{Side_Of_Track_Xmt} = \begin{cases} L, & \text{if } \mathbf{LEFT_Xmt} \bullet \mathbf{uXmt}_{COA} < 0 \\ R, & \text{otherwise} \end{cases}$$

(12) Compute the Doppler Cone Angle (DCA) measured from transmit APC velocity to the line-of-sight to the SCP. The DCA is on the interval $0 \leq DCA_Xmt_{COA} \leq 180$ degrees. See Figure 4.9-6.

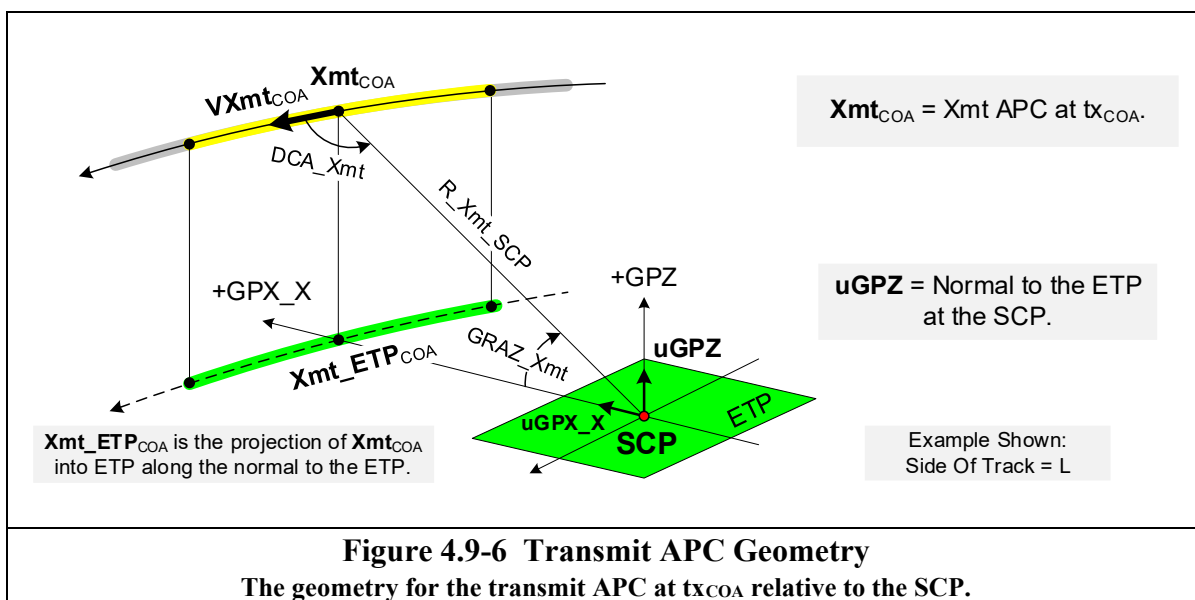
$$VXmt_M = | \mathbf{VXmt}_{COA} | \quad DCA_Xmt = \text{acos}(\frac{-1}{VXmt_M} \bullet \mathbf{Rdot_Xmt_SCP})$$

(13) Compute point $\mathbf{Xmt_ETP}_{COA}$ that is position of the transmit APC projected normal to the ETP at the SCP. Also compute unit vector $\mathbf{uGPX_X}$ points from the SCP to point $\mathbf{Xmt_ETP}_{COA}$.

$$\mathbf{Xmt_GPZ}_{COA} = (\mathbf{Xmt}_{COA} - \mathbf{SCP}) \bullet \mathbf{uGPZ}$$

$$\mathbf{Xmt_ETP}_{COA} = \mathbf{Xmt}_{COA} - \mathbf{Xmt_GPZ}_{COA} \bullet \mathbf{uGPZ}$$

$$\mathbf{uGPX_X} = \frac{1}{| \mathbf{Xmt_ETP}_{COA} - \mathbf{SCP} |} \bullet (\mathbf{Xmt_ETP}_{COA} - \mathbf{SCP})$$



(14) Compute the grazing angle of the line-of-sight from the transmit APC to the SCP relative to the ETP at the SCP. Also compute the incidence angle. The grazing angle is on the interval $-90 \leq \text{GRAZ_Xmt} \leq 90$ degrees and the incidence angle is on the interval $0 \leq \text{INCD_Xmt} \leq 180$ degrees.

$$\text{GRAZ_Xmt} = \text{asin}\left(\frac{1}{R_Xmt_SCP} \cdot Xmt_GPZ_{COA}\right) \quad \text{INCD_Xmt} = 90 - \text{GRAZ_Xmt}$$

(15) Compute the azimuth angle off of north at the SCP to Xmt_ETP. The computed angle, AZIM_Xmt, is on the interval $0 \leq \text{AZIM_Xmt} \leq 360$ degrees.

$$\text{AZ_Xmt_N} = \mathbf{uNORTH} \bullet \mathbf{uGPX_X} \quad \text{AZ_Xmt_E} = \mathbf{uEAST} \bullet \mathbf{uGPX_X}$$

$$\text{Azim_Xmt} = \text{atan}\left(\frac{\text{AZ_Xmt_E}}{\text{AZ_Xmt_N}}\right) \quad \text{Note: Four-Quadrant ArcTan}(y/x)$$

(16) The parameters of the /SCPCOA/Bistatic/TxPlatform branch of the XML shall be assigned as follows.

tx_{COA}	➔	XML: Time	seconds
Xmt_{COA}	➔	XML: Pos	meters
$VXmt_{COA}$	➔	XML: Vel	meters/sec
$AXmt_{COA}$	➔	XML: Acc	meters/sec ²
Side_Of_Track_Xmt	➔	XML: SideOfTrack	
R_Xmt_SCP	➔	XML: SlantRange	meters
Rg_Xmt_SCP	➔	XML: GroundRange	meters
DCA_Xmt	➔	XML: DopplerConeAng	degrees

GRAZ_Xmt	➔	XML: GrazeAng	degrees
INCD_Xmt	➔	XML: IncidenceAng	degrees
Azim_Xmt	➔	XML: AzimAng	degrees

RcvPlatform Parameters

The parameters that specify the receive APC position at SCP COA are included in the SCPCOA / Bistatic / RcvPlatform branch of the XML instance. The RcvPlatform parameters include the time, position, velocity, and acceleration computed in steps (1) & (4) above: tr_{COA} , Rcv_{COA} , $VRcv_{COA}$, and $ARcv_{COA}$. The other receive platform parameters are computed as shown in steps (10) through (15) above by replacing the transmit APC position and velocity with the receive APC position and velocity.

(17) The parameters of the /SCPCOA/Bistatic/RcvPlatform branch of the XML shall be assigned as follows.

tr_{COA}	➔	XML: Time	seconds
Rcv_{COA}	➔	XML: Pos	meters
$VRcv_{COA}$	➔	XML: Vel	meters/sec
$ARcv_{COA}$	➔	XML: Acc	meters/sec ²
Side_Of_Track_Rcv	➔	XML: SideOfTrack	
R_Rcv_SCP	➔	XML: SlantRange	meters
Rg_Rcv_SCP	➔	XML: GroundRange	meters
DCA_Rcv	➔	XML: DopplerConeAng	degrees
GRAZ_Rcv	➔	XML: GrazeAng	degrees
INCD_Rcv	➔	XML: IncidenceAng	degrees
Azim_Rcv	➔	XML: AzimAng	degrees

4.10 Radiometric Parameters

The radiometric parameters that may be included in an image product are summarized below. Refer to Table 3-10. The parameters available provide an estimate of the noise level in the image and a set of scale factors that convert pixel power level to radar cross section.

The complex image signal array is denoted $S_{II}(\text{row}, \text{col})$ where each pixel has a real and imaginary component. The pixel power value, sometimes referred to as the pixel intensity, is $P_{II}(\text{row}, \text{col})$, where $P_{II}(\text{row}, \text{col})$ is the sum of the squares of the real and imaginary components. A power level may also be expressed in decibels, $P_dB_{II}(\text{row}, \text{col})$, relative to power level 1.0.

$$S_{II}(\text{row}, \text{col}) = \text{Real}(S_{II}(\text{row}, \text{col})) + j \cdot \text{Imag}(S_{II}(\text{row}, \text{col}))$$

$$P_{II}(\text{row}, \text{col}) = \text{Real}(S_{II}(\text{row}, \text{col}))^2 + \text{Imag}(S_{II}(\text{row}, \text{col}))^2$$

$$P_dB_{II}(\text{row}, \text{col}) = 10 \cdot \log_{10}(P_{II}(\text{row}, \text{col}))$$

In the descriptions that follow, SCP pixel-centered indices ($irow, icol$) and image coordinates ($xrow, ycol$) are used to address the image signal array as a continuous two-dimensional space. The image pixel values $\{S_{II}(\text{row}, \text{col})\}$ are considered to be the discrete samples of a continuous, two-dimensional image signal at the integer values of indices ($irow, icol$). See Section 4.4.1.

For the parameters related to image signal power for inherently random signals, the term power refers to the expected value of the signal power. For a noise power value, the value is the expected value of the noise power at a given image location. For the clutter reflectivity scale factors, the scale factors relate the expected value of the clutter power to the clutter-based reflectivity.

4.10.1 Noise Power Level

The noise power level as a function of image location can be included in the image product. The noise power level is expressed as a two-dimensional polynomial of the image coordinates. At image grid location ($xrow, ycol$), the noise power level estimate $PN_dB(xrow, ycol)$ is computed as follows.

$$\text{NoisePoly}(m, n) = cNDB(m, n), \quad m = 0 \text{ to } M_NDB, \quad n = 0 \text{ to } N_NDB$$

$$PN_dB(xrow, ycol) = \sum_{m=0}^{M_NDB} \sum_{n=0}^{N_NDB} cNDB(m, n) \cdot (xrow)^m \cdot (ycol)^n$$

The noise level estimate may be one of two types – absolute or relative – as indicated by parameter `NoiseLevelType`. For an absolute noise level, a power estimate of $PN_dB(xrow, ycol) = 20.0$ dB indicates the expected noise power is $PN(xrow, ycol) = 100.0$. For a relative noise level, a noise power estimate $PN_dB(xrow, ycol) = 3.01$ dB indicates the

noise power is 3.01 dB higher than the noise level at the SCP pixel and the ratio $PN(xrow,ycol) / PN(0,0) = 2.0$.

4.10.2 Target RCS Scale Factor

The relationship between target radar cross section (RCS) and image signal power can be included in the image product. The relationship between point target RCS and image signal power is expressed as a two-dimensional polynomial of the image coordinates. At image grid location (xrow,ycol), the RCS scale factor $RCS_SF(xrow,ycol)$, is computed as follows.

$$RCSSFPoly(m,n) = cRCS(m,n), \quad m = 0 \text{ to } M_RCS, \quad n = 0 \text{ to } N_RCS$$

$$RCS_SF(xrow,ycol) = \sum_{m=0}^{M_RCS} \sum_{n=0}^{N_RCS} cRCS(m,n) \cdot (xrow)^m \cdot (ycol)^n$$

The RCS scale factor relates image signal power for an ideal point scatterer to RCS in square meters. For a point target with impulse response centered at (xrow,ycol) and peak image power $P_TGT_{II}(xrow,ycol)$, the RCS is computed by scaling the peak image power value.

$$\text{Target RCS}(m^2) = RCS_SF(xrow,ycol) \cdot P_TGT_{II}(xrow,ycol)$$

The scale factor $RCS_SF(xrow,ycol)$ is computed for a target located at the same Height Above the Ellipsoid (HAE) as the SCP. It is also computed for $P_TGT_{II}(xrow,ycol)$ being the peak signal response for the focused target with impulse response.

4.10.3 Clutter Reflectivity Scale Factors

The relationship between image signal power and clutter reflectivity can be included in the image product. Image power to clutter reflectivity scale factors may be included for clutter reflectivity parameters Sigma-Zero (σ^0), Beta-Zero (β_0) and/or Gamma-Zero (γ_0). For a given clutter reflectivity parameter, the image power to clutter reflectivity scale factor is a function of the image location. For example, at image location (xrow,ycol), the value of clutter reflectivity Sigma-Zero that produces image power $P_{II}(xrow,ycol)$ is as follows.

$$\text{Sigma-Zero} = \text{SigmaZero_SF}(xrow,ycol) \cdot P_{II}(xrow,ycol)$$

(1) Sigma-Zero Scale Factor

The Sigma-Zero scale factor is expressed as a two-dimensional polynomial of the image coordinates. At image grid location (xrow,ycol), the scale factor $\text{SigmaZero_SF}(xrow,ycol)$ is computed as follows.

$$\text{SigmaZeroPoly}(m,n) = cSIG(m,n), \quad m = 0 \text{ to } M_SIG, \quad n = 0 \text{ to } N_SIG$$

$$\text{SigmaZero_SF}(xrow,ycol) = \sum_{m=0}^{M_SIG} \sum_{n=0}^{N_SIG} cSIG(m,n) \cdot (xrow)^m \cdot (ycol)^n$$

The value of clutter reflectivity Sigma-Zero that produces a given image power is the image power $P_{II}(xrow,ycol)$ multiplied by the scale factor.

$$\text{Sigma-Zero} = \text{SigmaZero_SF}(\text{xrow}, \text{ycol}) \cdot P_{\Pi}(\text{xrow}, \text{ycol})$$

(2) Beta-Zero Scale Factor

The Beta-Zero scale factor is expressed as a two-dimensional polynomial of the image coordinates. At image grid location (xrow,ycol), the scale factor BetaZero_SF(xrow,ycol) is computed as follows.

$$\text{BetaZeroPoly}(m,n) = \text{cBETA}(m,n), \quad m = 0 \text{ to } M_BETA, \quad n = 0 \text{ to } N_BETA$$

$$\text{BetaZero_SF}(\text{xrow}, \text{ycol}) = \sum_{m=0}^{M_BETA} \sum_{n=0}^{N_BETA} \text{cBETA}(m,n) \cdot (\text{xrow})^m \cdot (\text{ycol})^n$$

The value of clutter reflectivity Beta-Zero that produces a given image power is the image power $P_{\Pi}(\text{xrow}, \text{ycol})$ multiplied by the scale factor.

$$\text{Beta-Zero} = \text{BetaZero_SF}(\text{xrow}, \text{ycol}) \cdot P_{\Pi}(\text{xrow}, \text{ycol})$$

(3) Gamma-Zero Scale Factor

The Gamma-Zero scale factor is expressed as a two-dimensional polynomial of the image coordinates. At image grid location (xrow,ycol), the scale factor GammaZero_SF(xrow,ycol) is computed as follows.

$$\text{GammaZeroPoly}(m,n) = \text{cGMMA}(m,n), \quad m = 0 \text{ to } M_GMMA, \quad n = 0 \text{ to } N_GMMA$$

$$\text{GammaZero_SF}(\text{xrow}, \text{ycol}) = \sum_{m=0}^{M_GMMA} \sum_{n=0}^{N_GMMA} \text{cGMMA}(m,n) \cdot (\text{xrow})^m \cdot (\text{ycol})^n$$

The value of clutter reflectivity Gamma-Zero that produces a given image power is the image power $P_{\Pi}(\text{xrow}, \text{ycol})$ multiplied by the scale factor.

$$\text{Gamma-Zero} = \text{GammaZero_SF}(\text{xrow}, \text{ycol}) \cdot P_{\Pi}(\text{xrow}, \text{ycol})$$

4.10.4 Clutter Reflectivity Details

For a given terrain, the clutter reflectivity parameters σ^0 , β_0 and γ_0 are dimensionless ratios that relate the area of an image resolution cell to the equivalent RCS of the cell containing ground clutter. The RCS of the ground clutter in the cell is equal to the area of the resolution cell times the clutter reflectivity. For a given reflectivity parameter, the area of the clutter cell is expressed after projection to a given plane. The plane of the projection is unique to each parameter.

Sigma-Zero(σ^0): Ratio between RCS and the area of the resolution cell projected to the ground plane.

Beta-Zero(β_0): Ratio of the RCS to the area of the resolution cell projected into the slant plane.

Gamma-Zero(γ_0): Ratio between RCS and the area of the clutter cell projected into plane of incidence which is normal to the line of sight.

For a given resolution cell in the image, the parameters have the following relationship. The grazing angle and the slope angle are dependent upon the ground plane selected for computing the area of the resolution cell. See Section 4.9.

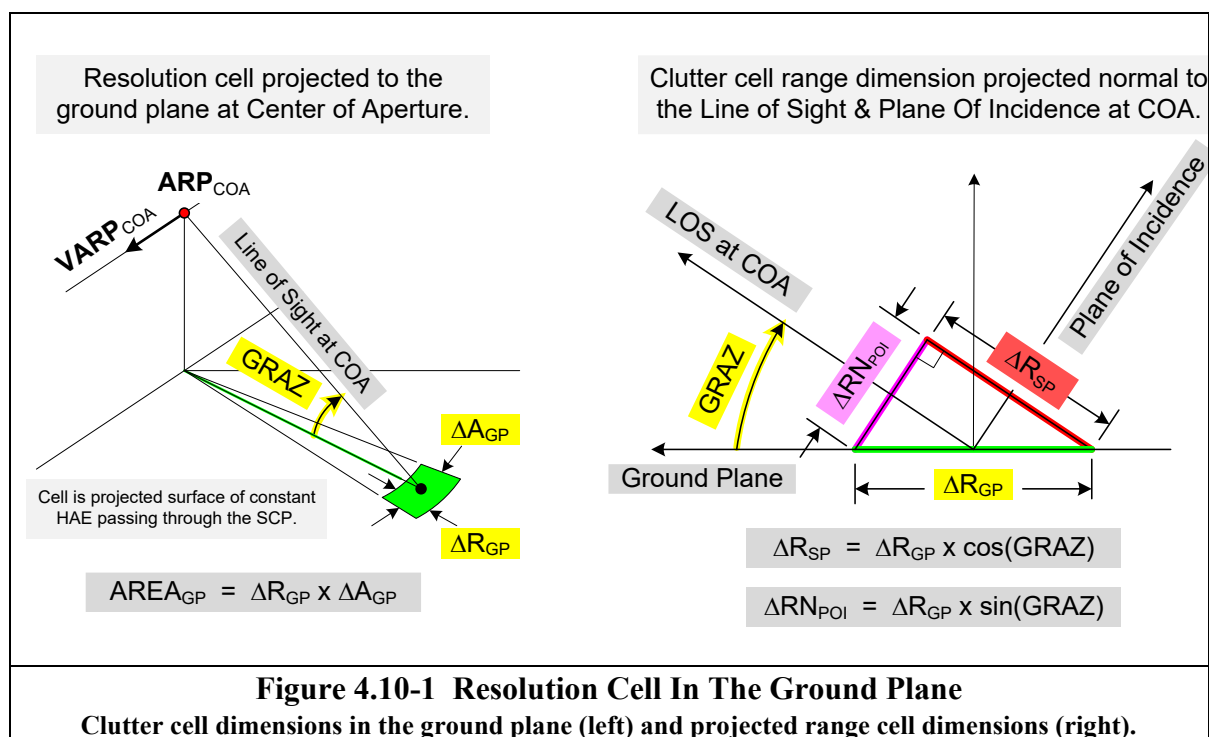
$$\text{Sigma-Zero} = \text{Beta-Zero} \cdot \cos(\text{SLOPE}) = \text{Gamma-Zero} \cdot \sin(\text{GRAZ})$$

The Sigma-Zero scale factor provided in an image product is computed based upon the projection of the image points onto the surface of constant Height Above Ellipsoid (HAE) passing through the SCP. Each image point is projected at the COA for the point. For each image point, the ground plane is the Earth Tangent Plane (ETP) that is tangent to the surface of constant HAE. The area of the resolution cell in the ETP is the ground plane area. The slope and grazing angles will vary slightly as the ETP geometry varies from image point to image point.

An example resolution cell projected to the ground plane is shown in the left half of Figure 4.10-1. The area in ground plane, AREA_{GP} , is computed from the resolution cell dimensions in the ground plane, ΔR_{GP} and ΔA_{GP} . The RCS for the ground clutter in the cell, RCS_{CLTR} , is the AREA_{GP} scaled by Sigma-Zero.

$$\text{AREA}_{\text{GP}} = \Delta R_{\text{GP}} \cdot \Delta A_{\text{GP}}$$

$$\text{RCS}_{\text{CLTR}} = \text{Sigma-Zero} \cdot \text{AREA}_{\text{GP}}$$



The projection to the slant plane and to the plane of incidence is shown in the right half of Figure 4.10-1. The resolution cell area projected into the slant plane, $AREA_{SP}$, is the ground plane area projected normal to the COA slant plane. The RCS for the ground clutter in the cell is $AREA_{SP}$ scaled by Beta-Zero. The cosine of the SLOPE angle may be written as the product of the cosine of the grazing angle times the cosine of the twist angle.

$$AREA_{SP} = AREA_{GP} \cdot \cos(SLOPE) = AREA_{GP} \cdot \underbrace{\cos(GRAZ) \cdot \cos(TWST)}_{\cos(SLOPE)}$$

$$AREA_{SP} = \Delta R_{SP} \cdot \Delta A_{SP} = \underbrace{\Delta R_{GP} \cdot \cos(GRAZ)}_{\Delta R_{SP}} \cdot \underbrace{\Delta R_{GP} \cdot \cos(TWST)}_{\Delta A_{SP}}$$

$$RCS_{CLTR} = BetaZero \cdot AREA_{SP}$$

The resolution cell area projected normal to the line of sight, $AREA_{POI}$, is the ground plane area projected normal to the Plane of Incidence (POI). The RCS for the ground clutter in the cell is $AREA_{POI}$ scaled by Gamma-Zero.

$$AREA_{POI} = AREA_{GP} \cdot \sin(GRAZ)$$

$$AREA_{POI} = \Delta R_{POI} \cdot \Delta A_{GP} = \underbrace{\Delta R_{GP} \cdot \sin(GRAZ)}_{\Delta R_{POI}} \cdot \Delta A_{GP}$$

$$RCS_{CLTR} = Gamma-Zero \cdot AREA_{POI}$$

4.11 Transmit & Receive Antenna Parameters

The parameters that describe the transmit and receive antenna patterns are summarized below. Refer to Table 3-11. The antenna parameters describe the mainlobe illumination patterns of the transmit and receive antennas of the SAR platform. The illumination patterns are functions of the time of the illumination and the pointing direction relative to the antenna orientation.

The antenna parameters allow the variation in received signal level due to the variation in illumination patterns to be computed for all points in the imaged scene. For a given point in the scene, the variation in received signal power and signal phase may be computed as a function of time and frequency. The variation in signal versus time may be used to compute compensation for undesired broadening of the azimuth impulse response. The variation in integrated signal level over time versus position in the image may be used to compute compensation for variation in overall image intensity.

For all SAR collections, the received signal level from a point in the scene is a function of the transmit antenna pattern at the time of transmit and the receive antenna pattern at the time of reception. Separate sets of antenna parameters may be provided for the transmit and receive antenna. For most collections, the combined effect of the two patterns may be accurately described by a single two-way antenna pattern. For these collections, a single set of antenna parameters may be provided that describes the two-way illumination pattern.

A set of antenna parameters describes the antenna's mechanical orientation versus time, the relative gain pattern versus pointing direction and the relative phase pattern versus pointing direction. The gain pattern is expressed in dB relative to the gain at boresight. The phase pattern is expressed in cycles relative to the phase at boresight. Variation in boresight gain versus frequency may also be provided. For electronically scanned antennas, the electronic steering of the mainlobe versus time is provided.

For a fixed point target in the imaged scene, denoted TGT, the relative gain and phase due to the antenna illumination patterns is $G(t, f)^{TGT}$ and $\Phi(t, f)^{TGT}$. Time t is the time the transmitted signal is incident at the scene being imaged. For an image product with only a two-way antenna pattern provided, the gain and phase are computed from the two-way pattern evaluated at the time of incidence.

$$G(t, f)^{TGT} = G_{2WAY}(t, f)^{TGT} \quad \Phi(t, f)^{TGT} = \Phi_{2WAY}(t, f)^{TGT}$$

For an image product with separate transmit and receive antenna patterns, the gain is the sum of the gains and the phase is the sum of the phases for the two patterns. For time of incidence t , the time of transmit $tx(t)$ and the time of reception is $trcv(t)$ may be accurately computed using the ranges from the TGT to aperture phase centers, $R_TX(t)^{TGT}$ and $R_RCV(t)^{TGT}$.

$$tx(t) = t - \frac{1}{c} \cdot R_TX(t)^{TGT} \quad trcv(t) = t + \frac{1}{c} \cdot R_RCV(t)^{TGT}$$

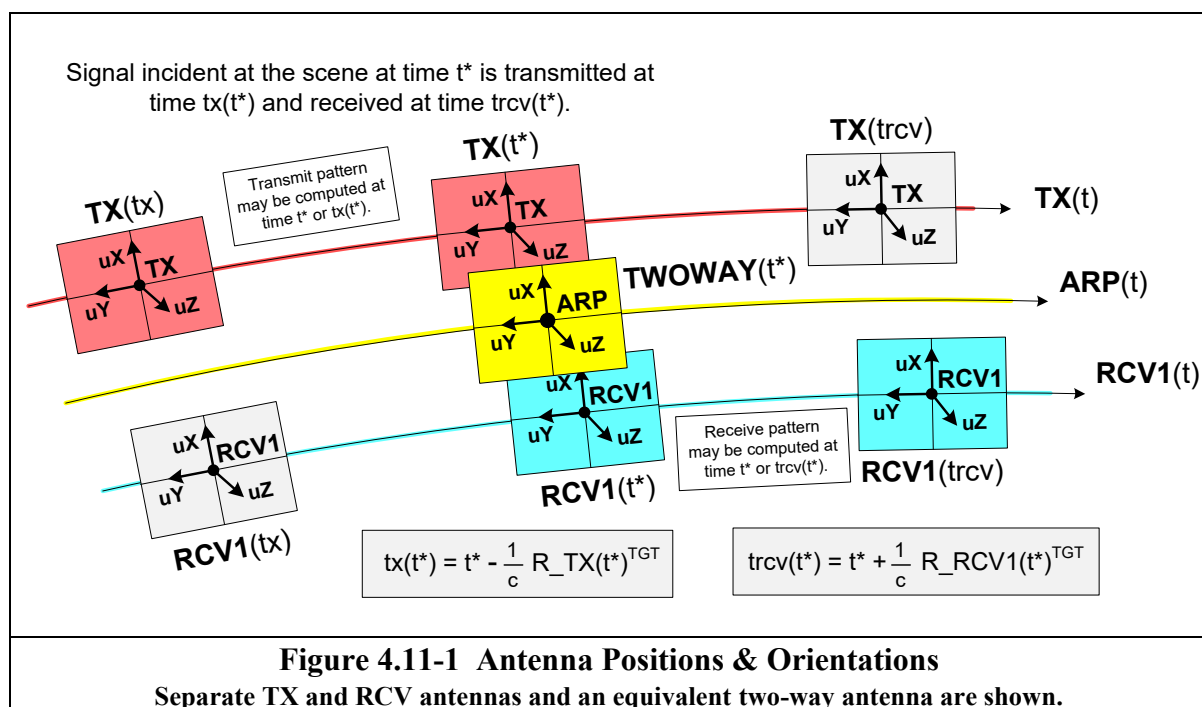
$$G(t, f)^{TGT} = G_{TX}(tx(t), f)^{TGT} + G_{RCV}(trcv(t), f)^{TGT}$$

$$\Phi(t, f)^{TGT} = \Phi_{TX}(tx(t), f)^{TGT} + \Phi_{RCV}(trcv(t), f)^{TGT}$$

For most imaging collections, the transmit and receive gains to a given point vary slowly with time. Computing the transmit and receive patterns at the time of incidence will be sufficiently accurate.

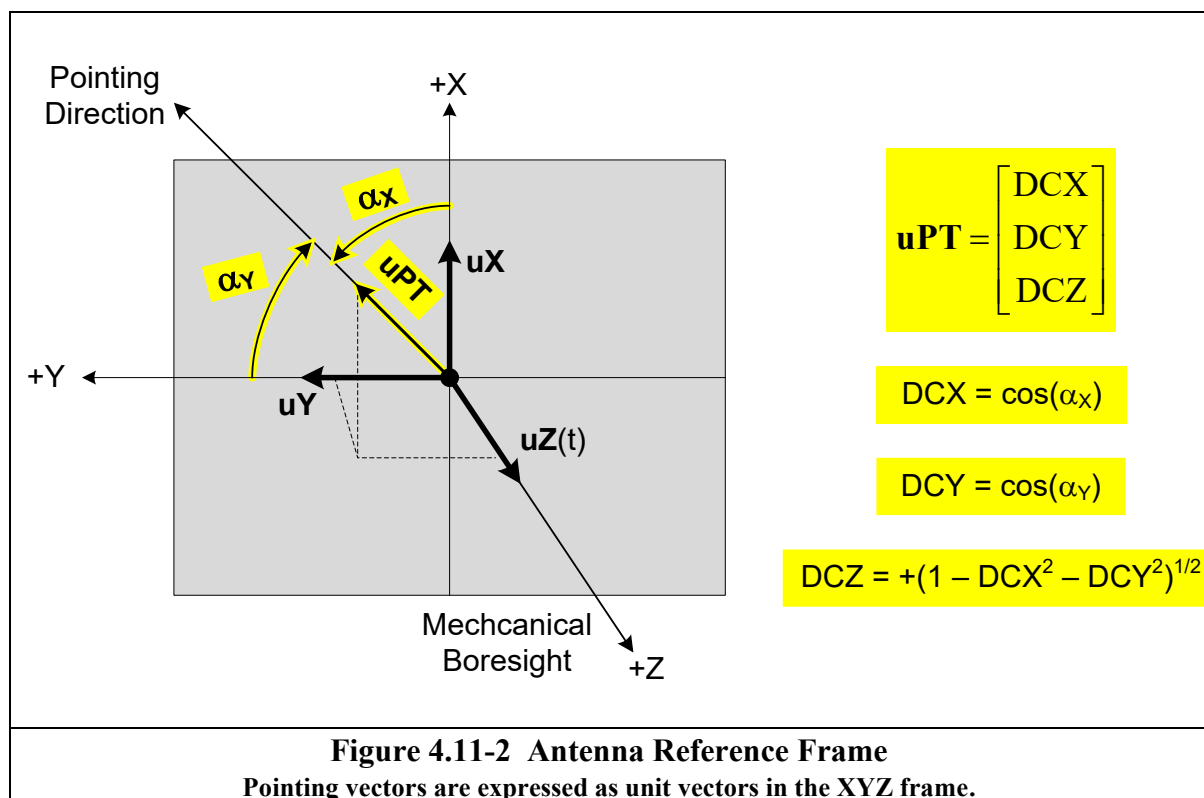
$$G(t, f)^{TGT} = G_{TX}(t, f)^{TGT} + G_{RCV}(t, f)^{TGT} \quad \Phi(t, f)^{TGT} = \Phi_{TX}(t, f)^{TGT} + \Phi_{RCV}(t, f)^{TGT}$$

Consider a collection from a platform with separate transmit and receive antennas. The transmit antenna phase center (APC) position versus time is $\mathbf{TX}(t)$. The receive antenna phase center position versus time is $\mathbf{RCV1}(t)$. A single receive data channel is collected and processed to form the image. The Aperture Reference Position versus time is $\mathbf{ARP}(t)$. Shown in Figure 4.11-1 is a portion the ARP and APC trajectories. For the signal incident at the scene at time t^* , the transmit time is $tx(t^*)$ and the receive time is $trcv(t^*)$. Shown is the ARP at time t^* and the APC positions at the three times. The transmit and receive antennas are shown at the three times. Also shown is an equivalent antenna located at the $\mathbf{ARP}(t^*)$ that can be used to compute the two-way illumination pattern. The mechanical orientation of each antenna is defined by a pair of orthogonal unit vectors, $\mathbf{uX}$ and $\mathbf{uY}$. The mechanical boresight is defined by unit vector $\mathbf{uZ} = \mathbf{uX} \times \mathbf{uY}$.



For a mechanically steered antenna, the mainlobe boresight is in the $+\mathbf{uZ}$ direction. For an electronically steered antenna, the mainlobe boresight is electronically steered within the hemisphere that is centered on $+\mathbf{uZ}$. Pointing directions in the antenna reference frame are

expressed as unit vectors. Shown in Figure 4.11-2 is an example antenna reference frame. A unit pointing vector, $\mathbf{uPT}$, is defined by the direction cosines relative to the X and Y axes, DCX and DCY. The ordered pair (DCX, DCY) is referred to as the direction cosine pair for unit pointing vector $\mathbf{uPT}$.



The following sections describe the antenna parameters that may be provided with an image product. The example collection shown in Figure 4.11-1 is used as a reference. For a target TGT illuminated time t^* , the relative gain and phase due to the antenna illumination patterns, $G(t^*, f)^{TGT}$ and $\Phi(t^*, f)^{TGT}$, are computed. The computation using an equivalent two-way antenna pattern is described in Section 4.11.1. The computation using separate transmit and receive antenna patterns is described in Section 4.11.2.

All antenna phase center positions and orientations are provided as functions of time t . All positions are in ECF coordinates. The aperture orientations are computed relative to the ECF coordinate frame as described below. For all computations, time t is in seconds relative to Collection Start.

4.11.1 Two-Way Antenna Pattern

The gain and phase for target TGT are computed using a set of Two-Way antenna pattern parameters as described in following section. Gain $G(t^*, f)^{TGT}$ is in dB and phase $\Phi(t^*, f)^{TGT}$ is in cycles.

$$G(t, f)^{TGT} = G_{2WAY}(t, f)^{TGT} \quad \Phi(t, f)^{TGT} = \Phi_{2WAY}(t, f)^{TGT}$$

(1) Frequency Dependent Parameters

The base set of parameters are provided for frequency f_0 . The parameters that vary with frequency are indicated by the inclusion of optional parameters.

$$f_0 = \text{Base Frequency} \quad \text{XML: Antenna.TwoWay.FreqZero}$$

Boresight gain as a function of frequency is specified by including the gain versus frequency polynomial. Gain term $GF_{2WAY}(f)$ is computed by evaluating the polynomial at $(f - f_0)/f_0$. Gain $GF_{2WAY}(f)$ is in dB. If polynomial is not included, set $GF_{2WAY}(f) = 0$.

$$GF_{2WAY}(f) = \text{GainBSPoly}((f - f_0)/f_0) \quad \text{XML: Antenna.TwoWay.GainBSPoly}$$

Mainlobe dilation versus frequency is indicated by optional parameter $MLFreqDilation$ set equal to TRUE. The modeled mainlobe dilation has the width vary inversely with frequency. If the parameter is not included or is set to FALSE, the mainlobe pattern does not vary with frequency. Set parameter $MLD(f)$ as follows.

$$MLD(f) = \begin{cases} \frac{f}{f_0}, & \text{If } MLDilation = \text{TRUE} \\ 1.0 & \text{Otherwise} \end{cases} \quad \text{XML: Antenna.TwoWay.MLFreqDilation}$$

For an electronically steered antenna, the mainlobe boresight shift versus frequency is indicated by optional parameter $EBFreqShift$ set equal to TRUE. The modeled boresight shift has the mainlobe boresight shift toward the mechanical boresight. If the parameter is not included or is set to FALSE, the mainlobe boresight does not shift with frequency. Set parameter $EBS(f)$ as follows.

$$EBS(f) = \begin{cases} \frac{f_0}{f}, & \text{If } EBFreqShift = \text{TRUE} \\ 1.0 & \text{Otherwise} \end{cases} \quad \text{XML: Antenna.TwoWay.EBFreqShift}$$

(2) Two-Way Antenna Phase Center & Orientation

For the equivalent Two-Way antenna pattern, the antenna phase center at time t^* is the $\mathbf{ARP}(t^*)$. The antenna orientation is also computed at time t^* . The antenna orientation vectors $\mathbf{XVP}(t^*)$ and $\mathbf{YVP}(t^*)$ are computed by evaluating the vector-time polynomials.

$$\begin{aligned}\mathbf{XVP}(t^*) &= \text{XAxisPoly}(t^*) & \text{XML: Antenna.TwoWay.XAxisPoly} \\ \mathbf{YVP}(t^*) &= \text{YAxisPoly}(t^*) & \text{XML: Antenna.TwoWay.YAxisPoly}\end{aligned}$$

Vectors $\mathbf{XVP}(t^*)$ and $\mathbf{YVP}(t^*)$ may not be unit length or precisely orthogonal due to slight errors in the polynomial fit. Vector $\mathbf{uZ}(t^*)$ is normal plane formed by vectors $\mathbf{XVP}(t^*)$ and $\mathbf{YVP}(t^*)$. Adjusted unit vectors $\mathbf{uX}(t^*)$ and $\mathbf{uY}(t^*)$ form an orthogonal basis. All vectors are in ECF coordinates. The antenna reference frame is shown in Figure 4.11-2.

$$\begin{aligned}\mathbf{uXVP}(t^*) &= \frac{1}{|\mathbf{XVP}(t^*)|} \cdot \mathbf{XVP}(t^*) & \mathbf{uYVP}(t^*) &= \frac{1}{|\mathbf{YVP}(t^*)|} \cdot \mathbf{YVP}(t^*) \\ \mathbf{Z}(t^*) &= \mathbf{uXVP}(t^*) \times \mathbf{uYVP}(t^*) & \mathbf{uZ}(t^*) &= \frac{1}{|\mathbf{Z}(t^*)|} \cdot \mathbf{Z}(t^*) \\ \alpha(t^*) &= \sin^{-1}(\mathbf{uXVP}(t^*) \bullet \mathbf{uYVP}(t^*)) & \mathbf{uN}(t^*) &= \mathbf{uZ}(t^*) \times \mathbf{uXVP}(t^*) \\ \mathbf{uX}(t^*) &= \cos\left(\frac{1}{2} \cdot \alpha(t^*)\right) \cdot \mathbf{uXVP}(t^*) - \sin\left(\frac{1}{2} \cdot \alpha(t^*)\right) \cdot \mathbf{uN}(t^*) \\ \mathbf{uY}(t^*) &= \sin\left(\frac{1}{2} \cdot \alpha(t^*)\right) \cdot \mathbf{uXVP}(t^*) + \cos\left(\frac{1}{2} \cdot \alpha(t^*)\right) \cdot \mathbf{uN}(t^*)\end{aligned}$$

(3) Mainlobe Boresight Pointing Vector

For an electronically steered antenna, the boresight pointing vector versus time is included with the optional boresight steering polynomials. The steering polynomials provide the boresight direction cosines at time t and frequency f_0 . The values are then scaled by parameter $\text{EBS}(f)$ to account for boresight shift versus frequency. For a mechanically steered antenna, the boresight is along $+\mathbf{uZ}(t^*)$ direction ($\text{DCX} = \text{DCY} = 0$). The direction cosine pair ($\text{DCX_EB}(t^*, f)$, $\text{DCY_EB}(t^*, f)$) is computed as follows.

For an electronically steered antenna:

$$\begin{aligned}\text{DCX_EB}_0(t^*) &= \text{DCX_EBPoly}(t^*) & \text{XML: Antenna.TwoWay.EB.DCXPoly} \\ \text{DCY_EB}_0(t^*) &= \text{DCY_EBPoly}(t^*) & \text{XML: Antenna.TwoWay.EB.DCYPoly}\end{aligned}$$

$$\text{DCX_EB}(t^*, f) = \text{EBS}(f) \cdot \text{DCX_EB}_0(t^*) \quad \text{DCY_EB}(t^*, f) = \text{EBS}(f) \cdot \text{DCY_EB}_0(t^*)$$

For a mechanically steered antenna: $\text{DCX_EB}(t^*, f) = 0$ and $\text{DCY_EB}(t^*, f) = 0$

(4) Target Pointing Vector

The unit pointing vector from the antenna phase center to the target TGT, $\mathbf{uTGT}(t^*)$, is computed in ECF coordinates. The direction cosine pair $(DCX(t^*)^{TGT}$ and $DCY(t^*)^{TGT})$ is then computed as follows.

$$\mathbf{uTGT}(t^*) = \frac{1}{|\mathbf{TGT} - \mathbf{ARP}(t^*)|} \cdot (\mathbf{TGT} - \mathbf{ARP}(t^*))$$

$$DCX(t^*)^{TGT} = \mathbf{uTGT}(t^*) \bullet \mathbf{uX}(t^*) \quad DCY(t^*)^{TGT} = \mathbf{uTGT}(t^*) \bullet \mathbf{uY}(t^*)$$

The difference in direction cosines for the target relative to the mainlobe boresight is also computed.

$$\Delta DCX(t^*, f)^{TGT} = DCX(t^*)^{TGT} - DCX_EB(t^*, f)$$

$$\Delta DCY(t^*, f)^{TGT} = DCY(t^*)^{TGT} - DCY_EB(t^*, f)$$

(5) Array Pattern Gain & Phase

The shape of the mainlobe of the illumination pattern is defined by the array pattern. The array pattern is specified by a gain polynomial and a phase polynomial. The gain polynomial, G_ARR_Poly , specifies the gain in dB relative to the gain at mainlobe boresight. The phase polynomial, Φ_ARR_Poly , specifies the phase in cycles relative to the phase at mainlobe boresight.

$$\begin{aligned} G_ARR_Poly(\Delta DCX, \Delta DCY) & \quad \text{XML: Antenna.TwoWay.Array.GainPoly} \\ \Phi_ARR_Poly(\Delta DCX, \Delta DCY) & \quad \text{XML: Antenna.TwoWay.Array.PhasePoly} \end{aligned}$$

For target TGT, the two-way array pattern gain and phase, $G_ARR_{2WAY}(t^*, f)^{TGT}$ and $\Phi_ARR_{2WAY}(t^*, f)^{TGT}$, are computed as follows. The array polynomials are for frequency f_0 and scale factor $MLD(f)$ accounts for mainlobe dilation versus frequency (if any).

$$G_ARR_{2WAY}(t^*, f)^{TGT} = G_ARR_Poly(MLD(f) \cdot \Delta DCX(t^*, f)^{TGT}, MLD(f) \cdot \Delta DCY(t^*, f)^{TGT})$$

$$\Phi_ARR_{2WAY}(t^*, f)^{TGT} = \Phi_ARR_Poly(MLD(f) \cdot \Delta DCX(t^*, f)^{TGT}, MLD(f) \cdot \Delta DCY(t^*, f)^{TGT})$$

For most mechanically scanned antennas, the phase within the mainlobe is constant and the phase polynomial is set to 0. For many antennas, the mainlobe gain is accurately described by a two-dimensional quadratic function based on the beamwidths.

(6) Element Pattern Gain & Phase

For an electronically steered antenna, the gain is reduced as the mainlobe is steered off mechanical boresight. The variation in gain as a function of steering is included with the optional element pattern gain polynomial, G_ELEM_Poly . An element pattern phase polynomial is also included, Φ_ELEM_Poly . For most antennas, the element pattern phase is constant and Φ_ELEM_Poly is set to zero. For a mechanically steered antenna, the element pattern polynomials are not included.

$G_ELEM_Poly(DCX, DCY)$	XML: Antenna.TwoWay.Elem.GainPoly
$\Phi_ELEM_Poly(DCX, DCY)$	XML: Antenna.TwoWay.Elem.PhasePoly

For target TGT, the two-way element pattern gain and phase, $G_ELEM_{2WAY}(t^*)^{TGT}$ and $\Phi_ELEM_{2WAY}(t^*)^{TGT}$, are computed as follows.

For an electronically steered antenna:

$$G_ELEM_{2WAY}(t^*)^{TGT} = G_ELEM_Poly(DCX(t^*)^{TGT}, DCY(t^*)^{TGT})$$

$$\Phi_ELEM_{2WAY}(t^*)^{TGT} = \Phi_ELEM_Poly(DCX(t^*)^{TGT}, DCY(t^*)^{TGT})$$

For a mechanically steered antenna:

$$G_ELEM_{2WAY}(t^*)^{TGT} = 0 \quad \Phi_ELEM_{2WAY}(t^*)^{TGT} = 0$$

(7) Two-Way Gain & Phase

For target TGT, the two-way gain and phase at time t^* and frequency f is the sum of the gains and the sum of the phases computed above.

$$G_{2WAY}(t^*, f)^{TGT} = G_{F_{2WAY}}(f) + G_ARR_{2WAY}(t^*, f)^{TGT} + G_ELEM_{2WAY}(t^*)^{TGT}$$

$$\Phi_{2WAY}(t^*, f)^{TGT} = \Phi_ARR_{2WAY}(t^*, f)^{TGT} + \Phi_ELEM_{2WAY}(t^*)^{TGT}$$

4.11.2 One-Way Antenna Patterns

The gain and phase for target TGT are computed using a pair of one-way antenna pattern parameter sets as summarized in following section. Gain $G(t^*, f)^{TGT}$ is in dB and phase $\Phi(t^*, f)^{TGT}$ is in cycles.

$$G(t^*, f)^{TGT} = G_{TX}(tx(t^*), f)^{TGT} + G_{RCV}(trcv(t^*), f)^{TGT}$$

$$\Phi(t^*, f)^{TGT} = \Phi_{TX}(tx(t^*), f)^{TGT} + \Phi_{RCV}(trcv(t^*), f)^{TGT}$$

The computations for the one-way patterns are nearly identical as those shown for the two-way pattern in Section 4.11.1. For the transmit pattern, the gain is computed as the sum of three gains and the phase the sum of two phases.

$$G_{TX}(t^*, f)^{TGT} = GF_{TX}(f) + G_ARR_{TX}(t^*, f)^{TGT} + G_ELEM_{TX}(t^*, f)^{TGT}$$

$$\Phi_{TX}(t^*, f)^{TGT} = \Phi_ARR_{TX}(t^*, f)^{TGT} + \Phi_ELEM_{TX}(t^*)^{TGT}$$

For the receive pattern, the gain is computed as the sum of three gains and the phase the sum of two phases.

$$G_{RCV}(t^*, f)^{TGT} = GF_{RCV}(f) + G_ARR_{RCV}(t^*, f)^{TGT} + G_ELEM_{RCV}(t^*)^{TGT}$$

$$\Phi_{RCV}(t^*, f)^{TGT} = \Phi_ARR_{RCV}(t^*, f)^{TGT} + \Phi_ELEM_{RCV}(t^*)^{TGT}$$

The only differences are the computation of the transmit APC position and the receive APC position. These differences are described below.

(1) Transmit APC & Orientation

The transmit APC position at time t^* is computed as shown below.

$$TX(t^*) = Tx_APC_Poly(t^*) \quad \text{XML: Position.TxAPCPoly}$$

The transmit antenna orientation is computed at t^* as shown below.

$$XVP(t^*) = XAxisPoly(t^*) \quad \text{XML: Antenna.Tx.XAxisPoly}$$

$$YVP(t^*) = YAxisPoly(t^*) \quad \text{XML: Antenna.Tx.YAxisPoly}$$

Unit vectors $\mathbf{uX}(t^*)$, $\mathbf{uY}(t^*)$ and $\mathbf{uZ}(t^*)$ are generated using the same normalization steps as shown above for the equivalent two-way antenna.

(2) Receive APC & Orientation

The receive APC position versus time polynomial is selected for the receive data channel processed. The Image Formation parameter block specifies which receive data channel is processed. The Radar Collection parameter block indicates which Receive APC is associated with the receive data channel. See Tables 3-6, 3-7 and 3-8.

The receive antenna APC and orientation are computed at t^* as shown below. Index idx is the receive APC index value for the processed receive data channel.

$$\mathbf{RCV}(t^*) = \text{Rcv_APC_Poly}(idx, t^*) \quad \text{XML: Position.RcvAPC.RcvAPCPoly}$$

$$\mathbf{XVP}(t^*) = \text{XAxisPoly}(t^*) \quad \text{XML: Antenna.Rcv.XAxisPoly}$$

$$\mathbf{YVP}(t^*) = \text{YAxisPoly}(t^*) \quad \text{XML: Antenna.Rcv.YAxisPoly}$$

Unit vectors $\mathbf{uX}(t^*)$, $\mathbf{uY}(t^*)$ and $\mathbf{uZ}(t^*)$ are generated using the same normalization steps as shown above for the equivalent two-way antenna.

The error statistics included with an image product are of three types: (1) composite error statistics, (2) component error statistics, and (3) unmodeled error statistics. The composite error statistics are the expected statistics for the combined effects of all components of geolocation error. The component error statistics are provided for the primary sources of geolocation errors in SAR image products. The unmodeled error statistics may be included to account for error sources that are not covered by the available component statistics.

4.12.1 Composite Error Statistics

The error sources that impact the accuracy of geo-location estimates from a SAR image are characterized by their contributions to the error in measured range and range rate. An error in range rate translates to an error in azimuth in the slant plane at COA. For a monostatic image, the composite effect of all error sources may be expressed as a range error and azimuth error. For a bistatic image, the composite effect of all error sources may be expressed as an average range error and an average range rate error. See SICD Volume 3, Section 12.

4.12.2 Component Error Statistics

The error statistics provided in an image product may be specified for the primary components that contribute to errors in geolocation estimates from a SAR image.

(1) Position & Velocity Covariance Matrix

For a monostatic image, the expected statistics for the sensor position error and velocity error are included. For a bistatic image, the expected statistics for the position error and velocity error for both sensors are included. For a given sensor, the error decorrelation versus time for position error may also be specified.

(2) Radar Sensor Errors

The primary error sources associated with the radar sensor are errors in the estimated sensor timing and frequency. For a monostatic image, errors in timing and frequency primarily result in a range bias. For a bistatic image, the errors in control clock timing and frequency on each sensor are sources of error in the estimated average range and average range rate.

(3) Propagation Model Errors

The error statistics associated with the predicted propagation delays due to the troposphere and the ionosphere may be included. The errors in estimated propagation delay contribute to an error in estimated range. Error decorrelation model parameters may also be included.

4.12.3 Unmodeled Error Statistics

The statistics for unmodeled error may be included for any product but are expected to be included for products that also include component error statistics. The unmodeled error is a two-dimensional error in image coordinates (xrow, ycol). The statistics of the error are assumed to be invariant across the image. The unmodeled error statistics may be included, for example, when the component error statistics do not accurately predict the statistics of empirically derived composite errors for a given class of image products. The method for

determining unmodeled error statistics and the image products for which they apply is specific to the sensors, imaging conditions, and data processing systems that form the image products.

4.12.4 Adjustable Parameter Offsets

For any product, the metadata provided may also include an optional set of parameters that may be used to correct errors in the computed projection parameters. The optional parameters are referred to as Adjustable Parameter Offsets (APOs). The APOs include corrections to the platform positions and velocities and parameters used to correct the computed ranges and range rates. The set of APOs is dependent upon the collect type for the image. For a monostatic image, the APO metadata consists of 8 parameters. For a bistatic image, the APO metadata consists of 16 parameters. See SICD Volume 3, Section 8, for guidance on how APOs are applied in image projections.

4.13 Matched Collection Parameters

The parameters that identify other imaging collections that are matched to the current collection are summarized below. The current imaging collection is the imaging collection from which the image contained in the product file was produced. Refer to Table 3-13. The use of these parameters is per program specific implementation and product design.

A pair of imaging collections may be “matched” based on many criteria. The criteria by which collections are matched is referred to as the Match Type. For example, a set of collections may be matched in that they are a time sequence collected from the same imaging geometry over a number of days or weeks in order to detect changes in the scene. A second set of collections may be matched in that they cover a set of adjacent scenes for which a large area composite image product may be produced.

The matched collection information provided is separated by Match Type. For each Match Type, a list of matched collections is provided. The number of match types and the match type identifiers are defined as part of the product design. The match type identifier, TypeID, is a text based parameter. For tasking and/or collection systems that assign each collection in the set an integer sequence index, the sequence index may also be provided. For the current collection, the sequence index is referred to as the Current Index. For a matched collection, the sequence index is referred to as the Match Index.

Additional parameters may be included with each matched collection that are relevant to the match criteria. For example, collections that are matched by collection geometry may include the parameters and their values that indicate the quality of the match. For collections matched for coherent image processing, the parameters included may be the differences in grazing angle and azimuth angle of the matched collection relative to the current collection. The choice of additional parameters is defined as part of the product design.

4.13.1 Matched Collection Example Structure

An example matched collection information structure is shown below. In the example shown, match collection lists are provided for two Match Types. For the first match type,TypeID = TIME_SEQUENCE, three match collections are listed for which the current collection is collection 2 of 4. Note that the number of matched collections does not include the current collection. For the second match type,TypeID = IMAGE_ANGLE, one matched collection is listed. For the matched collection, an additional parameter is provided with the name = “ANGLE” and value 10.0.

XML: MatchInfo

NumMatchTypes = 2

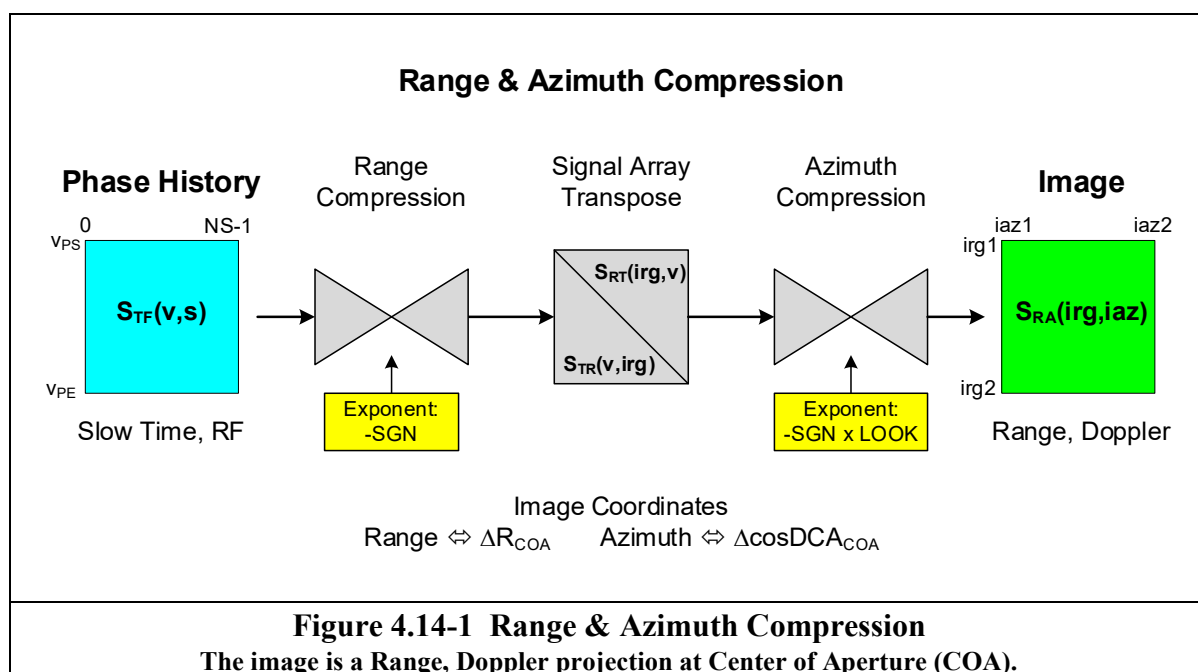
MatchType index = 1	⇔ Match Type 1
TypeID = “TIME_SEQUENCE”	⇔ TypeID 1
CurrentIndex = 2	⇔ Current Collection Index
NumMatchCollections = 3	
MatchCollection index = 1	⇔ Match 1
CoreName = “xxx”	⇔ Collection ID
MatchIndex = 1	⇔ Match Collection Index
MatchCollection index = 2	⇔ Match 2
CoreName = “xxx”	⇔ Collection ID
MatchIndex = 3	⇔ Match Collection Index
MatchCollection index = 3	⇔ Match 3
CoreName = “xxx”	⇔ Collection ID
MatchIndex = 4	⇔ Match Collection Index
MatchType index = 2	⇔ Match Type 2
TypeID = “IMAGE_ANGLE”	⇔ TypeID 2
NumMatchCollections = 1	
MatchCollection index = 1	⇔ Match 1
CoreName = “xxx”	⇔ Collection ID
Parameter name = “ANGLE” 10.0	⇔ Parameter ANGLE = 10.0

4.14 Range & Azimuth Compression Parameters

The parameters that describe the image formation processing for images formed with simple range and azimuth compression are described below. Refer to Table 3-14.

The range and azimuth compression algorithm is the simplest approach to transform SAR Phase History Data (PHD) into a complex image. The processing consists of two 1-dimensional Discrete Fourier Transforms (DFTs). See Figure 4.14-1. The resulting image is commonly referred to as a Range, Doppler image. This approach is only useful for coarse resolution and/or small image sizes. Simple compression is rarely (if ever) considered as a unique SAR image formation algorithm. Its limitations are what motivated the development of the other image formation algorithms. While simple compression is of limited practical utility, it is easily described in precise detail. The handful of parameters that are unique to the simple approach are provided for completeness.

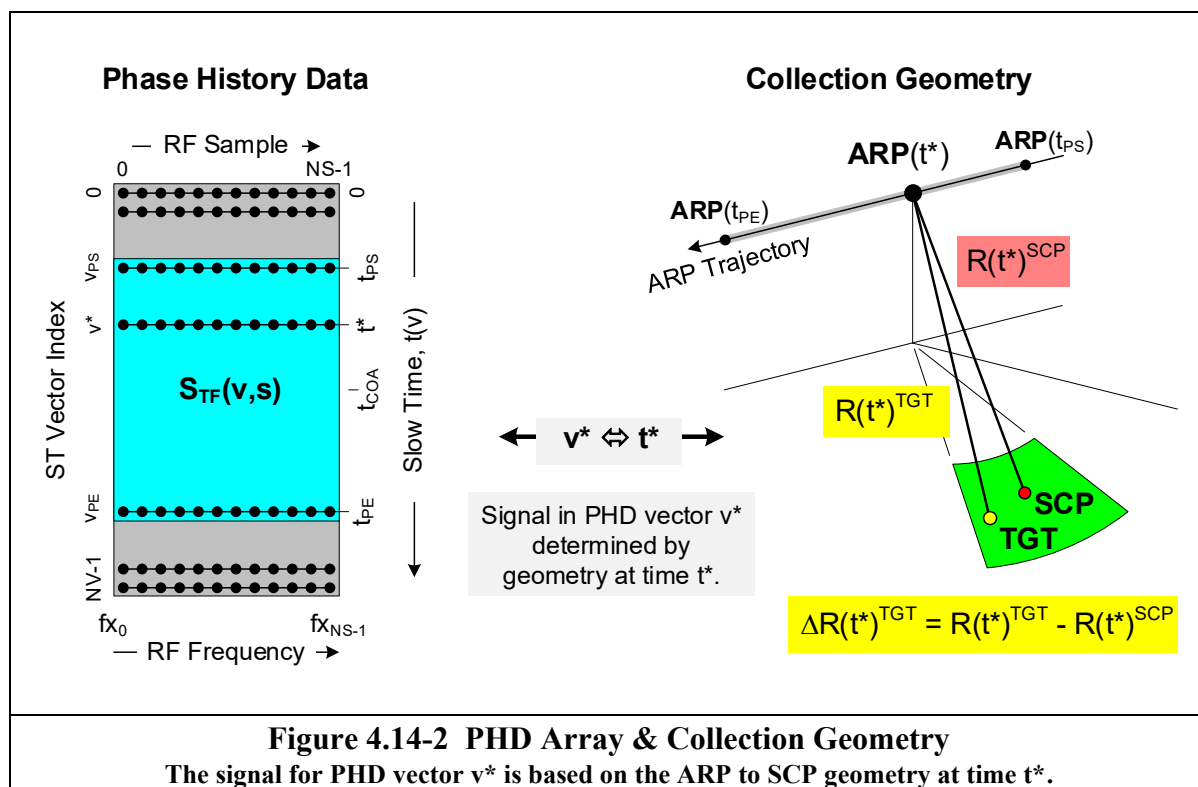
The discussion below covers two groups of parameters: SICD.RgAzComp parameters and a subset of the SICD.Grid parameters. The RgAzComp parameters relate image azimuth location to range rate at COA and slow time to the azimuth spatial frequency domain. The Grid parameters, provided for all images, are also described. For those less familiar with SAR image formation processing, this discussion may be helpful in understanding their use with the other image formation algorithms.



4.14.1 Input Phase History Data Array

An example input PHD array is shown in Figure 4.14-2. The PHD array, $S_{TF}(v,s)$, is composed of slow time vectors by frequency samples. A portion of the input vectors from $v = v_{PS}$ to v_{PE} are processed to form the image. Increasing vector index v corresponds to increasing slow time t . The set of RF samples from $s = 0$ to $NS - 1$ are processed to form the image. Increasing sample index s corresponds to increasing frequency value. For a given vector, the samples are the frequency aligned signals that represent the echoes from the scene that was imaged at reference time t .

Also shown in Figure 4.14-2 is the collection geometry at the time t^* associated with a single vector of the PHD array, v^* . The Aperture Reference Position (ARP) at time t^* is $ARP(t^*)$. The Scene Center Point (SCP) is a fixed ground point at the center of the imaged scene. The PHD has been motion compensated to the SCP. Also shown is an arbitrary fixed target (TGT) located in the imaged scene. Vector v^* is signal resulting from the received echoes of a single transmitted pulse. Time t^* is approximately the time the center of the transmitted pulse is reflected from the SCP.



The vector index v and slow time t , are assumed to be accurately related by a low order polynomial function. The mapping from vector index v to reference time t is given by the polynomial with coefficients denoted $cV2T(m)$ for $m = 0$ to M_V2T . The polynomial may be an exact representation or the result of a fit to a set of times that may include some slight jitter. Time t is in seconds relative to Collection Start time.

$$t(v) = \sum_{m=0}^{M_V2T} cV2T(m) \cdot v^m$$

For the set of vectors used to form the image, v_{PS} to v_{PE} , the Center of Aperture vector index, v_{COA} , is precisely in the center. For an even number of processed vectors, the value of v_{COA} will include a fractional portion of 0.5. The Center of Aperture time, t_{COA} , is the time computed for vector index $v = v_{COA}$. Times t_{PS} and t_{PE} are computed as well.

$$v_{COA} = \frac{1}{2} \cdot (v_{PS} + v_{PE})$$

$$t_{COA} = \sum_{m=0}^{M_V2T} cV2T(m) \cdot (v_{COA})^m$$

$$t_{PS} = \sum_{m=0}^{M_V2T} cV2T(m) \cdot (v_{PS})^m$$

$$t_{PE} = \sum_{m=0}^{M_V2T} cV2T(m) \cdot (v_{PE})^m$$

The vector index v is computed from slow time t using the polynomial shown below. The coefficients of the polynomial, denoted $cT2V(m)$ for $m = 1$ to M_T2V , are a series expansion about time t_{COA} . Parameter $ST_RATE_{COA} = cT2V(1)$ is the slow time sampling rate of the PHD in Hertz at time t_{COA} .

$$v(t) = v_{COA} + \sum_{m=1}^{M_T2V} cT2V(m) \cdot (t - t_{COA})^m$$

$$ST_RATE_{COA} = cT2V(1)$$

The samples of the PHD array are equally spaced samples of transmitted frequency, $fx(s)$. The frequency parameters associated with the PHD array are computed as shown below. Frequency parameters fx_0 and fx_SS are constant across vectors. The center index value, s_{COA} , and the center frequency value, fx_{COA} , are used to compute Center of Aperture parameters associated with the image. All frequency parameters are in Hertz.

$$fx(s) = fx_0 + fx_SS \cdot s$$

$$fx_BW = NS \cdot fx_SS$$

$$s_{COA} = \frac{1}{2} \cdot (NS - 1)$$

$$fx_{COA}(s) = fx_0 + fx_SS \cdot s_{COA}$$

The samples of the PHD array have been motion compensated to the SCP. The result of the motion compensation is to set the phase of the signal from the SCP to 0 for all samples of all vectors. For vector v^* , the phase of the signal for target TGT is a function of the geometry at time t^* . Ranges $R(t^*)^{TGT}$ and $R(t^*)^{SCP}$ are computed relative to the ARP at time t^* . Phase $\Phi(v^*, s)^{TGT}$ is in cycles.

$$\Phi(v^*, s) = SGN \cdot fx(s) \cdot \frac{2}{c} \cdot \Delta R(t^*)^{TGT}$$

$$\Delta R(t^*)^{TGT} = R(t^*)^{TGT} - R(t^*)^{SCP}$$

The phase sign parameter $SGN = +1$ or -1 is constant for all samples of the PHD array.

4.14.2 PHD Array To Image Array

The processing steps to compress the PHD signal array, $S_{TF}(v,s)$, to the image signal array, $S_{RA}(irg,iaz)$, are as follows. The size of the range compression DFT is NFRC. The sign of the exponent of the range compression DFT is set such that increasing index irg corresponds to increasing range. The size of the azimuth compression DFT is NFAC. The sign of the azimuth compression DFT is set to produce an image that is a view from above the earth. Optional amplitude weighting functions $w_{RG}(s)$ and $w_{AZ}(v)$ may be used to control image domain sidelobes. The DFT sizes are selected such that the image domain oversample ratios, Rg_OSR and Az_OSR , are between 1.0 and 2.0.

Range Compression For $irg = irg1, \dots, 0, \dots, irg2$:

$$S_{TR}(v, irg) = \sum_{s=0}^{NS-1} w_{RG}(s) \cdot S_{TF}(v, s) \cdot \exp \left\{ \underbrace{-SGN \cdot j2\pi \cdot \frac{(s-s_0) \cdot irg}{NFRC}}_{\text{Exponent: } -SGN} \right\}$$

$$Rg_OSR = \frac{NFRC}{NS} \quad 1.0 < Rg_OSR < 2.0$$

Signal Array Transpose

$$S_{RT}(irg, v) = S_{TR}(v, irg)$$

Azimuth Compression For $iaz = iaz1, \dots, 0, \dots, iaz2$:

$$S_{RA}(irg, iaz) = \sum_{v=v_{PS}}^{v_{PE}} w_{AZ}(v) \cdot S_{RT}(irg, v) \cdot \exp \left\{ \underbrace{-SGN \cdot LOOK \cdot j2\pi \cdot \frac{(v-v_0) \cdot iaz}{NFAC}}_{\text{Exponent: } -SGN \times LOOK} \right\}$$

$$Az_OSR = \frac{NFAC}{v_{PE} - v_{PS} + 1} \quad 1.0 < Az_OSR < 2.0$$

The resulting image signal array size is NumRows x NumCols.

$$NumRows = irg2 - irg1 + 1 \quad NumCols = iaz2 - iaz1 + 1$$

4.14.3 Center of Aperture Parameters

The imaging geometry parameters computed at time t_{COA} are used in the description below. The parameters to be computed are as follows. See Section 4.9 for precise equations.

SCP = SCP position in ECF coordinates.

ARP_{COA} = ARP position at time t_{COA} in ECF coordinates.

VARP_{COA} = ARP velocity at time t_{COA} in ECF coordinates.

VM_{COA} = Magnitude of the ARP velocity at time t_{COA} .

R_{COA}^{SCP} = Range to SCP at t_{COA} .

cosDCA_{COA}^{SCP} = Cosine of the Doppler Cone Angle (DCA) to the SCP at t_{COA} .

sinDCA_{COA}^{SCP} = Sine of the Doppler Cone Angle (DCA) to the SCP at t_{COA} .

LOOK = +1 or -1 where +1 ⇔ Left Looking and -1 ⇔ Right Looking

uLOS_{COA}^{SCP} = Line of sight unit vector from the **ARP**_{COA} to the **SCP**.

uSPN_{COA}^{SCP} = Unit vector normal to the SCP slant plane at t_{COA} .

The slant plane geometry at COA for two example collections, one left looking and one right looking, are shown in Figure 4.14-3. For the examples shown, the ARP trajectory is a straight line. Distance s is measured along the circle of constant range $R_{\text{COA}}^{\text{SCP}}$. Distance $s = 0$ at the SCP. The scale factor **AzSF** is the rate of change of the cosine of the DCA (**cosDCA**_{COA}) with respect to distance s at the SCP. The units of **AzSF** are meters⁻¹.

$$\text{AzSF} = \left. \frac{\partial}{\partial(s)} \left(\cos(\text{DCA}(s)_{\text{COA}}) \right) \right|_{s=0} \quad \text{AzSF} = -\text{LOOK} \cdot \frac{\sin(\text{DCA}_{\text{COA}}^{\text{SCP}})}{R_{\text{COA}}^{\text{SCP}}}$$

4.14.4 Range, Doppler Image

An example Range, Doppler image array is shown in Figure 4.14-4. The rows of the image are along contours of constant range relative to the ARP at COA. The rows are evenly spaced increments in range at COA. The columns of the image are along lines of constant $\Delta \cos \text{DCA}$ at COA. The columns are evenly spaced increments of $\Delta \cos \text{DCA}$ at COA which correspond to evenly spaced increments in Doppler frequency.

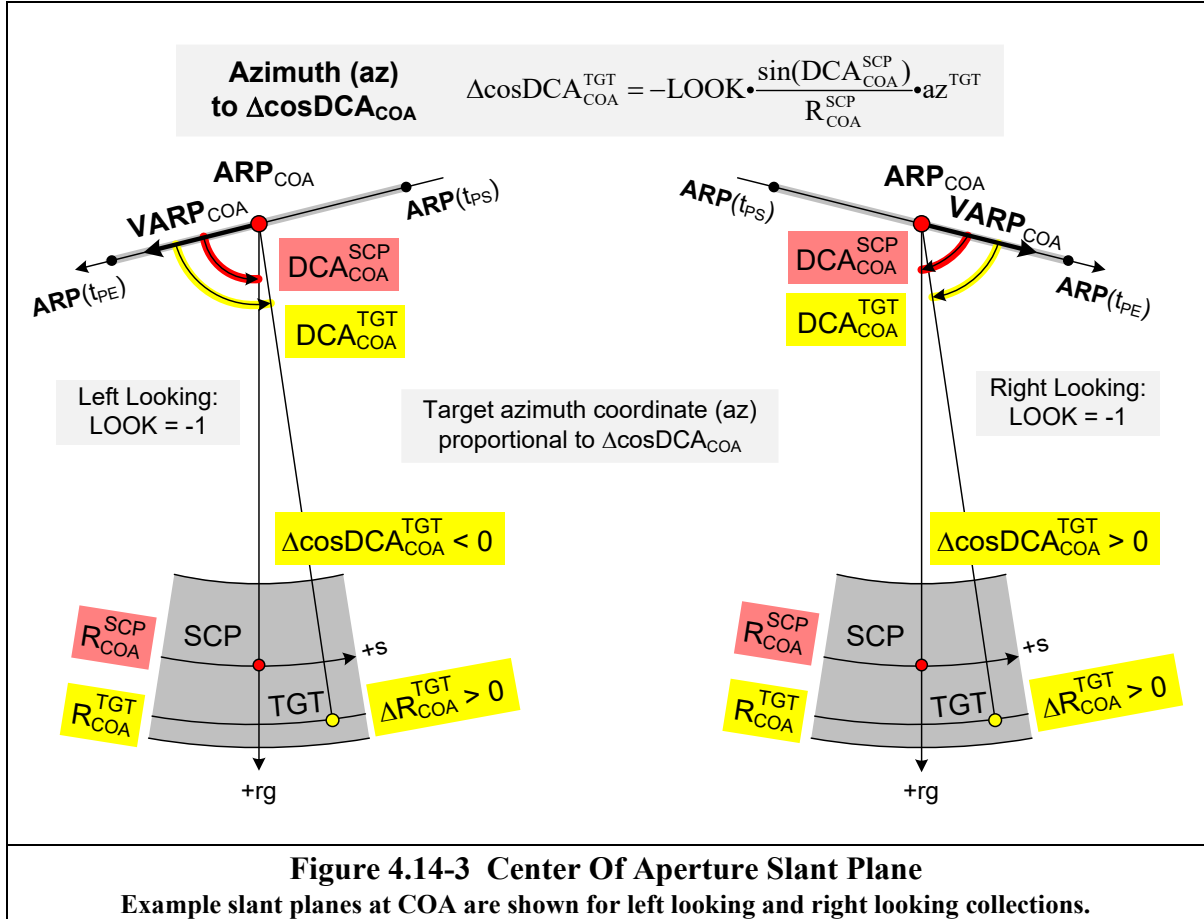
The image coordinates are range and azimuth (rg, az). The row coordinate is the range coordinate ($xrow = rg$) and the column coordinate is the azimuth coordinate ($ycol = az$). The range coordinate is range relative to the SCP at COA. The azimuth coordinate, expressed as a distance, is the difference in $\cos(\text{DCA})$ at COA relative to the SCP scaled by $1/\text{AzSF}$. For target TGT shown in Figure 4.14-3, the impulse response will be centered at image coordinates ($rg^{\text{TGT}}, az^{\text{TGT}}$). Image coordinates are computed as follows. For both example collections shown, the azimuth coordinate $az^{\text{TGT}} > 0$.

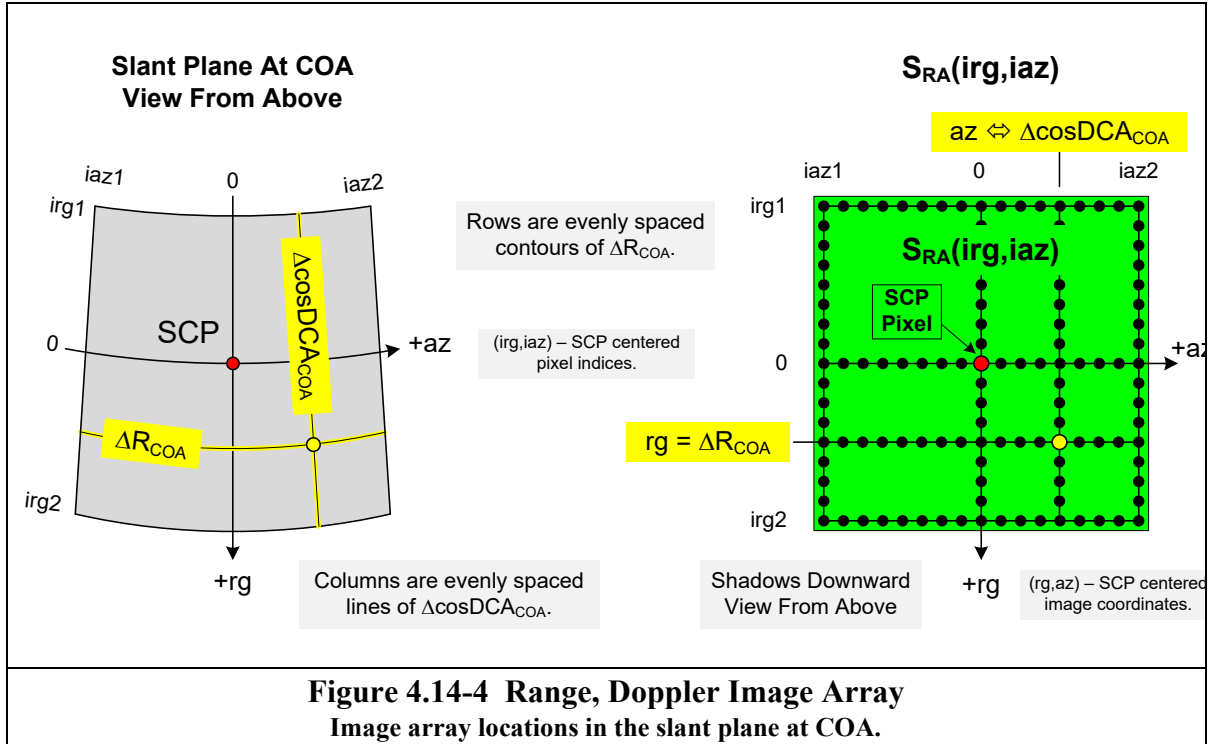
$$rg^{TGT} = \Delta R_{COA}^{TGT}$$

$$az^{TGT} = \frac{1}{AzSF} \cdot \Delta \cos DCA_{COA}^{TGT}$$

$$\Delta R_{COA}^{TGT} = R_{COA}^{TGT} - R_{COA}^{SCP}$$

$$\Delta \cos DCA_{COA}^{TGT} = \cos DCA_{COA}^{TGT} - \cos DCA_{COA}^{SCP}$$





4.14.5 RgAzComp Parameters

The parameters unique to simple range and azimuth compression are as follows. Scale factor AzSF (computed above) is the scale factor that scales the azimuth coordinate (in meters) to an increment in $\Delta \cos DCA$ at COA.

$$XML: RgAzComp.AzSF = AzSF$$

Scale factor $\Delta K_{az}/\Delta v$ is the increment in azimuth spatial frequency (K_{az}) per slow time vector. For left looking collections: $\Delta K_{az}/\Delta v > 0$. For right looking collections: $\Delta K_{az}/\Delta v < 0$.

$$\frac{\Delta K_{az}}{\Delta v} = LOOK \cdot \frac{2 \cdot f_{x_{COA}}}{c} \cdot \left(\frac{VM_{COA} \cdot \sin(DCA_{COA}^{SCP})}{R_{COA}^{SCP}} \right) \cdot \frac{1}{ST_RATE_{COA}}$$

For slow time vector v , the azimuth spatial frequency is: $K_{az}(v) = \frac{\Delta K_{az}}{\Delta v} \cdot (v - v_{COA})$.

The azimuth spatial frequency is computed from slow time t using the polynomial $K_{az}(t)$ with coefficients denoted $cT2K_{az}(m)$ for $m = 0$ to M_T2KAZ . The coefficients are computed by combining the polynomial for index v as a function t (coefficients $cT2V(m)$ and order M_T2V above) and the linear relationship for K_{az} as a function of index v . Order $M_T2KAZ = M_T2V$. For all images: $t_{COA} \Leftrightarrow v_{COA} \Leftrightarrow K_{az_{COA}} = 0$.

$$Kaz(t) = \sum_{m=0}^{M_T2KAZ} cT2Kaz(m) \cdot t^m \quad \text{XML: RgAzComp.KazPoly}$$

4.14.6 Image Grid Parameters

The image grid parameters for the image formed with simple range and azimuth compression are set as follows. See Table 3-4. Parameter formulas are provided with minimal description as they are common SAR processing parameters. The parameters are expressed in terms of the DFT expressions shown in Section 4.14.2 above.

The image formed is a slant plane image. The grid type is RGAZIM. The COA time for all pixels is t_{COA} . The COA time polynomial has only a constant coefficient.

XML: ImagePlane = SLANT

XML: Type = RGAZIM

XML: Grid.TimeCOAPoly - Constant coefficient set to t_{COA} . Order1 = Order2 = 0.

The image plane unit vectors **uRG** and **uAZ** lie in the SCP slant plane at COA.

$$\mathbf{uRG} = \mathbf{uLOS}_{COA}^{SCP}$$

XML: Row.UVectECF

$$\mathbf{uAZ} = \mathbf{uSPN}_{COA}^{SCP} \times \mathbf{uLOS}_{COA}^{SCP}$$

XML: Col.UVectECF

The phase sign parameters for row and column directions are set equal to the phase sign of the PHD array (i.e. parameter SGN in the equation below).

$$\Phi(v^*, s) = SGN \cdot f_x(s) \cdot \frac{2}{c} \cdot \Delta R(t^*)^{TGT}$$

XML: Row.Sgn & Col.Sgn

The parameters associated with the range coordinate (rg) and range spatial frequency (Krg) are computed from the parameters associated with the Range Compression DFT (NS, NFRC and s_0). Spatial frequency parameters (Krg_SS, Krg_COA, etc.) are in cycles/meter. Sample s_0 is the “zero” input sample index of the range compression DFT and will correspond to the center of the spatial frequency domain, Krg_Ctr. For parameter $s_0 = s_{COA}$, Krg_Ctr = Krg_COA and $\Delta Krg_COA = 0$. Optional polynomial parameter Row.DeltaKCOAPoly is omitted. For $s_0 \neq s_{COA}$, the optional polynomial is included with the constant coefficient equal to ΔKrg_COA and Order1 = Order2 = 0.

$$\begin{aligned} \text{Krg_COA} &= \frac{2}{c} \cdot \text{fx}_{\text{COA}} & \text{Krg_SS} &= \frac{2}{c} \cdot \text{fx_SS} \\ \text{Krg_IRBW} &= \text{NS} \cdot \text{Krg_SS} & \text{XML: Row.ImpRespBW} \\ \text{Krg_Ctr} &= \frac{2}{c} \cdot (\text{fx}_0 + \text{fx_SS} \cdot \text{s}_0) & \text{XML: Row.KCtr} \\ \Delta \text{Krg_COA} &= \text{Krg_COA} - \text{Krg_Ctr} & \text{XML: Row.DeltaKCOAPoly} \end{aligned}$$

Range spacing Rg_SS and impulse response width Rg_IRW are in meters. Impulse response broadening factor k_{RG} is associated with weighting function $w_{\text{RG}}(s)$. For uniform weighting, $k_{\text{RG}} = 0.886$.

$$\begin{aligned} \text{Rg_SS} &= \frac{1}{\text{NFRC} \cdot \text{Krg_SS}} & \text{XML: Row.SS} \\ \text{Rg_IRW} &= \frac{k_{\text{RG}}}{\text{Krg_IRBW}} & \text{XML: Row.ImpRespWid} \end{aligned}$$

Parameters ΔKrg1 and ΔKrg2 define support relative to Krg_Ctr of the range spatial frequency support of the image. Compute initial values of ΔKrg1 and ΔKrg2 as follows.

$$\begin{aligned} \Delta \text{Krg1_Init} &= \Delta \text{Krg_COA} - 0.5 \cdot \text{Krg_IRBW} \\ \Delta \text{Krg2_Init} &= \Delta \text{Krg_COA} + 0.5 \cdot \text{Krg_IRBW} \end{aligned}$$

The initial values are compared to the limits of $-0.5/\text{Rg_SS}$ and $+0.5/\text{Rg_SS}$ to detect a wrapped spectrum.

$$\text{IF } \Delta \text{Krg1_Init} < -\frac{0.5}{\text{Rg_SS}} \text{ OR } \Delta \text{Krg2_Init} > +\frac{0.5}{\text{Rg_SS}}, \Rightarrow \text{Krg Spectrum Wrapped.}$$

Set final values of ΔKrg1 and ΔKrg2 as follows.

$$\Delta \text{Krg1} = \begin{cases} \Delta \text{Krg1_Init}, & \text{Krg Spectrum Not Wrapped} \\ -\frac{0.5}{\text{Rg_SS}}, & \text{Krg Spectrum Wrapped} \end{cases} \quad \text{XML: Row.DeltaK1}$$

$$\Delta \text{Krg2} = \begin{cases} \Delta \text{Krg2_Init}, & \text{Krg Spectrum Not Wrapped} \\ +\frac{0.5}{\text{Rg_SS}}, & \text{Krg Spectrum Wrapped} \end{cases} \quad \text{XML: Row.DeltaK2}$$

The parameters associated with the azimuth coordinate (az) and azimuth spatial frequency (Kaz) are computed from the parameters associated with the Azimuth Compression DFT (v_{PS} , v_{PE} , NFAC and v_0). Spatial frequency parameters (Kaz_SS, Kaz_COA, etc.) are in cycles/meter. For all images, Kaz_COA = 0. Vector v_0 is the “zero” input vector index of the azimuth compression DFT and will correspond to the center of the spatial frequency domain, Kaz_Ctr. For parameter $v_0 = v_{COA}$, Kaz_Ctr = Kaz_COA = 0 and $\Delta\text{Kaz_COA} = 0$. Optional polynomial parameter Col.DeltaKCOAPoly is omitted. For $v_0 \neq v_{COA}$, the optional polynomial is included with the constant coefficient equal to $\Delta\text{Kaz_COA}$ and Order1 = Order2 = 0.

$$\text{Kaz_SS} = \text{LOOK}\left(\frac{\Delta\text{Kaz}}{\Delta v}\right)$$

$$\text{Kaz_IRBW} = (v_{PE} - v_{PS} + 1) \cdot \text{Kaz_SS} \quad \text{XML: Col.ImpRespBW}$$

$$\text{Kaz_Ctr} = \frac{\Delta\text{Kaz}}{\Delta v} \cdot (v_0 - v_{COA}) \quad \text{XML: Col.KCtr}$$

$$\Delta\text{Kaz_COA} = -\text{Kaz_Ctr} \quad \text{XML: Col.DeltaKCOAPoly}$$

Azimuth spacing Az_SS and impulse response width Az_IRW are in meters. Impulse response broadening factor k_{AZ} is associated with weighting function $w_{AZ}(s)$. For uniform weighting, $k_{AZ} = 0.886$.

$$\text{Az_SS} = \frac{1}{\text{NFAC} \cdot \text{Kaz_SS}} \quad \text{XML: Col.SS}$$

$$\text{Az_IRW} = \frac{k_{AZ}}{\text{Kaz_IRBW}} \quad \text{XML: Col.ImpRespWid}$$

Parameters ΔKaz1 and ΔKaz2 define support relative to Kaz_Ctr of the azimuth spatial frequency support of the image. Compute initial values of ΔKaz1 and ΔKaz2 as follows.

$$\Delta\text{Kaz1_Init} = \Delta\text{Kaz_COA} - 0.5 \cdot \text{Kaz_IRBW}$$

$$\Delta\text{Kaz2_Init} = \Delta\text{Kaz_COA} + 0.5 \cdot \text{Kaz_IRBW}$$

The initial values are compared to the limits of $-0.5/\text{Az_SS}$ and $+0.5/\text{Az_SS}$ to detect a wrapped spectrum.

$$\text{IF } \Delta\text{Kaz1_Init} < -\frac{0.5}{\text{Az_SS}} \text{ OR } \Delta\text{Kaz2_Init} > +\frac{0.5}{\text{Az_SS}}, \rightarrow \text{Kaz Spectrum Wrapped.}$$

Set final values of ΔK_{az1} and ΔK_{az2} as follows.

$$\Delta K_{az1} = \begin{cases} \Delta K_{az1_Init}, & \text{Kaz Spectrum Not Wrapped} \\ -\frac{0.5}{Az_SS}, & \text{Kaz Spectrum Wrapped} \end{cases} \quad \text{XML: Col.DeltaK1}$$

$$\Delta K_{az2} = \begin{cases} \Delta K_{az2_Init}, & \text{Kaz Spectrum Not Wrapped} \\ +\frac{0.5}{Az_SS}, & \text{Kaz Spectrum Wrapped} \end{cases} \quad \text{XML: Col.DeltaK2}$$

4.15 Polar Format Algorithm Parameters

The parameters that describe the image formation processing for images formed with the Polar Format Algorithm (PFA) are described below. Refer to Table 3-15.

The Polar Format Algorithm (PFA) is applicable to any SAR imaging collection. The algorithm was initially developed for fine resolution spotlight imaging but has been extended to handle all collections. The basic algorithm is well suited to imaging away from broadside and has been extended to handle accelerating platforms. The algorithm processes the motion compensated Phase History Data (PHD) as if the samples are located relative to a set of polar coordinates in the image spatial frequency domain. The key step in the PFA processing is a two-dimensional resampling of the PHD from polar coordinates to rectangular coordinates. A two-dimensional Discrete Fourier Transform then yields the complex image.

The Polar Format algorithm readily supports both Monostatic and Bistatic image formation processing. A common set of PFA-specific parameters are used to specify the collection geometry and processing applied to form the image. For monostatic collections, the parameters may be computed using the Aperture Reference Point position vs. time. For bistatic collections, the PFA parameters are computed using the transmit and receive APC positions vs. time. Refer to Table 3-6 and Section 4.6. In the description that follows, the computations are presented for both monostatic collections (using the ARP) and for bistatic collections (using two APC positions).

For an image formed with the Polar Format algorithm, the parameters that describe the image include both PFA-specific parameters as well as general image grid parameters used to describe all images. The discussion below addresses both sets of parameters. The PFA-specific parameters include the mapping of the PHD samples to polar aperture in the spatial frequency domain. Many of the image grid parameters, e.g., the image plane and sample grid orientation, are also determined as part of the PHD to spatial frequency mapping.

4.15.1 PFA Geometry Parameters

The PFA is based upon forming an image of the points that lie in a single geometric plane that contains the Scene Center Point (SCP). The geometric plane containing the points being imaged is referred to as the Focus Plane (FP). A typical choice for the Focus Plane is the best estimate of the ground plane for the scene being imaged. An example monostatic collection geometry is shown in Figure 4.15-1. Shown in the left portion of the figure is the ARP trajectory and the Focus Plane containing the SCP. The Focus Plane orientation is specified by the Focus Plane unit normal vector, denoted $\mathbf{uFPN}$. The $\mathbf{uFPN}$ always points away from the center of earth. The projection of the ARP position onto the FP is the ARP Focus Plane Nadir, denoted AFPN. The projection is normal to the FP. Shown in the right portion of Figure 4.15-1 is the two-dimensional Phase History Data array. The dimensions of the PHD are slow time vectors by RF samples with all vectors motion compensated to the SCP. The PHD is as described in Section 4.14-1. The portion of the PHD processed to form the image is from v_{PS} to v_{PE} corresponding to slow times t_{PS} and t_{PE} .

An image formation reference time, denoted t_{REF} , is used to establish the coordinate systems used in the PFA processing. Time t_{REF} is the time the polar angle is equal to zero and is referred to as the Polar Angle Reference time. The geometry at t_{REF} establishes the range direction in the image. The reference time is typically set equal to the SCP COA time but may be different. The AFPN at the reference time is used to establish the Focus Plane coordinate system (FPX, FPY, FPZ). The +FPZ direction is normal to the FP ($\mathbf{u}_{FPZ} = \mathbf{u}_{FPN}$). The +FPX direction is from the SCP to the AFPN at t_{REF} . The +FPY direction forms a right-handed coordinate system ($\mathbf{u}_{FPY} = \mathbf{u}_{FPZ} \times \mathbf{u}_{FPX}$).

$\mathbf{u}_{FPN}$ = Focus Plane Unit Normal

XML: PFA.FPN

t_{REF} = Polar Angle Reference Time

XML: PFA.PolarAngRefTime

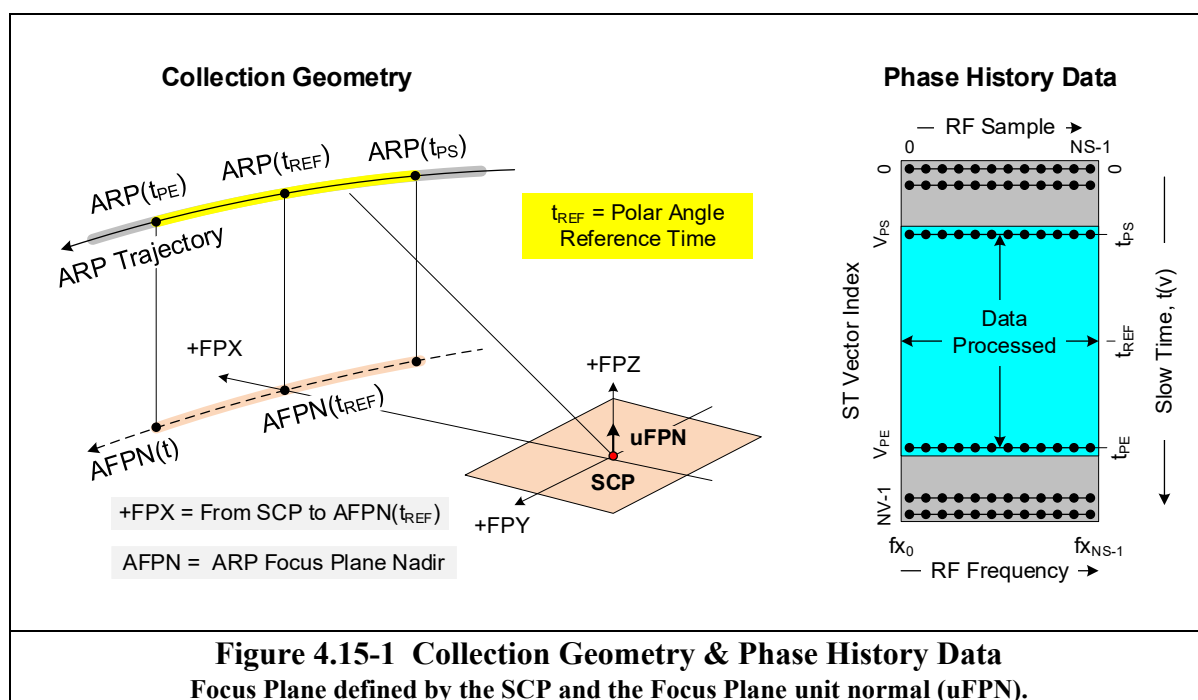
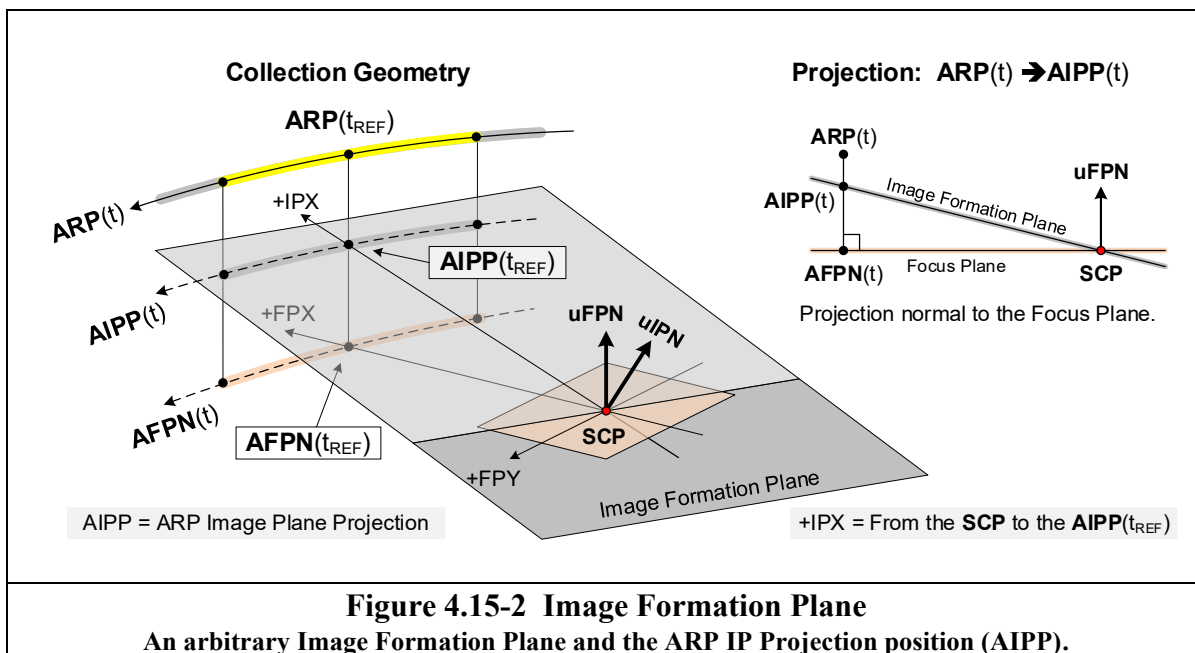


Figure 4.15-1 Collection Geometry & Phase History Data
 Focus Plane defined by the SCP and the Focus Plane unit normal ($\mathbf{u}_{FPN}$).

The PFA setup includes the selection of an Image Formation Plane or Image Plane (IP). The selection is independent of the selected Focus Plane. An arbitrary Image Formation Plane is shown in the Figure 4.15-2. The IP contains the SCP and is specified by the Image Plane unit normal, denoted $\mathbf{u}_{IPN}$. The $\mathbf{u}_{IPN}$ always points away from center of the earth. The projection of the ARP position onto the IP is the ARP Image Plane Projection, denoted AIPP. The projection is normal to the FP. The AIPP at the reference time is used to establish the Image Plane coordinate system (IPX, IPY, IPZ). The +IPZ direction is normal to the IP ($\mathbf{u}_{IPZ} = \mathbf{u}_{IPN}$). The +IPX direction is from the SCP to the AIPP at t_{REF} . The +IPY direction forms a right-handed coordinate system ($\mathbf{u}_{IPY} = \mathbf{u}_{IPZ} \times \mathbf{u}_{IPX}$). The SICD PFA parameters allow for the Image Formation Plane to be at any orientation (including the ground plane or the focus plane). A common choice for the IP is the instantaneous SAR slant plane containing the SCP at time t_{REF} . See Figure 4.15-3.

$\mathbf{u}_{IPN}$ = Image Formation Plane Unit Normal

XML: PFA.IPN



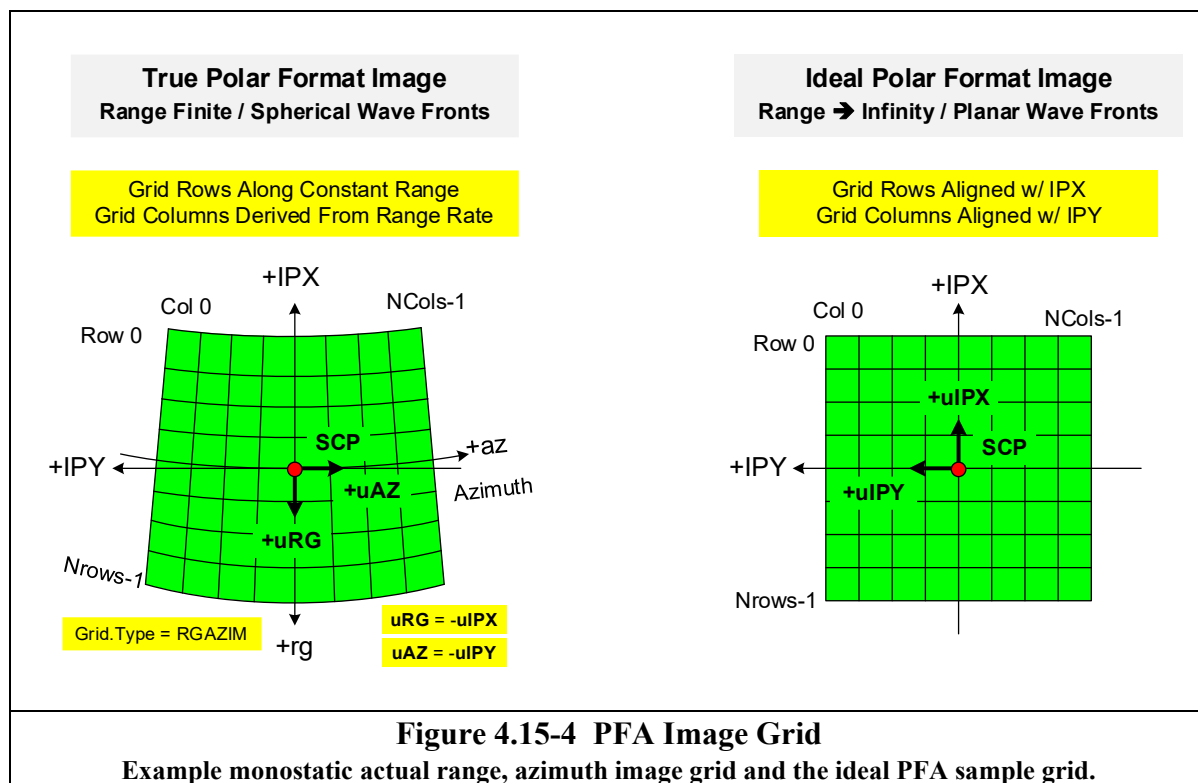
at t_{REF} goes to infinity and the radar wave fronts becomes planar. The rows and columns become an orthogonal grid. The image grid unit vectors, denoted $\mathbf{uRG}$ and $\mathbf{uAZ}$, are set as follows.

$$\mathbf{uRG} = -\mathbf{uIPX}$$

XML: Grid.Row.UVectECF

$$\mathbf{uAZ} = -\mathbf{uIPY}$$

XML: Grid.Col.UVectECF



Bistatic PFA Geometry

The method for computing the PFA geometry parameters for a bistatic collection is provided. The parameters are computed using the APC positions instead of the ARP position. The transmit APC position vs. time is denoted $\mathbf{Xmt}(t)$. The receive APC position vs. time is denoted $\mathbf{Rcv}(t)$. The method provided may also be used for a monostatic collection. For a monostatic collection, the results using the APC positions will be equal (or nearly equal) to those computed using the ARP position. Refer to Section 4.6.

For parameters that vary with slow time t , the following convention is used. For a signal incident at the SCP at time t , it was transmitted at time t_x and received at time t_r where $t_x < t < t_r$. For a geometry-dependent parameter, the value at time t is computed using the transmit APC position at time t_x and the receive position at time t_r . For the reference geometry parameters computed at slow time $t = t_{REF}$, the transmit APC position is computed at time $t = t_{xREF}$, and the receive APC position is computed at time $t = t_{rREF}$. Times t_{xREF} and t_{rREF} are estimated as follows.

$$t_{XREF} = t_{REF} - \frac{1}{c} \bullet |Xmt(t_{REF}) - SCP| \quad t_{RREF} = t_{REF} + \frac{1}{c} \bullet |Rcv(t_{REF}) - SCP|$$

The transmit APC position and velocity are computed at time t_{XREF} . Other parameters computed at time t_{XREF} include range R_Xmt_{REF} , range rate $Rdot_Xmt_{REF}$, unit vector $uXmt_{REF}$, and unit vector derivative with respect to time, $uXmtDot_{REF}$. Refer to Figure 4.15-5.

$$Xmt_{REF} = Xmt(t_{XREF})$$

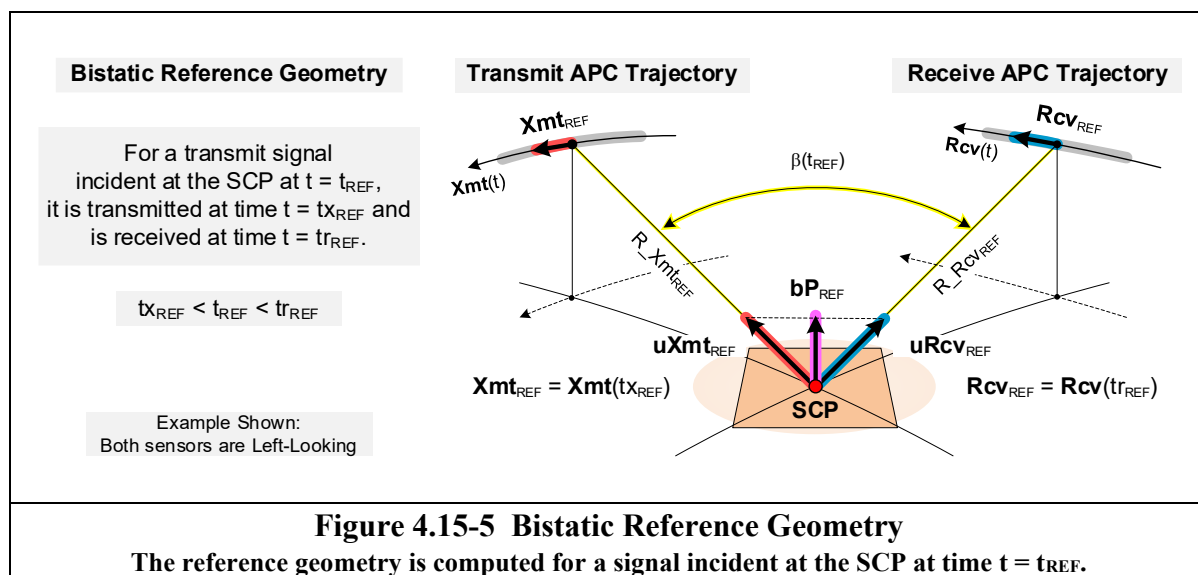
$$VXmt_{REF} = VXmt(t_{XREF})$$

$$R_Xmt_{REF} = |Xmt_{REF} - SCP|$$

$$uXmt_{REF} = \frac{1}{R_Xmt_{REF}} \bullet (Xmt_{REF} - SCP)$$

$$Rdot_Xmt_{REF} = VXmt_{REF} \bullet uXmt_{REF}$$

$$uXmtDot_{REF} = \frac{1}{R_Xmt_{REF}} \bullet (VXmt_{REF} - Rdot_Xmt_{REF} \bullet uXmt_{REF})$$



The receive APC position and velocity are computed at time t_{RREF} . Other parameters computed at time t_{RREF} include range R_Rcv_{REF} , range rate $Rdot_Rcv_{REF}$, unit vector $uRcv_{REF}$, and unit vector derivative with respect to time, $uRcvDot_{REF}$.

$$Rcv_{REF} = Rcv(t_{RREF})$$

$$VRcv_{REF} = VRcv(t_{RREF})$$

$$R_Rcv_{REF} = |Rcv_{REF} - SCP|$$

$$uRcv_{REF} = \frac{1}{R_Rcv_{REF}} \bullet (Rcv_{REF} - SCP)$$

$$Rdot_Rcv_{REF} = VRcv_{REF} \bullet uRcv_{REF}$$

$$uRcvDot_{REF} = \frac{1}{R_Rcv_{REF}} \bullet (VRcv_{REF} - Rdot_Rcv_{REF} \bullet uRcv_{REF})$$

The PFA-specific parameters are computed from the bistatic pointing vector, denoted $\mathbf{bP}(t)$. At time t , the pointing vector is computed from unit vectors $\mathbf{uXmt}(tx)$ and $\mathbf{uRcv}(tr)$. Unit vector $\mathbf{uXmt}(tx)$ is the unit vector that points from the SCP toward the transmit APC at time tx . Unit vector $\mathbf{uRcv}(tr)$ points from the SCP toward the receive APC at time tr . At the reference time, compute the bistatic pointing vector, $\mathbf{bP}_{REF} = \mathbf{bP}(t_{REF})$, and its derivative with respect to time, $\mathbf{bPDot}_{REF}$, as follows. Refer to Figure 4.15-6.

$$\mathbf{bP}_{REF} = \mathbf{bP}(t_{REF}) = \frac{1}{2} \cdot (\mathbf{uXmt}_{REF} + \mathbf{uRcv}_{REF})$$

$$\mathbf{bPDot}_{REF} = \frac{d}{dt} \mathbf{bP}(t_{REF}) = \frac{1}{2} \cdot (\mathbf{uXmtDot}_{REF} + \mathbf{uRcvDot}_{REF})$$

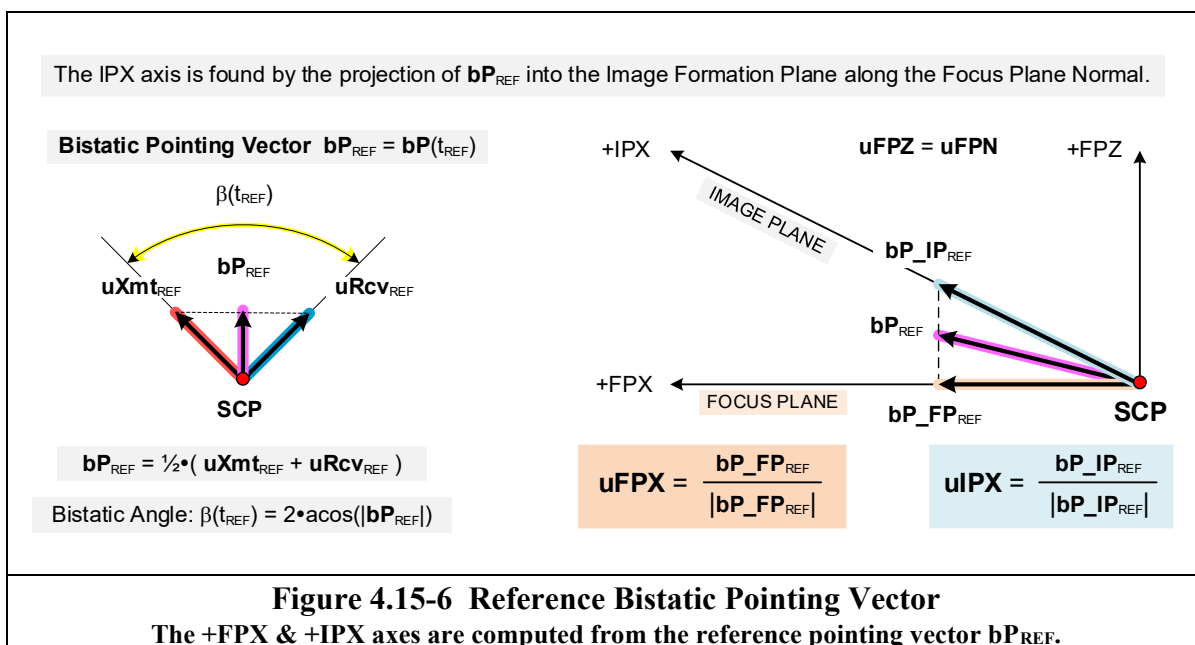
The Focus Plane +FPX axis is oriented by projecting the pointing vector $\mathbf{bP}_{REF}$ into the Focus Plane along the focus plane normal. Compute unit vectors $\mathbf{uFPX}$ and $\mathbf{uFPY}$ as follows.

$$\mathbf{uFPZ} = \mathbf{uFPN}$$

$$\mathbf{bP_FP}_{REF} = \mathbf{bP}_{REF} - (\mathbf{bP}_{REF} \bullet \mathbf{uFPZ}) \cdot \mathbf{uFPZ}$$

$$\mathbf{uFPX} = \frac{1}{|\mathbf{bP_FP}_{REF}|} \cdot \mathbf{bP_FP}_{REF}$$

$$\mathbf{uFPY} = \mathbf{uFPZ} \times \mathbf{uFPX}$$



The Image Plane +IPX axis is oriented by projecting vector $\mathbf{bP}_{REF}$ into the Image Formation Plane along the focus plane normal. Compute unit vectors $\mathbf{uIPX}$ and $\mathbf{uIPY}$ as follows.

$$\mathbf{uIPZ} = \mathbf{uIPN}$$

$$\mathbf{bP_IP}_{REF} = \mathbf{bP}_{REF} - \left(\frac{\mathbf{bP}_{REF} \bullet \mathbf{uIPZ}}{\mathbf{uFPZ} \bullet \mathbf{uIPZ}} \right) \cdot \mathbf{uFPZ}$$

$$\mathbf{uIPX} = \frac{1}{|\mathbf{bP_IP}_{REF}|} \cdot \mathbf{bP_IP}_{REF}$$

$$\mathbf{uIPY} = \mathbf{uIPZ} \times \mathbf{uIPX}$$

The resulting PFA image is Grid Type = RGAZIM. See Section 4.4.4. The rows and columns of the image nominally form a rectangular grid in the Image Plane aligned with the IPX and IPY axes. The image grid unit vectors, denoted **uRG** and **uAZ**, are set as follows.

$$\mathbf{uRG} = -\mathbf{uIPX}$$

XML: Grid.Row.UVectECF

$$\mathbf{uAZ} = -\mathbf{uIPY}$$

XML: Grid.Col.UVectECF

The instantaneous slant plane is computed using vector **bP(t)** and its derivative **bPDot(t)**. At the reference time, the slant plane unit normal, $\mathbf{uSPN}_{\text{REF}} = \mathbf{uSPN}(t_{\text{REF}})$, is computed as follows. The LOOK parameter that indicates the effective “side of track” of the bistatic geometry and is used to orient the SPN to oriented in the +FPZ direction.

$$\text{LOOK} = \begin{cases} +1, & \text{if } \mathbf{bPDot}_{\text{REF}} \bullet \mathbf{uFPY} > 0 \\ -1, & \text{otherwise} \end{cases}$$

$$\mathbf{SPN}_{\text{REF}} = \text{LOOK} \bullet (\mathbf{bP}_{\text{REF}} \times \mathbf{bPDot}_{\text{REF}}) \quad \mathbf{uSPN}_{\text{REF}} = \frac{1}{|\mathbf{SPN}_{\text{REF}}|} \bullet \mathbf{SPN}_{\text{REF}}$$

The orientation of the Image Formation Plane is qualitatively identified with the parameter Grid.ImagePlane. The allowed values are GROUND, SLANT, and OTHER. See Table 3-4. The criteria used for assigning the Image Plane identifier is a product design choice. A suggested criteria that may be used or modified is as follows.

If $\mathbf{uIPN} \bullet \mathbf{uSPN}_{\text{REF}} > 0.999$: IPN nearly aligned with reference SPN

ImagePlane = “SLANT”

Else If $\mathbf{uIPN} \bullet \mathbf{uFPN} > 0.999$: IPN nearly aligned with FPN

ImagePlane = “GROUND”

Else: The Image Plane is some other plane

ImagePlane = “OTHER”

4.15.2 PHD Mapping to Spatial Frequency Coordinates

The mapping of the Phase History Data coordinates to spatial frequency coordinates is what gives the polar format algorithm its name. The input PHD coordinates slow time and RF frequency, denoted (t, fx), are first mapped to polar coordinates $K_{\text{ap}}(t, f_x)$ and $\theta_{\text{PLR}}(t)$. Polar coordinates ($K_{\text{ap}}, \theta_{\text{PLR}}$) are mapped to rectangular coordinates ($K_{\text{rg}}, K_{\text{az}}$) via the standard polar to rectangular conversion. The rectangular coordinates ($K_{\text{rg}}, K_{\text{az}}$) are the spatial frequency coordinates of the image. The ARP position at time t determines the angle $\theta_{\text{PLR}}(t)$ and scale factor $\text{KSF}(\theta_{\text{PLR}}(t))$.

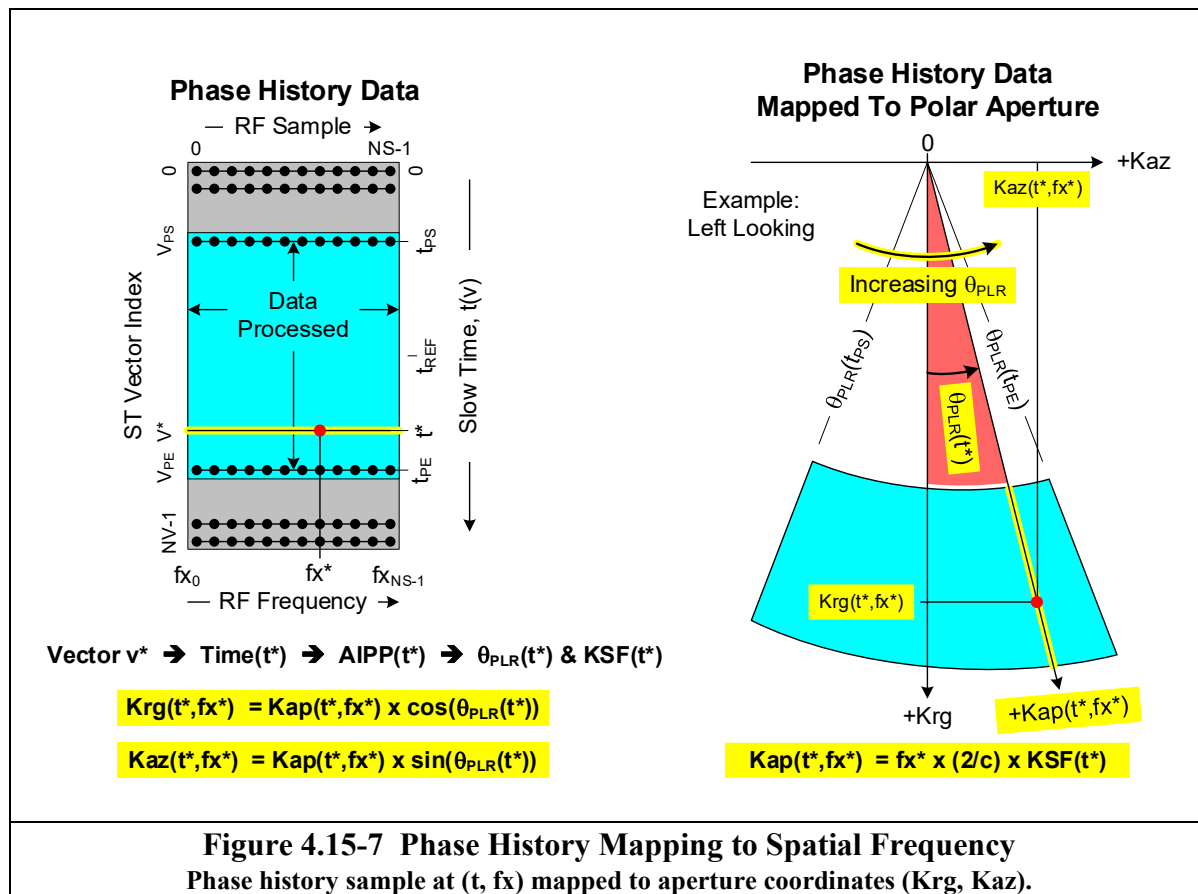
$$t \rightarrow \text{ARP}(t) \rightarrow \text{AIPP}(t) \rightarrow \theta_{\text{PLR}}(t) \text{ \& \& KSF}(\theta_{\text{PLR}}(t))$$

$$K_{\text{ap}}(t, f_x) = f_x \bullet (2/c) \bullet \text{KSF}(\theta_{\text{PLR}}(t))$$

$$K_{\text{rg}}(t, f_x) = K_{\text{ap}}(t, f_x) \bullet \cos(\theta_{\text{PLR}}(t))$$

$$K_{\text{az}}(t, f_x) = K_{\text{ap}}(t, f_x) \bullet \sin(\theta_{\text{PLR}}(t))$$

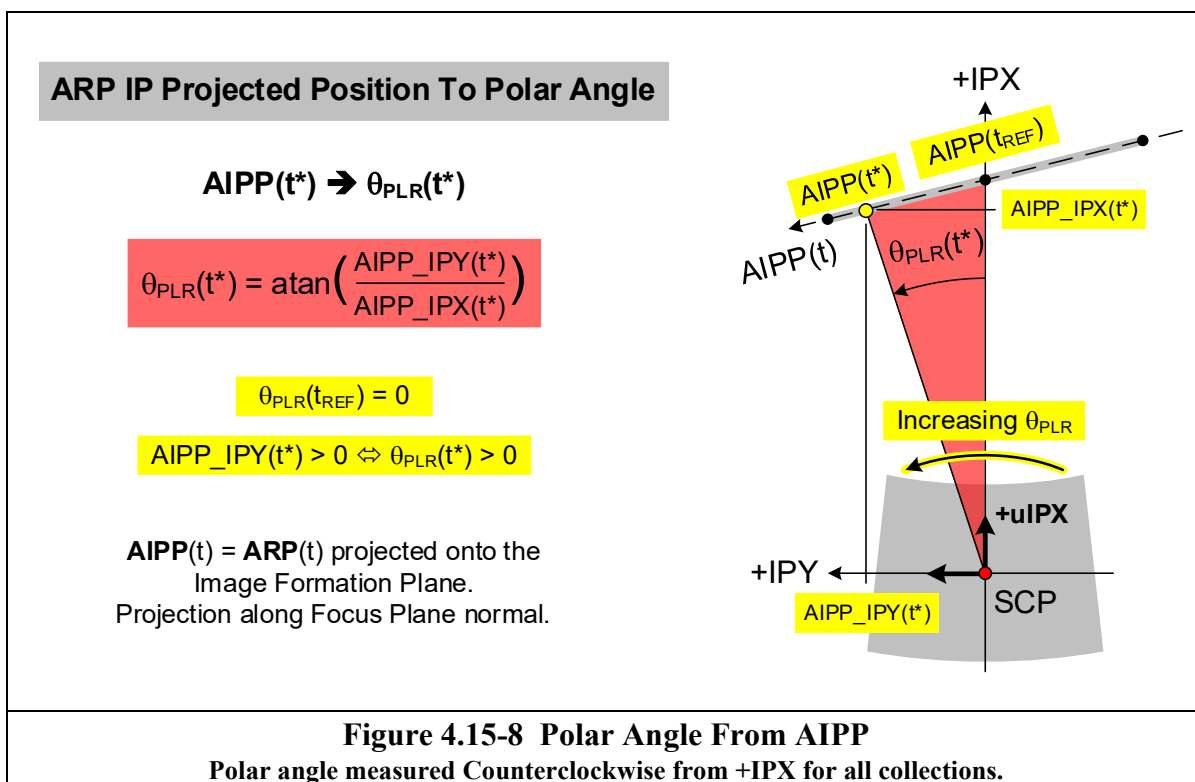
The mapping is shown graphically in Figure 4.15-7. For a given PHD data vector v^* collected with associated slow time t^* , the mapping of PHD coordinates (t^*, fx^*) to spatial frequency coordinates (Krg^*, Kaz^*) is shown.



The polar angle $\theta_{PLR}(t)$ is computed from the AIPP position at time t expressed in IP coordinates. The definition of the polar angle is shown below and graphically in Figure 4.15-8. By definition, $\theta_{PLR}(t_{REF}) = 0$. Figure 4.15-9 compares the polar angle versus time for left-looking and right-looking collection geometries.

$$\mathbf{AIPP}_{IPC}(t^*) = \begin{bmatrix} \text{AIPP_IPX}(t^*) \\ \text{AIPP_IPY}(t^*) \\ 0 \end{bmatrix} \quad \theta_{PLR}(t^*) = \text{atan} \left\{ \frac{\text{AIPP_IPY}(t^*)}{\text{AIPP_IPX}(t^*)} \right\}$$

Note: Component $\text{AIPP_IPX}(t^*) > 0$ for the period $t_{PS} \leq t^* \leq t_{PE}$.



Useful for image exploitation is having the polar angle expressed as a function of time. The polar angle polynomial, coefficients denoted $cPA(m)$, provides a convenient way for computing polar angle and polar angle rate. The scale factor KSF used to scale frequency value (f_x in Hz) to radial spatial frequency value (Kap in cycles/meter), is also used in image exploitation. The common method for computing the scale factor at a given time or polar angle is shown in Figure 4.15-10. In general, the scale factor varies as a function of polar angle. A convenient form is to have the scale factor as a function of polar angle, $KSF(\theta_{PLR})$. The spatial frequency scale factor polynomial, coefficients denoted $cKSF(m)$, provides KSF as function of polar angle. The value of the scale factor, $KSF(\theta_{PLR})$, is computed from the polynomial.

$$\text{XML: PFA.PolarAngPoly}(m) = cPA(m) \quad m = 0 \text{ to } M_PA$$

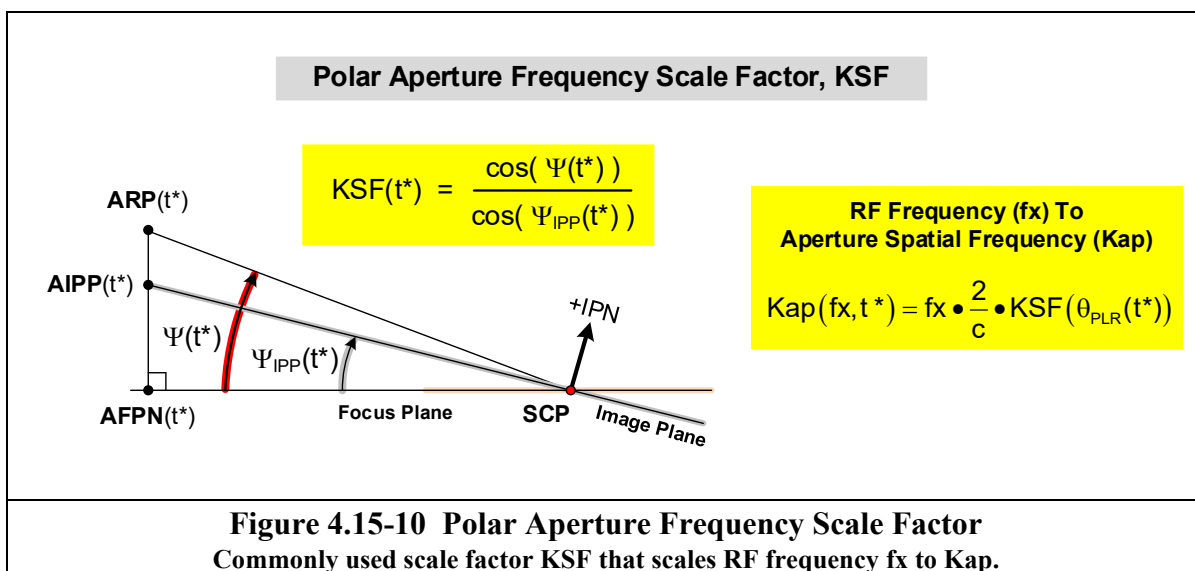
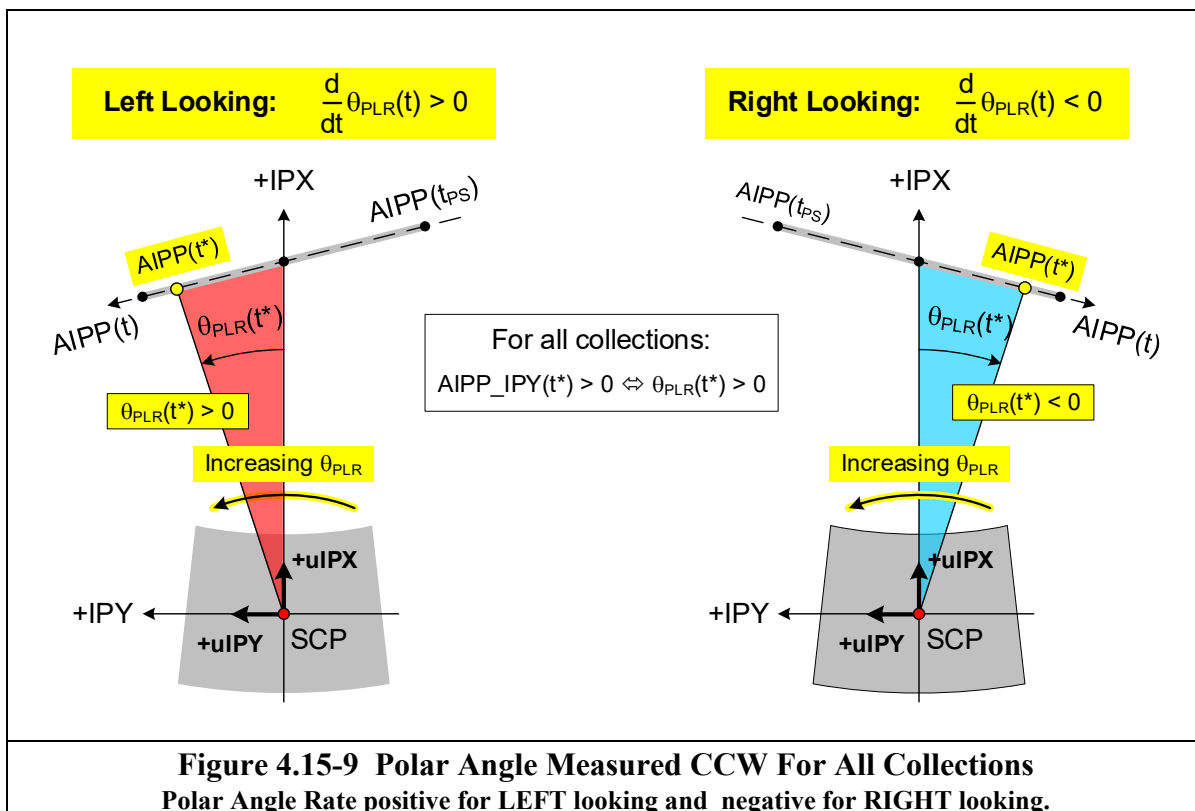
$$\text{XML: PFA.SpatialFreqSFPoly}(m) = cKSF(m) \quad m = 0 \text{ to } M_KSF$$

$$\theta_{PLR}(t) = \sum_{m=0}^{M_PA} cPA(m) \cdot t^m$$

$$d\theta_{PLR}(t)/dt = \sum_{m=1}^{M_PA} m \cdot cPA(m) \cdot t^{m-1}$$

$$KSF(\theta_{PLR}) = \sum_{m=0}^{M_KSF} cKSF(m) \cdot (\theta_{PLR})^m$$

$$\frac{d}{d\theta} KSF(\theta_{PLR}) = \sum_{m=1}^{M_KSF} m \cdot cKSF(m) \cdot (\theta_{PLR})^{m-1}$$



Bistatic PHD Mapping to Spatial Frequency Coordinates

The method to compute the mapping from PHD coordinates to image spatial frequency coordinates for a bistatic collection is specified. The key parameters to be computed are the polar angle, $\theta_{PLR}(t)$, and the polar aperture scale factor, $KSF(\theta_{PLR}(t))$. The parameters are computed using the bistatic pointing vector, denoted $\mathbf{bP}(t)$, instead of the ARP position. At time t , the bistatic pointing vector is computed based on the transmit APC position at time t_x and the receive APC position at time t_r where $t_r < t < t_x$. The method provided may also be used for a monostatic collection. For a monostatic collection, the results using the APC positions will be equal (or nearly equal) to those computed using the ARP position.

$$t \rightarrow t_x \ \& \ t_r \rightarrow \mathbf{Xmt}(t_x) \ \& \ \mathbf{Rcv}(t_r) \rightarrow \mathbf{uXmt}(t_x) \ \& \ \mathbf{uRcv}(t_r) \rightarrow \mathbf{bP}(t)$$

$$\mathbf{bP}(t) \rightarrow \mathbf{bP_IP}(t) \rightarrow \theta_{PLR}(t) \ \& \ KSF(\theta_{PLR}(t))$$

At time t^* , the method for computing the polar angle, $\theta_{PLR}(t^*)$, and the polar aperture scale factor, $KSF(\theta_{PLR}(t^*))$, is as follows. Compute the time t_x^* and t_r^* and unit vectors $\mathbf{uXmt}(t_x^*)$ and $\mathbf{uRcv}(t_r^*)$. All times are in seconds from CST and all positions in ECEF coordinates.

$$\begin{aligned} t_x^* &= t^* - \frac{1}{c} \cdot |\mathbf{Xmt}(t^*) - \mathbf{SCP}| & t_r^* &= t^* + \frac{1}{c} \cdot |\mathbf{Rcv}(t^*) - \mathbf{SCP}| \\ R_Xmt(t_x^*) &= |\mathbf{Xmt}(t_x^*) - \mathbf{SCP}| & \mathbf{uXmt}(t_x^*) &= \frac{1}{R_Xmt(t_x^*)} \cdot (\mathbf{Xmt}(t_x^*) - \mathbf{SCP}) \\ R_Rcv(t_r^*) &= |\mathbf{Rcv}(t_r^*) - \mathbf{SCP}| & \mathbf{uRcv}(t_r^*) &= \frac{1}{R_Rcv(t_r^*)} \cdot (\mathbf{Rcv}(t_r^*) - \mathbf{SCP}) \end{aligned}$$

Compute the bistatic pointing vector at time t^* , $\mathbf{bP}(t^*)$. Compute the projection of the $\mathbf{bP}(t^*)$ into the Image Formation Plane, $\mathbf{bP_IP}(t^*)$. The projection is along the normal to the Focus Plane. Refer to Figure 4.15-11.

$$\mathbf{bP}(t^*) = \frac{1}{2} \cdot (\mathbf{uXmt}(t_x^*) + \mathbf{uRcv}(t_r^*))$$

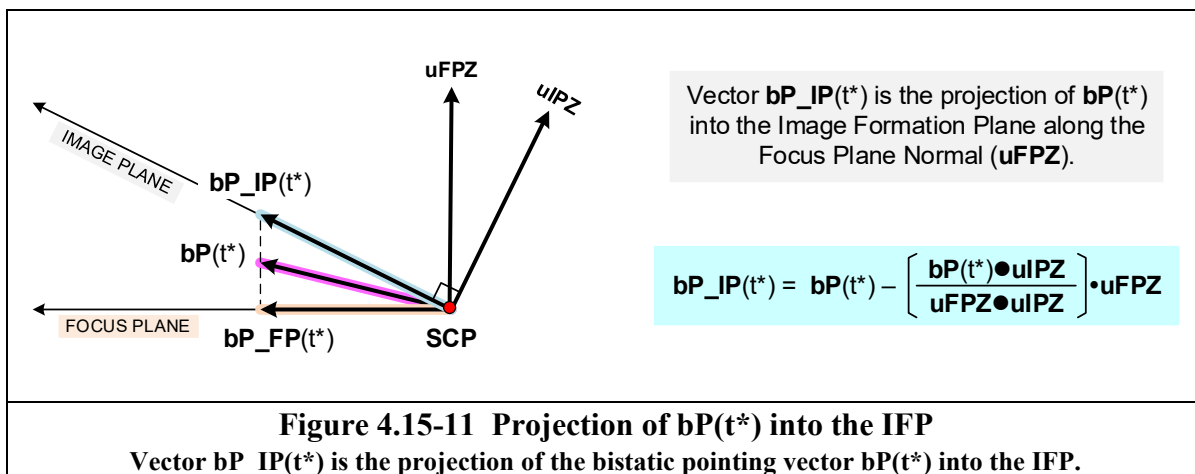
$$\mathbf{bP_IP}(t^*) = \mathbf{bP}(t^*) - \left(\frac{\mathbf{bP}(t^*) \bullet \mathbf{uIPZ}}{\mathbf{uFPZ} \bullet \mathbf{uIPZ}} \right) \cdot \mathbf{uFPZ}$$

Compute the components of the $\mathbf{bP_IP}(t^*)$ in Image Plane Coordinates, $\mathbf{bP_IP_IPX}(t^*)$, and $\mathbf{bP_IP_IPY}(t^*)$.

$$\mathbf{bP_IP_IPC}(t^*) = \begin{bmatrix} \mathbf{bP_IP_IPX}(t^*) \\ \mathbf{bP_IP_IPY}(t^*) \\ 0 \end{bmatrix}$$

$$\mathbf{bP_IP_IPX}(t^*) = \mathbf{bP_IP}(t^*) \bullet \mathbf{uIPX} \quad \mathbf{bP_IP_IPY}(t^*) = \mathbf{bP_IP}(t^*) \bullet \mathbf{uIPY}$$

Note: Component $\mathbf{bP_IP_IPX}(t^*) > 0$ for the period $t_{PS} \leq t^* \leq t_{PE}$.



Use the components of $\mathbf{bP_IP}_{IFC}(t^*)$ to compute the polar angle, $\theta_{PLR}(t^*)$, and the aperture scale factor, $KSF(\theta_{PLR}(t^*))$. Refer to Figure 4.15-12.

$$\theta_{PLR}(t^*) = \text{atan} \left\{ \frac{AIPP_IPY(t^*)}{AIPP_IPX(t^*)} \right\}$$

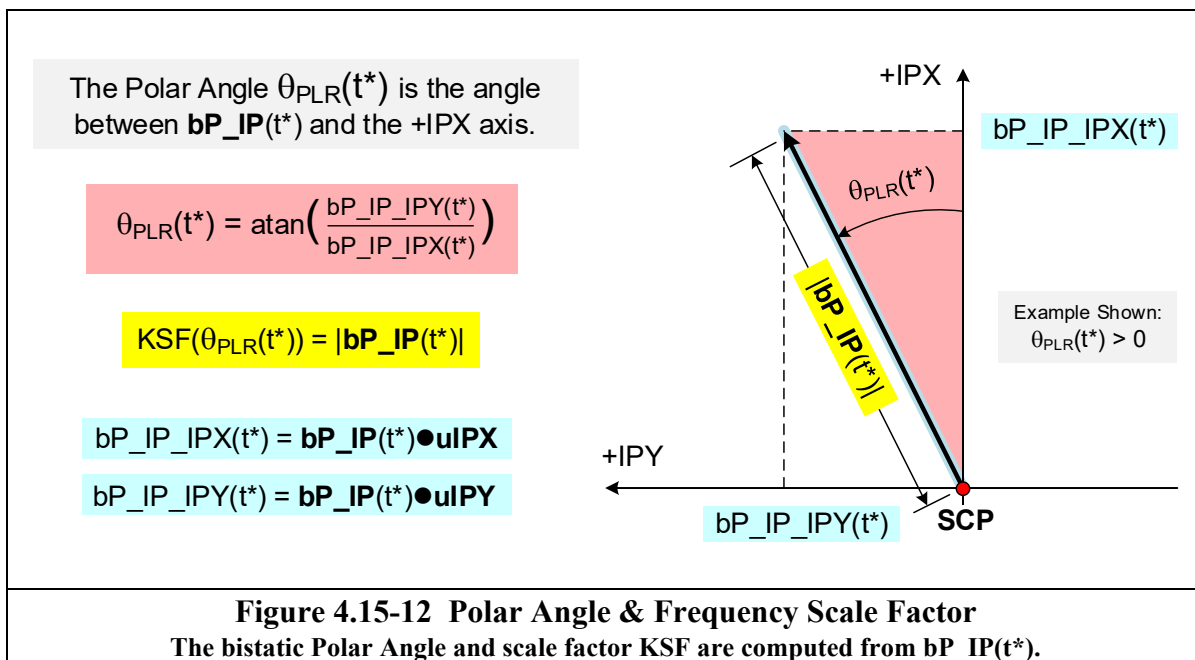
$$KSF(\theta_{PLR}(t^*)) = |\mathbf{bP_IP}(t^*)|$$

The mapping from (t, f_x) to (Krg, Kaz) is same as shown in Figure 4.15-7.

$$K_{ap}(t, f_x) = f_x \cdot (2/c) \cdot KSF(\theta_{PLR}(t))$$

$$K_{rg}(t, f_x) = K_{ap}(t, f_x) \cdot \cos(\theta_{PLR}(t))$$

$$K_{az}(t, f_x) = K_{ap}(t, f_x) \cdot \sin(\theta_{PLR}(t))$$



4.15.3 Image Spatial Frequency Parameters

The polar to rectangular resampling yields a rectangular aperture in the image spatial frequency domain. The rectangular aperture is aligned with the Krg and Kaz axes. The size of the rectangular aperture is a choice made in the image formation processing. The extent of the rectangular aperture used to form the image is given by range spatial frequency parameters Krg1 and Krg2 and azimuth spatial frequency parameters Kaz1 and Kaz2. These are the limits of the input PHD that contribute to the image. They do not include any additional zero-fill that may be needed for the two-dimensional DFT used to form the image.

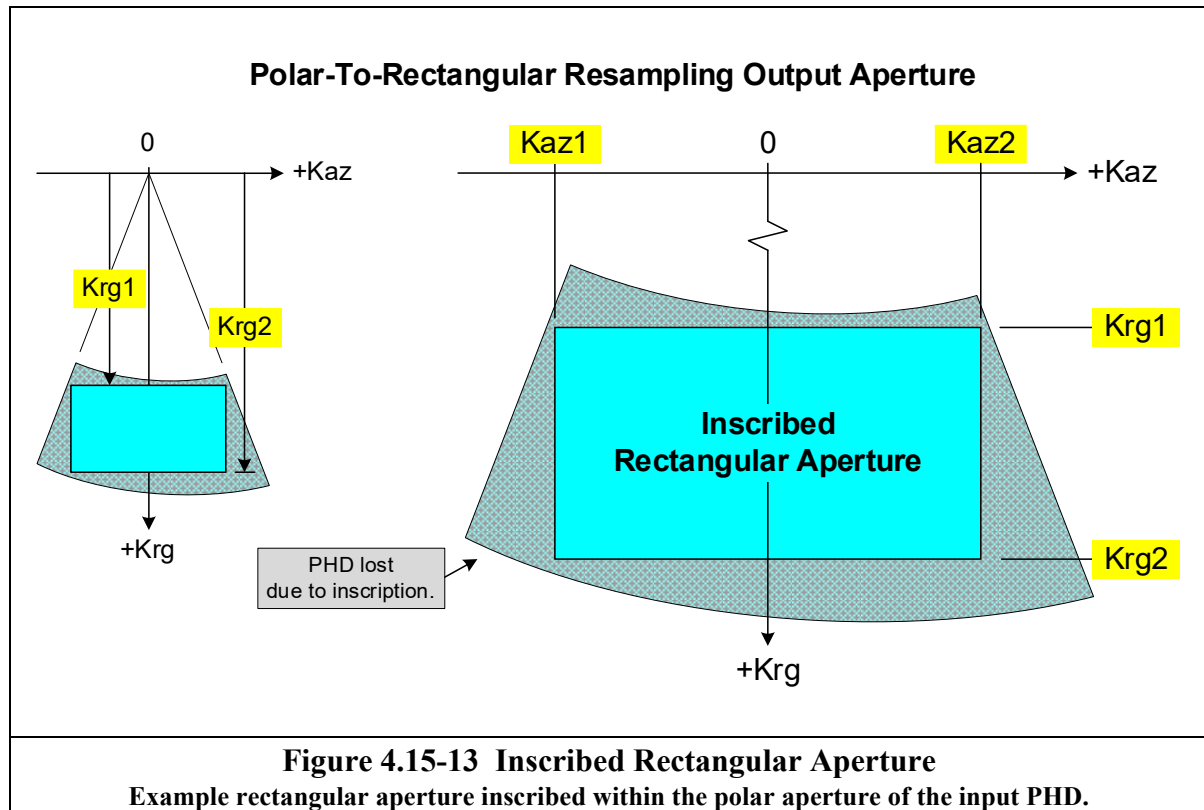
XML: PFA.Krg1

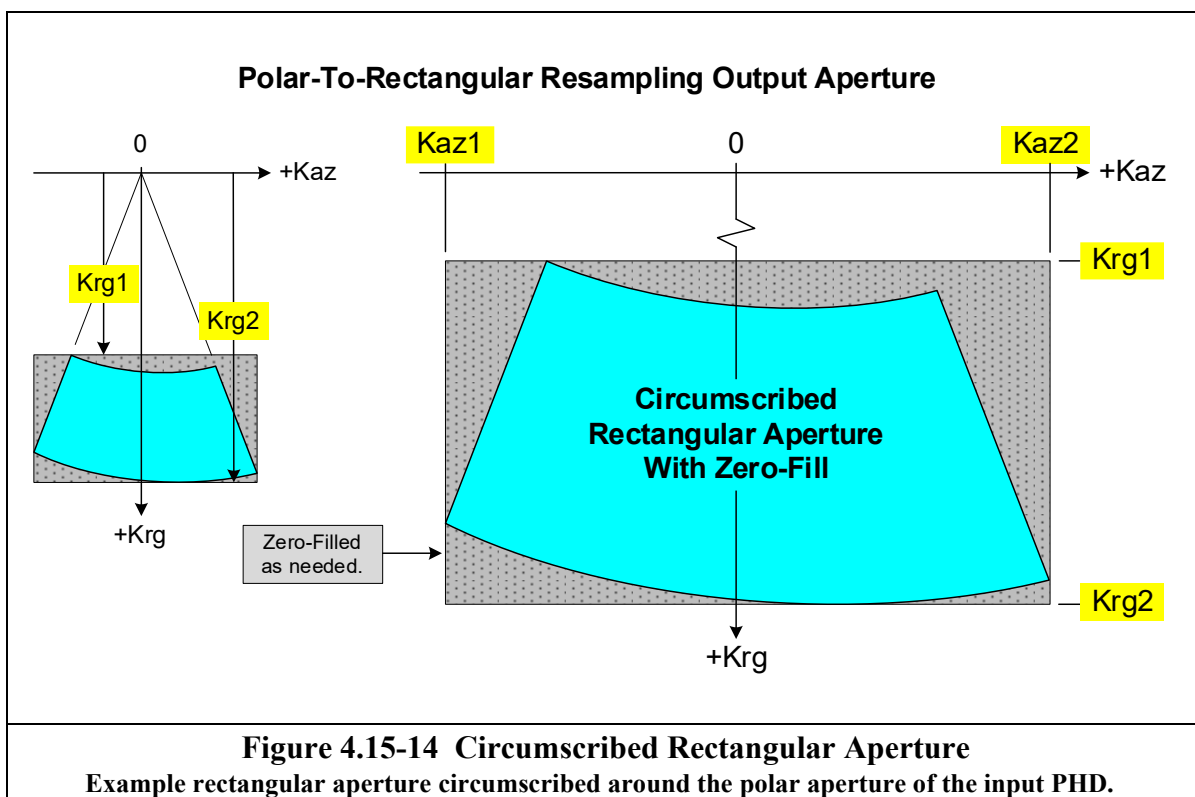
XML: PFA.Kaz1

XML: PFA.Krg2

XML: PFA.Kaz2

Figure 4.15-13 shows an example input PHD where the rectangular aperture is inscribed within the polar aperture of the input PHD. A portion of the collected PHD is discarded due to the choice of the inscribed aperture. Figure 4.15-14 shows the same input PHD where the rectangular aperture is circumscribed around the polar aperture of the input PHD. Portions of the circumscribed rectangular aperture are zero-filled as shown. For each aperture, the limits Krg1, Krg2, Kaz1 and Kaz2 are as shown.





The sampled image spatial frequency grid following the polar to rectangular resampling is denoted $S_{KK}(nrg, naz)$. The rows, indexed by nrg , are evenly spaced samples in the Krg dimension. The columns, indexed by naz , are evenly spaced samples in Kaz . An example grid is shown in Figure 4.15-15. Also shown is the expression for the image compression two-dimensional DFT. Spatial frequency point $(KrgCtr, KazCtr)$ corresponds to the center point of the DFT. The resulting image domain signal is $S_{II}(irg, iaz)$. The phase sign parameter SGN is associated with the motion compensated PHD (where $SGN = +1$ or -1). See Section 4.14-1. The sign of the image compression DFT is set as shown in the figure. Row and column phase sign parameters are set equal to the phase sign associated with PHD.

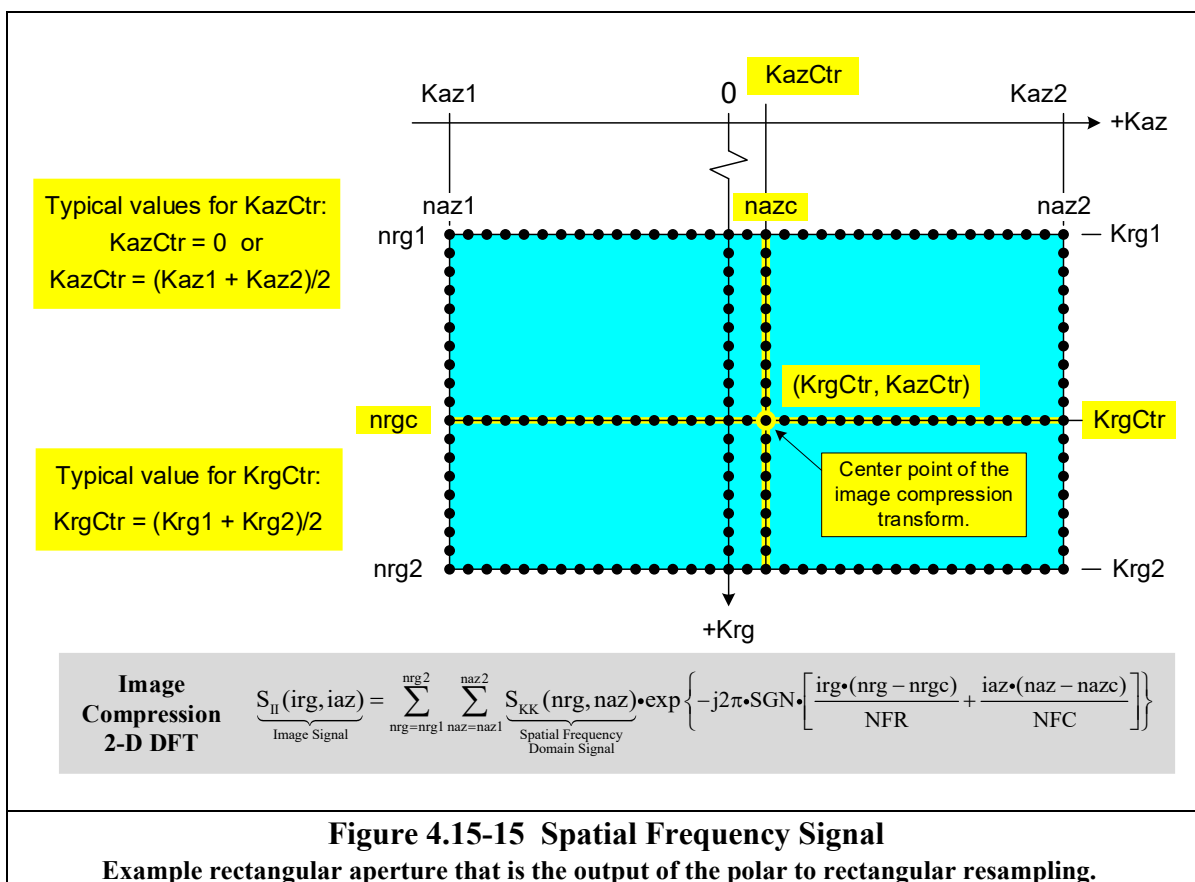
XML: Grid.Row.Sgn = SGN

XML: Grid.Col.Sgn = SGN

Spatial frequency $KrgCtr$ corresponding to index value $nrgc$ is typically set equal (or nearly equal) to $(Krg1 + Krg2)/2$. Spatial frequency $KazCtr$ corresponding to index value $nazc$ is typically set either to 0 or to $(Kaz1 + Kaz2)/2$. Row parameter $KCtr$ is set equal to $KrgCtr$ and column parameters $KCtr$ is set equal to $KazCtr$.

XML: Grid.Row.KCtr = KrgCtr

XML: Grid.Col.KCtr = KazCtr



4.15.4 Slow Time Deskew

The slow time deskew phase function is a phase function that is applied to the image domain signal to form an image with varying Center of Aperture times. Such images are typically formed from dynamic stripmap mode collections. The deskew function is used to align the center of aperture in azimuth spatial frequency for all points in the image. The slow time deskew function, denoted $\theta_{DS}(rg, az)$, is a two-dimensional polynomial function of image coordinates (rg, az). Also provided is the logical parameter, denoted APPLIED, to indicate if the deskew function has been applied to the image.

XML: PFA.STDeskew.Applied = APPLIED

APPLIED = TRUE $\Leftrightarrow$ ST Deskew polynomial has been applied.

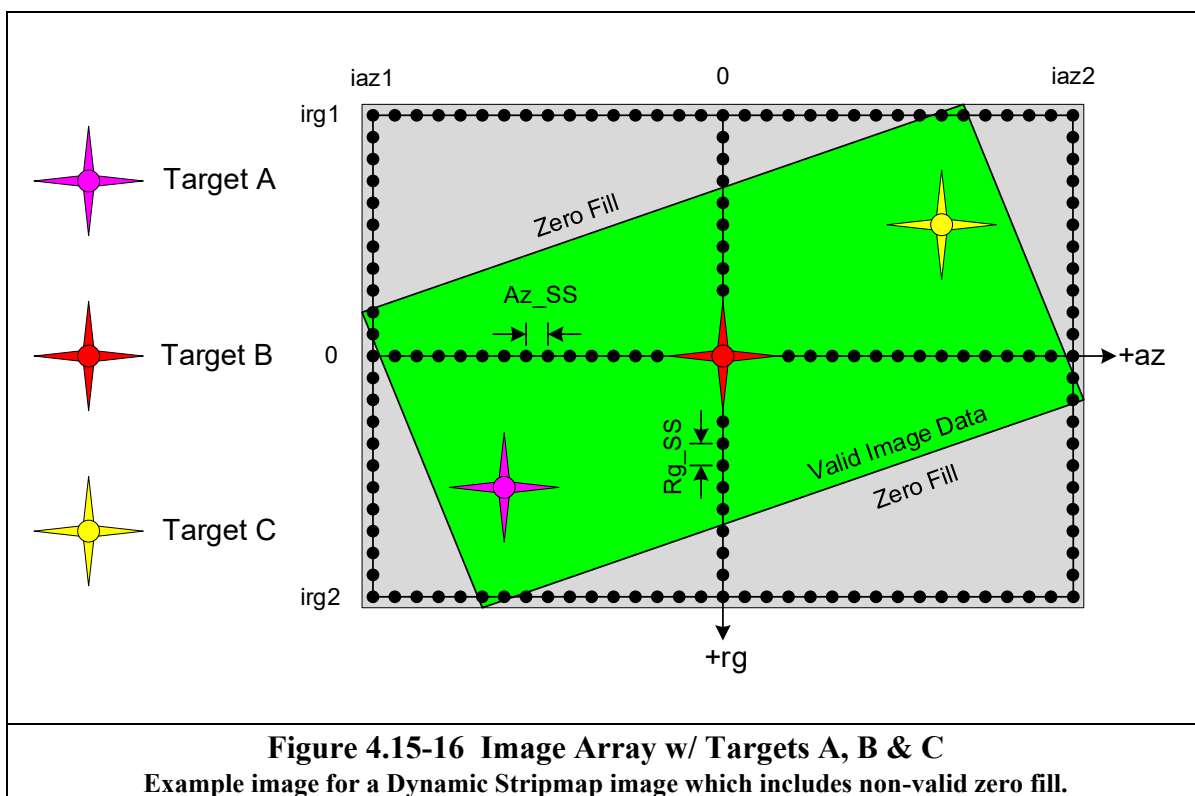
APPLIED = FALSE $\Leftrightarrow$ ST Deskew polynomial has NOT been applied.

XML: PFA.STDeskew.STDSPhasePoly(m,n) = cSTDS(m,n) m = 0 to M_DS, n = 0 to N_DS

$$\theta_{DS}(rg, az) = \sum_{m=0}^{M_DS} \sum_{n=0}^{N_DS} cSTDS(m,n) \cdot (rg)^m \cdot (az)^n$$

Consider the image signal array, $S1_{II}(\text{irg}, \text{iaz})$, that is the output of the image compression 2-D DFT and prior to applying the deskew function. An example image from a dynamic stripmap collection is shown in Figure 4.15-16. The image range index spans $\text{irg} = \text{irg1}$ to irg2 . The image azimuth index spans $\text{iaz} = \text{iaz1}$ to iaz2 . The image includes zero filled regions that correspond to area outside the imaged strip. The range sample spacing is Rg_SS and azimuth sample spacing is Az_SS . Also shown is the image response from targets A, B and C. Target B is located at the SCP and has its impulse response centered on the SCP pixel, $S1_{II}(0,0)$. The deskew phase function is then applied to signal $S1_{II}(\text{irg}, \text{iaz})$, as shown below. The resulting image signal array with deskew applied is $S2_{II}(\text{irg}, \text{iaz})$.

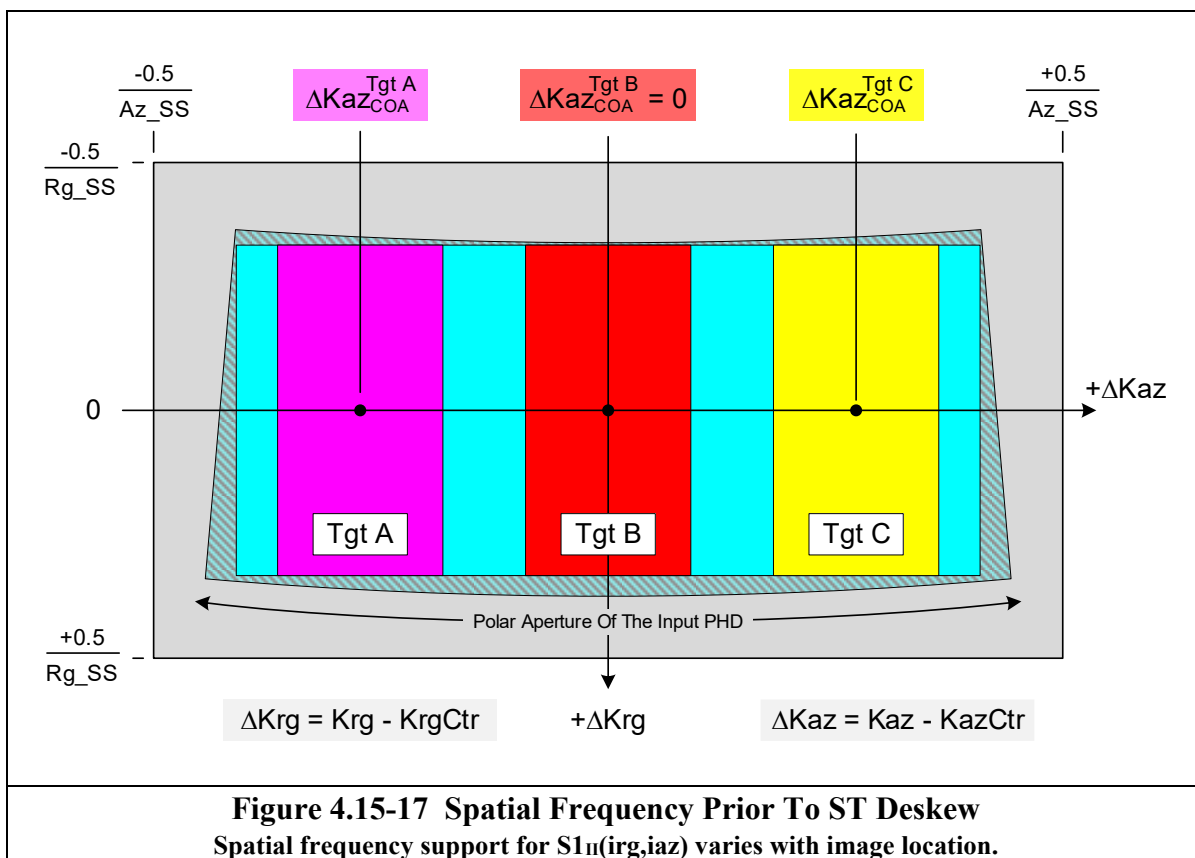
$$\underbrace{S2_{II}(\text{irg}, \text{iaz})}_{\substack{\text{Image Signal} \\ \text{With Deskew Applied}}} = \underbrace{S1_{II}(\text{irg}, \text{iaz})}_{\substack{\text{Image Signal} \\ \text{Prior To Deskew}}} \cdot \exp \left\{ j \cdot 2\pi \cdot \left(\sum_{m=0}^{M_DSN} \sum_{n=0}^{DS} cSTDS(m,n) \cdot (Rg_SS \cdot \text{irg})^m \cdot (Az_SS \cdot \text{iaz})^n \right) \right\}$$

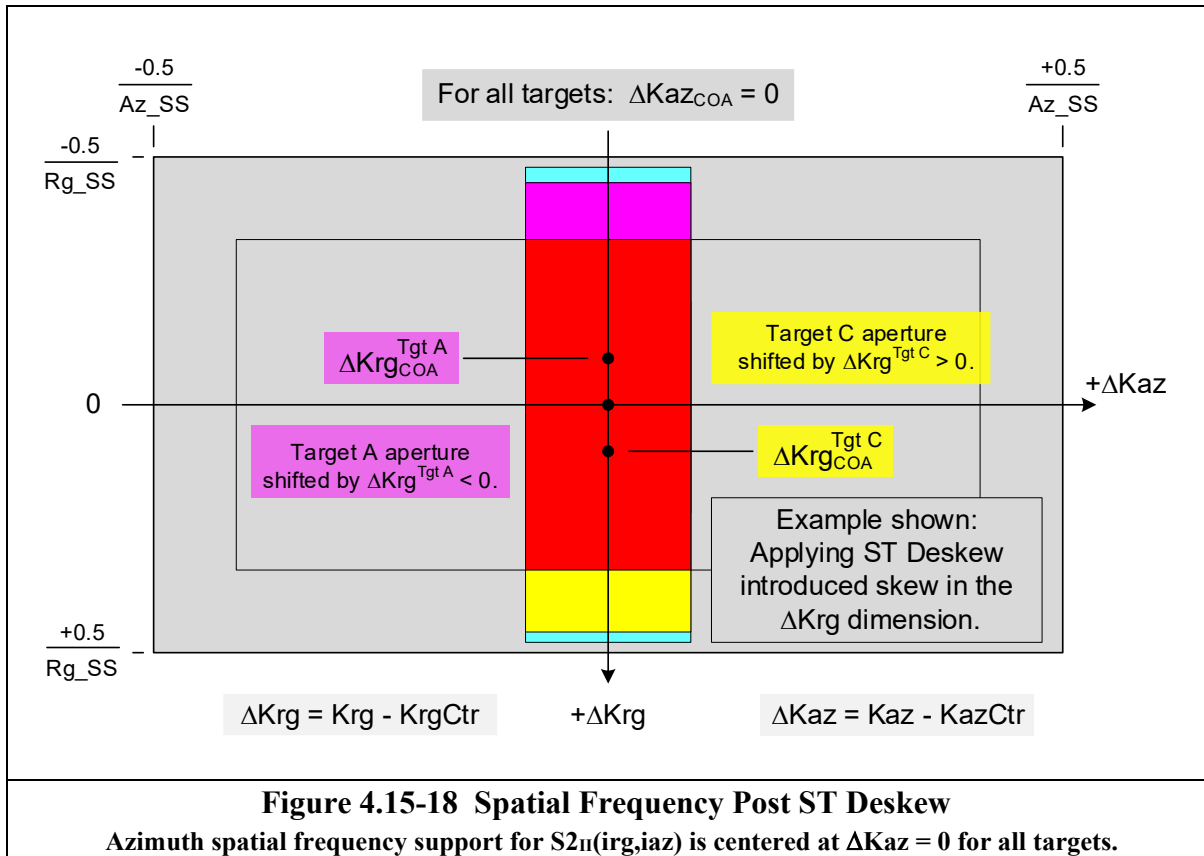


The spatial frequency domain for image signal $S1_{II}(\text{irg}, \text{iaz})$ is shown in Figure 4.15-17. The center point of spatial frequency domain at $\Delta Krg = 0$ and $\Delta Kaz = 0$ corresponds to center point $(KrgCtr, KazCtr)$ in the resampled rectangular aperture. For the example shown, an inscribed rectangular aperture was used. Targets A, B and C were illuminated at different times as the antenna beam was scanned along the imaged strip. The spatial frequency domain for image signal $S2_{II}(\text{irg}, \text{iaz})$ is shown in Figure 4.15-18. The effect of the deskew phase function is to shift the center of apertures to $\Delta Kaz = 0$ for all targets. Also shown is a secondary effect which is to introduce a “skew” in the range spatial frequency dimension.

This secondary effect, usually considered undesirable although unavoidable, is highly variable with imaging geometry and image size. For a given target located at image location (rg^{TGT}, az^{TGT}) , the range frequency shift due to the deskew, ΔKrg^{TGT} , is given by the following expression.

$$\Delta Krg^{TGT} = -SGN \cdot \left. \frac{\partial \theta_{DS}(rg, az)}{\partial(rg)} \right|_{TGT} = -SGN \cdot \sum_{m=1}^{M_{DS}} \sum_{n=0}^{N_{DS}} m \cdot cSTDS(m, n) \cdot (rg^{TGT})^{m-1} \cdot (az^{TGT})^m$$





4.16 Range Migration Algorithm Parameters

The parameters that describe the image formation processing for images formed with any one of several Range Migration Algorithms (RMAs) are described below. Refer to Table 3-16.

The Range Migration Algorithm is based on the principle that for straight line flight of the radar platform, the range of a fixed point as a function of platform displacement is hyperbolic. For a given fixed ground point, the shape of the hyperbola is defined by the range at closest approach (denoted R_{CA}). If the radar platform flies with constant speed, the range versus time is also described by a hyperbola. The development of the RM approach has resulted in several variations of the algorithm referred to here as RMA types. All RMA types (described below), are based on the hyperbolic shape of the range histories. The particular RMA type used is dependent upon the imaging geometry, the transmitted frequency band and the resolutions in range and azimuth of image. Also considered is the scene size to be processed in a single image formation operation and the extent to which the platform trajectory deviates from a precisely straight line.

The RMA is most commonly used for stripmap collections with imaging of the ground points as they pass through closest approach (i.e. broadside stripmap imaging). The RMA may also be used for imaging collections away from broadside (i.e. squinted imaging) by applying motion compensation to a moving point in the early stages of the processing. Use of RMA for squinted imaging is primarily used for spotlight collections.

The RMA approach is based upon the radar platform flying a straight line trajectory and collecting data at uniformly spaced positions along the flight path. A platform travelling with constant speed and operating with a fixed Pulse Repetition Frequency (PRF) will achieve this condition. For a typical airborne platform, the trajectory only approximates a straight line due to limitations in aircraft control (e.g., due to variable winds). Variation in platform displacement between radar pulses is also common when variations in platform speed are not compensated with variations in PRF. The early stages of the data processing will include motion compensation that corrects for deviation from straight line flight as well as vibration of the radar antenna. Compensation for deviations from uniform spacing in platform position may also be applied. The compensations for non-straight line flight will be precise at the center of the scene or strip being imaged. The compensated data is then processed as if were from a straight line trajectory.

In adapting the RM algorithm to process collections away from broadside, motion compensation is applied to account for the range rates that yield true Doppler frequencies that are many multiples of the operating PRF. Two approaches have been developed. The first approach applies motion compensation to a moving point in the along track direction. This results in an image that is still in the natural along track, cross track orientation. See `Grid.Type = XCTYAT`. A second approach applies motion compensation to a moving point in the cross range direction. This results in an image that is oriented range, cross range. See `Grid.Type = XRGYCR`. For collections at broadside, the two approaches become identical.

For platforms that only image near closest approach (within a few degrees of broadside), the RMA is easily adapted to accommodate non-straight line flight. For any platform trajectory, the range versus time for a point near closest approach is accurately approximated by a hyperbolic fit. For an orbital trajectory, the curve of the orbit is accounted for in the hyperbolic approximation. This allows the RMA to be readily extended to satellite platforms imaging near broadside.

(1) RMA Types

The RMA approach includes a transform from slow time to Doppler frequency or, more accurately, from platform displacement to along track spatial frequency. The processing of the data both before and after the transform differentiates the various RMA types. The choice of algorithm type depends upon the resolution to be achieved and the size of the scene to be processed in a single image formation operation. The three RMA types are summarized as follows. See parameter RMAIgoType.

Range-Doppler: RMAIgoType = RG_DOP. Prior to the slow time to Doppler transform, each fast time vector is range compressed. The output of the slow time to Doppler transform is the Range-Doppler domain. Following the slow time to Doppler transform, a correction for migration through range cells is performed. Following range cell migration correction, a one-dimensional transform yields the compressed image. The Range-Doppler algorithm is most commonly used for imaging near closest approach.

Omega-K: RMAIgoType = OMEGA_K. Prior to the slow time to Doppler transform, each fast time vector is range dispersed. The fast time dimension of the collection has been processed to align all echoes by radio frequency (RF). The RF dimension is often expressed in rad/sec and is the “Omega” dimension. The Doppler dimension is more accurately the along track spatial frequency dimension and is the “K” dimension (labeled the Kyt dimension below). A two-dimensional phase correction for the Scene Center Point is then applied (the first step in a matched filter). Following the phase correction, the RF/Omega dimension is resampled to yield cross track spatial frequency (labeled the Kxt dimension below). The resampling is commonly referred to as Stolt Interpolation. Following the resampling, a two-dimensional transform yields the compressed image.

Chirp-Scaling: RMAIgoType = CSA. Prior to the slow time to Doppler transform, each fast time vector is range dispersed and retains the differential in time of arrival associated with differences in range. This is the natural fast time signal for systems that use a linear FM waveform and do not employ de-ramp on receive. The response for any given target is a “chirp” in the fast time dimension. Following the slow time to Doppler transform, the signal history of any given point is a two-dimension chirp. The processing of the two-dimensional chirp signals uses a sequence of one-dimensional transforms and matched filters that yields the compressed image.

While the type of RMA algorithm is an important consideration for forming the image, the RMA type is not critical in describing or exploiting the resulting image. For finer image resolution, the image formation algorithm will require more accurate focusing of individual points. For larger scene sizes, the image formation algorithm will need to correct larger



4.16.1 RMA Images

The parameters associated with an image formed with the RMA and oriented cross track, along track are summarized below. For collections away from broadside, motion compensation to a reference point moving along track is applied. The collection geometry is defined by a straight line trajectory that approximates the actual trajectory of the platform. The straight line trajectory is referred to as the reference line trajectory. The reference line trajectory is defined by the reference position ($\mathbf{POS}_{\text{REF}}$) and the reference velocity vector ($\mathbf{VEL}_{\text{REF}}$). The reference line trajectory may be computed as the best fit straight line trajectory to the true ARP trajectory. The reference line trajectory may also be an extrapolated position and velocity determined during the collection setup computations.

$\mathbf{POS}_{\text{REF}}$ = Reference Position

XML: RMA.PosRef

$\mathbf{VEL}_{\text{REF}}$ = Reference Velocity

XML: RMA.VelRef

The image formation plane is the Reference Slant Plane (RSP). The RSP is the plane that contains the reference position and velocity vector and the Scene Center Point (SCP). Example RMA reference slant planes are shown in Figure 4.16.1-1. The reference Doppler Cone Angle, DCA_{REF} , is the DCA to the SCP relative to the reference position and velocity. The line passing through the SCP and parallel to the reference line trajectory is the Along Track reference line.

DCA_{REF} = Reference Doppler Cone Angle

XML: RMA.DCARef

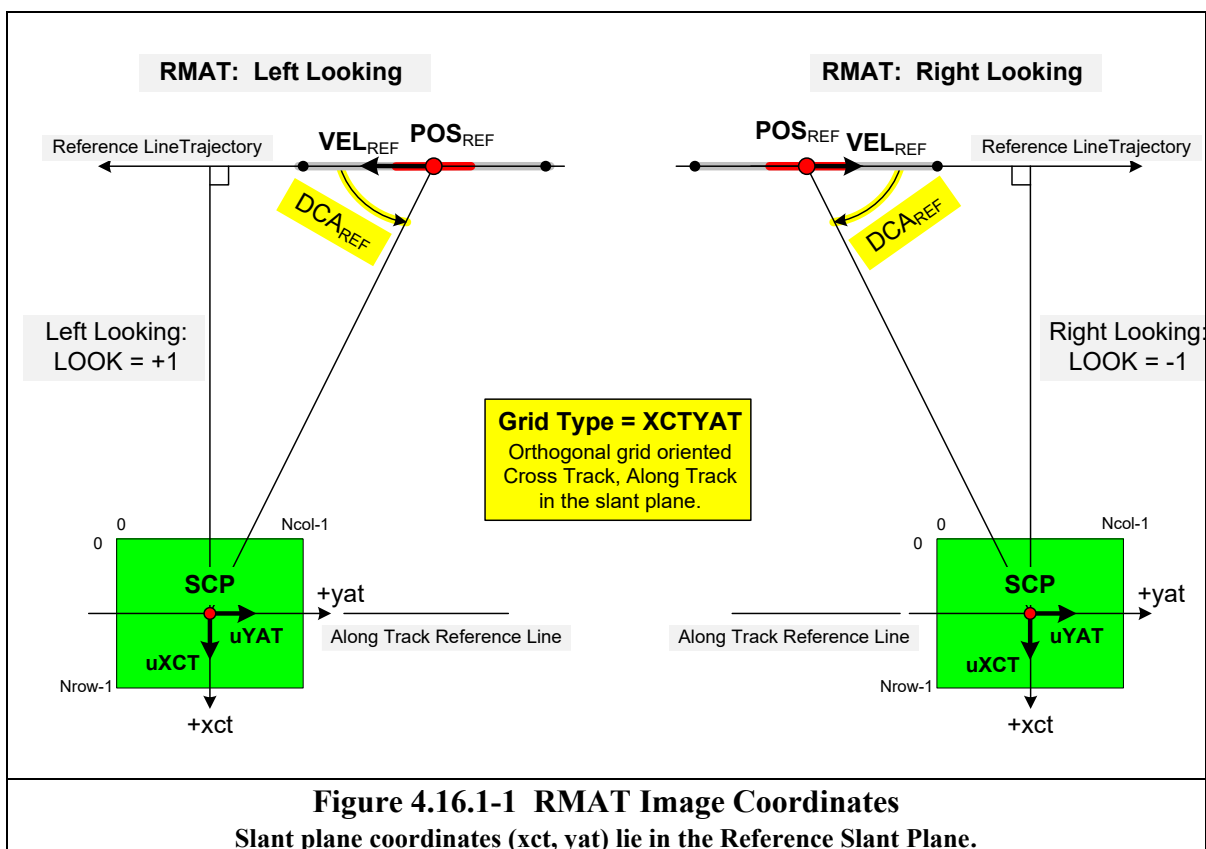
The image formed is a rectangular grid oriented as shown in Figure 4.16.1.1. For all cases, the image is a view from above the earth. The image plane basis vectors, $\mathbf{u}_{\text{XCT}}$ and $\mathbf{u}_{\text{YAT}}$, are shown in the figure. The row coordinate (xct) is increasing away from the reference line trajectory. For the left looking geometry, the image column coordinate (yat) is decreasing in the direction of $\mathbf{VEL}_{\text{REF}}$. For the right looking geometry, the image column coordinate is increasing in the direction of $\mathbf{VEL}_{\text{REF}}$. The side of track parameter $\text{LOOK} = +1$ for left looking collections and $\text{LOOK} = -1$ for right looking collections (see section 4.9 above).

$\mathbf{u}_{\text{XCT}}$ = Cross Track Unit Vector

XML: Grid.Row.UVectECF

$\mathbf{u}_{\text{YAT}} = -\text{LOOK} \cdot \frac{\mathbf{VEL}_{\text{REF}}}{|\mathbf{VEL}_{\text{REF}}|}$

XML: Grid.Col.UVectECF



For a point target in the imaged scene, the mapping from Phase History Data (PHD) coordinates (t, f_x) to the image spatial frequency coordinates is a function of the Doppler Cone Angle (DCA) to the point. For the grid type XCTYAT, the spatial frequency coordinates are (K_{xct}, K_{yat}) . At a time t , a target in the imaged scene, TGT, is at $DCA(t)^{TGT}$. Frequency f (in Hz) maps to aperture radial frequency K_f (in cycles/meter) by the normal $2/c$ scaling. For target TGT, spatial frequency K_f is mapped to spatial frequency along angle $KA(t)^{TGT}$ where $KA(t)^{TGT}$ is related to the $DCA(t)^{TGT}$ as follows.

$$K_f = f \cdot \frac{2}{c} \quad KA(t)^{TGT} = \begin{cases} \pi - DCA(t)^{TGT}, & \text{if LOOK} = +1 \\ DCA(t)^{TGT}, & \text{if LOOK} = -1 \end{cases}$$

The cosine and sine of the angle $KA(t)^{TGT}$, $\cos(KA(t)^{TGT})$ and $\sin(KA(t)^{TGT})$, are computed as follows.

$$\cos(KA(t)^{TGT}) = -\text{LOOK} \cdot \cos(DCA(t)^{TGT}) \quad \sin(KA(t)^{TGT}) = \sqrt{1 - \cos^2(KA(t)^{TGT})}$$

$$K_{xct}(t, f)^{TGT} = K_f \cdot \sin(KA(t)^{TGT}) \quad K_{yat}(t, f)^{TGT} = K_f \cdot \cos(KA(t)^{TGT})$$

An example spotlight mode collection is shown in Figure 4.16.1-2. The platform collected a fixed transmit bandwidth, BW (Hz), centered at frequency fx_C (Hz). The reference slant plane and image area are shown in the left portion of the figure. Shown are three reference slant plane locations, A, B and C, and the corresponding slant plane integration angles spanned by the collection. The spatial frequency domain following the equivalent of the Stolt interpolation of the Omega-K algorithm is shown in the right portion of the figure. The spatial frequency apertures are shown for targets located at points A, B and C. The variation in aperture size and location corresponds to the different DCA intervals spanned for each of the targets.

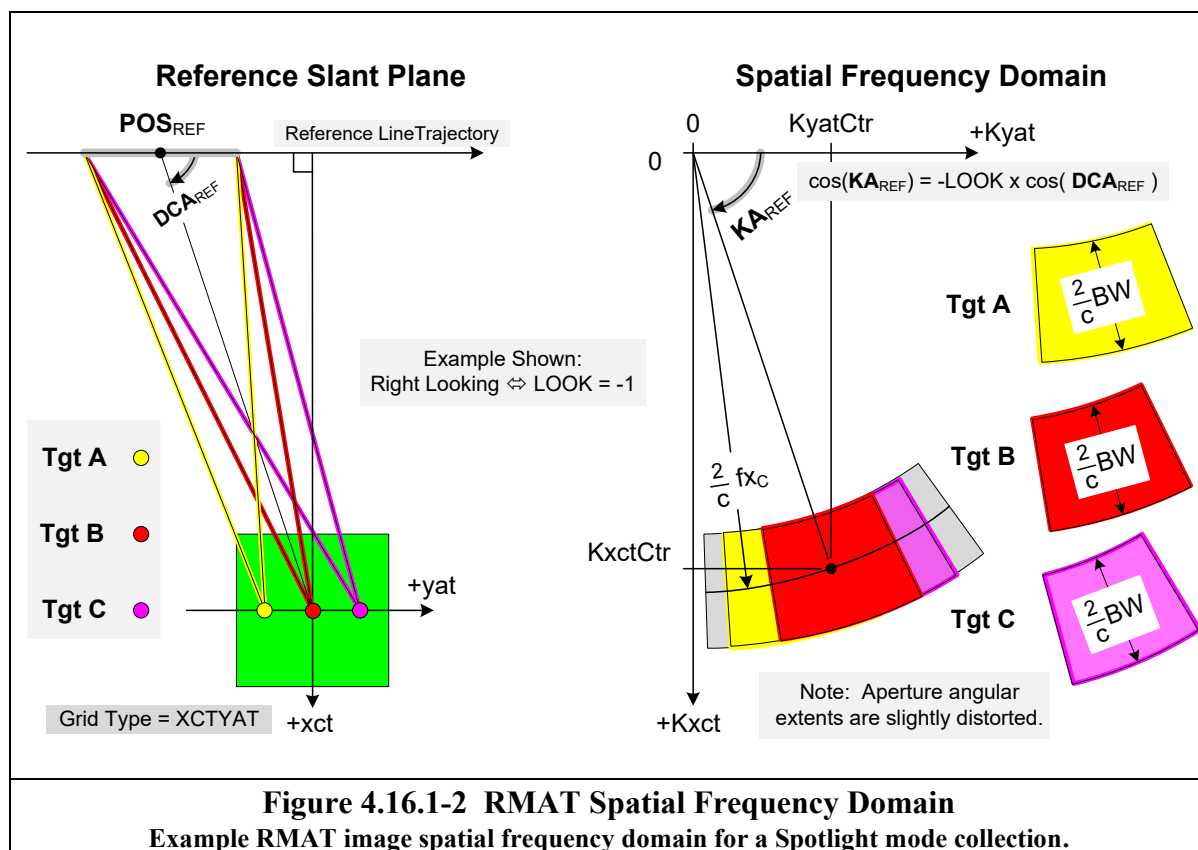
The center of the saved resampled aperture is located at spatial frequency location ($KxctCtr$, $KyatCtr$). The center of the spatial frequency domain is specified in the generic row and column center frequency parameters. For the example shown in Figure 4.16.1-2, the center of the saved spatial frequency corresponds to center frequency fx_C mapped along the radial corresponding to the $KA_{REF} = DCA_{REF}$.

$$KxctCtr = Kap_C \cdot \sin(KA_{REF})$$

$$\text{XML: Grid.Row.KCtr} = KxctCtr$$

$$KyatCtr = Kap_C \cdot \cos(KA_{REF})$$

$$\text{XML: Grid.Col.KCtr} = KyatCtr$$



4.16.2 RMCR Images

The parameters associated with an image formed with the RMA and oriented range, cross range are summarized below. For collections away from broadside, motion compensation to a reference point moving cross range is applied. The collection geometry parameters for RMCR images are identical to those used for the RMAT images. The collection geometry is defined by a reference line trajectory that approximates the actual trajectory of the platform. The reference line trajectory is defined by the reference position (**POS_{REF}**) and the reference velocity vector (**VEL_{REF}**).

POS_{REF} = Reference Position

XML: RMCR.PosRef

VEL_{REF} = Reference Velocity

XML: RMCR.VelRef

The image formation plane is the Reference Slant Plane (RSP). The RSP is the plane that contains the reference position and velocity vector and the Scene Center Point (**SCP**).

Example RMCR reference slant planes are shown in Figure 4.16.2-1. The reference Doppler Cone Angle, **DCA_{REF}**, is the DCA to the SCP relative to the reference position and velocity.

DCA_{REF} = Reference Doppler Cone Angle

XML: RMCR.DCARef

The image formed is a rectangular grid oriented as shown in Figure 4.16.2.1. For all cases, the image is a view from above the earth. The image plane basis vectors, **uXRG** and **uYCR**, are as shown in the figure. The row coordinate (xrg) is increasing in the increasing range direction. The column coordinate (ycr) is increasing as shown independent of the side of track.

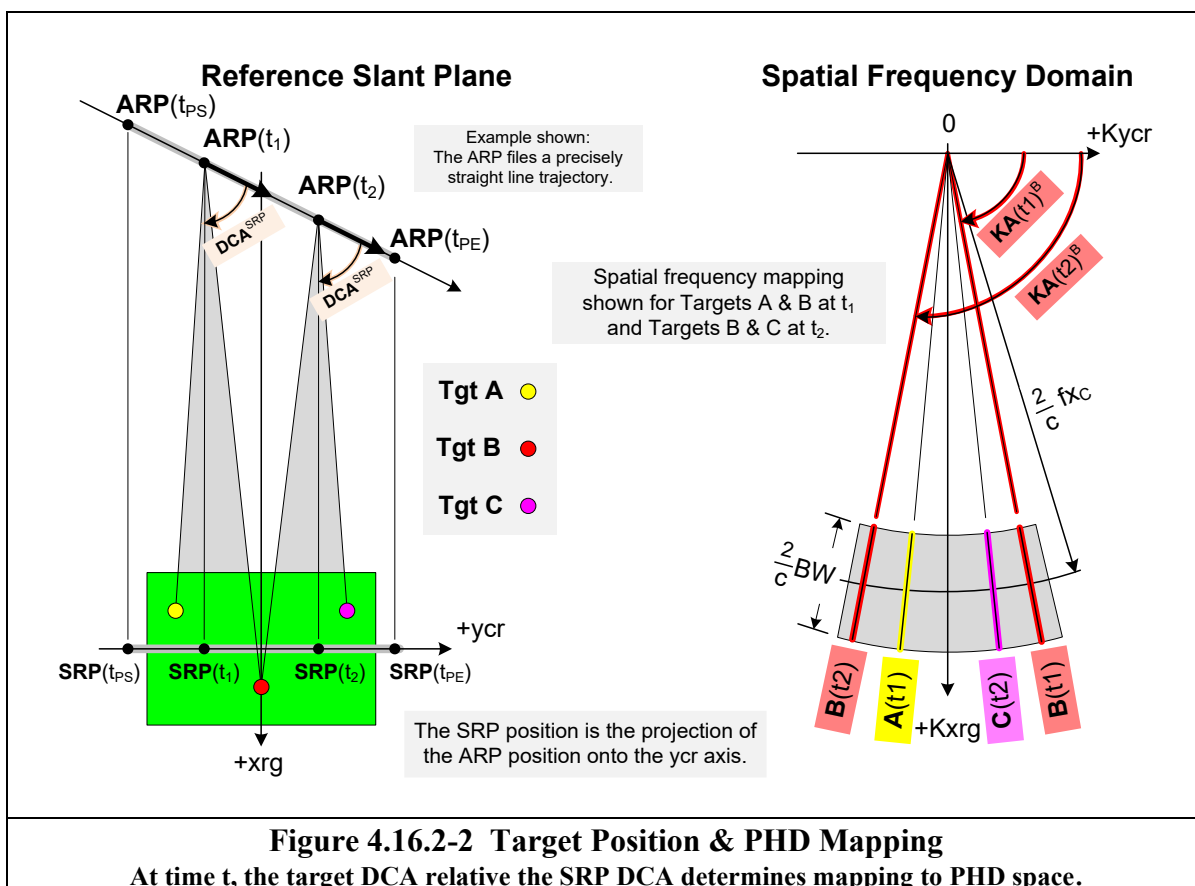
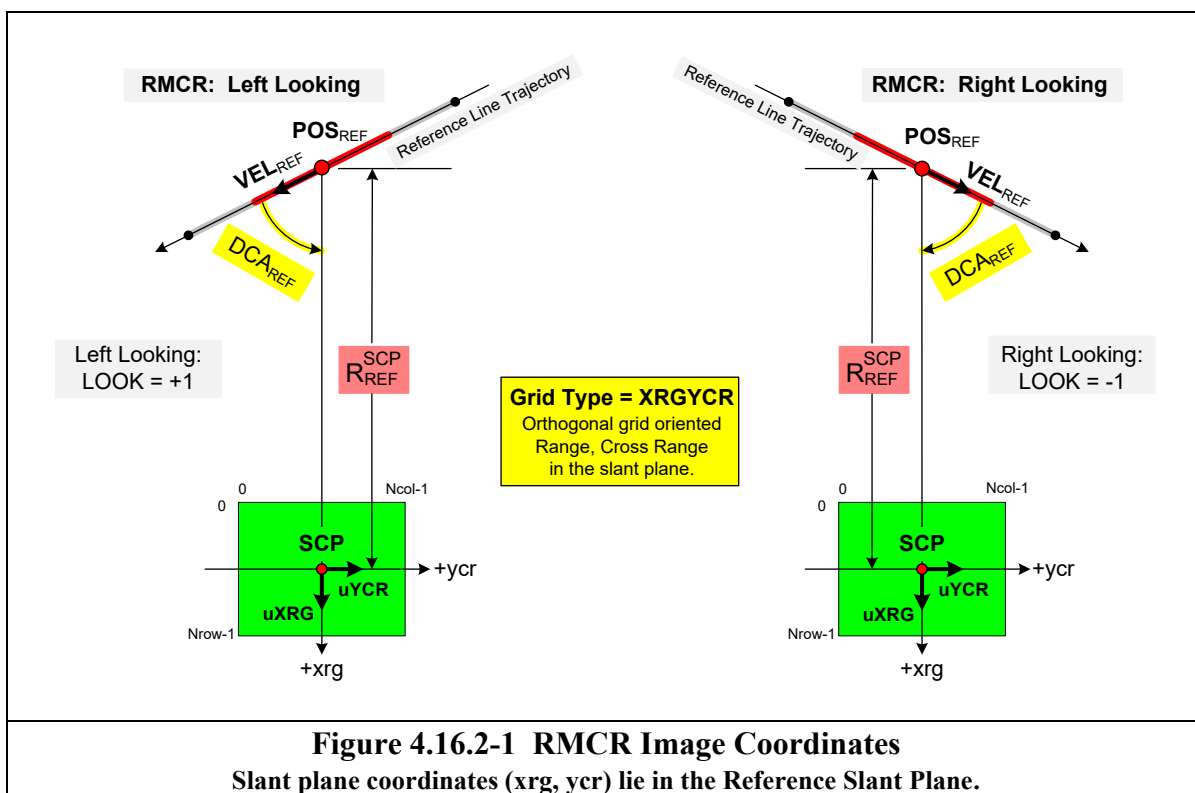
$$\mathbf{uXRG} = \frac{1}{|\mathbf{SCP} - \mathbf{POS}_{REF}|} \cdot (\mathbf{SCP} - \mathbf{POS}_{REF})$$

XML: Grid.Row.UVectECF

uYCR = Cross Range Unit Vector

XML: Grid.Col.UVectECF

During the initial steps of the processing, the PHD is motion compensated (also known as phase stabilized) to a point that moves in the cross range direction along the ycr axis. The stabilization point is referred to as the Stabilization Reference Point, denoted **SRP(t)**. An example right looking collection geometry is shown Figure 4.16.2-2. In the left portion of the figure, the ARP trajectory and the SRP track are shown. At all times, the SRP position is the projection of the ARP position onto the ycr axis. The portion the PHD processed is from slow time **t_{PS}** to **t_{PE}**. Shown in the figure are the ARP and SRP positions at time **t_{PS}** and **t_{PE}** and two additional times **t₁** and **t₂**, where **t_{PS} < t₁ < t₂ < t_{PE}**.



For a point target in the imaged scene (TGT), the mapping from PHD coordinates (t, fx) to image spatial frequency coordinates is a function of the DCA to the SRP and the DCA to the target. At a time t, the DCA to the SRP is $DCA(t)^{SRP}$ and DCA to the target is $DCA(t)^{TGT}$. Frequency f (in Hz) maps to aperture radial frequency Kf (in cycles/meter) by the normal 2/c scaling. For target TGT, spatial frequency Kf is mapped along spatial frequency angle $KA(t)^{TGT}$. The cosine and sine of $KA(t)^{TGT}$ are related to $DCA(t)^{SRP}$ and $DCA(t)^{TGT}$ as follows.

$$\cos(KA(t)^{TGT}) = \frac{-LOOK}{\sin(DCA(t)^{SRP})} \cdot (\cos(DCA(t)^{TGT}) - \cos(DCA(t)^{SRP}))$$

$$\sin(KA(t)^{TGT}) = +\sqrt{1 - \cos^2(KA(t)^{TGT})} \quad Kf = f \cdot \frac{2}{c}$$

For an RMCR image and grid type XRGYCR, the spatial frequency coordinates are (Kxrg, Kycr). The mapping to spatial frequency coordinates (Kxrg, Kycr) is as follows.

$$Kxrg(t, f)^{TGT} = Kf \cdot \sin(KA(t)^{TGT}) \quad Kycr(t, f)^{TGT} = Kf \cdot \cos(KA(t)^{TGT})$$

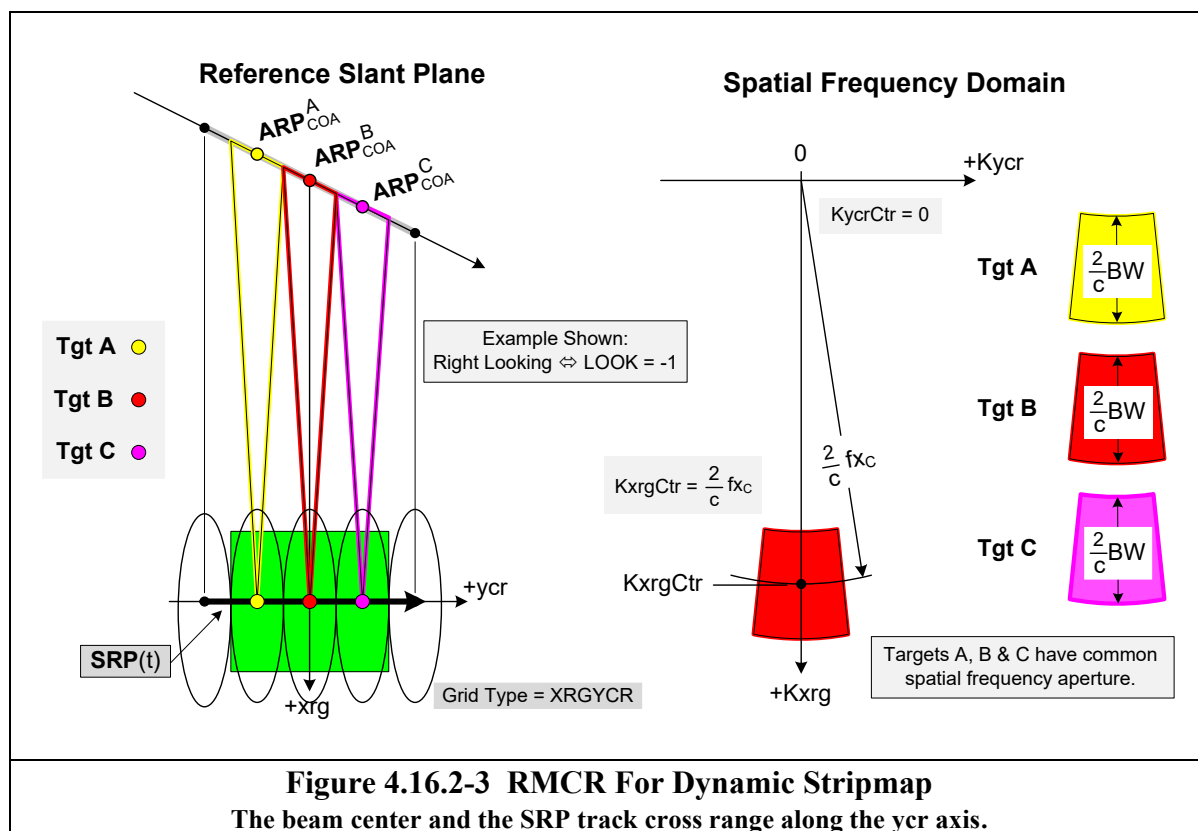
For small integration angles and/or imaging collections near closest approach, the expression for the $\cos(KA(t)^{TGT})$ is accurately approximated by the expression below.

$$\cos(KA(t)^{TGT}) \approx LOOK \cdot (DCA(t)^{TGT} - DCA(t)^{SRP})$$

For the example imaging collection shown in Figure 4.16.2-2, three slant plane target locations A, B and C are shown. The image spatial frequency domain is shown in the right half of the figure. The platform collected a fixed transmit bandwidth, BW (Hz), centered at frequency fx_C (Hz). For PHD collected at time t_1 , the mapping to (Kxrg, Kycr) is shown for targets located at points A and B. For PHD collected at time t_2 , the mapping to (Kxrg, Kycr) is shown for targets located at point B and C.

An example dynamic stripmap collection is shown in Figure 4.16.2-3 where the antenna footprint is scanned in the cross range direction. The beam center moves along the ycr axis at the same rate that the ARP moves in the cross range direction. The platform collected a fixed transmit bandwidth, BW (Hz), centered at frequency fx_C (Hz). The SRP tracks the beam center along the ycr axis. The image is formed such that the COA time for a point at (xrg^{TGT}, ycr^{TGT}) is when the beam center and the SRP are at $(0, ycr^{TGT})$. Shown are three target positions A, B and C located on the ycr axis. For each target, the integration angle and the ARP at COA are shown. For the points A, B and C, the apertures in slow time are non-overlapping. The apertures in the spatial frequency domain are shown in right portion of the

figure. For points A, B and C, the apertures in the spatial frequency domain are precisely overlapped and centered at $K_{ycr} = 0$. A typical RMCR image will have a spatial frequency domain aperture trim applied such that all points in the scene have the same aperture. Such a trimming is easily implemented as part of the Stolt interpolation processing.

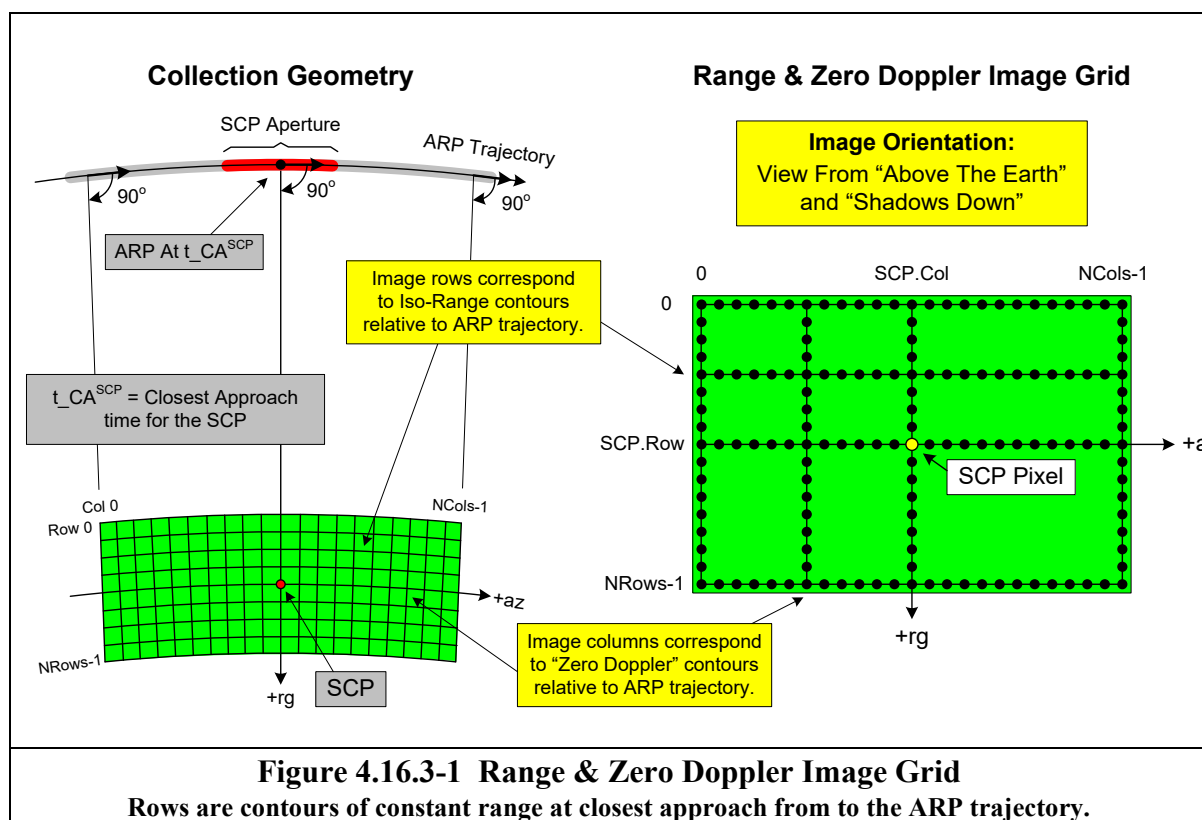


4.16.3 INCA Images

For the special case of imaging near closest approach, the RMA can be used to form the pixels of the image based on range at closest approach and time of closest approach. The image formed is of grid type RGZERO. The pixels of a given row are formed for points that have a common range at closest approach (R_{CA}). The columns of the image are formed for points that have a common time of closest approach (t_{CA}). See Figures 4.16.3-1 and 4.16.3-2. For straight flight trajectory, the image is a slant plane projection of the imaged scene with rows and columns forming an orthogonal grid (similar to grid type XCTYAT). For non-straight line trajectories (such as for an orbiting platform), the rows will be formed for points along curved contours of constant range from the trajectory. For all cases, the image is considered to be a slant plane image.

Grid.Type = RGZERO

Gird.ImagePlane = SLANT



The images formed with rows of constant R_{CA} and columns of constant t_{CA} are accurately described by the INCA parameters under the following conditions. For a given point in the imaged scene, the range as a function of time ($R(t)$) during the integration period for the point is accurately described by parameters computed at time of closest approach. For a ground point in the scene, the range as a function of time is approximated using the following parameters.

t_CA	Time of closest approach.
R_CA	Range at Closest Approach from the ARP trajectory.
VM_CA	Magnitude of the ARP velocity at t_CA.
DRSF	Doppler Rate Scale Factor used to compute the d^2R/dt^2 at t_CA.

The range versus time is approximated by the hyperbolic function RA(t) shown below. The Doppler Rate Scale Factor (DRSF) is computed such that the second derivative of RA(t) at closest approach is equal to the true second derivative (i.e. at $t = t_{CA}$, $d^2(RA)/dt^2 = d^2R/dt^2$).

$$RA(t) = \left[R_{CA}^2 + DRSF \cdot VM_{CA}^2 \cdot (t - t_{CA})^2 \right]^{1/2}$$

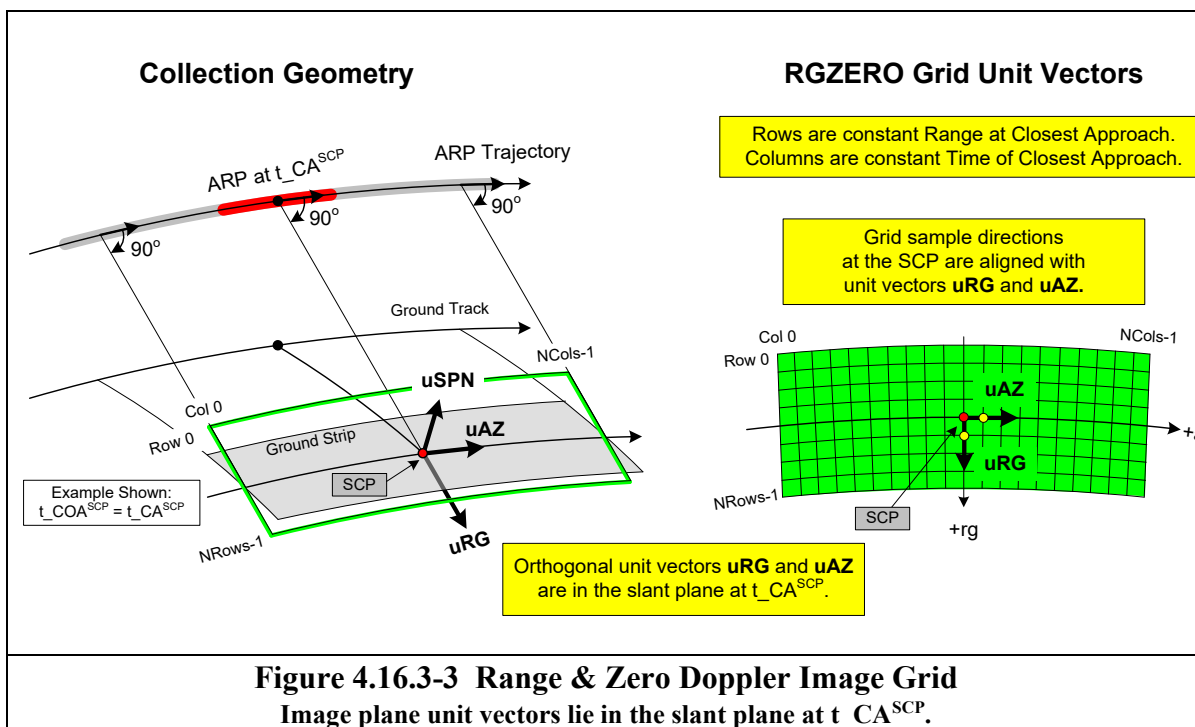
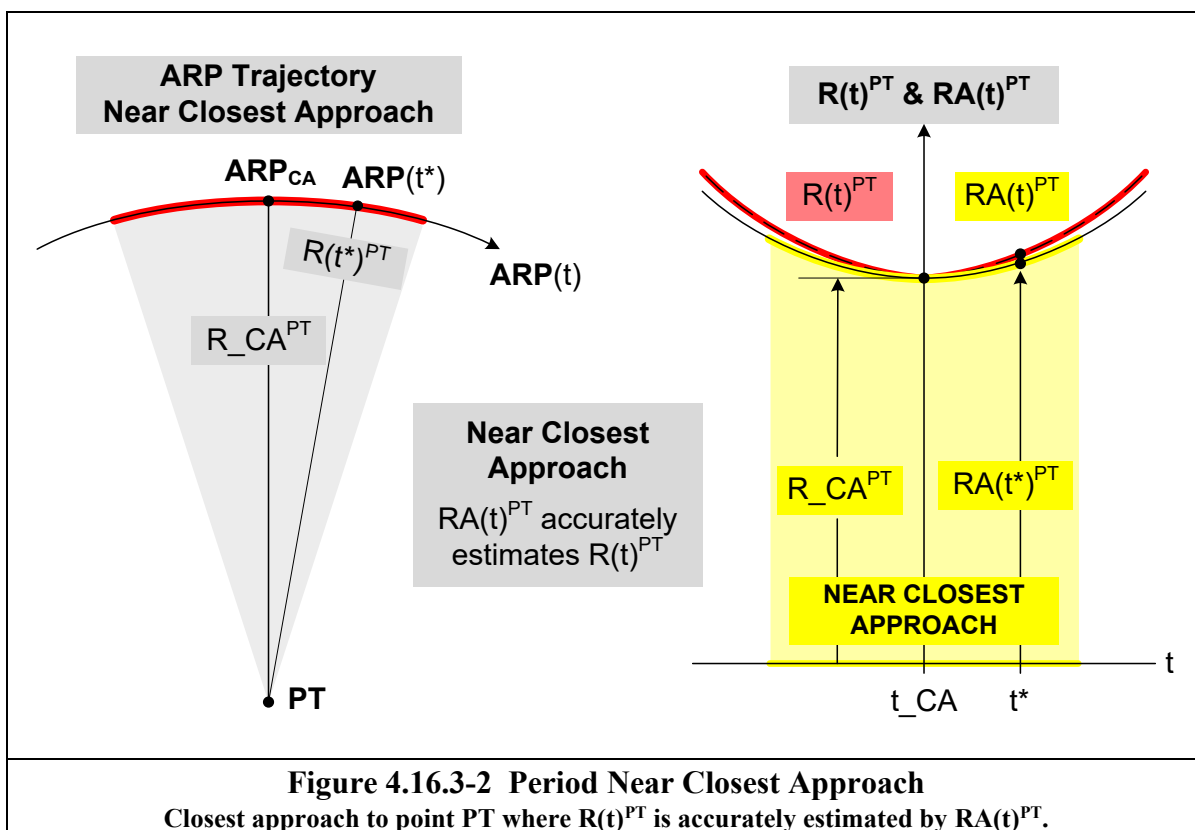
$$DRSF = \left. \frac{d^2R(t)}{dt^2} \right|_{CA} \cdot \frac{R_{CA}}{VM_{CA}^2} \Leftrightarrow \left. \frac{d^2RA(t)}{dt^2} \right|_{CA} = \frac{DRSF}{R_{CA}} \cdot VM_{CA}^2 = \left. \frac{d^2R(t)}{dt^2} \right|_{CA}$$

For a platform that flies a straight line at constant speed, the expression is exact for all time and parameter $DRSF = 1.0$ for all values of R_{CA} . An example trajectory and range versus time functions are shown in Figure 4.16.3-2. For ground point PT, the period “Near Closest Approach” is when the difference between RA(t) and R(t) is small. For accurate exploitation using INCA parameters, the error in RA(t) should be less than the smaller of 0.10 m or 0.25 of the range impulse response width (Rg_IRW). For an orbiting platform and $Rg_IRW = 1.0$ m, the period of closest approach for ground points spans approximately 6 degrees of Doppler Cone Angle (DCA) centered on closest approach ($DCA = 90$ deg).

$$\text{Near Closest Approach} \Leftrightarrow |R(t) - RA(t)| < \text{Min}(0.25 \cdot Rg_IRW, 0.10 \text{ m})$$

(1) SCP Position & SCP Pixel

The RMA may not explicitly define a ground point in the scene for image formation processing. For the INCA images, the SCP pixel and position are set by first selecting a pixel close to the center of the image. See Figure 4.16.3-3. The ground point in the scene that images to the SCP pixel is then computed. The computation is based upon the precise Center of Aperture used for the selected pixel. The SCP position lies along the Center of Aperture iso-range and iso-range rate contour for the selected pixel. The ground height should correspond to the estimated ground height used in the processing.



(2) Image Plane Parameters

The RGZERO image grid is oriented “shadows down” and “a view from above the earth”. Image plane unit vectors are computed based on the slant plane that contains the SCP and the ARP position and velocity at SCP closest approach (t_{CA}^{SCP}). See Figure 4.16.3-3. The row coordinate is referred to as the range coordinate (rg). The column coordinate is referred to as the azimuth coordinate (az). The unit row vector points from the ARP to the SCP and is in the increasing range direction (**uRG**). The unit column vector points in the azimuth direction (**uAZ**). The slant plane normal vector (**uSPN**) points away from the center of the earth. Vectors **uRG**, **uAZ** and **uSPN** always form a right-handed coordinate system.

$$INCA.R_CA_SCP = R_CA^{SCP} = \text{Range to the SCP at closest approach.}$$

$$\begin{aligned} \text{Grid.Row.UVectECF} &= \mathbf{uRG} & \mathbf{uRG} \times \mathbf{uAZ} &= \mathbf{uSPN} \\ \text{Grid.Col.UVectECF} &= \mathbf{uAZ} \end{aligned}$$

Image row spacing (Rg_SS) is the spacing in range at closest approach. The rows of the RGZERO grid are equally spaced in R_CA. For image pixel (row,col), the SCP pixel centered row index (irg), the row coordinate (rg) and the range at closest approach are computed as follows.

$$\begin{aligned} \text{Grid.Row.SS} &= Rg_SS \\ \text{irg} &= \text{row} - \text{SCPPixel.Row} & \text{rg} &= Rg_SS \cdot \text{irg} \\ R_CA &= R_CA^{SCP} + \text{rg} \end{aligned}$$

Image column spacing (Az_SS) is set equal to the spacing in azimuth measured at the SCP. For the RGZERO grid, the precise spacing between columns may vary slightly from column to column and along a column from near to far range. For image pixel (row,col), the SCP centered column index (iaz) and the column coordinate (az) are computed as follows.

$$\begin{aligned} \text{Grid.Col.SS} &= Az_SS = \text{Column spacing at the SCP pixel.} \\ \text{iaz} &= \text{col} - \text{SCPPixel.Col} & \text{az} &= Az_SS \cdot \text{iaz} \end{aligned}$$

The time of closest approach is given by a polynomial function of the azimuth coordinate. The t_{CA} polynomial (coefficients denoted $cT_{CA}(n)$, order N_{TCA}) gives the precise time of closest approach for all columns.

$$\begin{aligned} INCA.TimeCAPoly(n) &= cT_{CA}(n) \quad n = 0 \text{ to } N_{TCA} \\ t_{CA}(az) &= \sum_{n=0}^{N_{TCA}} cT_{CA}(n) \cdot (az)^n \end{aligned}$$

For platforms that fly at constant speed, the processing will form columns with constant spacing in t_CA and the image will have fixed azimuth spacing along a row. For platforms that fly with varying speed, the processing may still form columns with fixed spacing in t_CA or may form columns with varying spacing in t_CA to achieve fixed azimuth spacing along a row. For all images, the precise $t_CA(az)$ is used for exploitation.

(3) Image Impulse Response Widths

The impulse response widths in the range and azimuth dimensions are set as follows. Parameters Rg_IRW and Az_IRW are the slant plane half-power impulse response widths (IRWs) in range and azimuth measured at the SCP. For an INCA image, the range IRW is usually constant across the image. The azimuth IRW may vary slightly across the image.

$$\begin{aligned} \text{Grid.Row.ImpRespWid} &= Rg_IRW && \text{Slant plane, half-power IRWs} \\ \text{Grid.Col.ImpRespWid} &= Az_IRW && \text{at the SCP / SCP Pixel.} \end{aligned}$$

(4) DFT Sign Parameters & Doppler Frequency Convention

The DFT exponents for transforming from the image domain to the spatial frequency domain are set as follows. The parameter Row.Sgn is the sign of the DFT exponent (+1 or -1) that transforms the image row dimension (increasing range) to the increasing range spatial frequency (increasing Krg also corresponds to increasing RF). For a RGZERO grid, the same sign is used to transform the column dimension (increasing azimuth) to the increasing azimuth spatial frequency (increasing Kaz).

$$\begin{aligned} \text{Grid.Row.Sgn} &= \text{SGN} && \text{where SGN} = +1 \text{ or } -1 \\ \text{Grid.Col.Sgn} &= \text{SGN} \end{aligned}$$

The value of the SGN matches the sign of the phase versus range in the image formation processor. The SGN is initially set by the radar analog processing prior to analog-to-digital conversion but is ultimately set by the image formation processing. The phase versus range ($\Phi(f_0, R(t))$) and the Doppler frequency ($f_{dop}(f_0, R(t))$) versus range rate are as follows.

$$\Phi(f_0, R(t)) = \text{SGN} \cdot f_0 \cdot \frac{2}{c} \cdot R(t) \quad f_{dop}(f_0, R(t)) = \text{SGN} \cdot f_0 \cdot \frac{2}{c} \cdot \left(\frac{d}{dt} R(t) \right)$$

Frequency f_0 is the RF value considered, phase $\Phi(f_0, R(t))$ is in cycles and frequency $f_{dop}(f_0, R(t))$ is in Hz. The same value of SGN is used for relating closing/opening targets with positive/negative Doppler frequency values.

$$\begin{aligned} \text{For SGN} = -1: & \quad \text{Closing targets, } d(R(t))/dt < 0 \Leftrightarrow f_{dop} > 0 \\ & \quad \text{Opening targets, } d(R(t))/dt > 0 \Leftrightarrow f_{dop} < 0 \\ \text{For SGN} = +1: & \quad \text{Closing targets, } d(R(t))/dt < 0 \Leftrightarrow f_{dop} < 0 \\ & \quad \text{Opening targets, } d(R(t))/dt > 0 \Leftrightarrow f_{dop} > 0 \end{aligned}$$

For the Doppler frequency computations that follow, the frequency used (f_0) is the value that specifies the center of the range spatial frequency domain (KrgCTR). The center of the azimuth spatial frequency domain (KazCTR) is always 0.

INCA.FreqZero = f_0 = Frequency for computing all Doppler frequency values.

Grid.Row.KCtr = KrgCTR = $f_0 \cdot \frac{2}{c}$

Grid.Col.KCtr = KazCTR = 0

(5) Doppler Rate Scale Factor

The DRSF parameter sets the second derivative of the RA(t) function at closest approach ($d^2(RA)/dt^2$) to that used in the image formation. This also sets the first derivative of the Doppler frequency (i.e. Doppler Rate) at closest approach. The correct value of Doppler Rate is important to achieving correct azimuth focusing. The processing may vary the DRSF used across the image. It is provided as a function of image coordinates (rg,az). The DRSF polynomial (coefficients denoted cDRSF(m,n), orders M_DRSF and N_DRSF) gives the DRSF used for range function RA(t). For image location (rg,az), DRSF used to from the image is computed as follows.

INCA.DRateSFPoly(m,n) = cDRSF(m,n), $m = 0$ to M_DRSF, $n = 0$ to N_DRSF

$$DRSF(rg,az) = \sum_{m=0}^{M_DRSF} \sum_{n=0}^{N_DRSF} cDRSF(m,n) \cdot (rg)^m \cdot (az)^n$$

For image location (rg,az), the Doppler Rate at Closest Approach (DR_CA, in Hz/sec) is computed as follows.

$$DR_CA(rg,az) = SGN \cdot f_0 \cdot \frac{2}{c} \cdot \left(\frac{d^2}{dt^2} RA(t) \Big|_{CA} \right)$$

$$DR_CA(rg,az) = SGN \cdot f_0 \cdot \frac{2}{c} \cdot \left(\frac{DRSF(rg,az)}{R_CA(rg)} \cdot VM_CA(az)^2 \right)$$

For all points in scene, the Doppler Rate is nearly constant for period near closest approach. The Doppler frequency versus time may be accurately computed using the Doppler Rate at closest approach. For image location (rg,az), the Doppler frequency versus time is accurately approximated as a linear function.

$$fdop(t; rg,az) = DR_CA(rg,az) \cdot (t - t_CA(az))$$

(6) Center Of Aperture Time

The Center Of Aperture time (t_COA) is a key parameter needed for proper image exploitation. The time of Center of Aperture is provided as a function of image coordinates (rg, az). The t_COA polynomial (coefficients denoted $cT_COA(m,n)$, orders M_TCOA and N_TCOA) is specified as follows.

$$\text{Grid.TimeCOAPoly}(m,n) = cT_COA(m,n), \quad m = 0 \text{ to } M_TCOA, \quad n = 0 \text{ to } N_TCOA$$

$$t_COA(rg,az) = \sum_{m=0}^{M_TCOA} \sum_{n=0}^{N_TCOA} cT_COA(m,n) \cdot (rg)^m \cdot (az)^n$$

For a given image location, the difference between the COA time and the closest approach time ($\Delta t_COA(rg,az)$) is used for several computations.

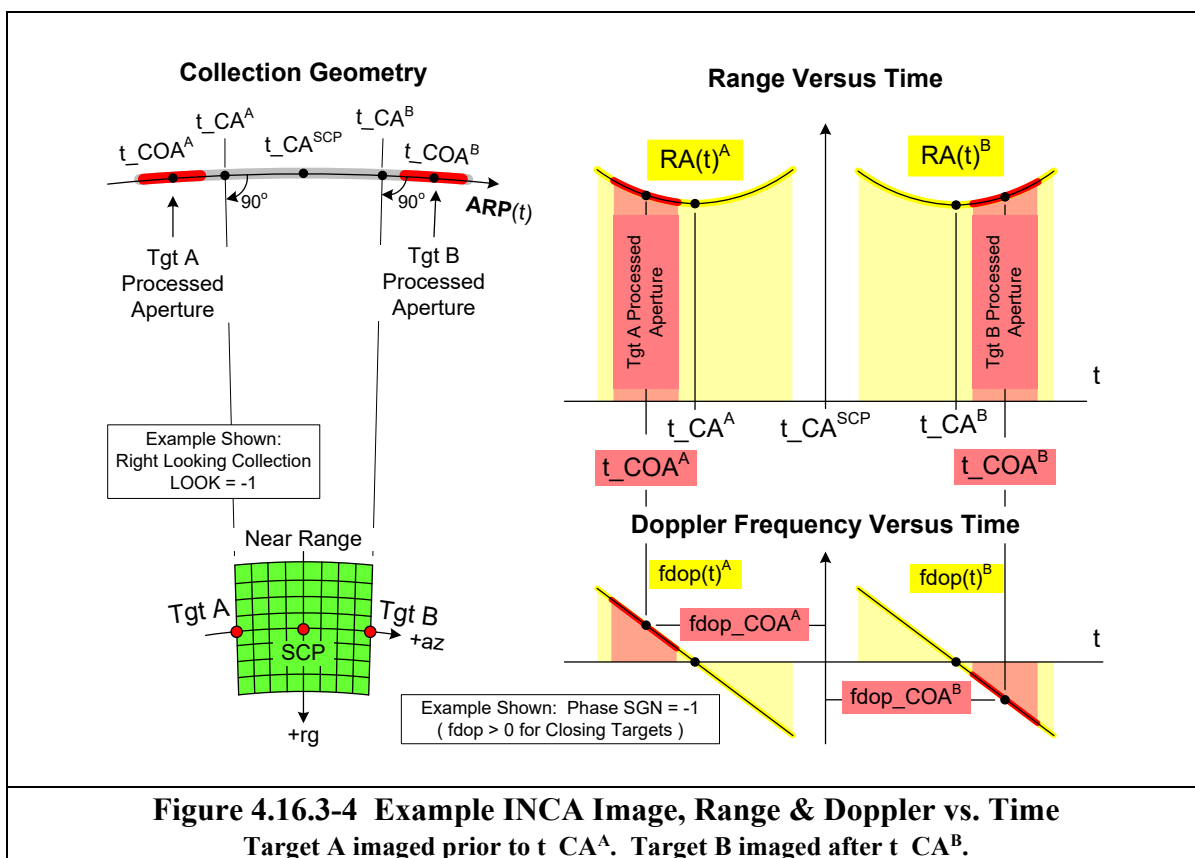
$$\Delta t_COA(rg,az) = t_COA(rg,az) - t_CA(az)$$

For image location (rg,az), the range and range rate at COA (R_COA and $Rdot_COA$) used for the projection from image location (rg,az) to a geo-located surface are computed as follows using the $RA(t)$ function and its derivative evaluated at $t = t_COA(rg,az)$.

$$R_COA(rg,az) = [R_CA(rg)^2 + DRSF(rg,az) \cdot VM_CA(az)^2 \cdot (\Delta t_COA(rg,az))^2]^{1/2}$$

$$Rdot_COA(rg,az) = \frac{1}{R_COA(rg,az)} \cdot DRSF(rg,az) \cdot VM_CA(az)^2 \cdot \Delta t_COA(rg,az)$$

An example INCA image is shown in Figure 4.16.3-4. The example shown is for a right looking collection and the phase $SGN = -1$. Two targets in the scene (Target A and Target B) are shown that have the same range at closest approach. The range function $RA(t)$ and the Doppler frequency function $fdop(t)$ are also shown for the two targets. The periods near closest approach do not overlap. Target A has its t_COA prior to t_CA ($\Delta t_COA^A < 0$). Target B has its t_COA after t_CA ($\Delta t_COA^B > 0$).



(7) Doppler Centroid

For stripmap and dynamic stripmap collections, the radar antenna is scanned across the scene that is imaged. For a given point in the scene, the received signal rises and then falls as the mainlobe of the radar antenna passes over the point. The Doppler frequency at the peak of the response is referred to as the Doppler Centroid ($fdop_DC$). The Doppler Centroid is provided as a function of image coordinates (rg, az). The Doppler Centroid polynomial (coefficients denoted $cDOPC(m,n)$, order M_DOPC and order N_DOPC) gives Doppler frequency at the peak signal.

$$INCA.DopCentroidPoly(m,n) = cDOPC(m,n), \quad m = 0 \text{ to } M_DOPC, \quad n = 0 \text{ to } N_DOPC$$

$$fdop_DC(rg,az) = \sum_{m=0}^{M_DOPC} \sum_{n=0}^{N_DOPC} cDOPC(m,n) \cdot (rg)^m \cdot (az)^n$$

The time the peak signal occurs, $t_DC(rg,az)$, is computed by as follows.

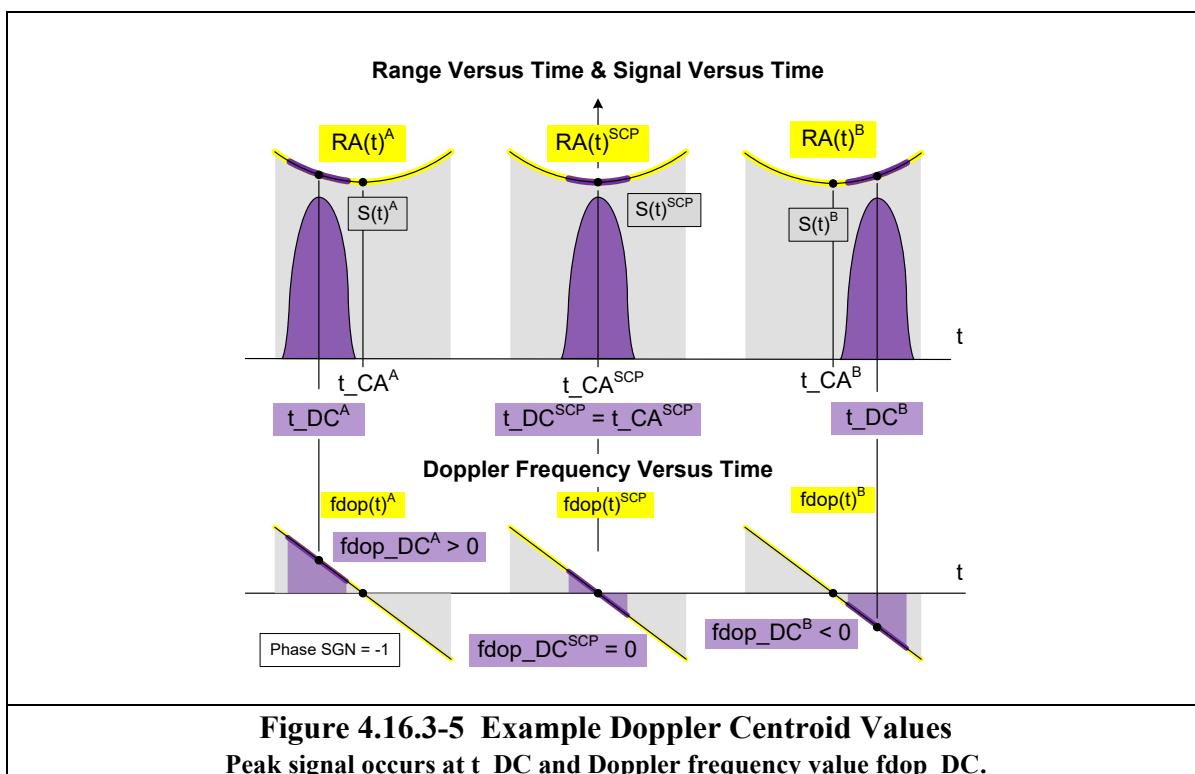
$$t_DC(rg,az) = t_CA(az) + \frac{1}{DR_CA(rg,az)} \cdot fdop_DC(rg,az)$$

For the example INCA image shown in Figure 4.16.3-4, example signal versus time functions are shown in Figure 4.16.3-5. Target A has peak signal occur before closest approach ($t_{DC}^A < t_{CA}^A$). Target B has peak signal occur after closest approach ($t_{DC}^B > t_{CA}^B$). Also in this example, the SCP target has its peak signal occur at closest approach ($t_{DC}^{SCP} = t_{CA}^{SCP}$). For Targets A and B, the COA times shown in Figure 4.16.3-4 (t_{COA}^A and t_{COA}^B) are equal to the peak signal times shown in Figure 4.16.3-5.

For a given imaging collection, the peak signal times are determined by the achieved platform attitude and radar antenna pointing. The COA times are chosen by the image formation processing. For many systems, the peak signal times are calculated in the early stages of the processing (referred to as Doppler Centroid estimation). Setting the COA times equal to the peak signal times maximizes image signal-to-noise ratio while also minimizing ambiguity levels. For most stripmap and dynamic stripmap images, the COA times are set equal to the peak signal times for all pixels. Optional parameter DopCentroidCOA can be used to indicate that the COA times are equal to the peak signal times for all pixels (the usual condition).

INCA.DopCentroidCOA = TRUE $\Leftrightarrow$ Indicates all COA times track peak signal times.

INCA.DopCentroidCOA = FALSE $\Leftrightarrow$ Indicates COA times do not track peak signal times.



(8) Spatial Frequency Domain Center Of Aperture

For all INCA images, the spatial frequency domain support as a function of image location is as follows. The range spatial frequency (Krg) support is centered at $Krg = Krg_{CTR}$ for image pixels (for all pixels, $\Delta Krg_{COA} = 0$). In general, the azimuth spatial frequency (Kaz) support varies as a function of image location. For image location (rg,az), the center of the azimuth spatial frequency support is as follows.

$$Kaz_{COA}(rg,az) = \underbrace{Kaz_{CTR}}_{\substack{\text{Equal to 0 for all} \\ \text{RGZERO images.}}} + \Delta Kaz_{COA}(rg,az)$$

The ΔKaz_{COA} polynomial (coefficients denoted $cKAZ_{COA}(m,n)$, orders M_KAZ and N_KAZ) is specified as follows.

$$\text{Grid.Col.DeltaKCOAPoly}(m,n) = c\Delta Kaz_{COA}(m,n), \quad m = 0 \text{ to } M_KAZ, \quad n = 0 \text{ to } N_KAZ$$

$$\Delta Kaz_{COA}(rg,az) = \sum_{m=0}^{M_KAZ} \sum_{n=0}^{N_KAZ} cKAZ_{COA}(m,n) \cdot (rg)^m \cdot (az)^n$$

An example spatial frequency domain plot is shown in Figure 4.16.3-6. The span of the spatial frequency domain is $1/Az_SS$. The Doppler frequency span corresponding to spatial frequency span $1/Az_SS$ is follows.

$$\Delta Kaz = \frac{1}{Az_SS} \Leftrightarrow \Delta fdop = \frac{1}{|cT_CA(1)| \cdot Az_SS}$$

Where $|cT_CA(1)| \cdot Az_SS = \Delta t_CA^{SCP} = \text{spacing in } t_CA \text{ at the SCP pixel.}$

